The AM-GM Boundary

A Spectral Realization of Geometric Composition Across Non-Transferable Blocks

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Abstract

The AM-GM boundary names a pattern recurring across decision theory, evolutionary biology, and non-equilibrium thermodynamics: arithmetic aggregation of resources inside a coherent unit, geometric aggregation of survival probabilities across non-transferable branches. This monograph establishes the boundary as a coupled pair of characterization theorems — Theorem II.1 forces the weighted geometric mean on the outer side from five compositional axioms, Theorem II.2 forces dependence on the arithmetic total on the inner side from complete internal transfer — with the *interface value* as the object connecting them. The synthesis is instantiated in the reversible killed birth-death process class with Gibbs lifts, where spectral compression makes the temporal interface sufficiency axiom (A5) operationally checkable for the induced geometric evaluator, and extends to slow-fast settings through a Feynman-Kac semigroup interface value on the coarse state space, with scalar outer evaluation recovered in the quasi-static and fast-coarse-mixing regimes. A three-part operational criterion — metastable coarse potential, uniformly open spectral gap, coarse motion slower than compression — makes the synthesis empirically falsifiable, and locates explicit boundary cases where the scalar theorem gracefully degrades into finite-mode vector structure. A final perturbative appendix shows that the clean block-local synthesis degrades quantitatively under weak coupling: scalar integrity is controlled by leakage-to-gap ratios, while composition integrity is controlled by Feynman-Kac tilted coupling sensitivities. In the broader AM-GM Boundary program, this is the scalar-realization document; the finite-band typology surrounding the scalar regime, and the empirical extraction of interface signatures from data, are treated separately.

Keywords: AM-GM boundary; geometric rationality; killed Markov processes; quasi-stationary distributions; metastability; spectral compression; Feynman-Kac semigroups; interface values; aggregation axioms.

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Part I. The puzzle

I.1. Four situations

Consider four situations. A biological cell is hit by a toxin that disables a fifth of its mitochondria. It does not die. The remaining mitochondria produce ATP, which diffuses to where it is needed. The cell is one thing. Two deer live in the same forest. One finds abundant food. The other starves. The first cannot feed the second. They are two things. An ant colony sends scouts. One scout finds food far from the nest. Worker ants regurgitate into the mouths of hungry others. The colony persists even when many individual ants die. Is the colony one thing, or many? A forest stand of trees is connected by mycorrhizal networks. Trees in shade receive sugars from trees in sun through the fungal web. When a pathogen kills one tree, the rest may or may not survive depending on what the pathogen does to the shared substrate. The stand is — what? The cell and the deer feel like clear cases; the colony and the forest feel like harder ones. But what exactly distinguishes them? What makes the cell a unified individual, the two deer separate individuals, and the colony and forest intermediate? The physical-boundary answer — the thing with its own membrane is the individual — is wrong. It fails immediately on the ant colony, which has no membrane. It fails on conjoined twins sharing a circulatory system. It fails on a forest stand. Something else is doing the work.

I.2. The conjecture: transferability over time

The claim of this paper is that the distinguishing feature is mathematical, and it is about transferability over time. A coherent unit is a region within which losses to one part can be compensated by gains to another part before the next viability-relevant event. The boundary of the unit is where this compensation breaks down. Inside the boundary, aggregation is arithmetic: parts sacrifice for each other, and only totals matter. Outside, aggregation is geometric: each unit faces its own absorbing states and cannot be zeroed out by its neighbors' successes.

We will call this the AM-GM boundary: the dynamical location where arithmetic aggregation gives way to geometric aggregation. The core thesis of this paper is that this boundary is well-defined, concretely realizable in a specific model class, and — under conditions we will make precise — the structural object that organizes what it means for a subsystem to count as a coherent unit. Three corollaries follow immediately if the conjecture is right. First, the boundary is not spatial. Two deer with no mechanism for resource transfer are two units even if they stand side by side. An ant colony is one unit across dispersed individuals because food can flow between them before hunger kills any of them. A mycorrhizal forest stand is one unit on the timescale of seasonal sugar redistribution and many units on the timescale of acute drought mortality. The boundary is scale-dependent; which scale is the right scale is determined by what viability-relevant question is being asked. Second, the boundary is dynamical, not structural. Two systems with identical static structure can be one unit or many depending on their dynamics — how fast resources redistribute, how often hazards arrive, how the fast and slow timescales relate. This is why membrane topology alone does not determine the boundary, and neither do Markov blankets in the information-theoretic sense. Third, the right mathematical object for aggregation changes at the boundary. Inside, where compensation is possible, arithmetic means capture what matters. Outside, where each branch faces its own fate, geometric means are forced. The AM-GM boundary is the specific dynamical location where one gives way to the other.

I.3. The central object: the interface value

Formalizing the conjecture requires a specific mathematical object. We call it the interface value. The interface value of a block is the positive scalar by which that block is represented once internal compensation has finished and external non-transferability begins. It is the object that mediates the transition between arithmetic aggregation (inside) and geometric aggregation (outside). If the interface value is well-defined — ratio-scale rather than merely affine, and produced by genuine internal compression rather than arbitrary labeling — then the block counts as one unit for the purposes of aggregation across blocks. If the interface value is not well-defined, or if it depends on internal microstate details that have not been absorbed into a single scalar, then the block has failed to cohere as a scalar unit for the present outer evaluator. It may still admit a richer interface object — for example a finite retained spectral band — but it is then outside the scalar regime developed in this monograph. This reframes "what is a coherent unit?" — and, by extension, "what is an agent?" — as a dynamical question about when the interface value exists. On the reading we will defend, a coherent unit is a region within which redistribution is possible on the relevant timescale, and its boundary is where redistribution becomes dynamically impossible — the place where the interface value, if it exists, lives. We use "agent" and "coherent unit" somewhat interchangeably in the informal setup, and we will refine this to a specific technical object through Parts II–V; until that refinement is in place, the agentic reading should be understood as motivation rather than as a derived claim. We should be honest about what kind of question this is. It is not a normative question about where to draw lines. It is a question about the causal structure of a system — about which losses can be compensated and which cannot, and whether that compensation produces a well-defined scalar summary on the relevant timescale. Given the viability question, partition, and timescale, the answer is not a free normative choice. It is a claim about the system's dynamics.

I.4. Relation to neighboring programs

Several research programs address adjacent questions. None of them answers this specific one. Friston's Free Energy Principle [Fri10; Kir+18] identifies systems by Markov blankets — conditional independence interfaces that separate internal from external states. This tells you where the blanket is, but not whether the internal states compose as a coherent unit for viability accounting. A cell and an ant colony and a rock all have Markov blankets in some sense. FEP is silent on the question of which blankets matter, or why. Krakauer and Flack's information-theoretic individuality [Kra+20] measures autonomy — how much of a system's future is determined by its own past. This captures something real about what makes an individual. But it does not tell you what the individual's aggregation rule is, or when the individual is coherent enough for its internal states to matter only through a scalar total. Autonomy is a property of information flow; coherence is a property of resource flow. Garrabrant's geometric rationality [Gar22] identifies geometric aggregation as the unique compositional rule across non-transferable branches. This is exactly the right aggregation rule for the "outside the boundary" regime. But Garrabrant does not have a theory of where the boundary comes from. Geometric rationality tells you what to do once the branches are specified; it does not tell you what counts as a branch. Peters's ergodicity economics [Pet19; PG16] makes the time-average versus ensemble-average distinction precise and derives log-optimal behavior from non-ergodicity. This is the mechanism that gives geometric aggregation its operational force for persistence: long-run growth is at the rate of the time-averaged log, not at the expected value. But Peters does not address subsystem structure. His domain is the single trajectory. Each of these programs provides something we need and leaves something we do not yet have. FEP and Krakauer identify candidate boundaries; they do not supply the aggregation rule. Garrabrant and Peters supply the aggregation rule; they do not derive the boundaries. The AM-GM program is the missing connection: a theory of when boundaries coincide with aggregation switches, with the interface value as the object that ties them together. This is the gap we are trying to fill. It is narrower than "what is an agent" and broader than "what is the right compositional aggregation rule." It is about when and why these two questions have the same answer.

Appendix H gives a quantitative version of the FEP/blanket distinction noted above. A blanket is structurally significant for viability aggregation not merely when it exists as a conditional-independence interface, but when its porosity is small in the coordinates relevant to persistence: projected through the block's spectral structure, it must be small relative to the compression gap, and under the Feynman-Kac survival-biased bridge, it must be small enough not to destroy product composition. In this sense the AM-GM criterion asks not simply where the blanket is, but which blankets support scalar interface values and geometric outer evaluation.

The previous comparisons situate the AM-GM program against external neighbors. There is also a within-program distinction worth naming explicitly. The present monograph develops the rank-one scalar regime: the case in which a block's continuation is compressed to a positive scalar interface value before outer evaluation. A companion typology document [Kim26b] surrounds this scalar regime with a finite-band classification of interface signatures, identifying the regions in which scalar compression remains adequate, basin-witnessed scalarization is supplied, or the interface object is vector- or cone-valued with only partial comparison structure. A further empirical-extraction document [Kim26a] treats how such signatures might be measured in real systems. Both are layers of the same program: the typology classifies the type of interface object on which any scalar or vector evaluation acts, and the empirical document treats how those types are detected in real systems. The present monograph supplies the scalar realization on which both build.

I.5. What this paper does

The rest of the paper gives the conjecture mathematical content. It characterizes the two edges of the AM-GM boundary (Part II), introduces the interface value formally and identifies the obstructions to its existence (Part III), constructs a concrete model class in which those obstructions can be overcome (Part IV), and exhibits a worked synthesis in which the outer geometric theorem is fully realized and the inner arithmetic-totality theorem is representationally compatible, both statically and under slow-fast coarse-state dynamics (Part V; the asymmetry between full outer realization and compatibility-only inner realization is developed in §V.5). Part VI returns to the motivating applications and articulates what remains unresolved.

Program placement. This paper is Document 1 of the AM-GM Boundary program. Its job is to realize the rank-one scalar case in detail, not to classify every possible interface type. The companion typology document [Kim26b] takes the boundary cases named here — weak scalarity, finite retained bands, candidate scalar reductions, and vector-valued continuation — as inputs to a broader classification of interface signatures, with separate regions for adequate scalarity, basin-witnessed scalarization, partially comparable cone-valued objects, and the formal cone-non-diagnostic case. The empirical extraction of those signatures from biological or other real systems is the methodological task of a third document [Kim26a]. The monograph's appendices remain proof-homes for its own theorems; the document-stack map is structural only.

How to read this document. The body parts (I–VI) are the argument. They state definitions, theorems, and their conceptual content, and sketch the key proofs in the text. The appendices are the proof-homes: they carry the full technical machinery needed to establish the body's theorems, together with the numerical verifications. The correspondence is: Appendix A proves Theorem II.1 (the outer characterization); Appendix B proves Theorem IV.1 (the diffusive baseline); Appendix C proves Theorem IV.2 (the three-factor Gibbs-lifted asymptotic); Appendix D is the proof-home for Part V's static synthesis (Theorem V.1); Appendix E proves Proposition III.2 (the abstract slow-fast bridge); Appendix F is the proof-home for Part V's dynamic synthesis (Theorems V.2 and V.3); Appendix G closes the heterogeneous n-block extension, establishing spatial regrouping invariance (Axiom A4) on the compressed interface-value class; Appendix H proves the finite-state weak-coupling extension of the scalar synthesis: scalar compression survives when leakage into nonprincipal modes is small relative to the surviving compression gap, and raw Feynman-Kac product defects are controlled by tilted coupling sensitivities under the survival-biased bridge, with two minimal worked examples (a leaky two-mode block and a two-state product Feynman-Kac model showing why equal raw coupling norms can have different AM-GM significance); Appendix I locates the boundary of the scalar compressed regime, proving a finite-band approximation theorem for the near-degenerate case $g\Delta = O(1)$ and diagnosing why Theorem II.1 does not automatically extend to vector interface structures; Appendix J is the representational coda, identifying the weighted geometric evaluator as the positive-domain image of linear pooling in an additive log-coordinate and clarifying the bridge to Peters-style and FEP-style formalisms; Appendix K is a canonical-assumptions glossary providing a reference lexicon for the monograph's load-bearing terms. Readers interested in the conceptual arc should follow Parts I–VI; readers checking the mathematics should follow body $\leftrightarrow$ appendix pairs as needed. The conceptual argument is self-contained; the spectral realization is self-contained as a formal model plus numerical support, with the metastable-capacity asymptotic of Theorem C.1 as the one explicit technical input not fully derived here.

Formal-status convention. Several results in this monograph carry *formal* status in the following sense: they are rigorous modulo one or more explicit offstage dependencies listed below, each of which relies on a technical step the present monograph does not independently derive. Four such dependencies exist, each flagged where it arises:

(F1) *Capacity-over-mass asymptotic with QSD/local-equilibrium qualification* (Theorem C.1). The Gibbs-lifted three-factor formula uses a Bovier/BEGK-style capacity-over-mass estimate. In the low-temperature regime where the quasi-stationary distribution (QSD) on the safe region approaches the local Gibbs equilibrium at the strip entrance, the stated three-factor prefactor is asymptotically sharp. At fixed β with $R \to \infty$, the leading barrier factor $e^{-\beta R \Delta h}$ and the $R^{-1/2}$ WKB scaling remain the relevant structure, but the unit prefactor may be modified by a β-dependent QSD boundary-layer factor. Affects Theorem IV.2, Theorem V.1 and its proof-home Theorem D.4, Theorem V.3 and its proof-home Theorem F.7, and Theorem G.10; downstream synthesis results use the leading scaling/window structure and are robust to this multiplicative prefactor correction.

(F2) *Adiabatic-elimination schematic estimate.* The dynamic persistence bound in Theorem V.2 / Theorem F.5 of the form $\epsilon_{\text{dyn}} \lesssim e^{-g_* \Delta} + q_*/g_*$ is a schematic remnant of adiabatic elimination rather than a fully derived perturbation bound. Affects Theorem V.2, Theorem V.3 and its proof-home Theorem F.7, Theorem G.10, and downstream log-scale statements that invoke them.

(F3) *Boundary-layer collapse for the diffusive baseline.* The $a(R) \sim \gamma\pi^2/R^2$ asymptotic of Theorem IV.1 / Theorem B.4 rests on a continuum argument plus exact numerics; a fully rigorous proof of the Robin-to-Dirichlet boundary-layer collapse is deferred. Affects Theorem IV.1 and Theorem B.4.

(F4) *Harmonic-well relaxation/gap asymptotic.* The compression-window claim $a_2(R) \asymp \gamma\beta h''(1/2)/R$ for the second-mode decay rate of the Gibbs-lifted block is heuristically motivated by Ornstein–Uhlenbeck restoring curvature in the metastable well and supported numerically (§D.3.2), but a full asymptotic proof is deferred. The compression *window* $\Delta \in [\tau_{\text{compress}}, \tau_{\text{kill}}]$ depends on this scaling: $\tau_{\text{compress}} \asymp 1/(a_2 - a_1)$ and $\tau_{\text{kill}} \asymp 1/a_1$. Affects Theorem D.3's window existence claim, Theorem V.1 / Theorem D.4 (downstream of compressed-class membership), and Theorem V.3 / Theorem F.7 (whose dynamic compression window inherits the same heuristic).

Results carrying formal status: Theorem IV.1 and Theorem B.4 (via F3); Theorem IV.2 (via F1); Theorem V.1 and Theorem D.4 (via F1, F4); Theorem V.2 and Theorem F.5 (via F2); Theorem V.3 and Theorem F.7 (via F1, F2, F4); and Theorem G.10 (via F1, F2, F4). Their theorem titles carry the word "formal" to signal this. Results elsewhere (Theorem II.1, Theorem II.2, Theorem A.8, the structural results of Appendices E, I, and J) are rigorous without qualification. The finite-state perturbation results of Appendix H are likewise rigorous under their stated finite-dimensional hypotheses; they do not depend on the Gibbs capacity-over-mass asymptotic, and they do not prove stability of that asymptotic under perturbation.

We want to state clearly what is claimed and what is not. What is claimed: the AM-GM boundary is a mathematically coherent object with concrete realization in a specific killed-process model class. In that class, we can point at interface values explicitly, verify the conditions under which arithmetic and geometric aggregation compose, and identify the thermodynamic structure that makes the realization work. What is not claimed: that biological or agentic systems in general satisfy the conditions we identify. The bridge from the mathematical realization to real systems — which real boundaries induce the required coarse potential, spectral compression, and timescale ordering — is the program's open frontier, sharpened by the present work but not dissolved by it.

Part II. What the edges must look like

This part characterizes the two edges of the AM-GM boundary conditionally: assuming the interface value exists and is well-behaved, what aggregation rule is forced on each side? Part III will identify the obstructions to constructing one; Parts IV and V will overcome those obstructions in a concrete class.

Reading note (conditional characterization). Both edge results below are *pre-theorem* in a specific methodological sense: they say what the answer must look like *if* a suitable interface value exists and the compositional axioms hold. They do not establish existence of such a value in any concrete system. An existence theorem for a scalar interface value in a specific dynamical class is a separate, substantive question, and it is the subject of Parts IV and V. The present part's role is to characterize what we are looking for, not to find it.

II.1. Setting

We consider systems in which viability is absorbing. A unit either survives a viability-relevant interface or it does not; survival is a probability, and repeated survival over time is multiplicative along any branch. We will use the word block for a candidate coherent unit: a region of the system whose internal dynamics we are trying to aggregate arithmetically, and whose external interactions with other blocks are to be aggregated geometrically. The AM-GM boundary is the interface between the arithmetic regime inside each block and the geometric regime across blocks. Three premises are carried throughout. (P1) Absorbing viability. Each branch or block is either viable or extinct at any given time. There is no intermediate state; viability is a binary event structure. (P2) Branch non-transferability. Viability resources cannot be transferred across the AM-GM boundary. Inside a block, resources may move freely; across blocks, they may not. *(P3) Positive interface value.* Each block admits a positive scalar $c \in (0, \infty)$ — the interface value — that summarizes its viability at the moment the boundary is crossed. (P3) is the structural assumption that Part III will interrogate. For now, treat it as granted.

Definition II.0 (Interface value). An *interface value* for a block is a positive scalar $c \in (0, \infty)$ representing the block's viability at the moment the AM-GM boundary is crossed, satisfying three structural requirements:

- *(i) scalar compression* — the block's state is represented, for the purpose of outer aggregation, by the single positive number c, and its future evolution under outer aggregation depends on this number alone and not on further microstate detail;
- *(ii) ratio-scale* — meaningful comparisons across blocks are ratios rather than differences, with the zero point $c \to 0^+$ corresponding to loss of viability and shared across blocks (so the outer evaluator is homogeneous in $(c_1, \ldots, c_n)$);
- *(iii) dynamical production* — c arises from the block's own internal dynamics (in a sense made precise by Parts III–V) rather than from external labeling.

Premise (P3) asserts the existence of such a c as a standing assumption for the edge theorems below. Part III interrogates when an interface value exists and catalogs the obstructions to its construction. Parts IV and V construct it explicitly in the reversible killed-process class. Appendices F and I generalize Definition II.0 to parameterized continuation families and finite-mode profiles respectively, extending it to settings where the type of the object shifts under spectral or dynamic structure.

Type-shift preview. The object called "interface value" changes type as the dynamical setting grows richer: bare scalar (Definition II.0, this section); cycle-parameterized scalar $c_i(\Delta)$ once spectral compression is in play (§V.2); scalar-valued function $\mathcal{A}_i(\Delta, z)$ of a coarse state under slow-fast dynamics (§V.3); finite-mode retained band near the scalar-compression boundary (Appendix I). Each type extends the previous one by adding a parameter or lifting the codomain. Definition II.0's three structural requirements can be restored after a scalar readout choice in each row; through the third row the readout is canonical (evaluation at the current coarse state, forced by the outer question), but in the finite-mode case the readout is additional structure — no single scalar readout is privileged by the geometry alone. A full taxonomy table opens Part V (§V.1); readers who want the full roadmap before entering the concrete constructions of Parts IV–V can look ahead there.

II.2. The outer edge: geometric aggregation is forced

On the outer side of the AM-GM boundary, we aggregate across blocks (or branches) whose viabilities are non-transferable. Each block has, by the standing assumption (P3), produced its own positive scalar interface value c_i; the question is what form the *aggregator over those interface values* must take. We should pause on what kind of question this is. It is not a question about the raw joint survival probability of several blocks, which — for absorbing-failure branches with independent hazards — is simply the product $\prod_i p_i$ of their survival probabilities. That raw product is unnormalized and is not the object the outer theorem produces. What the outer theorem produces is a *normalized evaluator* on interface values: a map from $(c_1, \ldots, c_n)$ to a single positive scalar that satisfies certain compositional axioms, including idempotence. The distinction matters. The unnormalized product of survival probabilities tells you the literal probability that all blocks survive; the normalized evaluator tells you what single positive scalar represents the aggregate, under axioms that let us treat the aggregate as a new unit for further composition.

Given this framing, what aggregator satisfies the compositional axioms we need?

The answer, under axioms we will state presently, is the weighted geometric mean. The five axioms below are the minimal compositional requirements that a normalized evaluator on positive interface values must satisfy if we want it to behave as an aggregator *of units for further composition*. Four of the five are familiar (continuity, monotonicity, idempotence, regrouping invariance); the fifth — temporal interface sufficiency — is the axiom that does the real work and that connects the outer theorem to the interface-value problem of Part III.

Theorem II.1 (Outer characterization). *Suppose $\{A_w\}$ is a coherent all-arity family of outer evaluators: for each integer $n \geq 1$ and each strictly positive weight vector $w = (w_1, \ldots, w_n)$ with $w_i > 0$ and $\sum_i w_i = 1$, A_w acts on positive interface values $(c_1, \ldots, c_n) \in (0, \infty)^n$. If $\{A_w\}$ satisfies the following axioms across all arities n:*

- *(A1)* continuity in $(c_1, \ldots, c_n)$;
- *(A2)* strict monotonicity in each coordinate;
- *(A3)* idempotence: $A_w(c, c, \ldots, c) = c$;
- *(A4)* spatial regrouping invariance: for any partition of the coordinates into clusters $C_1, \ldots, C_k$ with cluster outer weights $W_\alpha := \sum_{i \in C_\alpha} w_i$, direct evaluation agrees with two-step evaluation,

  $$A_w(c_1, \ldots, c_n) \;=\; A_W\big(B_1, \ldots, B_k\big), \qquad B_\alpha := A_{w/W_\alpha}\big((c_i)_{i \in C_\alpha}\big),$$

  where the cluster value B_α is computed by applying the same evaluator to the coordinates of C_α with renormalized inner weights w_i/W_α (which now sum to 1 over C_α), and $W = (W_1, \ldots, W_k)$ carries the outer weights across clusters;
- *(A5)* temporal interface sufficiency: if $c, c' \in (0, \infty)^n$ are two profiles corresponding to successive temporal segments, their *temporally concatenated profile* is defined coordinatewise as $(c \odot c')_i := c_i c'_i$ (each coordinate's two-segment survival is the product of its one-segment survivals). The axiom is that there exists a fixed function $F_w : (0, \infty)^2 \to (0, \infty)$ with

  $$A_w(c \odot c') \;=\; F_w\big(A_w(c),\, A_w(c')\big) \qquad \text{for all } c, c' \in (0, \infty)^n,$$

  so that the evaluator's value on a concatenated profile depends on the two segment profiles only through their separately-evaluated scalar values, *then for every arity* $n \geq 1$ *and every strictly positive weight vector* $w \in (0,1)^n$ *with* $\sum_i w_i = 1$*,*

$$A_w(c_1, \dots, c_n) = \prod_i c_i^{w_i}.$$

Note on coherence and arity. "Coherent all-arity family" means that the axioms hold not for a single fixed n but for every arity simultaneously — (A4)'s regrouping invariance, in particular, relates A_w at arity n to A_W at arity k where k is the number of clusters. The exponent-pinning argument in Theorem A.8's proof requires (A4) at $n = 3$ specifically: at $n = 2$ alone, the axioms force the form $A(c_1, c_2) = c_1^{\alpha} c_2^{1-\alpha}$ for some continuous strictly increasing $\alpha : (0,1) \to (0,1)$ with $\alpha(p) + \alpha(1-p) = 1$, but do not pin $\alpha = \mathrm{id}$. The strict-positivity requirement $w_i > 0$ is similarly load-bearing: zero weights are incompatible with strict monotonicity (A2). Boundary extensions to zero weights or zero interface values are addressed separately in Appendix A's Theorem A.9.

II.2.1. Evaluator vs conjunction survival

Before proceeding to the axiom discussion, we flag a distinction that is easy to miss at this point and hard to unlearn if the wrong mental model forms. The weighted geometric mean $\prod_i c_i^{w_i}$ looks superficially similar to a raw joint-survival probability $\prod_i c_i$ under independent branches. These are different objects answering different questions. The theorem characterizes an evaluator, not a conjunction.

What the distinction is. Given two branches with positive interface values (c_1, c_2) and weights (w_1, w_2) summing to 1, two constructions are available. The *unnormalized product* $c_1 \cdot c_2$ is the literal probability that both branches survive if they are independent; it measures raw joint survival. The *normalized evaluator* $c_1^{w_1} c_2^{w_2}$ is a weighted scalar summary that behaves as a unit for further composition. They answer different questions:

Worked contrast. Consider two identical branches, $c_1 = c_2 = c$. The unnormalized product gives $c \cdot c = c^2$, which is strictly less than c whenever $c < 1$. This violates idempotence (A3): aggregating two identical units does not return the unit. It does give the correct literal survival probability under independence, but that is a different object. The normalized evaluator gives $c^{w_1} c^{w_2} = c^{w_1 + w_2} = c$, which satisfies idempotence exactly. A reader who thinks the outer theorem is characterizing raw joint survival will read (A3) as wrong — "of course c and c together is c^2, not c" — and correctly so, but for the wrong object.

The axioms (A1)–(A5) are conditions on a *normalized* evaluator, an aggregator whose output is itself admissible as an interface value for further composition. They are not conditions on a raw conjunction. If one wants the raw conjunction probability, that is $\prod_i c_i$ under independence and $\mathbb{E}[\prod_i c_i]$ under post-interface correlation; neither is what Theorem II.1 characterizes.

Why this matters beyond idempotence. The distinction sharpens in the dynamic setting: at $n = 2$ the two objects differ only by a normalization, but under slow-fast coarse-state coevolution a Jensen gap opens between the evaluator summary (which aggregates a weighted *average* of rates) and the unnormalized joint-survival summary (which aggregates a *sum* of rates). Appendix G.7 develops the n-block and dynamic forms of this distinction explicitly and tracks the Jensen gap; Appendix G's three-layer calibration separates the evaluator layer from the raw-product layer with explicit error bookkeeping. The present distinction at $n = 2$ positive scalars is the seed; the machinery that grows from it is in Appendix G.

With the evaluator-vs-conjunction distinction in hand, we can return to what the axioms (A1)–(A5) do.

Two aspects of this result deserve emphasis. First, the multiplicativity that appears in $\prod_i c_i^{w_i}$ is not inserted by hand, but it also is not a restatement of raw conjunction survival. The axioms (A_1)–(A4) constrain the evaluator's form substantially but do not by themselves force a specific functional family. What forces the geometric mean is temporal interface sufficiency (A5): the requirement that the evaluator's output, once produced, be itself admissible as an interface value for further temporal composition. Combined with the other axioms, (A5) forces the evaluator to be homogeneous and multiplicative in a specific way, and the weighted geometric mean is the unique resulting form.

Second, temporal interface sufficiency is a strong axiom, and it is where the interface-value assumption

(P3) does its conceptual work on the outer side. If (A5) holds, the outer theorem follows. If it fails — if the block's future behavior depends on microstate details that the scalar c_i has not absorbed — then no scalar aggregator of the form we are characterizing can exist, because the block has not yet compressed enough for the axiom to hold. Absorbing viability alone (P1) does not force geometry. Non-transferability alone (P2) does not force geometry. What forces geometry is absorbing viability plus non-transferability plus the temporal scalar-compression axiom on the outer evaluator. The relationship to Garrabrant's geometric rationality is close: Garrabrant derives geometric aggregation from a related but distinct axiom set, and the present theorem is most naturally read as a version of that characterization adapted to the survival-probability setting with temporal interface sufficiency replacing some of Garrabrant's symmetry axioms [Gar22].

Relation to quasi-arithmetic means. The Kolmogorov-Nagumo [Kol30; Nag30] / Hardy-Littlewood-Pólya [HLP52] theory characterizes aggregators of the form $\phi^{-1}(\sum_i w_i\phi(x_i))$ for continuous strictly monotone ϕ; the weighted geometric mean corresponds to $\phi = \log$. Theorem II.1 is not literally a quasi-arithmetic characterization with $\phi = \log$: axioms (A1)–(A4) do not by themselves force the ϕ-choice. What selects $\phi = \log$ is the temporal interface sufficiency axiom (A5), which specializes here to staged survival composition. Appendix A's Theorem A.8 develops this connection; the one-line summary is that (A5) is what the classical quasi-arithmetic framework lacks, and what makes survival-probability aggregation a different question from generic mean-value theory.

A note on the exogenous weights. The weights w appear in the theorem statement as exogenous data. Theorem II.1 characterizes, for each fixed strictly positive weight vector w at each arity n, the unique evaluator on interface values satisfying the all-arity (A1)–(A5); it does not tell us where w comes from. This is worth lingering on because several distinct readings of w are operationally plausible — a scheduling fraction (time-share of block i), a prevalence (branch frequency across a population), a cohort weighting (statistical reweighting across subgroups), or a priority (a normative weight assigned by an agent or designer) — and these readings are not equivalent. A schedule or prevalence is in principle a measurable property of a system; a priority or cohort-reweighting is a choice that does not live inside the system. The theorem is agnostic between them.

The natural next question is whether some of these readings can be *derived* from the same thermodynamic or spectral structure that forces the geometric form on the evaluator itself. The answer is: not from (A1)–(A5) alone. The axioms operate on the evaluator, with $(c_1, \ldots, c_n, w)$ as its inputs. A derivation of w from, say, stationary coarse-state distributions, Gibbs-weighted exposure times, or branch-wise mixing rates would be a distinct result — a theorem supplementary to II.1, additional to it rather than an extension, and requiring hypotheses about the system-level provenance of weights that the present axiomatic characterization deliberately abstracts away. No such derivation is attempted in this monograph.

The synthesis of Parts IV–V is therefore conditional on an external source of w, whatever its operational interpretation. The open question — which real applications admit an internal derivation of weights, versus which require weights to be specified exogenously by the designer or modeler — is taken up in §VI.4 as part of the program's open-problem statement. The reader should carry the acknowledgment forward: the weighted geometric evaluator characterized here is weighted, and the weights are inputs the characterization itself does not produce.

II.3. The inner edge: total-resource sufficiency under complete transfer

On the inner side, we aggregate within a block. Now the assumption of non-transferability is reversed: inside the block, a conserved scalar viability resource can move freely between components. The question is: what is the aggregator that takes the within-block allocation and returns the block's viability?

The answer, under assumptions that we will make precise, is that the block value depends only on the total R, not on the specific allocation. *Linearity* in R is a separate condition required only for the lottery extension. The two conditions below are the strong form of complete internal transfer; as with Theorem II.1, the value of the theorem is as much in the conditions under which it fails as in the conditions under which it holds.

Before stating the theorem, define one-step block viability. Given a pre-hazard allocation $u = (u_1, u_2, \ldots)$ with total $R = \sum_i u_i$, let $s(u')$ denote the one-step survival probability at post-transfer allocation u', and let $\mathcal{T}(u)$ denote the set of admissible reallocations of u before the next viability-relevant hazard. The *induced block value* is

$$V(u) \;:=\; \sup_{u' \in \mathcal{T}(u)} s(u').$$

The theorem below asks when $V(u)$ depends on u only through R.

Theorem II.2 (Inner sufficiency: total-resource sufficiency under complete transfer). *Suppose a block consists of components holding viability resources $u_1, u_2, \ldots$ summing to a fixed total $R = \sum_i u_i$, with induced block value $V(u)$ as defined above. Suppose further that:*

- *(B1) complete internal transfer: the admissible-reallocation set is $\mathcal{T}(u) = \{u' : u'_i \geq 0,\ \sum_i u'_i = R\}$, containing every allocation consistent with the total-conservation constraint;*
- *(B2) survival depends on allocation alone: $s(u')$ is a function of the post-transfer allocation u' only, with no further dependence on microstate history or internal dynamics prior to the reallocation.*

Then $V(u)$ depends on u only through R: two allocations with the same total give the same induced one-step viability. Deterministic allocations collapse to this total — the block behaves, for purposes of one-step survival, as if its resources were pooled into a single fungible scalar.

Moreover, if lotteries over allocations are evaluated by expected one-step viability, and if the pooled value map $R \mapsto V(R)$ is affine on the attainable-total domain, then rankings over lotteries coincide with rankings by expected total resource:

$$\mathbb{E}[V(R)] = V(\mathbb{E}[R]).$$

Proof. For the deterministic half: by (B1), the feasible post-transfer set $\mathcal{T}(u)$ depends on u only through R. By (B2), s is a function of u' alone. Setting

$$V(R) := \sup\left\{s(u') : u'_i \geq 0,\ \textstyle\sum_i u'_i = R\right\},$$

we obtain $V(u) = V(R)$: two allocations with the same total give the same induced one-step viability. For the lottery extension, let X be a lottery over allocations with total $R(X) = \sum_i X_i$ supported on the attainable-total domain. Under expected-viability evaluation, X is ranked by $\mathbb{E}[V(R(X))]$. If V is affine on that domain, $V(R) = aR + b$ with $a > 0$, then

$$\mathbb{E}[V(R(X))] = a\,\mathbb{E}[R(X)] + b = V(\mathbb{E}[R(X)]),$$

an order-preserving affine transform of $\mathbb{E}[R(X)]$. Rankings therefore coincide with rankings by expected total resource. □

The first half of the theorem (deterministic total-sufficiency) establishes *dependence on the total* only: under complete internal transfer, the specific allocation is irrelevant, and only R matters. Crucially, this does *not* force the block value to be *linear* in R — a nonlinear function $\Phi(R)$ still satisfies the deterministic claim. The second half (expected-total coherence) is the extension to lotteries, and *does* require linearity in R (the affinity condition on the pooled value map). The unguarded phrase “arithmetic aggregation” can elide this distinction; we use “total-resource sufficiency” for the deterministic half and reserve “expected-total coherence” for the additional affine-pooled-value lottery extension.

As with the outer theorem, the assumptions that do real work deserve attention, because the theorem is narrow. Complete internal transfer (B1) is strong: it requires the block’s internal dynamics to be fast enough and unconstrained enough that any allocation consistent with the total is reachable before the next hazard. Real blocks do not satisfy this in full generality. Real blocks have internal transport costs, dynamical constraints, and limited time between hazards.

The failure modes matter for understanding what the theorem does and does not give us.

Failure mode 1: closure alone is insufficient. A block whose internal dynamics preserve some scalar quantity (an energy, a population size, a conserved charge) does not automatically satisfy total-sufficiency. Closure is about what cannot leave; total-sufficiency is about what can move freely within. Without complete transfer, one-step viability may still depend on how the resource is distributed, not just on its total.

Failure mode 2: incomplete internal transfer destroys total-sufficiency. If the block's internal dynamics are slow relative to hazard arrival, then between hazards there is not enough time for the allocation to equilibrate. The block retains memory of its initial allocation, and one-step survival depends on that allocation, not just on R. Part III will show that this failure mode is the generic case; (B1) describes a limit rather than a property of real systems.

Failure mode 3: nonlinear pooled value leaves Jensen effects. Even when deterministic total-sufficiency holds, the extension to lotteries requires the pooled value map to be affine. If V is strictly convex or strictly concave in R, then Jensen's inequality separates $\mathbb{E}[V(R)]$ from $V(\mathbb{E}[R])$, and the ranking of lotteries does not reduce to expected total resource.

The inner theorem is exact where it holds and narrow in what it requires. It is not a general statement about arithmetic aggregation; it is a statement about the specific circumstances under which the arithmetic total is the sufficient statistic.

II.4. What we have and what we do not yet have

At this point we have characterized both edges conditionally. On the outer side, Theorem II.1 tells us that if the interface value exists and satisfies the axioms, the aggregator is the weighted geometric mean. On the inner side, Theorem II.2 tells us that if complete internal transfer and affine pooled value hold, the aggregator is the arithmetic total. What we do not yet have is any guarantee that these two characterizations compose. The outer theorem presumes that each block already admits a positive scalar interface value c. The inner theorem presumes that the block has equilibrated internally before the interface. Neither theorem tells us whether real blocks produce an interface value, what structure that value must have, or whether the composition of "arithmetic inside, geometric outside" gives a well-defined nested evaluator. In particular, it is tempting to write a naive nested formula of the form

$$V_{\text{total}} = G(E[U_1], E[U_2], \dots)$$

where G is the weighted geometric mean and E[Ui] is the expected arithmetic total of block i's resource. Part III will show that this formula, in general, does not hold. The reasons are structural rather than computational: the interface value is a more subtle object than either the inner or outer theorem alone suggests, and the conditions under which it exists as a ratio-scale positive scalar are not automatically implied by the conditions in Theorems II.1 and II.2. We turn now to those obstructions.

II.5. Additive primitives and the log-rate coordinate

Before proceeding to the interface value problem, we note one structural observation that will become central later. Both edge theorems have a natural coordinate in which aggregation is arithmetic.

On the outer side, taking logarithms of the weighted geometric mean gives

$$\log A_w(c_1, \dots, c_n) = \sum_i w_i \log c_i.$$

Geometric aggregation in the positive-survival coordinate is arithmetic aggregation in the log-survival coordinate.

On the inner side, Theorem II.2 already states aggregation as arithmetic in the resource coordinate:

$$V(u_1, u_2, \dots) = V(R), \qquad R = \sum_i u_i.$$

Arithmetic aggregation in the resource coordinate. If a block's internal dynamics produce a scalar interface value that connects these two coordinates — if, for instance, $c = e^{-a\Delta}$ for an effective decay rate a that decreases with a resource-like quantity (more resource, slower decay) — then the outer geometric mean becomes linear in a coordinate that is itself additive in something resource-like. Outer geometry, inner arithmetic, and a unified log coordinate connecting them.

This observation is structural, not yet a theorem. The question of whether the log coordinate admits this role requires knowing something about what the interface value actually is — an observation Part V will make concrete once the interface value has been constructed. We record the observation here because it foreshadows the synthesis: the AM-GM boundary, when the synthesis works, is not a boundary between two incommensurate aggregation rules. It is a boundary in a single additive coordinate, where the coordinate changes meaning at the interface.

Part III. The interface value problem

Part II characterized the edges assuming that the interface value exists and behaves well. This part takes the assumption seriously: when does the interface value exist, what structure must it have, and what obstructions arise in constructing one?

III.1. What the interface value has to do

Recall the structural role of the interface value. It is the positive scalar by which a block is represented once internal dynamics have finished producing their summary and external aggregation takes over. Three things have to be true of it. First, it must be a scalar. The block's state at the interface must compress to a single positive number — or more precisely, must be represented by a single positive number for the purposes of outer aggregation. Whatever internal microstate the block occupies, its future evolution under outer aggregation must depend only on that number. *Second*, it must be ratio-scale. The outer evaluator in Theorem II.1 is the weighted geometric mean, which is homogeneous: $A_w(\lambda c_1, \ldots, \lambda c_n) = \lambda \cdot A_w(c_1, \ldots, c_n)$. For this to be meaningful, the interface values must be comparable on a ratio scale; the zero point matters, and affine representations are insufficient. Third, it must be produced by genuine internal compression, not by arbitrary labeling. If a block's "interface value" is just a label attached to its internal state by external fiat, then the compositional structure is fictitious. The interface value has to emerge from the block's own dynamics. These three requirements — scalar, ratio-scale, dynamically produced — are the conditions the interface value must satisfy for the outer and inner characterizations to compose meaningfully.

III.2. Three obstructions

There are at least three obstructions to constructing such a value from generic block dynamics.

III.2.1. Affine versus ratio-scale representation. Even when a block's state compresses to a scalar summary, that summary may be affine rather than ratio-scale. Consider a block whose one-step survival probability, as a function of internal allocation, takes the form

$$V(u_1, u_2, \ldots) = \alpha + \beta \cdot f(u_1, u_2, \ldots).$$

for some scalar function f and constants α, β. This block has a scalar summary in the sense that V depends only on f, but the representation is affine: only differences in f matter, not ratios. Affine representations are incompatible with geometric aggregation. The weighted geometric mean is a ratio-scale operation; applying it to affine summaries gives meaningless results because the zero point is not shared. A block with an affine scalar representation has not produced an interface value in the sense the outer theorem requires.

The resolution, in specific cases, is zero-resource normalization. If we can identify a natural zero — a state at which the block has no viability left — and if the block's internal dynamics make the representation homogeneous with respect to that zero, then the affine intercept vanishes and the representation becomes ratio-scale. But this is a structural condition on the block's dynamics, not an automatic consequence of scalar compression.

Lemma III.A (Zero-resource normalization)**.** *Let $\Phi(z, R)$ be a pooled block value that is affine in the within-block total R at fixed coarse state z, i.e., $\Phi(z, R) = a(z)R + b(z)$ with $a(z) > 0$. Suppose that $R = 0$ lies in the block's attainable-total domain, and that the block's dynamics satisfy the zero-normalization condition $\Phi(z, 0) = 0$. Then $b(z) = 0$ identically, $\Phi(z, R) = a(z)R$ is ratio-scale in R, and the ranking-reversal pathology of the example above cannot occur through an affine intercept.*

The lemma is elementary: setting $R = 0$ in the affine form yields $b(z) = \Phi(z, 0) = 0$. Its content is not the algebra but the two structural preconditions that licence it: *(a)* the attainable-total domain must include $R = 0$, i.e., the block must be capable of being reduced to zero resource by its own dynamics, and *(b)* the pooled value must vanish there. Both are structural assumptions about the block, not about the interface value's compression. Absorbing-state constructions of the killed-process type (Parts IV–V) make $R = 0$ or a zero-viability boundary naturally attainable and force $\Phi(z, 0) = 0$ automatically, which is why the ratio-scale form is recovered in those classes. In more general settings neither condition holds for free, and the affine intercept must be handled explicitly.

> ***Example (ranking reversal from a nonzero intercept).*** (Reminder: throughout this monograph, interface values live in the positive scalar domain $(0, \infty)$, not necessarily $(0, 1]$. The example below uses values like $c = 5$ that are legal in the theorem's domain but would not be raw survival probabilities; the example illustrates a ratio-scale pathology, not a literal survival statement.) Fix equal outer weights $w = 1/2$. Suppose each block $C \in \{A, B\}$ admits an affine interface value of the form $c_C = 1 + \mathbb{E}[R_C]$, where R_C is the within-block total resource. Inside each block, lotteries are then ranked by expected total resource. Consider two profiles of expected totals, $P = (\mathbb{E}[R_A], \mathbb{E}[R_B]) = (0, 4)$ and $Q = (1, 1)$. The correct outer geometric aggregation of interface values gives $V(P) = \sqrt{(1 + 0)(1 + 4)} = \sqrt{5} \approx 2.236$ and $V(Q) = \sqrt{2 \cdot 2} = 2$, so P is preferred. The naive literal form $G(\mathbb{E}[R])$ gives $\sqrt{0 \cdot 4} = 0$ for P and $\sqrt{1 \cdot 1} = 1$ for Q, ranking Q above P. A nonzero affine intercept can survive the inner representation and *reverse* outer rankings after geometric aggregation — the problem is not magnitude, but ratio scale.

III.2.2. Timing-sensitive nesting. Suppose we have an outer evaluator and an inner evaluator, and we want to compose them. The naive composition

$$V_{\text{total}} = G(V_1, V_2, \ldots, V_n)$$

requires the Vi to be well-defined at the moment the outer evaluator acts. But "when the outer evaluator acts" is itself a temporal statement, and the block's internal state may not have equilibrated to a scalar summary at that moment. If the outer evaluator is applied *before* the block has compressed to its interface value, the composition fails: the scalar Vi does not exist yet, and the outer evaluator has no admissible input. If it is applied *after* — but too long after — the block may have already begun decaying in ways that the interface value should have captured but didn't. The nesting is timing-sensitive. A related issue is whether the outer aggregator should be written as G(E[U]) or E[G(U)] under uncertainty. These are the same when G is linear but different when G is nonlinear, and which one is correct depends on when the uncertainty is resolved relative to the interface. If within-block uncertainty is resolved before the interface, the block acts on its realized value and G(E[U]) is wrong; if cross-block uncertainty persists past the interface, then E[G(U)] is wrong. The right formula depends on the timing of uncertainty resolution relative to the interface, and the formula changes as that timing changes.

III.2.3. Post-interface common uncertainty. Even when a block has compressed to a scalar at the interface, uncertainty in the external environment may persist past the interface. Two blocks facing correlated post-interface hazards cannot be treated as independent; the outer aggregator's weighted geometric mean presumes branch-level independence in the absorbing-failure setting. This is a structural issue about what the outer theorem can and cannot absorb. The clarification worth making here is that Theorem II.1's axioms (A1)–(A5) operate on a *normalized evaluator* of interface values, not on raw conjunction survival probabilities; at that axiomatic level, no probabilistic independence is required. Where independence enters is in the physical interpretation of the interface values themselves: if two blocks share post-interface uncertainty, the scalars c_1, c_2 do not represent truly non-transferable branches, and the evaluator we compute $A_w(c_1, c_2)$ does not correspond to any meaningful aggregate viability. So the issue is not that Theorem II.1 fails when correlation is present — it characterizes the evaluator regardless — but that its premises (P1)–(P3) are no longer satisfied, and the application is unwarranted. In practice, this obstruction is often addressed by absorbing the common uncertainty into a modified model where the (P3) interface values represent properly separated blocks under a coarser conditioning. But it is a structural complication that the bare formulation of the interface value does not resolve.

> ***Example (common post-interface shock).*** Fix equal weights $w = 1/2$, and let a common external shock η take two values with probability $1/2$ each: $(c_A(\eta), c_B(\eta)) = (4,1)$ or $(1,4)$. Then $\mathbb{E}[G(c_A, c_B)] = \mathbb{E}[\sqrt{c_A c_B}] = \mathbb{E}[\sqrt{4}] = 2$, whereas $G(\mathbb{E}[c_A], \mathbb{E}[c_B]) = \sqrt{2.5 \cdot 2.5} = 2.5$. The two orderings of expectation and geometric aggregation can diverge sharply when common uncertainty survives the interface. Whether $\mathbb{E}[G(c)]$, $G(\mathbb{E}[c])$, or neither is correct depends on how the residual common uncertainty is evaluated after the outer aggregation.

III.3. Why the naive nested formula fails

The three obstructions together show why a naive nested formula

$$V_{\text{total}} = G(V_{\text{inner}}(R_1), \ldots, V_{\text{inner}}(R_n))$$

is not in general valid, even when both the outer and inner theorems hold separately. The outer theorem requires ratio-scale scalar inputs. The inner theorem, in its basic form, gives an affine expected-total representation, which may or may not be ratio-scale. The timing of the interface is left unspecified by either theorem; they simply assume their respective compositional axioms hold at the relevant moment. And post-interface correlations are outside the scope of both. What we need, to compose the two edges, is a construction that produces a ratio-scale positive scalar at a specific moment in the block's dynamics, under conditions that prevent premature or excessive compression. The rest of the paper is about such constructions.

III.4. Bridges: approximate interface values from timescale separation

The first approach to overcoming the obstructions is to weaken the strongest assumptions of the inner theorem and recover an approximate version.

Recall that Theorem II.2 assumed complete internal transfer (B1). Real blocks do not satisfy this. But many real blocks do satisfy a weaker condition: internal mixing is fast relative to hazard arrival, even if it is not complete before every hazard.

Under this weaker assumption — call it *fast mixing on a fixed-total fiber* — the block's state equilibrates within constant-total slices of its configuration space on a timescale much shorter than the hazard timescale. Between hazards, the block's internal distribution relaxes toward the stationary distribution of its fast dynamics, conditioned on the current total resource.

In this regime, the one-step survival probability becomes approximately a function only of total resource, with error controlled by the mismatch between the mixing timescale and the hazard timescale. To state this precisely, four objects live on the fiber at fixed total R: the stationary distribution π_R of the fast within-fiber dynamics; the t-step fiber transition kernel K_R^t, so $K_R^t(u,\cdot)$ is the law at time t starting from allocation u; the one-step survival kernel $s(u)$, giving the probability of surviving a hazard from allocation u; and the hazard-time law $\nu_R(t)$, the distribution of the time to the next viability-relevant hazard at total R. The induced one-step value from allocation u is the hazard-time-averaged t-step propagation:

$$V(u) \;=\; \sum_{t\geq 0} \nu_R(t) \int s(x)\, K_R^t(u, dx).$$

When fiber mixing is complete before typical hazard arrival, $K_R^t(u,\cdot) \to \pi_R$ before ν_R places significant mass, and $V(u) \to \bar{s}(R) := \int s\, d\pi_R$ uniformly in u. The scalar $\bar{s}(R)$ is our first candidate for a dynamically constructed interface value; the proposition below formalizes this.

Proposition III.1 (Frozen-state bridge). *Suppose a block's fast internal dynamics, at fixed total R, admit a stationary distribution π_R on the fiber $\{u : \sum u_i = R\}$, with one-step survival kernel s and hazard-time law ν_R as above. If fast mixing is complete on the fiber before the next hazard, then the effective one-step survival is*

$$\bar{s}(R) = \langle s, \pi_R \rangle = \int s(u)\, \pi_R(du),$$

and this serves as an approximate scalar interface value of the block at total resource R.

In the ideal complete-mixing limit, $\varepsilon_{\text{mix}} = 0$ and the equality above is exact; more generally, the same argument gives approximation error bounded by the hazard-weighted mixing error.

Qualification as a candidate interface value. The scalar $\bar{s}(R)$ is a plausible candidate for an interface value in the sense of Definition II.0, but whether it satisfies Definition II.0's full requirements depends on modeling choices we flag here rather than defer. Scalar compression and dynamical production are clear: $\bar{s}(R)$ is a positive scalar arising from the block's own fast mixing via the stationary-fiber average. Ratio-scale admissibility is the delicate one. In the present finite-window setup with killing rate κ_0 over a cycle of length Δ, the natural scalar proxy is closer to $e^{-\kappa_0 \Delta}$ — a survival probability bounded away from zero even when resource is depleted. For $\bar{s}(R)$ to behave as a ratio-scale interface value with zero point at resource exhaustion, one needs either (i) an absorbing/zero-normalized survival convention in which survival genuinely vanishes on the hazardous boundary, or (ii) a limiting regime in which the zero-viability boundary is actually approached — for instance, the large-window limit or a model with strictly absorbing sub-threshold states. Under either supplementary condition, $\bar{s}(R) \to 0$ as R approaches the value below which the stationary fiber mass concentrates on sub-threshold allocations, and ratios $\bar{s}(R_1)/\bar{s}(R_2)$ become the natural comparisons between blocks. Without such a condition, $\bar{s}(R)$ is a scalar proxy that satisfies two of Definition II.0's three requirements and is "ratio-scale-in-spirit" rather than ratio-scale literally; one of Part V's deliveries is that the killed-process synthesis supplies the absorbing-survival structure needed to make the ratio-scale reading unambiguous on the compressed block class. The caveat is that $\bar{s}(R)$ is *frozen* at fixed total R: the proposition does not accommodate the coarse state changing over time. Proposition III.2 extends the construction to a moving coarse state.

Proof sketch. Conditional on the hazard time H, with $t = H$, the induced value at allocation u is $V(u)$ as defined in the setup. The stationary-fiber average is $\bar{s}(R) = \int s\, d\pi_R$. For each t, since $s \in [0, 1]$, the standard total-variation bound for a function bounded in $[0, 1]$ gives $\left| \int s\, dK_R^t(u, \cdot) - \int s\, d\pi_R \right| \le d_R(t)$, where $d_R(t) := \sup_u \|K_R^t(u, \cdot) - \pi_R\|_{\text{TV}}$ and we adopt the convention $\|\mu - \nu\|_{\text{TV}} = \sup_A |\mu(A) - \nu(A)|$ (so the bound is by d_R rather than $2d_R$; the alternative convention with $\|\mu - \nu\|_{\text{TV}} = 2 \sup_A |\mu(A) - \nu(A)|$ rescales constants but does not change asymptotic rates). Averaging over t with weights $\nu_R(t)$ yields $|V(u) - \bar{s}(R)| \le \varepsilon_{\text{mix}}$, where

$$\varepsilon_{\text{mix}} := \sum_{t \ge 0} \nu_R(t)\, d_R(t)$$

is the *hazard-weighted mixing error*. This quantity is small exactly when typical hazard arrival waits long enough for the fiber chain to nearly equilibrate — the quantitative regime $\tau_{\text{mix}} \ll \tau_{\text{haz}}$. □

This is still narrow: it assumes fast mixing is complete (not merely fast relative to something), and it works at a frozen total R. But it is a concrete step toward a dynamically constructed scalar.

The second step weakens the frozen-state assumption. In many systems, the block's coarse state — the macroscopic variables that determine the landscape of fast mixing — is not static. It evolves on a timescale comparable to hazard arrival. The question becomes: can we still construct a scalar interface value when the coarse state is co-evolving with the hazard?

The answer is yes, but the interface value is now dynamically constructed. It is the value function of an averaged coarse process, rather than a static fiber average.

Proposition III.2 (Slow-fast bridge). *Suppose a block has a fast within-fiber dynamics that mixes rapidly relative to hazard arrival, and a slow coarse state z that evolves on the hazard timescale. If the fast-fiber mixing is fast enough, the block's pre-interface value admits an approximate representation* *as the value function of a coarse process with averaged transition rates and averaged survival payoffs, both integrated against the stationary fiber law at each z.*

The value function of this averaged coarse process is the block's interface value in the slow-fast regime. It is no longer a static scalar; it is a function of the current coarse state z. But at each z, it is a positive scalar, and it serves the structural role the interface value is supposed to play.

Propositions III.1 and III.2 are abstract. They give existence results for approximate interface values under timescale-separation assumptions, but they do not tell us whether those assumptions hold in any specific dynamical system, what the resulting interface values look like concretely, or whether the three obstructions of Section III.2 are actually overcome in any case we can exhibit.

The rest of the paper is about instantiating these abstract bridges in a concrete model class. Part IV constructs the killed-process framework that will carry explicit dynamics; Part V shows that, in that framework with a suitable thermodynamic structure, the interface values required by Propositions III.1 and III.2 become computable objects, and the outer theorem of II.1 is realized concretely on the compressed block class. The abstract bridges are the target; Parts IV–V are the instantiation.

III.5. What we need next

The abstract bridges give us a template: if the block's dynamics have the right timescale structure, a scalar interface value exists and satisfies the requirements for the outer evaluator to act on it. But the template is not self-checking. To know whether it applies, we need a concrete model class in which the relevant dynamics can be exhibited, the interface value computed, and the three obstructions demonstrably resolved. The rest of the paper constructs such a class. Part IV introduces the killed-process framework that will carry the concrete dynamics: a reversible birth-death chain with threshold killing on a fixed-total fiber. This framework provides the survival semigroups, quasi-stationary distributions, and principal decay rates that will serve as explicit interface-value candidates. The killed-process framework alone, in its simplest form, will not produce a satisfactory interface value. The naive uniform baseline gives only diffusive protection — polynomially slow decay, not the extensive suppression we would expect from a thermodynamic regime. This is a useful negative result: it tells us what structural ingredient is still missing.

The missing ingredient turns out to be a Gibbs weight on the fast fiber: a reversible equilibrium measure that concentrates mass away from the dangerous configurations. Once that Gibbs structure is added, the killed-process class acquires extensive suppression, a spectral compression window, and — under slow-fast coarse-state dynamics — a timescale-ordering structure that lets the compression persist dynamically. Part V then returns to the outer side and shows that, in the Gibbs-lifted killed-process class, the outer weighted geometric mean of Theorem II.1 is realized on the compressed block class. The interface value thereby acquires a specific mathematical construction — the principal decay rate of a reversible killed chain with a Gibbs fiber measure — and the three-factor asymptotic assigns it concrete structure, albeit with the capacity step of Appendix C remaining formal rather than fully rigorous (so the downstream Part V synthesis inherits "formal" status at that juncture). And the log-rate coordinate foreshadowed in Section II.5 becomes the concrete meeting point between inner arithmetic and outer geometric aggregation. This is the synthesis the program is after. The next two parts build it.

Part IV. A concrete model class: the killed-process framework

IV.1. Opening: why this class

Notation reminder. Δ denotes the outer cycle length throughout this part and the synthesis that follows; $\Delta h := h(0) - h(1/2)$ denotes the barrier height of the potential h (appearing in the Gibbs-lift concentration factor $e^{-\beta R \Delta h}$ in IV.3 and its proof-home Appendix C). The two symbols are related only through the shared letter Δ; they measure different things.

Part III identified the interface value as the central object and showed that approximate interface values can be constructed under timescale separation, but left the existence of concrete examples open. This part introduces the model class in which the abstract bridges of Propositions III.1 and III.2 become computable objects. The choice of class is motivated by three requirements. First, the class must support an explicit survival semigroup — a family of operators Qt that track how survival probability evolves over time — because the interface value will turn out to be a specific spectral object of this semigroup. Second, the class must have a natural notion of "fixed-total fiber" on which fast internal dynamics live, so that resource redistribution between components of a block can be modeled explicitly. Third, the class must be simple enough that the three obstructions of Section III.2 — affine-vs-ratio-scale, timing-sensitive nesting, post-interface correlations — can be diagnosed and, where possible, resolved by direct calculation rather than by appeal to further abstraction. A canonical toy class satisfying these requirements is a continuous-time reversible birth-death chain on a fixed-total fiber, with per-component threshold killing. We will call this the *killed-process class*. A block in this class consists of a finite resource R distributed among components; the state space is the set of admissible allocations summing to R; the dynamics are a reversible chain that redistributes resource between components; and the killing potential represents the viability hazard — the block "dies" (is absorbed) when any component drops below a threshold. Part IV proceeds in two stages, reflecting a real structural observation: the bare killed-process class, with uniform fiber dynamics, does not produce a satisfactory interface value. This negative result is not a flaw of the approach; it is a diagnostic that identifies the structural ingredient still missing. Section IV.2 exhibits the diffusive baseline and shows what fails. Section IV.3 adds a Gibbs weight to the fast fiber and shows how the missing structure appears.

IV.2. The diffusive baseline: what killed-process machinery alone does not buy

We begin with the simplest reversible fiber dynamics. The state space is the fixed-total fiber

$$\Omega_R = \{0, 1, \dots, R\},$$

where $k \in \Omega_R$ records how much of the total resource R sits at site 1, with the remainder $R-k$ at site 2. The fast redistribution dynamics is the continuous-time symmetric birth-death chain with exchange rate γ and reflecting endpoints,

$$(L_R f)(k) = \begin{cases} \gamma\big(f(1) - f(0)\big), & k = 0, \\ \gamma\big(f(k+1) - 2f(k) + f(k-1)\big), & 1 \le k \le R-1, \\ \gamma\big(f(R-1) - f(R)\big), & k = R, \end{cases}$$

with uniform stationary law

$$\pi_R(k) = \frac{1}{R+1}.$$

The killing potential is the same threshold strip used throughout the killed-process class,

$$\kappa_R(k) = \kappa_0 \mathbf{1}_{\{k<m\}} + \kappa_0 \mathbf{1}_{\{R-k<m\}},$$

so the killed generator is

$$G_R = L_R - \mathrm{diag}(\kappa_R).$$

We focus on the nontrivial regime $R \geq 2m$, where the left and right reactive strips are disjoint. The cases $R < 2m$ reduce to constant killing rates and hence to trivial spectral shifts.

Theorem IV.1 (Diffusive baseline asymptotics; formal)**.** *In regime $R \geq 2m$, the principal decay rate $a(R) = -\lambda_{\max}(G_R)$ satisfies*

$$a(R) = \frac{\gamma\pi^2}{R^2} + o(R^{-2}), \qquad R \to \infty.$$

Proof sketch; the full argument is in Appendix B. Writing $x = k/R$, the interior eigenvalue equation $\gamma(\phi_R(k+1) - 2\phi_R(k) + \phi_R(k-1)) = -a(R)\phi_R(k)$ rescales to the heat-equation generator $\gamma\,\partial_{xx}$ on $[0,1]$ on the diffusive time scale $\tau = t/R^2$. The content of the argument is the effective boundary condition. Strip elimination does not remove the hazardous strip; it produces a κ_0-dependent discrete Robin relation at the first safe site. For $m = 2$, solving the left strip equations at leading order (using only that $a(R) = O(R^{-2})$ in the strip) gives

$$\phi_1 \;=\; r(\kappa_0/\gamma)\,\phi_2, \qquad r(\kappa_0/\gamma) = \frac{\gamma(\gamma+\kappa_0)}{\gamma^2 + 3\gamma\kappa_0 + \kappa_0^2},$$

which is equivalent to the microscopic Robin relation $(\phi_2 - \phi_1)/(1/R) = \beta_R\,\phi_2$ with $\beta_R := R(1 - r(\kappa_0/\gamma))$. Since $0 < r < 1$ for every fixed $\kappa_0 > 0$, the Robin coefficient diverges: $\beta_R \sim c(\kappa_0/\gamma)\,R \to \infty$. Under the macroscopic rescaling the continuum limit therefore sees an effectively Dirichlet boundary $\phi(0) = \phi(1) = 0$, not because Robin structure is absent but because it becomes infinite. The limiting eigenproblem $\gamma\phi'' = -\Lambda\phi$ on $(0,1)$ with Dirichlet conditions has principal mode $\sin(\pi x)$ and eigenvalue $\gamma\pi^2$; rescaling back gives the stated asymptotic.

At small κ_0/γ the crossover to effectively Dirichlet behaviour is slow: $1 - r = O(\kappa_0/\gamma)$, so $\beta_R = O(R\kappa_0/\gamma)$ and Robin corrections remain visible when $\kappa_0/\gamma \lesssim 1/R$. The convergence is direct to verify numerically. For the choice $\gamma = \kappa_0 = 1$ and $m = 2$, exact diagonalization of G_R gives:

R	$a(R)$	$R^2 a(R)$
10	0.1115	11.153
50	0.00405	10.134
100	0.00100	10.002
500	3.96×10^{-5}	9.896

$R^2 a(R) \to \pi^2 = 9.8696\ldots$ *in the uniform baseline with $\gamma = \kappa_0 = 1$, $m = 2$. Full table and quasi-stationary law in Appendix B.*

The simplest one-step interface proxy is the stationary-fiber average of survival over a fixed observation window Δ,

$$c_\Delta^{\text{frozen}}(R) := \left\langle e^{-\kappa_R\Delta}, \pi_R \right\rangle.$$

In regime $R \geq 2m$ this is exactly

$$c_\Delta^{\text{frozen}}(R) = 1 - \frac{2m}{R+1}\left(1 - e^{-\kappa_0\Delta}\right),$$

so $c_\Delta^{\text{frozen}}(R) \to 1$ as $R \to \infty$ at fixed Δ. The hazardous strip carries vanishing stationary mass under the uniform measure, so this proxy reads as near-perfect survival on fixed windows even though the block still dies on diffusive long timescales.

The structural point is that mixing and killing are controlled by the same interior Dirichlet-type Laplacian. The mixing time satisfies $\tau_{\text{mix}} \asymp R^2/\gamma$, while the survival time set by the principal decay rate is

$$\frac{1}{a(R)} \asymp \frac{R^2}{\gamma\pi^2}.$$

These scales are of the same order. The baseline therefore supplies no parametric separation between internal equilibration and effective hazard, which is the specific way the control model fails for the present program. Full derivations, the exact numerical convergence, and the stationary-fiber vs spectral interface-value comparison are given in Appendix B.

Closing interpretation of IV.2. The diffusive baseline is doing the role of a control experiment. It shows that the bare killed-process machinery supplies the right *kinds* of mathematical objects — survival semigroup, quasi-stationary distribution, principal decay rate — but does not supply the right *behavior* of those objects for the interface value to be useful. The interface value candidate, whether defined as the stationary-fiber one-step proxy or as $\exp(-a(R)\Delta)$ for the principal decay rate $a(R)$, does not exhibit the extensive suppression that the Appendix C thermodynamic analogy would require. The block is "safe" at large R, but only in a diffusive sense: reaching the hazardous configuration takes time of order R^2, not because dangerous configurations are exponentially rare.

This is a specific failure to diagnose. The stationary measure on the fiber is uniform, which puts non-negligible mass on configurations near the killing boundary. To produce extensive suppression, we need the stationary measure to concentrate mass *away* from the dangerous region. That is the role of the Gibbs weight in the next section.

IV.3. The Gibbs lift: extensive suppression and the three-factor decomposition

We now keep the same fixed-total fiber as in the baseline,

$$\Omega_R = \{0, 1, \ldots, R\},$$

but change the equilibrium measure seen by the fast redistribution dynamics. Let $h : [0,1] \to \mathbb{R}$ be a smooth, symmetric one-well potential with a unique minimum at $x = 1/2$, strictly monotone on each half of the interval, and with higher endpoint values:

$$h(0) = h(1) > h(1/2), \qquad h(x) = h(1-x), \qquad \Delta h := h(0) - h(1/2) > 0.$$

Symmetry is what makes the two boundary channels equivalent; the asymmetric case is treated in the same framework but with distinct left/right reactive factors and distinct endpoint slopes $h'(0)$, $h'(1)$. Appendix C, where the derivation is carried out, does the symmetric case. The Gibbs weight on the fiber is then

$$\pi_R(k) = Z_R^{-1} e^{-\beta R h(k/R)}, \qquad Z_R := \sum_{j=0}^{R} e^{-\beta R h(j/R)}.$$

The qualitative effect is immediate. In the uniform baseline, the stationary law assigned order-one mass to configurations near the dangerous strip, so safety could come only from diffusive travel time. Under the Gibbs weight, equilibrium mass is concentrated near balanced allocations around $k/R = 1/2$, while the boundary layer becomes exponentially unlikely in R. The scaling regime therefore changes at the level of the invariant measure itself: the block is protected by concentration in configuration space before one even asks about escape or killing.

The reversible nearest-neighbor fast dynamics are chosen to have π_R as stationary law. A convenient choice is

$$r_k^+ = \gamma \exp\left(-\frac{\beta R}{2}\big(h((k+1)/R) - h(k/R)\big)\right),$$

$$r_k^- = \gamma \exp\left(-\frac{\beta R}{2}\big(h((k-1)/R) - h(k/R)\big)\right),$$

with generator

$$(L_R f)(k) = r_k^+\big(f(k+1) - f(k)\big) + r_k^-\big(f(k-1) - f(k)\big),$$

and with the same threshold-strip killing potential κ_R used in Section IV.2. The killed generator is therefore

$$G_R = L_R - \mathrm{diag}(\kappa_R).$$

By construction,

$$\pi_R(k) r_k^+ = \pi_R(k+1) r_{k+1}^-,$$

so the fast dynamics satisfy detailed balance with respect to π_R. This is the point where the killed-process class acquires a thermodynamic structure: redistribution is now organized by a reversible Gibbs landscape with a metastable well in the interior of the fiber.

The principal decay rate $a(R) = -\lambda_{\max}(G_R)$ then enters a qualitatively different asymptotic regime from the diffusive baseline. Before stating the asymptotic, define the three factors that will organize it. At the *macroscopic* scale is the Gibbs concentration factor $e^{-\beta R\,\Delta h}$, where $\Delta h := h(0)-h(1/2) > 0$ is the endpoint-to-minimum potential difference; this captures the exponentially small stationary mass of endpoint-near configurations relative to the well bulk, and is the extensive suppression the uniform baseline lacked. At the *mesoscopic* scale is the WKB transport prefactor $A_{\mathrm{WKB}}(\beta, h, R)$, an Eyring-Kramers-type one-dimensional endpoint-exit prefactor for reaching the reactive strip from the well interior (Theorem C.5 gives the derivation; explicit form in (1) below). At the *microscopic* scale is the local strip-reactivity factor $A^{\mathrm{Gibbs}}_{\mathrm{react}}$, an R-independent limit of the local boundary conductance computed from Gibbs-biased birth-death dynamics inside the hazardous strip of fixed width m (Theorem C.9 works it out explicitly for $m = 2$).

Theorem IV.2 (Formal three-factor asymptotic, with QSD-prefactor qualification)**.** *For the Gibbs-lifted killed generator G_R in the one-well setting described above, the principal decay rate has leading fixed-β structure*

$$\log a(R) = -\beta R\Delta h - \frac{1}{2}\log R + O_\beta(1),$$

and in the low-temperature/local-equilibrium regime satisfies the sharper three-factor form

$$a(R) \sim 2\, A^{\mathrm{Gibbs}}_{\mathrm{react}}\, A_{\mathrm{WKB}}\, e^{-\beta R\Delta h}.$$

At fixed β as $R \to \infty$, the displayed three-factor prefactor may be multiplied by a β-dependent QSD boundary-layer correction; this correction tends to 1 in the low-temperature regime and does not affect the exponential barrier factor or the $R^{-1/2}$ WKB scaling.

Each factor belongs to a distinct scale, and that separation is what makes the formula useful. The barrier term $e^{-\beta R\Delta h}$ comes from Laplace asymptotics for the well normalization Z_R, expressing the exponentially small stationary mass of endpoint-near configurations relative to the bulk of the well.

The prefactor A_{WKB} carries the mesoscopic transport cost. In the present one-dimensional endpoint-exit geometry,

$$A_{\mathrm{WKB}}(\beta, h, R) = \gamma \exp\left(-\frac{\beta(2m-1)}{2} h'(0)\right) \sqrt{\frac{\beta h''(1/2)}{2\pi R}}, \tag{1}$$

up to the asymptotic normalization used in Appendix C. The factors involving $h''(1/2)$ and $R^{-1/2}$ come from the local width of the well and the Laplace normalization near its minimum, while the endpoint-slope term records the cost of transport into the boundary layer. This is the endpoint analogue of an Eyring-Kramers prefactor for escape from a convex well [Bov+04].

The microscopic factor $A^{\mathrm{Gibbs}}_{\mathrm{react}}$ comes from the local death problem inside the hazardous strip after the process has already reached it. Because the strip width m is fixed while $R \to \infty$, this factor is computed from a finite Gibbs-biased birth-death problem near the endpoint and remains independent of R. It records the probability of being killed before returning to the safe region, with the local bias inherited from detailed balance. In Appendix C this factor is worked out explicitly for $m = 2$ and shown to reduce to the uniform-strip expression when the endpoint Gibbs bias is removed.

The prefactor depends on the endpoint slope $h'(0)$ rather than on an interior saddle curvature because the escape set is the boundary layer itself. The relevant transport problem is endpoint entry, not passage over an interior saddle. That is the correct one-dimensional endpoint-exit analogue of Eyring-Kramers in this geometry.

There is also a useful continuum-limit refinement behind this asymptotic. Eliminating the reactive strip produces a κ_0-dependent discrete Robin relation at the first safe site. Under the macroscopic rescaling, the corresponding Robin coefficient grows like $R(1-r)$, so the continuum limit seen by the well interior is Dirichlet. The local strip physics therefore survives in subleading boundary-layer data, while the leading macroscopic mode is governed by an effectively absorbing endpoint condition.

The derivation of Theorem IV.2 is formal, but the leading asymptotic structure is numerically well-supported. Appendix C verifies the predicted log-linear decay and the finite-β QSD prefactor correction in the canonical test family

$$h(x) = 2\left(x - \frac{1}{2}\right)^2,$$

for which $\Delta h = 1/2$, by direct diagonalization of the symmetrized tridiagonal generator. For the purposes of the present paper, the important point is the structural one: the principal decay rate is now an explicit positive scalar with computable asymptotics, and its leading behavior is organized by the product of a macroscopic concentration factor, a mesoscopic transport factor, and a microscopic reactivity factor. Full derivations and numerical details are in Appendix C.

Closing interpretation of IV.3. The Gibbs lift resolves the failure of the diffusive baseline by making dangerous configurations exponentially improbable under the stationary measure. The three-factor decomposition — microscopic reactivity times mesoscopic transport times macroscopic concentration — is a structural statement about how the thermodynamic leg of the program separates into physically distinct scales. The macroscopic factor $e^{-\beta R \Delta h}$ is the extensive suppression we were looking for. The mesoscopic factor is the Eyring-Kramers prefactor for endpoint escape from the convex well. The microscopic factor captures the Gibbs-biased local dynamics of killing within the hazardous strip.

This gives an interface value *candidate* with the right asymptotic structure: $c(\Delta) \approx e^{-a(R)\Delta}$ with $a(R)$ exponentially small in R. On timescales $\Delta \ll 1/a(R)$, the block is overwhelmingly likely to survive; on timescales $\Delta \sim 1/a(R)$, the block decays at a rate whose leading scaling is given by the three-factor structure. The result is *formal* in the sense of §I.5 (rigorous modulo Theorem C.1's capacity step). What remains to be established is whether this candidate is *admissible* to the outer evaluator of Theorem II.1 — whether axioms (A1)–(A5), and temporal interface sufficiency in particular, hold on a class of block summaries constructed from $a(R)$. Part V answers this affirmatively on the compressed block class; Appendix D carries the full verification.

But the interface value's *admissibility* to the outer evaluator of Theorem II.1 is not yet established. The outer theorem requires temporal interface sufficiency, which in Part III we identified as a strong structural condition on the block. Whether the Gibbs-lifted killed-process blocks actually satisfy that condition — whether their future behavior, at the outer cycle timescale, is genuinely captured by the scalar $c(\Delta)$ — is the question Part V addresses.

IV.4. What Part IV has established

At this point the paper has a concrete model class in which the interface value is a specific mathematical object. We have exhibited a negative control (diffusive baseline) showing that the bare framework is insufficient, and a positive construction (Gibbs lift) in which extensive suppression appears with a clean three-scale structure, up to the QSD prefactor qualification of Theorem IV.2. What is not yet established: whether this interface value satisfies the compositional axioms of Theorem II.1 when we aggregate across multiple blocks, and whether the construction survives when the coarse state of each block evolves in time rather than remaining static. These are the questions Part V addresses.

Part V. The synthesis

V.1. Opening: what the synthesis requires

Part IV exhibited a concrete interface value: the principal-mode survival factor $c(\Delta) \approx e^{-a(R)\Delta}$ of a Gibbs-lifted killed-process block with extensive suppression of $a(R)$. Part V shows that this interface value is admissible to the outer evaluator of Theorem II.1 — but only under a specific additional condition that turns out to be the concrete content of temporal interface sufficiency in this class. The condition is *spectral compression*: on the outer cycle timescale Δ, the block must already be dominated by its principal killed mode. Operationally, this requires the gap between the principal decay rate a_1 and the second decay rate a_2 to be large compared to $1/\Delta$: $(a_2 - a_1)\Delta \gg 1$.

Notation reminder. Throughout Part V, Δ denotes the outer cycle length. The barrier height from Part IV remains $\Delta h := h(0) - h(1/2)$; the two quantities are unrelated except for the shared letter.

Before this compression occurs, the block's future behavior depends on microstate details beyond the scalar $c(\Delta)$, and temporal interface sufficiency fails. After compression, the future is well-summarized by the scalar, and the outer theorem applies. This observation does the work that Part III's obstructions were raising. The affine-vs-ratio-scale obstruction is resolved because the Gibbs structure produces a genuinely multiplicative interface value (the ratio $c(\Delta_1 + \Delta_2)/c(\Delta_1)$ depends only on Δ_2, not on the state at time Δ_1).

The timing-sensitive nesting obstruction is resolved because spectral compression identifies precisely when the scalar summary is admissible. The post-interface correlation obstruction is resolved by a pair of distinct assumptions the realization imposes: (i) externally non-transferable blocks (no admissible inter-block transfer of viability-relevant resource), which is what (P2) requires, and (ii) independent fast dynamics between blocks (block-local killed generators with no coupling), which is additional to non-transferability and is where the product structure that Theorem II.1 will act on comes from. These are separate assumptions. Non-transferability blocks resource flow; independent fast dynamics blocks stochastic coupling. The realization uses both. Part V has two main results. The first (Section V.2) is static: two Gibbs-lifted killed-process blocks satisfying (i) and (ii), observed on a cycle long enough for spectral compression, admit scalar interface values and an induced outer evaluator that is the weighted geometric mean. The second (Section V.3) extends this to slow-fast coarse-state dynamics: if each block's Gibbs landscape depends on a slow coarse variable z that evolves alongside the fast fiber, the block interface value generalizes to a Feynman-Kac expectation of the averaged coarse process, and the outer weighted geometric mean survives. Section V.4 then brings out the structural observation that unifies both results: outer geometric aggregation, in this class, is literally arithmetic in the log-rate coordinate — or, under slow-fast dynamics, in the log of the Feynman-Kac expectation. This is the concrete meeting point between the outer theorem and the additive-primitive perspective foreshadowed in Section II.5.

Exact limit, perturbative extension. The synthesis in this part is the exact scalar limit: block-local generators, scalar compression before outer evaluation, and external non-transferability at the interface. Appendix H asks how this exact limit degrades under weak coupling. It separates two perturbative questions: whether each block still forms a scalar interface value, and whether the resulting Feynman-Kac product objects still factor approximately across blocks. The exact synthesis below should not be read as a claim that real biological boundaries are perfectly sealed; the perturbative diagnostics of Appendix H are what convert the clean-limit assumptions into quantitatively falsifiable conditions.

Type-shift roadmap for Part V. Before entering V.2's static realization, we record the types the interface-value object will take across this part and the related appendices. Definition II.0 fixed the structural requirements for a bare positive scalar. The constructions below successively extend the object by adding a parameter or lifting the codomain:

Type	Where defined	Notation	Scalar readout
Bare positive scalar	Def II.0 (Part II)	$c \in (0, \infty)$	(identity)
Cycle-parameterized scalar (frozen landscape)	Def V.2, Def D.3	$c_i(\Delta') = e^{-a_1^{(i)}\Delta'}(1 + r_i)$	$c_i(\Delta)$ at outer cycle Δ
Feynman-Kac continuation vector	Def V.3, Def F.2, Def F.4	$\mathcal{A}_i : \Delta' \mapsto e^{\Delta' M_i}\mathbf{1} \in \mathbb{R}_{>0}^{Z^{(i)}}$	$\mathcal{A}_i(\Delta, z) = [e^{\Delta M_i}\mathbf{1}]_z$ at current z
Finite-mode retained band (near-degenerate static)	Def I.2, Thm I.3	$\mathbf{c}_i^{(K)} \in \mathbb{R}_{>0}^K$ with readout ρ_i	$\rho_i^\top \mathbf{c}_i^{(K)}$ (choice-dependent)

The third-row readout is canonical as a state-conditioned scalar value $\mathcal{A}_i(\Delta, z)$, but scalar multiplicative factorization for the outer theorem applies only in the quasi-static and fast-coarse-mixing regimes; the intermediate regime retains genuine continuation-family/vector structure (§V.3).

Two structural patterns are worth flagging in advance. First, each row extends the previous by adding a parameter or lifting the codomain, while Definition II.0's three requirements continue to be satisfiable after an appropriate scalar readout. The outer theorem (Theorem II.1) always acts on scalars; what changes is how those scalars are produced from the richer underlying object. Second, readout choice is canonical through the first three rows (evaluation-at-current-z in the Feynman-Kac case is forced by the coarse-state conditioning of the outer question). It becomes a genuine choice at the finite-mode level, where no single scalar readout is privileged by the geometry alone: Definition II.0's requirements can be *restored* after a readout choice, but the readout is itself no longer canonical. This is the scalar-to-vector boundary Appendix I maps.

Notation note on the third row. We write $\mathcal{A}_i$ for both the continuation-family map $\Delta' \mapsto e^{\Delta' M_i}\mathbf{1}$ (a vector-valued function) and for its scalar readout at a particular (Δ, z), denoted $\mathcal{A}_i(\Delta, z)$ in the readout column. This is deliberate type-punning: we read $\mathcal{A}_i$ as "the continuation family of block i" and evaluating it at (Δ, z) produces the scalar summary. Context makes the intended type clear — $\mathcal{A}_i(\Delta_1 + \Delta_2, z)$ is a positive scalar; $\mathcal{A}_i$ alone (as in §F.6's continuation-family semigroup identity) is the vector-valued function. No single-letter rename seemed to repay the global cost.

Evaluator vs physical product, at cluster scale. The two-block evaluator-vs-conjunction distinction of §II.2.1 extends to n-block regrouping at cluster scale, and the distinction sharpens in the dynamic setting where a Jensen gap opens between the normalized evaluator and the physical product-space Feynman-Kac summaries. This calibration is not cosmetic: an evaluator super-block is the object that satisfies idempotent regrouping, while a physical product super-block is a literal joint survival object and need not obey the same normalization. Appendix G.7 develops the cluster-scale and dynamic versions explicitly with the three-layer calibration of Theorem G.1; we do not reproduce that machinery in the body. Readers focused on the synthesis can take the two-block seed from §II.2.1 and the taxonomy above; readers interested in the n-block regrouping question should consult Appendix G directly.

V.2. Static realization: the outer theorem on the compressed block class

Fix two blocks $i \in \{1, 2\}$, each a fixed-fiber Gibbs-lifted killed-process system of the Section IV.3 class. Each block i carries its own total resource $R^{(i)}$, threshold width $m^{(i)}$, exchange rate $\gamma^{(i)}$, killing rate $\kappa_0^{(i)}$, inverse temperature $\beta^{(i)}$, and one-well potential $h^{(i)}$. Write the ordered decay rates of block i's killed generator $G^{(i)}$ as

$$0 < a_1^{(i)} < a_2^{(i)} \le a_3^{(i)} \le \cdots .$$

The two blocks are externally non-transferable: after block formation, there are no admissible dynamics moving viability-relevant resource from one block to the other. Each block thus has its own internal dynamics and its own principal-mode survival factor $c_i(\Delta) \approx e^{-a_1^{(i)}\Delta}$ on any observation window Δ.

The outer composition adds one new structural element: an exogenous schedule.

Definition V.1 (Exogenous outer schedule). Fix an outer cycle of duration $\Delta > 0$. An *exogenous outer schedule* is a weight vector $w = (w_1, w_2)$ with $w_1, w_2 > 0$ and $w_1 + w_2 = 1$.

The weights are outer bookkeeping. They may represent scheduling, exposure fractions, branch prevalence, or any other exogenously assigned measure of relative importance across the two blocks; the realization is agnostic between these readings. What matters is that they are not derived from block thermodynamics. They are outer evaluator weights in the formal sense of Theorem II.1 — inputs to the outer evaluator, not outputs of it.

This warrants a brief aside, because it determines what Theorem V.1 below is and is not saying. It is tempting to read the weighted geometric evaluator as the literal joint survival probability of "both blocks survive a shared process." For absorbing-failure branches with independent hazards, that joint probability is the unnormalized product c_1c_2, which is not idempotent. Theorem V.1 concerns a different object: the normalized outer evaluator on block interface values, parameterized by exogenous weights w. The weighted geometric mean appears because the outer question is evaluator-valued and takes exogenous weights as input. An exposure-schedule reading of w is operationally consistent but not required; nothing in the realization pins the weights uniquely to fractions of block runtime.

The next ingredient is the class of block summaries on which the outer evaluator will act.

Intuition: the two-sided compression window. Before the definition, the motivating picture. Spectral compression is a *two-sided* window on the outer cycle length Δ: it must be long enough for the block's killed semigroup to collapse onto its principal mode ($\Delta \gg 1/(a_2 - a_1)$, so the block has a genuine scalar summary), but short enough to sit inside the metastable regime where the scalar reading is meaningful ($\Delta \ll 1/a_1$, so the block has not yet decayed). The Gibbs lift of Section IV.3 is what makes this window exist: it separates within-well relaxation (polynomial in R) from inter-well escape (exponentially large in R). The diffusive baseline of Section IV.2 lacks this separation — both scales are of order R^2/γ — and this is the precise sense in which the baseline fails to produce an admissible interface value. Definition V.2 below encodes this two-sided window as the class on which Theorem V.1 will verify the outer axioms.

Definition V.2 (Compressed block class). Fix an outer cycle length $\Delta > 0$ and a compression parameter $C > 0$ (chosen large enough that e^{-C} is small; $C \gtrsim 5$ suffices for most purposes). For each block i, set $\Delta_{\text{comp}} := C/(a_2^{(i)} - a_1^{(i)})$ (block-implicit; read with index i in context). Block i belongs to the *compressed class at scale* Δ if the following four conditions hold.

(i) Ideal compressed scalar. The block admits an *ideal* (principal-mode) scalar interface value

$$c_i(\Delta') := e^{-a_1^{(i)}\Delta'}$$

defined for every $\Delta' \in [\Delta_{\text{comp}}, \Delta]$. This is the post-transient principal-mode survival factor; it is exact in the sense that $c_i(\Delta_1 + \Delta_2) = c_i(\Delta_1)\, c_i(\Delta_2)$ holds identically.

(ii) Exact continuation with residual. The block also admits an *exact* finite-time continuation factor

$$\hat{c}_i(\Delta') = c_i(\Delta')\big(1 + r_i(\Delta')\big) = e^{-a_1^{(i)}\Delta'}\big(1 + r_i(\Delta')\big)$$

where $r_i(\Delta')$ records the residual nonprincipal-mode contribution.

(iii) Uniform subleading control. The relative error satisfies

$$|r_i(\Delta')| \lesssim e^{-(a_2^{(i)} - a_1^{(i)})\Delta'}$$

with a fixed implicit constant, uniformly for $\Delta' \in [\Delta_{\text{comp}}, \Delta]$.

(iv) Spectral compression. $(a_2^{(i)} - a_1^{(i)})\Delta_{\text{comp}} = C$, so that $|r_i(\Delta')| \lesssim e^{-C} \ll 1$ uniformly on the admissible range $[\Delta_{\text{comp}}, \Delta]$. (At the working value $C \gtrsim 5$ this gives $|r_i(\Delta')| \lesssim e^{-5} \approx 7 \times 10^{-3}$, in particular well below the regularity bound $|r_i| \leq 1/2$ that Appendix D's Proposition D.5 uses as an explicit smallness hypothesis for its division step; the body definition's condition (iv) thus implies Appendix D's regularity clause and we do not state the latter separately here.)

The *compressed block class at scale* Δ is the set of pairs (c_1, c_2) — equivalently $(\hat{c}_1, \hat{c}_2)$ via (ii) — from two independent blocks each satisfying (i)–(iv).

Notation convention going forward. From this point onward we adopt the discipline already used in Appendix G: $c_i(\Delta')$ denotes the *ideal* principal-mode scalar (exact temporal composition); $\hat{c}_i(\Delta')$ denotes the *exact* finite-time continuation factor (approximate temporal composition with residual r_i). Theorem V.1's (A5) verification will be stated for $\hat{c}_i$ — the actual model output — and reduces to exact (A5) for c_i in the $C \to \infty$ limit. This matches the closure/error bookkeeping in Appendix G's n-block treatment, where regrouping is exact on c_i as evaluator inputs and carries log-error bound when applied to $\hat{c}_i$.

Two observations about Definition V.2. First, in the sense of Definition II.0, each $c_i(\Delta')$ is a *parameterized* interface value: Definition II.0's three structural requirements (scalar compression, ratio-scale, dynamical production) hold for each admissible Δ' in the compressed range, but the interface value is now cycle-parameterized rather than a bare scalar. The object has changed type from scalar to scalar-valued function of Δ'. Appendices F and I extend this type further still — to continuation families and finite-mode profiles — as the dynamical setting grows richer.

Second, the compressed class is a proper subset of positive pair summaries. Not every pair of positive scalars obtained from a killed-process block belongs to it, because compression is a timing condition (as §V.1 noted).

Proposition V.0 (Block-level scalar factorization underlying (A5)). *Fix blocks $i = 1, 2$ both in the compressed class of Definition V.2 at scale Δ. For any partition $\Delta = \Delta_1 + \Delta_2$ with $\Delta_1, \Delta_2 \in [\Delta_{\text{comp}}, \Delta]$, the ideal scalars compose exactly,*

$$c_i(\Delta_1 + \Delta_2) = c_i(\Delta_1)\, c_i(\Delta_2),$$

and the exact continuations satisfy the same relation up to controlled error,

$$\hat{c}_i(\Delta_1 + \Delta_2) = \hat{c}_i(\Delta_1)\, \hat{c}_i(\Delta_2)\big(1 + \varepsilon_i(\Delta_1, \Delta_2)\big), \qquad |\varepsilon_i| \lesssim e^{-C},$$

uniformly on the admissible range.

Proof sketch. The first identity is immediate from the closed form $c_i(\Delta') = e^{-a_1^{(i)}\Delta'}$. For the second, direct substitution of Definition V.2(ii) gives

$$\frac{\hat{c}_i(\Delta_1 + \Delta_2)}{\hat{c}_i(\Delta_1)\, \hat{c}_i(\Delta_2)} = \frac{c_i(\Delta_1 + \Delta_2)\big(1 + r_i(\Delta_1 + \Delta_2)\big)}{c_i(\Delta_1)\, c_i(\Delta_2)\big(1 + r_i(\Delta_1)\big)\big(1 + r_i(\Delta_2)\big)} = \frac{1 + r_i(\Delta_1 + \Delta_2)}{\big(1 + r_i(\Delta_1)\big)\big(1 + r_i(\Delta_2)\big)} = 1 + \varepsilon_i,$$

using the first identity to cancel the ideal-scalar ratio, with $\varepsilon_i = O(r_i(\Delta_1) + r_i(\Delta_2) + r_i(\Delta_1 + \Delta_2)) = O(e^{-C})$ by Definition V.2(iv). □

What Proposition V.0 does and does not do. The proposition establishes that on the compressed class, the *inputs* to any outer evaluator — the block-level exact interface values $(\hat{c}_1, \hat{c}_2)$ — admit coordinate-wise multiplicative factorization under temporal concatenation up to subleading error, and that the ideal scalars satisfy the same factorization exactly. This makes the functional-equation of axiom (A5) operationally well-posed on the class: there is now a concrete notion of "temporal concatenation of profiles" for the evaluator to act on. What the proposition does *not* establish is that A5 then holds for every outer evaluator on this class. The arithmetic-mean evaluator of Theorem A.7 satisfies (A1)–(A4) on positive-scalar profiles with coordinate-wise multiplicative composition and nonetheless fails (A5); compression does not change this. A5 remains a condition on the evaluator, not a derivation from (A1)–(A4) plus class structure. What Theorem V.1 below establishes is that the specific *induced geometric* *evaluator* satisfies (A1)–(A5) on the class — exactly on the ideal scalars c_i, approximately to subleading error on the exact continuations $\hat{c}_i$. Uniqueness in the sense of Theorem II.1 requires the all-arity (A4) regrouping family on the compressed class, which is closed by Theorem G.7; the two-block theorem alone does not pin the exponent.

Theorem V.1 (Weighted geometric outer evaluation for compressed blocks; formal)**.** *Consider two independent, externally non-transferable, fixed-fiber Gibbs-lifted killed-process blocks in the compressed class of Definition V.2 at scale* Δ*, with exogenous weights* $w = (w_1, w_2)$*,* $w_1 + w_2 = 1$*. The induced geometric evaluator*

$$A_w^{\text{geom}}(c_1, c_2) := c_1^{w_1} c_2^{w_2}$$

satisfies axioms (A1)–(A5) of Theorem II.1 on this class, with subleading error $O(e^{-C})$ *in the (A5) functional equation. (A4) is vacuous at* $n = 2$ *on its own; the all-arity (A4) realization on the compressed class is verified in Theorem G.7.*

Proof. (A1)–(A3) are immediate for A_w^{geom} on positive scalars: continuity, strict monotonicity in each coordinate for $w_i > 0$, idempotence from $w_1 + w_2 = 1$. (A4) is vacuous at $n = 2$. For (A5) we verify both readings supplied by Definition V.2:

Exact (A5) on ideal scalars. For the ideal compressed scalars c_i, Proposition V.0 gives $c_i(\Delta_1 + \Delta_2) = c_i(\Delta_1)\, c_i(\Delta_2)$ identically, so

$$A_w^{\text{geom}}\big(c(\Delta_1 + \Delta_2)\big) = \prod_i c_i(\Delta_1 + \Delta_2)^{w_i} = \prod_i c_i(\Delta_1)^{w_i} c_i(\Delta_2)^{w_i} = A_w^{\text{geom}}\big(c(\Delta_1)\big) \cdot A_w^{\text{geom}}\big(c(\Delta_2)\big),$$

which is axiom (A5) with functional-equation map $F_w(s,t) = st$ exactly.

Approximate (A5) on exact continuations. For the exact continuations $\hat{c}_i$, Proposition V.0 gives $\hat{c}_i(\Delta_1 + \Delta_2) = \hat{c}_i(\Delta_1)\, \hat{c}_i(\Delta_2)(1 + \varepsilon_i)$ with $|\varepsilon_i| \lesssim e^{-C}$, so

$$A_w^{\text{geom}}\big(\hat{c}(\Delta_1 + \Delta_2)\big) = \prod_i \big(\hat{c}_i(\Delta_1)\, \hat{c}_i(\Delta_2)(1 + \varepsilon_i)\big)^{w_i},$$

and expanding $(1 + \varepsilon_i)^{w_i} = 1 + O(e^{-C})$ uniformly yields

$$A_w^{\text{geom}}\big(\hat{c}(\Delta_1 + \Delta_2)\big) = A_w^{\text{geom}}\big(\hat{c}(\Delta_1)\big) \cdot A_w^{\text{geom}}\big(\hat{c}(\Delta_2)\big) + O(e^{-C}),$$

which is (A5) with $F_w(s,t) = st$ to uniform subleading error. The first reading is exact and is what Theorem II.1 applies to in the $C \to \infty$ limit; the second is the actual model output, which is exact only up to compression error. □

Formal status. Theorem V.1's title carries "formal" per the convention of §I.5: rigorous modulo Theorem C.1's capacity step in Theorem IV.2's three-factor asymptotic.

On the structure of this realization theorem and what it does not *say about uniqueness.* Theorem V.1 establishes that the induced geometric evaluator A_w^{geom} satisfies (A1)–(A5) on the two-block compressed class, with (A5) holding to subleading error $O(e^{-C})$ and (A4) vacuous at $n = 2$. This is a *realization* statement: it exhibits a specific evaluator on the class. It is *not* a uniqueness statement among approximately-axiom-satisfying evaluators on the class.

The reason is structural: Theorem II.1's exponent-pinning argument requires the all-arity (A4) regrouping family. At $n = 2$ alone, the axioms force the form $A(c_1, c_2) = c_1^{\alpha} c_2^{1-\alpha}$ for some continuous strictly increasing $\alpha : (0,1) \to (0,1)$ with $\alpha(p) + \alpha(1-p) = 1$, but do not pin $\alpha(p) = p$; the $n = 3$ regrouping step in Theorem A.8's proof (§A.6) is exactly the step that forces $\alpha = \text{id}$. To invoke Theorem II.1 for uniqueness on the compressed class, one therefore needs the all-arity (A4) realization, which is the content of Theorem G.7. Together with the present two-block theorem and its n-block generalizations across the compressed class, that closure verifies the all-arity (A4) realization for the induced geometric evaluator on the compressed class; Theorem II.1 then applies to the ideal scalar evaluator at the exact-axiom limit.

We make the corresponding approximate-uniqueness claim only conditionally. At finite C, the induced geometric evaluator satisfies (A1)–(A5) with controlled subleading error, but we do not prove an independent finite-C stability theorem: deriving that any evaluator satisfying (A1)–(A4) exactly and (A5) up to $O(e^{-C})$ must coincide with A_w^{geom} up to comparable error would require a Hyers-Ulam-style stability argument in the space of evaluators, which we do not carry out. Exact uniqueness in the sense of Theorem II.1 is recovered in the $C \to \infty$ limit (perfect compression), assuming the all-arity package is in place. Readers who want a sharp finite-C stability statement — the strongest natural conjecture being that the space of evaluators satisfying (A1)–(A5) up to $O(\varepsilon)$ contracts to the weighted-geometric-mean ball of radius $O(\varepsilon)$ — should treat that as a natural extension rather than as established here. The same structural caveat (n=2 realization plus all-arity (A4) realization plus exact-axiom limit) applies to Theorem V.3 below, with ϵ_{dyn} in place of e^{-C} and Theorem G.12 in place of Theorem G.7.

Closing interpretation of V.2. The static realization resolves a gap that was open at the end of Part III and narrowed but not closed at the end of Part IV. Part III identified the interface value as the missing object connecting the inner and outer theorems. Part IV constructed a specific interface value in the Gibbs-lifted killed-process class but did not verify its admissibility to the outer evaluator. V.2 closes that verification by realization: on the compressed class, the induced geometric evaluator satisfies (A1)–(A5) to the stated subleading error. Together with the all-arity (n-block) realization of Theorem G.7 on the same compressed class, this verifies the full package required for Theorem II.1 to apply at the exact-axiom limit, where the weighted geometric form is the unique evaluator. Compression does not derive (A5) from (A1)–(A4) (that would contradict Appendix A's arithmetic-mean counterexample); it makes the (A5) functional equation *well-posed* on the class, and the realization theorem then exhibits a specific evaluator satisfying it.

One structural observation worth flagging before proceeding. The thermodynamic leg of the program now carries a structural role beyond extensive suppression. Before V.2, the Gibbs structure was understood as the mechanism for making hazard-reaching exponentially rare. V.2 shows that the same Gibbs structure also supplies the compression window that makes the outer evaluator well-defined. Two distinct services from one structural ingredient. This is the sort of consolidation that tells us we are looking at the right object.

Exact-vs-approximate theorem discipline. A general principle worth flagging once, here, that governs the realization theorems of §V.2 and §V.3 below and the all-arity package in Appendix G: *exact characterization theorems are invoked only on exact evaluator families.* Spectral compression and adiabatic elimination establish that the model-produced interface values lie near the ideal compressed scalar class, with subleading errors $O(e^{-C})$ statically and $O(\epsilon_{\text{dyn}})$ dynamically. Without a separate stability theorem in the space of evaluators (a Hyers-Ulam-style result we do not establish here), this near-membership does *not* imply uniqueness among approximately-axiom-satisfying evaluators on the class. The realization theorems exhibit the induced geometric evaluator as satisfying the axioms to subleading error; uniqueness in the sense of Theorem II.1 is recovered only at the exact-axiom limit, after combining the two-block realization with the all-arity (n-block) realization closed by Theorem G.7 and Theorem G.12. This discipline applies throughout Part V and Appendix G; we will not re-flag it at each invocation.

V.3. Slow-fast realization: persistence under coarse-state dynamics

The realization in V.2 fixed each block's Gibbs landscape for the duration of the outer cycle. In realistic settings, the coarse state that determines the landscape evolves over time. V.3 asks whether the static realization survives when the landscape itself moves while the fast fiber tries to compress.

We extend each block $i \in \{1, 2\}$ from the fixed-fiber Section IV.3 class to a block with a discrete slow coarse state $z \in Z^{(i)}$ and, for each such z, a z-dependent Gibbs-lifted killed generator $G_z^{(i)}$ on the fast fiber $\Omega^{(i)}_{R^{(i)}}$ of the Appendix C type. The coarse state evolves according to a slow Markov generator $Q^{(i)}$ on $Z^{(i)}$. For each fixed z, write the ordered decay rates of $G_z^{(i)}$ as

$$0 < a_1^{(i)}(z) < a_2^{(i)}(z) \le a_3^{(i)}(z) \le \cdots .$$

Two scalar parameters control the interaction between coarse motion and fast compression:

$$g_*^{(i)} := \min_{z \in Z^{(i)}} \big(a_2^{(i)}(z) - a_1^{(i)}(z)\big), \qquad q_*^{(i)} := \max_{z \in Z^{(i)}} \sum_{z' \neq z} Q_{zz'}^{(i)}, \qquad \lambda_Z^{(i)} := \text{spectral gap of } Q^{(i)}.$$

The quantity $g_*^{(i)}$ is the *uniform fiber-spectral gap*: a strictly positive $g_*^{(i)}$ means the fast fiber at every coarse state has at least compression-rate $g_*^{(i)}$. The quantity $q_*^{(i)}$ is the *maximal coarse jump rate*: a small $q_*^{(i)}$ means the coarse landscape changes slowly. The third quantity $\lambda_Z^{(i)}$ is the *coarse-chain spectral gap*: it controls the rate at which the coarse chain $Z^{(i)}$ converges toward its stationary distribution. In general $\lambda_Z^{(i)} \neq q_*^{(i)}$ — q_* measures maximum jump rate, while λ_Z measures convergence rate to stationary. For a two-state coarse chain they coincide up to a factor of order 1, but for larger coarse state spaces they can differ substantially. Both quantities matter and play distinct roles below.

The key modeling assumption is that coarse motion is slow relative to compression:

$$q_*^{(i)} \ll g_*^{(i)}.$$

This is the moving-landscape analogue of V.2's spectral compression condition. V.2 required $(a_2-a_1)\Delta \gg 1$; V.3 adds the requirement that the landscape itself not move on the compression scale. The inequality $q_* \ll g_*$ is what makes adiabatic elimination of the fast fiber legitimate in the dynamic setting.

Under adiabatic elimination, the block's effective survival trajectory is determined by the integrated principal rate along the coarse-state path. This motivates the following definition.

Definition V.3 (Dynamic block interface value). For block i, cycle length $\Delta > 0$, and initial coarse state $z \in Z^{(i)}$, the *dynamic block interface value* is the Feynman-Kac expectation

$$\mathcal{A}_i(\Delta, z) := \mathbb{E}_z\left[\exp\left(-\int_0^\Delta a_1^{(i)}\big(Z_s^{(i)}\big)\, ds\right)\right],$$

or equivalently in matrix-exponential form,

$$\mathcal{A}_i(\Delta, z) = \left[e^{\Delta(Q^{(i)} - \mathrm{diag}(a_1^{(i)}))}\,\mathbf{1}\right]_z,$$

where $\mathbf{1}$ is the all-ones vector on $Z^{(i)}$.

This is the continuous-time instantiation of the averaged-coarse-process value function of Proposition III.2. In V.2, the interface value was $e^{-a_1^{(i)}\Delta}$: a frozen principal-mode survival factor. Here it is the expected survival factor along a coarse-state trajectory under the killing potential $a_1^{(i)}(\cdot)$. The interface value now depends on the law of $Z_s^{(i)}$ over $[0, \Delta]$, and the expectation carries the Jensen content that V.4 will make explicit in the log-coordinate picture.

Type-shift flag. The object $\mathcal{A}_i(\Delta, z)$ has a different type from the static interface value of Definition II.0 (a bare positive scalar) and from the cycle-parameterized scalar of Definition V.2. Here the underlying object is a Feynman-Kac continuation family $\Delta' \mapsto e^{\Delta' M_i}\mathbf{1}$ taking values in $\mathbb{R}_{>0}^{Z^{(i)}}$; evaluation at the current coarse state z produces a scalar for the outer evaluator to act on. This corresponds to the third row of the Part V taxonomy table (§V.1). The readout-at-current-z is canonical — it is forced by the coarse-state conditioning of the outer evaluator's question — but the underlying data is now vector-valued. Proposition V.0' below records that despite this type shift, scalar multiplicative composition (which is what axiom (A5) requires) is preserved in the two limiting regimes.

Theorem V.2 (Dynamic persistence of compressed interface values; schematic error bound, formal)**.** *For block i, assume: (i) for each $z \in Z^{(i)}$, the fixed-z killed generator $G_z^{(i)}$ lies in the Appendix C metastable class; (ii) the uniform spectral gap is strictly positive, $g_*^{(i)} > 0$; (iii) coarse motion is slow relative to compression, $q_*^{(i)} \ll g_*^{(i)}$; and (iv) the outer cycle length satisfies $g_*^{(i)}\Delta \gg 1$ while remaining short relative to the metastable killing scale. Then the exact block survival over one cycle, starting from coarse state z and any initial fiber microstate in the metastable well, is approximated by the dynamic interface value $\mathcal{A}_i(\Delta, z)$, with error controlled schematically by*

$$\varepsilon_i^{\mathrm{dyn}}(\Delta) \lesssim e^{-g_*^{(i)}\Delta} + \frac{q_*^{(i)}}{g_*^{(i)}}.$$

Proof sketch. For fixed z, the fast survival semigroup has a spectral decomposition whose principal mode dominates after times of order $1/(a_2^{(i)}(z) - a_1^{(i)}(z))$. Assumption (ii) makes this dominance uniform across $Z^{(i)}$, and (iv) ensures the cycle is long enough for compression to occur. Assumption (iii) prevents the coarse landscape from moving on the compression scale, so adiabatic elimination of the fast fiber is legitimate. The eliminated process is a slowly changing amplitude attached to the current coarse state, evolving under the coarse generator with z-dependent killing potential $a_1^{(i)}(z)$: $\partial_t u = (Q^{(i)} - \mathrm{diag}(a_1^{(i)}))\, u$ with $u(0) = \mathbf{1}$. Its solution is the matrix exponential of Definition V.3. The two displayed error terms are schematic remnants of the elimination: the first is the subleading spectral term from principal-mode dominance, the second is the penalty for coarse motion on the compression scale. Full derivation in Appendix F. □

With each block represented by a dynamic scalar summary, V.2's outer theorem extends to the dynamic compressed class — once we identify how axiom (A5) operates on it. The type of the object has changed: an interface value is now a scalar-valued function of the coarse state z rather than a bare scalar, and temporal concatenation of interface values is Feynman-Kac semigroup composition on the continuation vector rather than coordinate-wise multiplication. The realization structure parallels V.2: Proposition V.0' below records the block-level scalar factorization that makes (A5) well-posed in the dynamic setting, and Theorem V.3 is the corresponding realization theorem for the induced dynamic geometric evaluator.

Proposition V.0' (Dynamic block-level scalar factorization underlying (A5)). *Under the hypotheses of Theorem V.2 (uniform compression $g_*^{(i)}\Delta \gg 1$ and slow coarse motion $q_*^{(i)} \ll g_*^{(i)}$), the Feynman-Kac semigroup satisfies the exact identity*

$$e^{(\Delta_1+\Delta_2)M_i}\,\mathbf{1} = e^{\Delta_1 M_i}\, e^{\Delta_2 M_i}\,\mathbf{1}, \qquad M_i := Q^{(i)} - \mathrm{diag}\big(a_1^{(i)}\big),$$

at the continuation-vector level. Scalar dynamic interface values $\mathcal{A}_i(\Delta, z) = [e^{\Delta M_i}\mathbf{1}]_z$ admit approximate multiplicative composition

$$\mathcal{A}_i(\Delta_1+\Delta_2, z) = \mathcal{A}_i(\Delta_1, z)\,\mathcal{A}_i(\Delta_2, z)\big(1 + O(\epsilon_{\text{dyn}})\big),$$

with two-source error decomposition

$$\epsilon_{\text{dyn}} \lesssim \underbrace{e^{-g_*^{(i)}\Delta} + \frac{q_*^{(i)}}{g_*^{(i)}}}_{\text{fiber compression + adiabatic separation}} + \underbrace{C_Z^{(i)}\, e^{-\lambda_Z^{(i)}\min(\Delta_1,\Delta_2)}}_{\text{coarse mixing per segment, fast-}z\text{ regime only}},$$

where the second term applies only in the fast-z regime, with $C_Z^{(i)}$ a prefactor depending on the coarse state space and the range of $a_1^{(i)}(\cdot)$ across coarse states. The factorization holds only in two regimes: the quasi-static *regime (z nearly frozen over the cycle, $q_*^{(i)}\Delta \ll 1$) and the* fast-z *regime (z mixing to its stationary law but slowly relative to fiber compression, $q_*^{(i)} \ll g_*^{(i)}$), with the additional segment-scale condition in the fast-z case that each sub-interval satisfies $\lambda_Z^{(i)}\Delta_j \gg 1$ for $j = 1, 2$ (so each segment is long enough for coarse-chain self-averaging; without this the scalar factorization fails on a short sub-interval even while the vector-level semigroup identity holds). In the genuinely intermediate regime ($q_*^{(i)}$ comparable to* $\max_z a_1^{(i)}$ *but still small relative to $g_*^{(i)}$), the exact semigroup identity above continues to hold at the continuation-vector level but scalar multiplicative factorization fails: a Jensen gap opens between $[e^{\Delta M_i}\mathbf{1}]_z$ and any scalar product of segmentwise values. Consequently the displayed scalar factorization should be read as regime-restricted, not as a generic consequence of the semigroup identity.*

What Proposition V.0' does and does not do. Like Proposition V.0, the dynamic version establishes that the *inputs* to the outer evaluator — scalar dynamic interface values $\mathcal{A}_i(\Delta, z)$ at the current coarse state — admit approximate multiplicative factorization under temporal concatenation, in the stated regimes. This makes (A5) operationally well-posed on those regimes. It does not establish A5 for arbitrary outer evaluators on the class; the arithmetic-mean counterexample of Theorem A.7 continues to apply. Theorem V.3 below verifies that the induced geometric evaluator satisfies (A1)–(A5) to the stated dynamic subleading error in those same regimes. §V.4 analyzes the intermediate-regime Jensen gap explicitly, and Appendix F supplies the rigorous treatment including a continuation-family analogue that extends into the intermediate regime on the vector object.

Theorem V.3 (Dynamic geometric outer evaluation; quasi-static and fast-z regimes, formal). *Consider two independent, externally non-transferable blocks satisfying the hypotheses of Theorem V.2, with* *exogenous weights* $w = (w_1, w_2)$*,* $w_1 + w_2 = 1$*. The induced dynamic geometric evaluator*

$$A_w^{\mathrm{geom}}(\mathcal{A}_1, \mathcal{A}_2) := \mathcal{A}_1^{w_1}\, \mathcal{A}_2^{w_2}$$

acting on scalar dynamic interface values $\mathcal{A}_i(\Delta, z_i)$ *satisfies axioms (A1)–(A5) of Theorem II.1 on the dynamic compressed-block class, to within* $O(\epsilon_{\mathrm{dyn}})$ *in both the quasi-static and fast-*z *regimes of Proposition V.0'. (A4) is vacuous at* $n = 2$ *on its own; the all-arity dynamic (A4) realization is verified in Theorem G.12. The one-cycle value is*

$$V_\Delta \;=\; \mathcal{A}_1(\Delta, z_1)^{w_1}\, \mathcal{A}_2(\Delta, z_2)^{w_2}.$$

Proof. (A1)–(A3) are immediate for A_w^{geom} on positive scalar inputs (continuity, strict monotonicity, idempotence from $w_1 + w_2 = 1$). (A4) is vacuous at $n = 2$. (A5) follows from Proposition V.0': expanding

$$A_w^{\mathrm{geom}}\big(\mathcal{A}(\Delta_1 + \Delta_2, z)\big) = \prod_i \mathcal{A}_i(\Delta_1 + \Delta_2, z)^{w_i} = \prod_i \big(\mathcal{A}_i(\Delta_1, z)\mathcal{A}_i(\Delta_2, z)(1 + O(\epsilon_{\mathrm{dyn}}))\big)^{w_i}$$

gives the multiplicative composition to within $O(\epsilon_{\mathrm{dyn}})$ in the stated regimes. Appendix F carries the full verification and the continuation-family extension to the intermediate regime. As at $n = 2$ in V.1, (A4) is vacuous here on its own; the dynamic all-arity (A4) realization on the compressed class is closed in Theorem G.12. □

Formal status and regime restriction. The scalar theorem applies only in the quasi-static and fast-z regimes (title-level); the intermediate regime is not within its scope, though the continuation-family extension in Appendix F covers it on the vector object with additional error. Reading the abstract's claim "extends to slow-fast settings through a Feynman-Kac semigroup interface value" in this light: the exact semigroup structure is available generically; the scalar outer theorem applies in the two limiting regimes.

The Feynman-Kac expression has two instructive limits. In the *quasi-static* limit, where the coarse state is nearly frozen over the cycle,

$$\mathcal{A}_i(\Delta, z) \approx \exp\!\left(-\int_0^\Delta a_1^{(i)}(z(t))\, dt\right),$$

reducing to $e^{-a_1^{(i)}(z)\Delta}$ when z is truly static, which recovers V.2. In the *fast-*z limit, where the coarse chain mixes to its stationary law $\pi_Z^{(i)}$ rapidly relative to killing but still slowly relative to fast-fiber compression,

$$\mathcal{A}_i(\Delta, z) \approx e^{-\bar{a}_1^{(i)}\Delta}, \qquad \bar{a}_1^{(i)} := \sum_{z \in Z^{(i)}} \pi_Z^{(i)}(z)\, a_1^{(i)}(z),$$

a simple exponential survival factor with stationary-averaged rate. These are approximation regimes; the generic case sits between them. Since e^{-x} is convex, Jensen's inequality gives

$$\mathcal{A}_i(\Delta, z) \geq \exp\!\left(-\mathbb{E}_z\left[\int_0^\Delta a_1^{(i)}(Z_s)\, ds\right]\right),$$

with equality only when the integrated rate is effectively non-random. The intermediate regime therefore sits strictly above the exponential of the mean integrated rate: the effective decay is slower than naive time-averaging suggests.

Appendix F verifies the expected scaling numerically in a discrete curvature-modulated family $h_z(x) = c(z)(x - 1/2)^2$ with $c(z) > 0$ varying across a finite coarse state space. Direct diagonalization confirms that the principal rate $a_1^{(i)}(z)$ is barrier-controlled and decreases rapidly with curvature, while the compression gap $a_2^{(i)}(z) - a_1^{(i)}(z)$ remains uniformly open across coarse states. In a two-state sub-family, the Feynman-Kac interface value is also computed explicitly for several coarse jump rates, interpolating between the quasi-static and fast-z approximations as predicted. The full numerical development is in Appendix F.

Closing interpretation of V.3. The slow-fast realization extends the static synthesis in a non-trivial way. Before V.3, the interface value was a static scalar; the outer evaluator acted on those scalars via the weighted geometric mean. V.3 shows that when the coarse state of each block evolves in time, the interface value generalizes to a Feynman-Kac expectation of the averaged coarse process; the scalar weighted-geometric evaluator survives in the quasi-static and fast-z regimes (with the segment-scale condition of Proposition V.0' in the latter), while the intermediate regime retains only the continuation-family structure on the vector object, with an explicit Jensen gap between the scalar geometric evaluator and the exact Feynman-Kac semigroup. The regime restriction here is sharper than the original statement of the slow-fast dynamic theorem in the canonical memo corpus: that earlier statement read the Feynman-Kac semigroup identity as directly licensing scalar (A5) across slow-fast regimes, and the present section corrects that reading by separating continuation-vector semigroup composition (which holds generically) from scalar multiplicative factorization (which holds only in the two limiting regimes above).

The structural upgrade here is in what the interface value *is*. Statically, it was $e^{-a\Delta}$: the survival factor of a frozen principal mode. Dynamically, it is $\mathbb{E}_z\left[\exp\left(-\int_0^\Delta a_1(Z_s)\,ds\right)\right]$: the expected survival factor along a coarse-state trajectory. The generalization is not uniform — Jensen's inequality tells us that in the intermediate regime (neither quasi-static nor fast-z), the Feynman-Kac expectation is strictly larger than the exponential of the expected integrated rate, so the effective decay is *slower* than naive averaging would suggest.

The thermodynamic leg now carries a third structural role. Beyond extensive suppression (macroscopic scale) and compression window opening (mesoscopic scale), the Gibbs structure must also supply *timescale ordering* — coarse motion slow relative to compression ($q_* \ll g_*$) — for the synthesis to survive coarse dynamics. When this ordering fails, compression does not form in time, and temporal interface sufficiency breaks down for a structurally new reason: not because the cycle is too short (as in Part III) but because the landscape itself moves too fast.

V.4. The log-rate coordinate: where the synthesis lives

The log-rate observation of Section II.5 now has concrete content. In the compressed block class, taking logarithms of the outer weighted geometric mean gives

$$\log V_\Delta = \sum_i w_i \log c_i(\Delta) \approx -\Delta \sum_i w_i a_1^{(i)}.$$

This is arithmetic in the log-survival coordinate, which — because

$$c_i(\Delta) = e^{-a_1^{(i)}\Delta}$$

— is also arithmetic in the decay-rate coordinate. Under slow-fast dynamics, the same observation generalizes:

$$\log V_\Delta = \sum_i w_i \log \mathcal{A}_i(\Delta, z_i).$$

Now the additive primitive is no longer a frozen integrated rate but the log of a Feynman-Kac expectation. The Jensen relation

$$\log \mathcal{A}(\Delta, z) \neq -\mathbb{E}_z\left[\int_0^\Delta a_1\,ds\right]$$

tells us that the log-Feynman-Kac coordinate is strictly more general than the log-integrated-rate coordinate, coinciding with it only in the quasi-static and fast-z boundary limits (where either the path is effectively frozen so $\int_0^\Delta a_1(Z_s)\,ds$ is deterministic, or the integrated rate self-averages to a deterministic effective rate $\bar{a}_1\Delta$). This is the concrete meeting point between the outer theorem and the additive-primitive perspective. In the Gibbs-lifted killed-process class, outer geometric aggregation is arithmetic in the log-rate (or log-Feynman-Kac) coordinate; the compatibility with an additive representation is a literal coordinate identity, not a loose analogy. The two sides of the AM-GM boundary meet in a single additive coordinate whose meaning changes — in a precise way — at the interface. This observation is representational rather than constructive: it does not produce new theorems beyond V.2 and V.3. But it tells us something about the shape of the synthesis that the separate theorems do not. The AM-GM boundary, where the synthesis works, is not a boundary between incommensurate aggregation rules. It is a boundary in a single coordinate where the object being summed changes: inside the block, we sum resources; at the interface, we sum log decay rates (or log Feynman-Kac expectations); across blocks, the sum is weighted by the exogenous schedule w. One additive structure, three levels of abstraction.

V.5. What Part V has established

The synthesis has been realized in a concrete model class. The outer theorem of Part II has been instantiated concretely: it is the induced evaluator on a specific compressed block class with computable interface values. The inner theorem's arithmetic-over-resource is *compatible with* the outer theorem in the log-rate coordinate — both are represented alongside each other in a single additive coordinate, with the inner theorem contributing resource-additivity and the outer theorem contributing rate-additivity — but the synthesis does not instantiate Theorem II.2 with the same force as Theorem II.1. The inner theorem's full arithmetic-totality structure (complete-transfer pooling within a block under externalized hazard) is not directly realized by the Gibbs-lifted killed-process synthesis; it is preserved in representational compatibility, not in dynamical co-realization. The interface value itself is the principal-mode survival factor of a killed reversible chain with Gibbs-lifted fast dynamics, which becomes a Feynman-Kac expectation under slow-fast coarse-state evolution. The thermodynamic leg of the program — the Gibbs-lifted killed-process structure — does three structural services simultaneously: extensive suppression of the hazard rate, a spectral compression window that makes the interface value admissible to the outer evaluator, and a timescale-ordering condition that preserves that admissibility under coarse-state dynamics. These three services are not independent; they are three faces of the same structural ingredient. What the synthesis does not yet give is a bridge from the mathematical realization to real biological or agentic systems. Which real boundaries induce the right coarse potential, spectral gap, and timescale ordering? That question is sharpened by Part V — we now know specifically what structure the synthesis requires — but it is not solved.

Robustness under weak coupling. The exact synthesis above should not be read as a claim that real biological boundaries are perfectly sealed. Its finite-state perturbative extension is given in Appendix H. There, boundary leakage threatens scalarity by injecting amplitude into nonprincipal modes, while cross-block coupling threatens physical product factorization through Feynman-Kac tilted product defects. These perturbative diagnostics leave the exact theorem unchanged but make its clean-limit assumptions quantitatively falsifiable: scalar compression survives weak coupling when the leakage-to-gap ratio is small, and physical product Feynman-Kac factorization survives when the tilted coupling sensitivity is small under the survival-biased bridge.

Scope of the scalar regime. The synthesis above operates in the *rank-one* compressed regime $g\Delta \gg 1$, where $g := a_2 - a_1$ and each block's killed semigroup collapses onto its principal mode before the outer evaluator is applied. At the boundary $g\Delta = O(1)$, near-principal modes remain visible on the outer cycle timescale and the appropriate interface object is a finite-dimensional retained spectral band rather than a positive scalar; the weighted geometric evaluator of Theorem II.1 is canonical only on the scalar side of that transition, and extending it to vector interface structures requires either a scalar readout choice or a new vector-valued aggregation theory. Appendix I develops the static finite-band approximation theorem (Theorem I.3) and scalarity-defect bounds of the form $\sum_{r=2}^{K} \rho_r \, e^{-(a_r - a_1)\Delta}$, and deliberately defers the dynamic vector problem — which requires matrix-valued Feynman-Kac or adiabatic-mode-tracking machinery, together with fixed-rank and non-crossing hypotheses ruling out eigenvalue crossings that change the retained band's identity — as its own arc distinct from both the static scalar synthesis and the moving-landscape failure mode of §V.3.

The scalar synthesis established in Part V is therefore one regime of the broader program. Appendix G closes it under heterogeneous scalar regrouping; Appendix H asks how it degrades under weak coupling; Appendix I locates the near-degenerate boundary at which scalar interface values thicken into finite-mode vector structure. The companion typology document [Kim26b] classifies the interface signatures that appear once the scalar regime is no longer adequate, separating cases where additional supplied structure — basin witnesses, admissible readout classes, cone orders — restores some form of comparison from cases where the formal cone is non-diagnostic.

Part VI returns to the motivating applications and articulates what the synthesis does and does not tell us.

Part VI. What this gives and what remains

VI.1. Opening: what the synthesis does and does not tell us

The synthesis in Part V is a specific mathematical claim realized in a specific model class. This final part considers what the synthesis tells us — and, equally important, what it does not tell us — about the motivating questions that opened the paper. Three kinds of claims can be distinguished. First, claims internal to the mathematical framework: the interface value exists, the outer evaluator on compressed blocks is the weighted geometric mean, the thermodynamic leg supplies compression and timescale-ordering. These are established by Parts IV–V. Second, claims about the shape of the framework's application: the synthesis provides a testable structural criterion for when a subsystem counts as a coherent unit. These are consequences of the framework that follow if the framework's conditions are satisfied. Third, claims about biological or agentic systems satisfying the framework's conditions: that real cells, ant colonies, forest stands, or agents exhibit the thermodynamic structure the synthesis requires. These claims are not established by this paper and remain conjectural. The discipline of separating these three kinds of claims is essential. The mathematical synthesis is genuine (modulo the formal metastable-capacity asymptotic of Theorem C.1, in the sense of §I.5); we have exhibited a worked realization. The structural criterion is a direct consequence of the synthesis and is operationally concrete: systems that satisfy the synthesis's conditions admit scalar interface values and aggregate according to the weighted geometric mean. Whether specific real systems satisfy these conditions is an empirical question the paper does not answer. The remainder of this part is organized around this separation: Section VI.2 states what the synthesis gives, Section VI.3 identifies falsification targets, and Section VI.4 closes with open problems organized by proximity to current results.

VI.2. What the synthesis gives

The synthesis produces a structural criterion for when a subsystem counts as a rank-one scalar coherent unit. The criterion has three measurable components, each inherited from the model-class realization of Parts IV–V:

(i) a coarse potential with metastable well structure on the relevant viability timescale,

(ii) a spectral gap at each coarse state, large enough that the principal mode dominates on the outer cycle timescale,

(iii) coarse-state dynamics slow enough to respect the compression window the gap defines.

When all three components hold and the background outer hypotheses of Theorem II.1 are in force (absorbing viability, external non-transferability across blocks, exogenous weights, and the scalar-compression axioms), the block admits a scalar interface value and the outer aggregator across non-transferable blocks is forced to be the weighted geometric mean. When any of the three components fails, the block has not compressed to a scalar on the relevant timescale, and the AM-GM composition the synthesis requires is not licensed on that timescale. The criterion is operational: each component can in principle be measured by spectral and timescale analysis of the block's dynamics on the viability scale of interest.

The first domain where this criterion earns its keep is evolutionary biology. The question "is this group a unit of selection?" has a long history of productive ambiguity in that literature — colonies, species, and multicellular organisms present candidate higher-level units, and the transferability intuition ("losses to one member can be compensated by gains elsewhere in the group") is part of the informal justification for treating a group as one. But the intuition has resisted sharp formalization. The synthesis makes the question measurable: a group counts as a scalar AM-GM unit of selection on a given generational timescale if its aggregate viability has ratio-scale interface structure on that timescale — that is, if the group satisfies conditions (i)–(iii) on the scale where selection acts. Within the framework's scalar regime, "unit of selection" carries measurable structural content; outside that regime, finite-band/vector interface signatures may still apply but the scalar AM-GM theorem does not directly license group-level aggregation.

The synthesis does not tell us that any particular group — ant colony, forest stand, coalition of cooperating individuals — satisfies the three conditions on the evolutionary timescale. It tells us what the group would have to satisfy for "unit of selection" to carry structural content. Price-equation and contextual-analysis machinery are natural targets for corollaries of the log-rate synthesis of Section V.4 under the spectral compression hypothesis, with the timescale-ordering condition specifying the regimes in which group-level selection is coherent. The present paper does not develop those corollaries; their availability is part of what the synthesis opens up.

A second domain is collective entities more generally — ecosystems, coalitions, institutions, firms, other organizations with candidate unit-level behavior but without the biological absorbing-failure structure the outer theorem assumes. Here the criterion applies with a significant caveat. The outer theorem relies on branch viability being a survival probability with absorbing failure: a block either survives or it does not, and once it fails, it is gone. Biological units satisfy this cleanly. Non-biological collectives often do not. A firm that goes bankrupt is not the same kind of absorbing state as a cell that dies, because the constituent humans persist, re-coalesce into new firms, and carry with them viability-relevant resources (skills, relationships, capital) that transfer across the ostensible boundary.

What the framework offers in this regime is structural measurements. For a given collective on a given timescale, one can in principle measure whether the three conditions hold: whether the collective's aggregate viability has a coarse potential with metastable structure, whether there is a spectral gap on the relevant timescale, whether the timescale ordering respects the compression window. If the measurements come out positive, the weighted-geometric aggregation is coherent on that scale. If they do not, the framework is silent — the synthesis's scope is the regime where the three conditions hold. Extending the outer theorem to collectives with non-absorbing failure modes, where "dissolution" means something weaker than death, is a separate research question.

A third domain, tentatively, is the philosophical question of personal identity — of when a persisting system has survived a gradual replacement of its components versus become something else. The framework gives no answer to the metaphysical question. The scale-relativity of the boundary means there is no unique fact about which persisting system counts as "the same" across arbitrary timescales and component replacements; different scales give different boundary locations, none privileged over the others. What the framework does give is a reductive structure for the empirical parts of the question. On a specified viability timescale, with a specified coarse potential and spectral gap, the question of whether a persisting physical structure counts as a single scalar coherent unit across its components' replacement has a determinate answer: yes if and only if the three conditions continue to hold across the replacement.

These are facts about coherence. Whether they amount to facts about identity — whether "the same coherent unit on every viability-relevant scale" is what it means to be the same person — is a further philosophical claim, and the framework does not speak to it. The reductive structure is available to anyone who wants to build a theory of personal identity on it; the framework itself makes no such claim.

Closing framing of VI.2. The synthesis tells us what structural measurements distinguish scalar coherent units from systems whose interface is finer than a scalar (or absent altogether) on the relevant timescale. It does not tell us which real systems will turn out, when measured, to be coherent by this standard. That distinction matters. A biologist asking "is this group a unit of selection?" can, in principle, translate the question into measurements on the group's coarse potential, spectral relaxation, and timescale ordering. Whether those measurements give coherence-answers that match the biologist's prior intuitions is an empirical matter. The framework's value is in providing a specific thing to measure, not in pre-determining the answer.

VI.3. Falsification targets

The synthesis makes a specific structural prediction: coherent units, in the worked framework, are systems whose viability dynamics admit the three-part structure named in Section VI.2. Three falsifications of that prediction deserve naming explicitly.

First, a coherent system that lacks the spectral structure. Suppose a biological or agentic system is clearly one unit by every reasonable criterion other than this paper's — autonomy in the sense of Krakauer-Flack [Kra+20], evolutionary individuality in the sense of Godfrey-Smith [God09], organismality in the sense of Queller-Strassmann [QS09], causal integration, developmental coherence — but its viability dynamics do not exhibit the three-part spectral structure on any relevant timescale. If such systems are the rule in biology, the AM-GM synthesis is a mathematical curiosity that does not track what individuality actually amounts to in real systems. If they are rare, the synthesis has genuine explanatory claim on the concept. The question is empirical: pick real candidate individuals, measure their coarse potentials and spectral gaps on their viability timescales, and see whether the three conditions hold robustly or fail generically. If the synthesis has explanatory claim on individuality, one would expect the conditions to hold robustly for paradigm cases (well-integrated cells, tightly bonded colonies) and admit failure on marginal ones. That expectation is measurable.

Second, a system with the spectral structure but the wrong aggregation rule. Suppose a real system exhibits a coarse potential with metastable wells, a spectral gap on the cycle timescale, and slow coarse motion, but its aggregate viability across non-transferable branches does not factor as the weighted geometric mean of block interface values. This would indicate that one of the axioms (A1)–(A5) fails on the system — most likely temporal interface sufficiency, which the synthesis realizes as spectral compression within the killed-process class. If real systems satisfy spectral compression in the sense Section V.2 requires but nonetheless aggregate non-geometrically, the identification of spectral compression with temporal interface sufficiency is wrong, or temporal interface sufficiency itself is the wrong axiom for the viability-accounting application. Either conclusion would force a structural revision of the outer theorem's applicability, though not of the outer theorem itself.

Third, a systematic failure of rank-one scalar compression on the relevant interface timescale. The scalar synthesis operates in the regime $g\Delta \gg 1$ at fixed coarse state, with slow-coarse-motion $q_* \ll g_*$ dynamically; rank-one compression can fail in two structurally distinct ways. *(i) No useful compression*: the block retains high-dimensional microstate dependence and no scalar summary is available — this overlaps with the first target, since systems lacking the three-part spectral structure also lack compression. *(ii) Finite-band compression*: the block compresses to a retained spectral band of near-principal modes rather than to a scalar, producing vector interface structure; the outer evaluator of Theorem II.1 is no longer canonically applicable until a scalar readout or a vector-valued aggregation theory is supplied. This is the static *near-degenerate* regime developed in Appendix I; it is graceful degradation rather than collapse, but the scalar theorem does not apply without extra structure. The same boundary has a dynamic face: real systems may exhibit spectral compression at each frozen coarse state but systematically violate $q_* \ll g_*$ across biological scales, placing the typical situation in the moving-landscape regime of Section V.3 rather than the adiabatic window the synthesis requires. The falsifying observation takes one of two forms: in a representative sample of candidate coherent units across biological scales, either the empirical distribution of $(g\Delta)^{-1}$ is concentrated around or above unity (static near-degenerate failure) or the empirical distribution of q_*/g_* is concentrated around or above unity (moving-landscape failure). The program currently assumes, without empirical evidence, that real systems sit in the rank-one adiabatic window. If they typically do not, the synthesis degrades either to vector interface structure or to non-adiabatic coarse motion, each of which is a separate research program.

Fourth, weak-coupling targets identified by Appendix H. Scalar boundary integrity fails when the leakage-to-gap ratio is not small,

$$\lambda_\perp / \widetilde{g} \not\ll 1,$$

where $\lambda_\perp$ measures coupling-induced leakage from the principal interface mode into the nonprincipal complement and $\widetilde{g}$ is the surviving compression gap. Composition integrity fails when the Feynman-Kac tilted coupling sensitivity over one outer cycle is not small,

$$\int_0^\Delta \|B\|^{\ell,r}_{\mathrm{FK},\epsilon,s}\, ds \not\ll 1.$$

The second condition is not a raw-coupling norm condition: it weights coupling by the survival-biased path ensemble, so coupling where surviving paths concentrate matters more than coupling in high-killing regions that rarely reach the interface. Empirically, these targets demand spectrally- and path-weighted measurements of porosity rather than raw cross-boundary coupling.

Closing framing of VI.3. The falsification targets identified above are genuine. The program is not unfalsifiable: it makes a specific structural prediction about what coherence should look like, and that prediction is open to empirical challenge. Whether the challenges materialize is a matter for future work, both theoretical (extending the framework to richer settings where the current conditions may fail) and empirical (identifying real systems and measuring their structural properties). The soundness of the synthesis does not depend on the empirical verdict; the synthesis's status as a theory of agency does.

VI.4. Open problems

Open problems divide cleanly into proximate and deep.

Proximate problems extend the synthesis within the killed-process class. *Heterogeneous n-block composition*: Appendix G closes this at the evaluator level, showing that heterogeneous scalar interface values are stable under weighted n-block regrouping with componentwise compression hypotheses. What remains are its extensions *beyond* the scalar compressed class: physical product-space interpretations of clusters (which differ from normalized evaluator super-blocks by the Jensen gap of Proposition G.13), and failures of the common outer-cycle window that Appendix G requires. *Weakly coupled boundaries*: finite-state weak coupling in the scalar regime is treated in Appendix H, which controls scalar compression integrity via the leakage-to-gap ratio $\lambda_\perp/\widetilde{g}$ and raw Feynman-Kac product factorization via tilted coupling sensitivities under the survival-biased bridge. What remains open is stronger: stability of the Gibbs three-factor asymptotic of Appendix C under perturbation, weak-coupling stability of finite-band cone or basin interpretations in the regime of Appendix I, and the non-perturbative search for new partitions once an old scalar partition fails (an empirical/algorithmic problem in spectral clustering, approximate lumpability, and large-deviation landscape reconstruction). *Near-degenerate / vector interface theory*: the synthesis of Appendices D, F, G operates in the rank-one compressed regime $g\Delta \gg 1$. Static systems at the boundary $g\Delta = O(1)$ retain a finite-dimensional near-principal spectral band rather than a scalar interface value, and Appendix I develops the static finite-band approximation theorem (Theorem I.3) and scalarity-defect bounds, together with a diagnosis of why Theorem II.1 does not automatically extend to vector interface objects (readout and regrouping compatibility must be supplied from outside the block dynamics). The natural continuation is a *dynamic* vector theory — a matrix-valued Feynman-Kac or adiabatic-mode-tracking problem for retained spectral bands, requiring fixed-rank and non-crossing hypotheses to rule out eigenvalue crossings that change the retained band's identity — which is deferred as its own arc, distinct from both the static near-degenerate regime of Appendix I and the dynamic non-adiabatic (moving-landscape) regime handled within Appendix F. *Non-reversible or driven dynamics*: the reversible capacity formulas of Appendix C break once detailed balance is dropped. Chetrite-Touchette-style path-ensemble methods [CT15] become the natural tool here, at the cost of losing the clean reversible machinery. *State-dependent scheduling $w(z)$*: the outer weights in Appendices D, F, and G are strictly exogenous. Allowing schedule-state coupling introduces a control problem the current outer theorem does not address. *Higher-dimensional fiber geometry*: for $d > 2$, the fiber becomes a simplex and the boundary layer acquires multi-dimensional structure, making the transport and reactivity problems non-trivially dimensional. *Multi-well and non-convex potentials*: competing metastable channels would enrich the quasi-stationary structure and introduce new outer summaries absent from the single-well framework.

Deep problems concern what the synthesis does not yet attempt. *The three-part $Q_1 \to Q_2$ bridge*: the synthesis has sharpened the open bridge into three sub-problems — which real boundaries induce a coarse potential with metastable structure, which induce spectral compression on the relevant timescale, and which induce coarse motion slow enough to respect the compression window. None of these has a general answer. Substantive progress on any one of them would reduce the bridge problem but not dissolve it. *Biological grounding*: Section VI.3's empirical predictions require measuring coarse potentials, spectral gaps, and timescale ratios on candidate coherent units. This demands first identifying which biological quantities and mechanisms instantiate the three structural roles — a modeling question the paper does not address — and then specifying observational protocols capable of measuring them. *Endogenizing boundaries and weights*: the outer theorem takes branch boundaries and evaluator weights as given. A more complete theory would derive them from symmetry principles, causal prevalence, posterior uncertainty, or a dedicated boundary-detection theory. This would turn Question 1 (what counts as a block) from a prerequisite of the synthesis into a consequence of it.

These are the real frontier of a framework that has just finished proof-of-concept. Each is a structural direction the current paper opens but does not close.

Document-stack handoff. Several of these open problems are partitioned across the broader AM-GM Boundary program. The finite-band/vector boundary diagnosed in Appendix I is classified structurally in the typology document [Kim26b], with separate analyses for adequate scalarity, basin-witnessed scalarization, partially comparable cone-valued objects, and the formally non-diagnostic case. Weak-coupling scalar integrity is treated perturbatively in Appendix H; the stability of cone orders, basin witnesses, and mixed-type composition under coupling remains a typology-level future arc. The empirical question — which real systems supply which signatures under which partitions, timescales, and viability readouts — is the territory of the empirical extraction document [Kim26a]. This monograph does not pre-empt those efforts; it supplies the scalar realization on which they build.

VI.5. Closing

The four situations that opened this paper — cell, deer, colony, forest — are not, yet, resolved by this work. We have not demonstrated that the cell exhibits the three-part spectral structure the synthesis requires on the relevant viability timescales. We have not characterized exactly what thermodynamic structure makes an ant colony cohere on the foraging timescale but perhaps not on the drought timescale. We have not specified the coarse potential and spectral gap that distinguish a tightly integrated forest stand from a collection of separately-viable trees. What we have done is provide a mathematical framework in which these questions become specific rather than vague. We have exhibited a model class in which the arithmetic-inside / geometric-outside story works concretely, with interface values, outer evaluators, and thermodynamic structure all explicitly realized. We have identified the structural conditions — coarse potential with metastable well, uniformly open spectral gap, coarse motion slower than compression — that the synthesis requires. And we have shown that the log-rate coordinate unifies the inner and outer aggregation rules into a single additive structure. The program that began with the conjecture that coherence is about transferability over time has, after this paper, reached the point where the conjecture has concrete mathematical content in a worked model class. Whether the content extends to real biological and agentic systems — whether the cell, the colony, and the forest stand satisfy the structural conditions the synthesis identifies — is the open frontier. The paper is a step toward that frontier, not a resolution of it.

In the broader program, this monograph supplies the scalar rank-one realization. The typology document [Kim26b] asks what interface type remains when scalar compression is weak, finite-band, cone-valued, or readout-dependent; the empirical extraction document [Kim26a] asks how those signatures can be measured in real systems. The monograph closes its own scalar arc; the program continues.

Appendix A. Outer Characterization: Theorem II.1 in Full

A.1. Opening

This appendix supplies the formal statement and proof of Theorem II.1. The body of Part II presents the theorem in five axioms (A1)–(A5) and sketches the force of the temporal-sufficiency axiom; the proof is log-linearization via the standard Cauchy functional-equation argument, but it depends on structural content beyond the log-linearization step, and we make those dependencies explicit here. Specifically: the branchwise multiplicative law (Theorem A.5) is derived from the chain rule for nested survival events, not assumed; temporal interface sufficiency is shown to force multiplicative temporal composition (Theorem A.3), together with a counterexample (Theorem A.7) verifying that the axiom is not redundant; and the main theorem (Theorem A.8) extracts the weighted geometric mean from these ingredients via log-linearization. A staged-survival corollary and brief continuous-time and soft-multiplicative extensions close the appendix.

A.2. Setup

For each arity $n \geq 1$, let

$$I_n := \{1, \dots, n\}, \qquad \Delta_n^\circ := \Big\{ w \in (0,\infty)^n : \sum_{i=1}^{n} w_i = 1 \Big\}.$$

For each $n \geq 1$ and each $w = (w_1, \dots, w_n) \in \Delta_n^\circ$, let

$$A_w : (0,\infty)^n \to (0,\infty)$$

be an outer aggregation rule over n branch viabilities. When the arity is fixed, we write $I = I_n = \{1, \dots, n\}$. The weight w_i is the weight assigned to branch i.

For each branch i, let $(\Omega_i, \mathcal{F}_i, \mathbb{P}_i)$ be a probability space supporting a discrete-time process with absorbing death time $\tau_i \in \{0, 1, 2, \dots\} \cup \{\infty\}$. Define the survival event at stage t by $S_{i,t} := \{\tau_i \geq t\}$. Since death is absorbing,

$$S_{i,t} \subseteq S_{i,t-1} \qquad (t \geq 1).$$

Definition A.1 (Branch viability)**.** For each branch i and horizon $T \geq 0$, define branch viability by

$$v_i(T) := \mathbb{P}_i(S_{i,T}) = \mathbb{P}_i(\tau_i \geq T).$$

Remark A.2 (Positivity conventions and boundary cases)*.* The core characterization is stated on the strictly positive simplex and the strictly positive orthant. This avoids two boundary problems that would otherwise have to be treated separately: first, if some weight $w_i = 0$, strict monotonicity in coordinate i is incompatible with the geometric formula; second, if some viability coordinate is exactly zero, the logarithmic reduction used in the proof is unavailable. Likewise, whenever conditional survival factors $q_{i,t} = \mathbb{P}_i(S_{i,t} \mid S_{i,t-1})$ are used below, we tacitly restrict to horizons for which $\mathbb{P}_i(S_{i,t-1}) > 0$ at each stage under discussion. Zero-weight and zero-viability boundary cases can be handled by deleting irrelevant coordinates or by limiting arguments, but those extensions are not part of the theorem as stated.

A.3. Axioms

The five axioms below constitute the full content of Theorem II.1's hypotheses, matching the body's (A1)–(A5) enumeration.

Axiom A.1 ((A1) Continuity)**.** *For each $n \geq 1$ and each $w \in \Delta_n^\circ$, the map A_w is continuous on $(0,\infty)^n$.*

Axiom A.2 ((A2) Strict monotonicity)**.** *For each $n \geq 1$ and each $w \in \Delta_n^\circ$, the map A_w is strictly increasing in each coordinate.*

Axiom A.3 ((A3) Idempotence). *For each $n \geq 1$, each $w \in \Delta_n^\circ$, and every $c \in (0, \infty)$,*

$$A_w(c, \dots, c) = c.$$

Axiom A.4 ((A4) Spatial interface sufficiency (regrouping invariance)). *Fix $n \geq 1$ and $w \in \Delta_n^\circ$, and write $I = I_n$. Let $\{C_1, \dots, C_k\}$ be a partition of I into nonempty clusters. Define cluster outer weights and renormalized inner weights*

$$W_\alpha := \sum_{i \in C_\alpha} w_i, \qquad \frac{w_i}{W_\alpha} \quad (i \in C_\alpha).$$

Then for every $x \in (0, \infty)^n$,

$$A_w(x_1, \dots, x_n) = A_W\Big(A_{w/W_1}(x_{C_1}), \dots, A_{w/W_k}(x_{C_k})\Big).$$

Here x_{C_α} denotes the subvector indexed by C_α, the inner evaluator A_{w/W_α} acts on C_α with renormalized weights $(w_i/W_\alpha)_{i \in C_\alpha}$ (which sum to 1 over C_α), and $W = (W_1, \dots, W_k) \in \Delta_k^\circ$ is the vector of cluster outer weights. The same cluster/cluster-weight notation is used in Theorem II.1's (A4) (§II.2) and in Appendix G's n-block realization.

Axiom A.5 ((A5) Temporal interface sufficiency). *For each $n \geq 1$ and each $w \in \Delta_n^\circ$, there exists a map*

$$F_w : (0, \infty)^2 \to (0, \infty)$$

such that for every $x, y \in (0, \infty)^n$,

$$A_w(x \odot y) = F_w\big(A_w(x), A_w(y)\big),$$

where $(x \odot y)_i := x_i y_i$. Equivalently, once two temporal sub-intervals have each been compressed to their scalar interface values under A_w, whole-horizon evaluation depends on those sub-intervals only through the resulting pair of scalars.

A.4. Temporal sufficiency forces multiplicative composition

Proposition A.3 (Temporal interface sufficiency implies temporal compositionality). *Assume Axioms A.3 and A.5. Then for every $n \geq 1$, every $w \in \Delta_n^\circ$, and every $x, y \in (0, \infty)^n$,*

$$A_w(x \odot y) = A_w(x) A_w(y).$$

Proof. Fix $n \geq 1$ and $w \in \Delta_n^\circ$, and let F_w be as in Axiom A.5. For arbitrary $a, b > 0$, apply the axiom to the constant profiles $x = a\mathbf{1}$ and $y = b\mathbf{1}$, where $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^n$. By Axiom A.3,

$$A_w(a\mathbf{1}) = a, \qquad A_w(b\mathbf{1}) = b.$$

Also $(a\mathbf{1}) \odot (b\mathbf{1}) = ab\,\mathbf{1}$, so again by Axiom A.3,

$$F_w(a, b) = A_w(ab\,\mathbf{1}) = ab.$$

Since $a, b > 0$ were arbitrary, it follows that $F_w(s, t) = st$ for all $s, t \in (0, \infty)$. Therefore, for arbitrary $x, y \in (0, \infty)^n$,

$$A_w(x \odot y) = F_w\big(A_w(x), A_w(y)\big) = A_w(x) A_w(y). \qquad \square$$

Remark A.4. Axiom A.5 is the temporal analogue of Axiom A.4: both say that a substructure admits scalar compression and that later evaluation factors through that scalar summary. By Theorem A.3, temporal interface sufficiency plus idempotence forces the multiplicative functional equation. It is nevertheless *not* implied by the existence of absorbing states alone; Theorem A.7 below exhibits a concrete aggregator that satisfies (A1)–(A4) yet fails (A5).

A.5. Branchwise survival composes multiplicatively

Proposition A.5 (Branchwise survival composes multiplicatively). *Fix a branch i and a horizon $T \geq 1$.*

Set $S_{i,0} := \Omega_i$ (the full sample space on branch i), and assume $\mathbb{P}_i(S_{i,t-1}) > 0$ for each $1 \le t \le T$. Then

$$v_i(T) = \prod_{t=1}^{T} q_{i,t}, \qquad q_{i,t} := \mathbb{P}_i(S_{i,t} \mid S_{i,t-1}).$$

Proof. Since $S_{i,T} \subseteq S_{i,T-1} \subseteq \cdots \subseteq S_{i,1}$,

$$\mathbb{P}_i(S_{i,T}) = \mathbb{P}_i(S_{i,T} \mid S_{i,T-1})\mathbb{P}_i(S_{i,T-1}).$$

Iterating this identity gives

$$\mathbb{P}_i(S_{i,T}) = \prod_{t=1}^{T} \mathbb{P}_i(S_{i,t} \mid S_{i,t-1}).$$

No independence assumption is required. □

Remark A.6. Theorem A.5 is a statement about each branch individually. It does not yet imply any multiplicative law for the *outer* evaluator A_w. That additional step is exactly the content of temporal interface sufficiency together with idempotence via Theorem A.3.

A.6. Arithmetic aggregation fails (A5): a counterexample

Example A.7 (Arithmetic aggregation fails temporal interface sufficiency)**.** Let $n = 2$, $w = (1/2, 1/2)$, and define

$$A_w(x_1, x_2) = \frac{x_1 + x_2}{2}.$$

Then A_w is continuous, strictly increasing, idempotent, and spatially regrouping-invariant, but it fails Axiom A.5. Indeed, if temporal interface sufficiency held, there would exist a map F_w such that

$$A_w(x \odot z) = F_w\big(A_w(x), A_w(z)\big)$$

for all $x, z \in (0, \infty)^2$. Now take

$$x = (0.9, 0.1), \qquad y = (0.1, 0.9), \qquad y' = (0.5, 0.5).$$

Then

$$A_w(y) = A_w(y') = 0.5,$$

but

$$A_w(x \odot y) = A_w(0.09, 0.09) = 0.09, \qquad A_w(x \odot y') = A_w(0.45, 0.05) = 0.25.$$

Two second-stage profiles with the same scalar summary interact differently with the first stage, contradicting the existence of such an F_w. Equivalently, by Theorem A.3, temporal interface sufficiency would force multiplicativity, which arithmetic aggregation does not satisfy. Absorbing states and branchwise survival multiplication do not by themselves force geometric aggregation.

A.7. Uniqueness of the outer aggregator

Theorem A.8 (Unique aggregation of branch survival probabilities)**.** *Assume Axioms A.1 to A.5 hold for all arities. Then for every $n \ge 1$ and every $w \in \Delta_n^\circ$,*

$$A_w(x_1, \ldots, x_n) = \prod_{i=1}^{n} x_i^{w_i} \qquad \text{for all } (x_1, \ldots, x_n) \in (0, \infty)^n.$$

In particular, for survival probabilities,

$$A_w\big(v_1(T), \ldots, v_n(T)\big) = \prod_{i=1}^{n} v_i(T)^{w_i}$$

whenever the survival profile is strictly positive.

Proof. For $n = 1$, $\Delta_1^\circ = \{1\}$ and the unique weight is $w = (1)$; idempotence (Axiom A.3) on the constant profile $(x) \in (0,\infty)^1$ gives $A_{(1)}(x) = x = x^1$, which is the claimed form. Assume $n \geq 2$ throughout the rest of the proof. Define, for $u = (u_1, \dots, u_n) \in \mathbb{R}^n$,

$$H_w(u_1, \dots, u_n) := \ln A_w(e^{u_1}, \dots, e^{u_n}).$$

By Theorem A.3,

$$A_w(e^{u_1+v_1}, \dots, e^{u_n+v_n}) = A_w(e^{u_1}, \dots, e^{u_n})A_w(e^{v_1}, \dots, e^{v_n}),$$

so

$$H_w(u+v) = H_w(u) + H_w(v) \qquad (u, v \in \mathbb{R}^n).$$

Since H_w is continuous (Axiom A.1) and additive on $\mathbb{R}^n$, the standard Cauchy-functional-equation argument implies that H_w is linear. Hence there exist coefficients $\lambda_i(w) \in \mathbb{R}$ such that

$$H_w(u) = \sum_{i=1}^{n} \lambda_i(w) u_i.$$

Because A_w is strictly increasing in each coordinate by Axiom A.2, each $\lambda_i(w)$ is strictly positive.

To normalize the coefficients, write $c = e^t$ in Axiom A.3. Then

$$H_w(t, \dots, t) = \ln A_w(e^t, \dots, e^t) = \ln(e^t) = t.$$

Using linearity,

$$t = \sum_{i=1}^{n} \lambda_i(w) t, \qquad \text{so} \qquad \sum_{i=1}^{n} \lambda_i(w) = 1.$$

Next, the coefficient on coordinate i depends only on w_i. Let e_i denote the ith standard basis vector in $\mathbb{R}^n$, and fix $t \in \mathbb{R}$. Apply Axiom A.4 to the partition $\{\{i\}, I \setminus \{i\}\}$. Since the singleton cluster has renormalized inner weight (1) and idempotence gives $A_{w/(1-w_i)}(1, \dots, 1) = 1$ on the remaining cluster, we obtain

$$A_w(1, \dots, 1, e^t, 1, \dots, 1) = A_{(w_i, 1-w_i)}(e^t, 1).$$

Taking logarithms yields

$$H_w(te_i) = H_{(w_i, 1-w_i)}(t, 0).$$

The right-hand side depends only on w_i, so there exists a function $c : (0,1) \to (0,\infty)$ such that $\lambda_i(w) = c(w_i)$.

Now derive an additive functional equation for c using only regrouping across arities. Fix $p, q \in (0,1)$ with $p+q < 1$ and consider the weight vector $w = (p, q, 1-p-q) \in \Delta_3^\circ$. On the one hand, direct evaluation gives

$$H_w(t, t, 0) = \big(c(p) + c(q)\big)t.$$

On the other hand, regrouping the first two coordinates into a block of total weight $p+q$ and normalized inner weights $\big(p/(p+q), q/(p+q)\big)$, Axiom A.4 gives

$$A_w(x_1, x_2, x_3) = A_{(p+q, 1-p-q)}\Big(A_{\left(\frac{p}{p+q}, \frac{q}{p+q}\right)}(x_1, x_2), x_3\Big).$$

Evaluating at $(x_1, x_2, x_3) = (e^t, e^t, 1)$, the inner term is e^t by Axiom A.3, so taking logarithms yields

$$H_w(t, t, 0) = H_{(p+q, 1-p-q)}(t, 0) = c(p+q)t.$$

Therefore

$$c(p+q) = c(p) + c(q) \qquad \text{whenever } p, q, p+q \in (0,1).$$

Because $c(r) > 0$ for every $r \in (0,1)$, this additivity implies c is strictly increasing on $(0,1)$: if $0 < a < b < 1$, then $c(b) - c(a) = c(b-a) > 0$. A monotone additive function on an interval is linear, so there exists $k > 0$ such that $c(p) = kp$ for $p \in (0,1)$. For any two-branch weight vector $(p, 1-p) \in \Delta_2^\circ$,

$$1 = \lambda_1(p, 1-p) + \lambda_2(p, 1-p) = c(p) + c(1-p) = kp + k(1-p) = k,$$

so $k = 1$. Therefore $\lambda_i(w) = w_i$, and hence $H_w(u) = \sum_{i=1}^n w_i u_i$. Exponentiating gives

$$A_w(x_1, \ldots, x_n) = \exp\Big(\sum_{i=1}^n w_i \ln x_i\Big) = \prod_{i=1}^n x_i^{w_i}. \qquad \square$$

Remark A.9 (Boundary extension). If one additionally wants zero-weight boundary cases, the same geometric formula can be recovered either by deleting zero-weight coordinates from the outset or by adjoining a continuity-in-weights assumption and passing to the boundary of the simplex. That extension is straightforward but logically separate from the strictly positive theorem stated here.

Remark A.10 (Relation to Kolmogorov-Nagumo / quasi-arithmetic means). Theorem A.8 sits in the neighborhood of the Kolmogorov-Nagumo [Kol30; Nag30] / Hardy-Littlewood-Pólya [HLP52] theory of quasi-arithmetic means, which characterize aggregators of the form $\phi^{-1}\big(\sum_i w_i \phi(x_i)\big)$ for continuous strictly monotone ϕ. The present theorem is not literally a quasi-arithmetic-mean result with $\phi = \log$: axioms (A1)–(A4) do not in themselves force the ϕ-choice, and it is the temporal interface sufficiency axiom (A5), specialized to staged survival composition, that pins down the multiplicative temporal structure required for $\phi = \log$ to be the correct primitive. The derivation of multiplicative temporal composition from (A5) and the staged-survival setup (Theorems A.3 and A.5) is the substantive work this classical tradition does not do.

A.8. Staged-survival corollary

Corollary A.11 (Weighted geometric aggregation of staged survival)**.** *Assume Theorem A.8. Fix a horizon $T \geq 1$ such that, for every branch i, the positivity condition in Theorem A.5 holds through stage T. Then*

$$A_w\big(v_1(T), \ldots, v_n(T)\big) = \prod_{i=1}^n \Big(\prod_{t=1}^T q_{i,t}\Big)^{w_i} = \prod_{t=1}^T \prod_{i=1}^n q_{i,t}^{w_i}.$$

Proof. Apply Theorem A.5 branchwise, substitute into Theorem A.8, and rearrange finite products. $\square$

Remark A.12. Theorem A.11 shows that the aggregate viability of a multistage absorbing process factorizes into the product over stages of the weighted geometric mean of stagewise conditional survival factors. This is the form in which the outer theorem makes contact with the log-rate coordinate of Section V.4 in the body: taking logarithms converts the product-over-stages into a sum-over-stages of weighted log survival rates.

A.9. Continuous-time hazard form

Assume each branch i has an absolutely continuous survival function with hazard rate λ_i, so that

$$v_i(T) = \mathbb{P}_i(\tau_i \geq T) = \exp\Big(-\int_0^T \lambda_i(s)\, ds\Big), \qquad \ln v_i(T) = -\int_0^T \lambda_i(s)\, ds.$$

Theorem A.8 then yields

$$A_w\big(v_1(T), \ldots, v_n(T)\big) = \exp\Big(-\sum_{i=1}^n w_i \int_0^T \lambda_i(s)\, ds\Big).$$

This is the continuous-time hazard form of the outer characterization, and it is the specific form used in Parts IV and V when the block interface value is identified with the principal-mode survival factor $e^{-a_1\Delta}$.

A.10. Soft multiplicative extension

The preceding argument extends from literal survival to any positive multiplicative viability process.

Definition A.13 (Positive multiplicative viability)**.** A branchwise viability process is *positive multiplicative* if, for each branch i,

$$V_{i,t+1} = V_{i,t} R_{i,t+1}, \qquad R_{i,t+1} > 0.$$

Equivalently, $V_{i,T} = V_{i,0} \prod_{t=1}^T R_{i,t}$.

Examples include discounted survivability, reliability, multiplicative decay, and exponential-of-cumulative-cost functionals $V_{i,T} = \exp(-\sum_{t=1}^{T} c_{i,t})$.

Corollary A.14 (Geometric aggregation for general multiplicative viability)**.** *Suppose branch viability is given by a positive multiplicative process $V_{i,T}$ in the sense of Theorem A.13, and the outer evaluator satisfies Axioms A.1 to A.5. Then*

$$A_w(V_1, \dots, V_n) = \prod_{i=1}^{n} V_i^{w_i}$$

for every positive viability profile $(V_1, \dots, V_n) \in (0, \infty)^n$.

Proof. This is Theorem A.8 applied on the positive orthant. Only the multiplicative structure of the branchwise viability variables is used. □

Remark A.15*.* The theorem is not fragile to replacing hard survival by soft multiplicative degradation. What it does *not* automatically cover are additive observables such as expected remaining lifetime, expected stock, or distance-to-threshold. In such cases one must either work with a different theorem or first identify a transformation that converts the relevant observable into a multiplicative viability functional.

Appendix B. Diffusive Baseline: Theorem IV.1 in Full

B.1. Opening

This appendix supplies the proof-home for Theorem IV.1 — the diffusive baseline asymptotics $a(R) = \gamma\pi^2/R^2 + o(R^{-2})$ of the uniform killed-process class described in Part IV. The body stated the theorem and outlined the continuum-limit argument; here we give the full setup, the explicit Robin-to-Dirichlet derivation of the effective boundary condition, the Dirichlet-ground-state mode that supplies the leading constant, the exact numerical convergence, and the two interface-value proxies (stationary-fiber one-step, long-time spectral) with explicit expressions in this class.

The appendix is intentionally a *control result*. Its role in the monograph is diagnostic: it exhibits the simplest killed-process class for which the objects we will need later (survival semigroup, principal decay rate, quasi-stationary law, fiber stationary average) all exist and can be computed exactly, *and* it shows concretely that this simplest class fails to realize a thermodynamic regime. The mixing time and the survival time are both $\Theta(R^2/\gamma)$, so there is no timescale separation between internal equilibration and hazard. Appendix C lifts this control to the Gibbs-weighted case and recovers extensive suppression.

B.2. Setup

Fix a total resource $R \in \mathbb{Z}_{\geq 0}$, a threshold width $m = u_{\min} \in \mathbb{Z}_{\geq 1}$, an exchange rate $\gamma > 0$, and a killing rate $\kappa_0 > 0$. We work in the $d = 2$ occupation-number picture. The fiber is

$$\Omega_R = \{0, 1, \dots, R\},$$

where $k \in \Omega_R$ means site 1 holds $u_1 = k$ units of resource and site 2 holds $u_2 = R - k$.

B.2.1. Fast generator. The fast redistribution dynamics is the continuous-time symmetric birth-death chain with reflecting endpoints:

$$(L_R f)(k) = \begin{cases} \gamma\,[f(1) - f(0)], & k = 0, \\ \gamma\,[f(k+1) - 2f(k) + f(k-1)], & 1 \leq k \leq R-1, \\ \gamma\,[f(R-1) - f(R)], & k = R. \end{cases}$$

This chain is reversible with respect to the uniform stationary law

$$\pi_R(k) = \frac{1}{R+1}.$$

B.2.2. Threshold killing. Define the per-site threshold hazard by

$$\kappa_R(k) = \kappa_0\,\mathbf{1}_{\{k<m\}} + \kappa_0\,\mathbf{1}_{\{R-k<m\}}.$$

The left strip $\{0, \dots, m-1\}$ is hazardous because site 1 is under threshold; the right strip $\{R-m+1, \dots, R\}$ is hazardous because site 2 is under threshold. The killed generator is

$$G_R := L_R - \mathrm{diag}(\kappa_R).$$

The survival semigroup is $Q_t^{(R)} := e^{tG_R}$, so $q_t^{(R)}(k) := (Q_t^{(R)}\mathbf{1})(k)$ is the probability of surviving to time t starting from state k. Writing $\lambda_{\max}(R) \leq 0$ for the top eigenvalue of G_R, the principal decay rate is

$$a(R) := -\lambda_{\max}(R) \geq 0.$$

The long-time survival law is controlled by $e^{-a(R)t}$.

B.2.3. Three regimes. The threshold geometry splits the model into three regimes: (A) $R < m$, where both sites are always under threshold; (B) $m \le R < 2m$, where exactly one site is always under threshold; (C) $R \ge 2m$, where the two hazardous strips are disjoint and the interior $\{m, \dots, R-m\}$ is safe. Only regime (C) is nontrivial.

B.3. Easy regimes: spectral shifts

The degenerate regimes (A) and (B) reduce to spectral shifts of the reflecting birth-death generator.

Lemma B.1 (Regime A: universal double killing)**.** *If $R < m$, then $\kappa_R(k) \equiv 2\kappa_0$ on Ω_R. Consequently $G_R = L_R - 2\kappa_0 I$, $\lambda_{\max}(R) = -2\kappa_0$, $a(R) = 2\kappa_0$, and $Q_t^{(R)}\mathbf{1} = e^{-2\kappa_0 t}\mathbf{1}$. The quasi-stationary distribution is the uniform law π_R.*

Proof. If $R < m$, both $k < m$ and $R - k < m$ hold for every $k \in \Omega_R$, so the killing rate is constantly $2\kappa_0$. Since L_R has top eigenvalue 0 with eigenvector $\mathbf{1}$, the claims follow. □

Lemma B.2 (Regime B: universal single killing)**.** *If $m \le R < 2m$, then $\kappa_R(k) \equiv \kappa_0$ on Ω_R. Consequently $G_R = L_R - \kappa_0 I$, $a(R) = \kappa_0$, and $Q_t^{(R)}\mathbf{1} = e^{-\kappa_0 t}\mathbf{1}$. The quasi-stationary distribution is the uniform law π_R.*

Proof. Exactly one of $k < m$ or $R - k < m$ holds for each $k \in \Omega_R$, so the killing rate is constantly κ_0. The rest is the same spectral shift. □

B.4. Regime C: formal continuum limit and the main theorem

We now fix $R \ge 2m$, so the killing potential is

$$\kappa_R(k) = \begin{cases} \kappa_0, & 0 \le k \le m-1, \\ 0, & m \le k \le R-m, \\ \kappa_0, & R-m+1 \le k \le R. \end{cases}$$

B.4.1. Interior diffusive limit. Inside the safe interior, the eigenvalue equation for the principal right eigenvector ϕ_R is

$$\gamma(\phi_R(k+1) - 2\phi_R(k) + \phi_R(k-1)) = -a(R)\phi_R(k), \qquad m \le k \le R-m.$$

Introduce the macroscopic coordinate $x = k/R \in [0,1]$. For a smooth profile $f(x)$,

$$R^2(L_R f)(k) = \gamma\, \frac{f(x+1/R) - 2f(x) + f(x-1/R)}{(1/R)^2} \longrightarrow \gamma f''(x).$$

On the diffusive time scale $\tau = t/R^2$, the interior converges to the heat equation $\partial_\tau u = \gamma\, \partial_{xx} u$.

B.4.2. Strip elimination and the microscopic Robin relation. The boundary condition requires care. Strip elimination does not remove the microscopic killing strip by magic; it produces a discrete Robin relation at the first safe site. We illustrate this at $m = 2$; for general fixed m the same finite-strip elimination gives $\phi_m - \phi_{m-1}$ with a κ_0- and m-dependent ratio $r_m(\kappa_0/\gamma) \in (0,1)$, hence the same linearly divergent Robin coefficient. For $m = 2$ explicitly, solving the left strip equations at leading order (using only that $a(R) = O(R^{-2})$ inside the strip) gives

$$\phi_0 = \frac{\gamma}{\gamma + \kappa_0}\phi_1, \qquad \phi_1 = r(\kappa_0/\gamma)\,\phi_2,$$

with

$$r(\kappa_0/\gamma) = \frac{\gamma(\gamma + \kappa_0)}{\gamma^2 + 3\gamma\kappa_0 + \kappa_0^2}.$$

Equivalently,

$$\frac{\phi_2 - \phi_1}{1/R} = \beta_R \phi_2, \qquad \beta_R := R\big(1 - r(\kappa_0/\gamma)\big).$$

So the strip induces a κ_0-dependent microscopic Robin coefficient. For every fixed $\kappa_0 > 0$ one has $0 < r < 1$, hence

$$\beta_R \sim c(\kappa_0/\gamma)\, R \to \infty.$$

Under the macroscopic rescaling the Robin coefficient therefore diverges. The continuum limit is Dirichlet not because Robin structure is absent, but because it becomes effectively infinite:

$$\phi(0) = \phi(1) = 0.$$

B.4.3. Dirichlet ground state. The limiting boundary-value problem is

$$\gamma\phi''(x) = -\Lambda\phi(x) \qquad \text{on } (0,1), \qquad \phi(0) = \phi(1) = 0.$$

The principal mode is $\phi(x) = \sin(\pi x)$, with principal eigenvalue $\Lambda_1 = \gamma\pi^2$. Rescaling back to the original time variable gives the asymptotic decay rate.

Remark B.3 (Crossover to Dirichlet behaviour)*.* At small κ_0/γ, the convergence can be slow: since $1 - r(\kappa_0/\gamma) = O(\kappa_0/\gamma)$ as $\kappa_0/\gamma \to 0$, one has $\beta_R = O(R\kappa_0/\gamma)$. So the crossover to effectively Dirichlet behaviour occurs only when $(\kappa_0/\gamma)R \gg 1$; Robin corrections remain visible when $\kappa_0/\gamma \lesssim 1/R$. Appendix C returns to this point explicitly when reconciling the baseline with the Gibbs lift.

Theorem B.4 (Diffusive baseline asymptotics; Theorem IV.1 restated, formal)**.** *Fix $m \in \mathbb{Z}_{\geq 1}$ and $\kappa_0, \gamma > 0$. In regime (C),*

$$a(R) = \frac{\gamma\pi^2}{R^2} + o(R^{-2}) \qquad (R \to \infty).$$

Equivalently, $R^2 a(R) \to \gamma\pi^2$. The dependence on (m, κ_0) survives only in the subleading boundary-layer correction.

Remark B.5 (Why this is a control result)*.* The leading-order constant is independent of the threshold width and kill strength as long as both remain fixed while $R \to \infty$, because the microscopic Robin coefficient diverges under the macroscopic rescaling. Once $\beta_R \to \infty$, the interior only "sees" an effectively Dirichlet boundary, so the leading mode is the universal Dirichlet ground state of the interval. This is the control behaviour of a model whose dominant mechanism is diffusive first passage rather than large-deviation suppression: the baseline is asymptotically safe because reaching the hazardous strip takes diffusive time of order R^2, not because dangerous configurations are exponentially disfavoured.

B.5. Mixing and hazard are on the same scale

For the reflecting birth-death chain on a segment of length R, the mixing time satisfies

$$\tau_{\text{mix}}(R) \asymp \frac{R^2}{\gamma}.$$

The survival timescale set by the principal decay rate is

$$\frac{1}{a(R)} \sim \frac{R^2}{\gamma\pi^2}.$$

In the diffusive baseline there is no parametric separation between internal mixing and effective hazard — the two scales are of the same order.

This is not a quirk of parameters. In the uniform baseline, both mixing and killing are governed by the same interior Dirichlet-type Laplacian on the interval. Mixing is slow because diffusive transport across an $O(R)$ domain takes time R^2/γ; killing is slow for the same reason, because reaching the reactive boundary layer from a typical interior point also takes diffusive time. The degeneracy is structural: in the baseline, hazard-reaching is not rare by any mechanism distinct from ordinary interior mixing.

Any lift that breaks this degeneracy must make hazard-reaching rare by a different mechanism than interior equilibration. Two natural routes: a scaling lift (making the hazardous region occupy a nonvanishing fraction of the fiber, $m = \alpha R$) or a Gibbs lift (making dangerous configurations exponentially improbable under a nontrivial equilibrium measure). Appendix C works out the Gibbs lift in detail and shows it is the route that yields genuine thermodynamic suppression.

B.6. Numerical confirmation

For the parameter choice $\gamma = 1$, $\kappa_0 = 1$, $m = 2$, exact diagonalization of the symmetric tridiagonal matrix G_R for representative values of R gives the following convergence.

R	$a(R)$	$R^2 a(R)$
5	0.467911113762	11.697777844051
10	0.111528140454	11.152814045396
20	0.026328968771	10.531587508318
50	0.004053725995	10.134314988287
100	0.001000164338	10.001643382949
200	0.000248388011	9.935520441652
500	0.000039583772	9.895943047036

Convergence of $R^2 a(R)$ to $\pi^2 = 9.8696\ldots$ in the uniform baseline with $\gamma = \kappa_0 = 1$ and $m = 2$.

The limiting value is $\pi^2 = 9.869604401089\ldots$, matching the prediction of Theorem B.4 very closely already at $R = 100$.

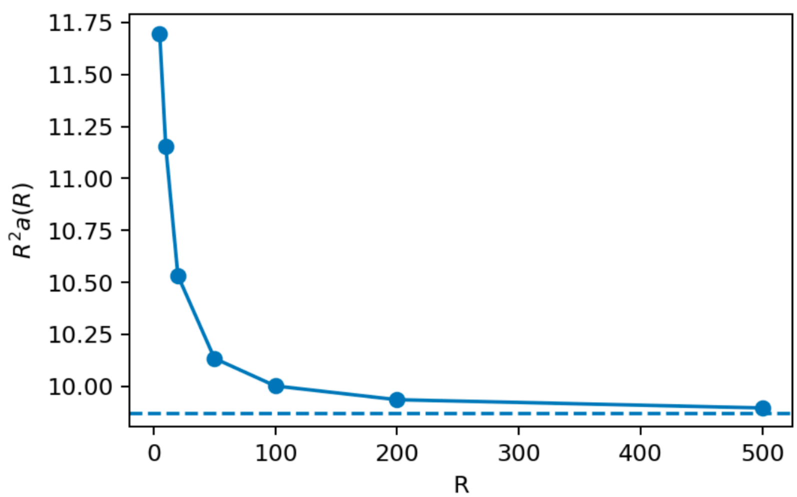

Figure 1: Convergence of $R^2 a(R)$ to π^2 in the uniform baseline with $\gamma = \kappa_0 = 1$ and $m = 2$. The asymptotic limit $\pi^2 = 9.8696\ldots$ (dashed) is approached as R grows: the residual at $R = 100$ is about 1.3% in the dimensionless ratio and falls to about 0.27% by $R = 500$, consistent with the $o(R^{-2})$ correction of Theorem B.4.

The quasi-stationary distribution behaves exactly as the continuum picture predicts. For $R = 50$ the QSD is sharply depressed near the hazardous strips and elevated in the interior. In particular

$$\mu^*_{50}(0) = 0.0006765, \qquad \mu^*_{50}(1) = 0.0013503, \qquad \mu^*_{50}(25) = 0.0317932,$$

whereas the uniform equilibrium law assigns $1/51 \approx 0.0196078$ to every state. The QSD shape is the discrete Dirichlet ground state, a sine-like bulge away from the boundary (Figure 2).

B.7. One-step vs spectral interface values

The baseline makes it possible to compute two distinct interface proxies explicitly.

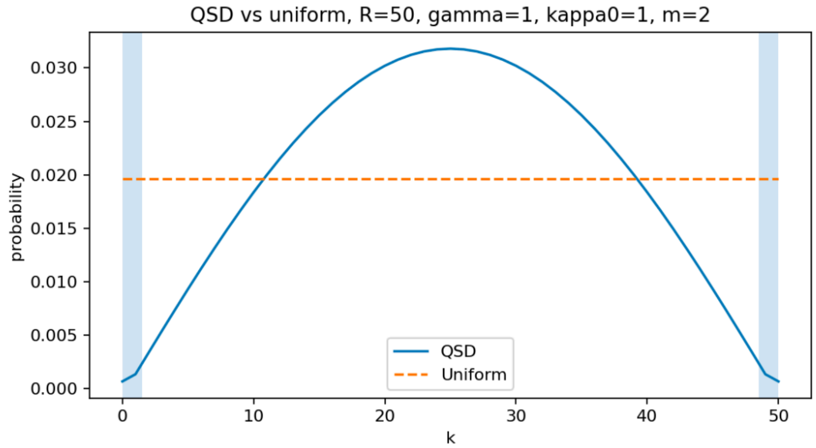

Figure 2: Quasi-stationary distribution for $R = 50$ in the uniform baseline with $\gamma = \kappa_0 = 1$ and $m = 2$. The QSD (solid) is suppressed near the two killing strips (shaded) and elevated in the safe interior, matching the discrete Dirichlet ground-state picture. The uniform equilibrium law (dashed) assigns $1/51$ to every state.

B.7.1. Stationary-fiber one-step proxy. For a fixed observation window $\Delta > 0$, define the frozen-hazard one-step proxy by

$$s_\Delta(k) := e^{-\kappa_R(k)\Delta}, \qquad c_\Delta^{\text{frozen}}(R) := \langle s_\Delta, \pi_R \rangle.$$

When $R \geq 2m$, this admits the exact closed form

$$c_\Delta^{\text{frozen}}(R) = \frac{R + 1 - 2m + 2m\, e^{-\kappa_0 \Delta}}{R + 1} = 1 - \frac{2m}{R+1}\big(1 - e^{-\kappa_0 \Delta}\big).$$

Hence

$$c_\Delta^{\text{frozen}}(R) \to 1 \qquad (R \to \infty,\ \Delta \text{ fixed}).$$

The one-step stationary-fiber interface value converges to perfect survival simply because the hazardous strip has vanishing stationary mass under the uniform measure.

B.7.2. Long-time spectral interface value. The spectral interface object is the principal decay rate $a(R)$. By Theorem B.4, $a(R) \sim \gamma\pi^2/R^2 \to 0$ as $R \to \infty$. The one-step and long-time interface values therefore tell different stories on fixed windows: $c_\Delta^{\text{frozen}}(R) \to 1$, while $a(R) \to 0$. Both say that large-R blocks are "safe," but for different reasons. The one-step proxy is safe because typical stationary configurations rarely lie in the killing strip; the spectral rate is safe because it takes diffusive time to reach that strip from the interior.

B.7.3. When do they agree? The two objects become comparable only on the spectral time scale $\Delta \sim 1/a(R) \sim R^2/\gamma$. Only on that scale does one expect a relation of the form $c_\Delta(R) \approx e^{-a(R)\Delta}$ to become meaningful. But this is exactly the scale at which mixing and killing both become order one in the baseline. There is no asymptotic window

$$\tau_{\text{mix}}(R) \ll \Delta \ll \frac{1}{a(R)}$$

because both endpoints are of the same order. This is the precise sense in which the baseline is degenerate from the point of view of the hazard-weighted mixing story of Part III: the frozen one-step interface proxy and the spectral decay rate are both legitimate objects, but the model does not create a clean parameter regime in which one approximates the other through pre-interface equilibration alone.

This timing mismatch is not an annoyance. It is the concrete control-model manifestation of Part III's warning that interface-value type depends on when compression happens. A finite-window stationary average and an asymptotic exponential rate are not interchangeable unless the stage structure and timescales make them interchangeable.

B.8. Relation to Appendix C

In the baseline of this appendix, the reactive strip carries $O(m/R)$ stationary mass and reaching it takes diffusive time $O(R^2)$. The next step — Appendix C — replaces the uniform fiber law by a reversible Gibbs measure $\pi_R(k) \propto e^{-\beta R h(k/R)}$ with h a symmetric one-well metastable potential minimized in the interior. The reactive strip then carries *exponentially small* Gibbs mass, and the principal decay rate acquires the leading Gibbs-lifted structure

$$\log a(R) = -\beta R \Delta h - \frac{1}{2} \log R + O_\beta(1),$$

with a sharper three-factor prefactor in the low-temperature/local-equilibrium regime. The baseline's R^{-2} Dirichlet scaling (Theorem B.4) is replaced, in the Gibbs lift, by an exponentially small large-deviation factor $e^{-\beta R \Delta h}$ together with a separate $R^{-1/2}$ Laplace/WKB prefactor; the two scalings arise from distinct physical mechanisms (continuum-diffusion boundary mode vs. well-normalization Laplace asymptotics) and should not be read as a continuous deformation of one into the other. The discrepancy between the one-step and spectral proxies remains, but it becomes exponentially sharp rather than polynomially sharp. Appendix C is the proof-home for Theorem IV.2, including the finite-β QSD prefactor qualification.

Appendix C. Gibbs Lift and the Three-Factor Asymptotic: Theorem IV.2 in Full

C.1. Opening

This appendix supplies the proof-home for Theorem IV.2 — the Gibbs-lifted three-scale asymptotic for the principal decay rate of the killed birth-death generator. The leading fixed-β statement is

$$\log a(R) = -\beta R \Delta h - \frac{1}{2} \log R + O_\beta(1),$$

and in the low-temperature/local-equilibrium metastable regime the sharper prefactor form is

$$a(R) \sim 2\, A_{\text{react}}^{\text{Gibbs}}(\kappa_0, \gamma, m; \beta, h'(0)) \cdot A_{\text{WKB}}(\beta, h, R) \cdot e^{-\beta R \Delta h}.$$

The body stated the three factors as local (reactivity), mesoscopic (transport), and macroscopic (concentration), and wrote out the explicit form of A_{WKB}; here we give the formal derivation and its numerical checks. The result is formal in the sense that it combines a capacity-over-mass approximation at the metastable-well scale with exact Laplace asymptotics and a finite local strip problem. The numerical checks below support the exponential barrier factor and the $R^{-1/2}$ WKB term at fixed β, while also showing a finite-β QSD boundary-layer correction to the unit prefactor.

The derivation proceeds in three stages. First, a structural strip-elimination argument reduces the killed generator on the full fiber to an effective killed process on the safe region, with killing concentrated at the two entrance sites; this produces a capacity-over-mass formula [Bov+04] together with a QSD/local-equilibrium qualification. Second, Laplace asymptotics and a one-dimensional conductance estimate extract the macroscopic barrier factor $e^{-\beta R \Delta h}$ and the endpoint Eyring-Kramers prefactor A_{WKB}. Third, the Gibbs-biased local strip problem supplies the microscopic factor $A_{\text{react}}^{\text{Gibbs}}$, for which we give the explicit closed form at $m = 2$. A numerical verification in the canonical test potential $h(x) = 2(x - 1/2)^2$ confirms the predicted log-linear suppression with slope $-\Delta h = -1/2$ and identifies the finite-β boundary-layer correction that vanishes in the low-temperature regime.

C.1.1. Capacity-over-mass primer

Readers not familiar with Bovier-style potential-theoretic metastability may find the “capacity-over-mass” language opaque; this primer gives a working translation. Readers fluent in the machinery can skip to §C.2.

Dirichlet form and variational principle. For a reversible continuous-time Markov chain on a finite state space, with rates $r(x,y)$ and stationary law π satisfying detailed balance $\pi(x)r(x,y) = \pi(y)r(y,x)$ (so the conductances $c(x,y) := \pi(x)r(x,y) = c(y,x)$ are symmetric, even though the raw rates need not be), the Dirichlet form is

$$\mathcal{E}(f) := \tfrac{1}{2} \sum_{x,y} \pi(x)\, r(x,y)\, \big(f(x) - f(y)\big)^2.$$

It is the quadratic form associated to the generator via $\mathcal{E}(f) = -\langle f, Lf \rangle_\pi$. When killing is added on a subset $\mathcal{K}$ of states (the “hazardous” set), the relevant modified form accounts for escape into $\mathcal{K}$. The principal decay rate $a = -\lambda_{\max}(G)$ of the killed generator admits a variational characterization: roughly,

$$a \asymp \frac{\text{cap}(\mathcal{K})}{Z_{\text{safe}}},$$

where $\text{cap}(\mathcal{K})$ is the *capacity* of the hazardous set relative to the safe region, and Z_{safe} is the effective mass of the metastable well. This is the “capacity over mass” formula.

Capacity: operational definition. The capacity $\text{cap}(\mathcal{K})$ is the minimum of $\mathcal{E}(f)$ over a specific class of test functions (those that are 1 on $\mathcal{K}$ and 0 on a reference metastable point in the safe region). It measures how expensive it is, in Dirichlet-form units, to have a function interpolate from 0 in the well to 1 at the hazardous boundary. In the discrete 1D setting of this appendix, this is the *effective conductance* of the boundary layer: a sum of reciprocal edge resistances, weighted by Gibbs factors. An explicit formula is worked out in Theorem C.5.

Mass. The mass Z_{safe} is the integrated Gibbs weight over the safe region, $Z_{\text{safe}} = \sum_{k\in\mathcal{W}} e^{-\beta R h(k/R)}$. By Laplace's method, $Z_{\text{safe}} \sim e^{-\beta R h_{\min}} \sqrt{2\pi/(\beta R h''_{\min})}$; this is where the $R^{-1/2}$ in A_{WKB} enters.

QSD and the principal right eigenfunction. For a reversible killed chain, let ϕ_1 denote the positive right eigenfunction of the killed generator G associated with the largest (least-negative) eigenvalue $-a_1$. The *quasi-stationary distribution* (QSD) — the distribution on the safe region invariant under the killed dynamics conditioned on survival — is then

$$\nu(x) = \frac{\phi_1(x)\,\pi(x)}{\sum_y \phi_1(y)\,\pi(y)},$$

where π is the reversible reference measure (the Gibbs law π_R in the lifted setting of §C.2 below). In the symmetric representation $S = DGD^{-1}$ used throughout this appendix — with $D = \operatorname{diag}\big(\sqrt{\pi(x)}\big)$, so S is a symmetric tridiagonal matrix — if $\widetilde{\phi}_1$ is the principal eigenvector of S, then $\phi_1(x) = \widetilde{\phi}_1(x)/\sqrt{\pi(x)}$, and the QSD has mass proportional to $\widetilde{\phi}_1(x)\sqrt{\pi(x)}$, equivalently $\phi_1(x)\pi(x)$.

For a reader arriving from the rationality/economics side: the QSD ν is "the shape of the block's state once it has survived long enough that its initial condition no longer matters." The decay rate a_1 is the rate at which probability mass leaks out of the safe region from this quasi-stationary shape. In the uniform-baseline setting of Appendix B, where π is uniform and the generator is already symmetric, the distinction between ϕ_1 (right eigenfunction) and ν (probability measure) collapses — they differ only by a normalization. In the Gibbs-lifted setting of this appendix, the distinction is non-trivial: ν is biased by π_R relative to ϕ_1, and conflating the two would produce an incorrect QSD.

What is and is not rigorous. The Dirichlet form characterization of a, the Laplace asymptotics for Z_{safe}, and the discrete-to-continuous conductance estimate are standard and rigorous [Bov+04; BH15]. The "capacity over mass" formula as a sharp *asymptotic identity* $a \sim \text{cap}/Z_{\text{safe}}$, with controlled unit prefactor, is a low-temperature/local-equilibrium statement: it requires the QSD on the safe region to approach the local Gibbs equilibrium near the strip entrance. At fixed β with $R \to \infty$, the identity still gives the leading exponential barrier and polynomial-in-R structure, but the prefactor can carry a β-dependent boundary-layer correction coming from the QSD shape adjacent to the killing strip. It is this qualified capacity-over-mass input that gives Theorem IV.2 its formal status. Everything downstream — Theorem V.1, Theorem V.3, Theorem G.10 — inherits this single point of conditionality, but only through the scaling/window structure used to build scalar compression. No other step in the monograph depends on metastability folklore; all other arguments are rigorous given the qualified capacity-over-mass input. This is the single "trust the metastability literature" bet in the paper, and it is concentrated in Theorem C.1 below.

C.2. Setup

Fix $R \in \mathbb{Z}_{\geq 0}$, threshold width $m \in \mathbb{Z}_{\geq 1}$, exchange rate $\gamma > 0$, killing rate $\kappa_0 > 0$, inverse temperature $\beta > 0$, and a smooth potential $h : [0,1] \to \mathbb{R}$. The one-well conditions are:

(i) $h \in C^2([0,1])$;

(ii) h has a unique minimum at $x_* = 1/2$;

(iii) h is strictly decreasing on $[0, 1/2]$ and strictly increasing on $[1/2, 1]$;

(iv) $h(0) = h(1) > h(1/2)$;

(v) h is symmetric: $h(x) = h(1-x)$ for all $x \in [0,1]$.

Set $\Delta h := h(0) - h(1/2) > 0$. The fiber is $\Omega_R = \{0, 1, \ldots, R\}$ as in Appendix B. Symmetry (v) is used to equate the left and right boundary channels in Theorem C.1; the asymmetric case requires distinct reactive factors $A^{\text{Gibbs}}_{\text{react},L}$, $A^{\text{Gibbs}}_{\text{react},R}$ and endpoint slopes $h'(0)$, $h'(1)$ but the derivation is otherwise unchanged.

C.2.1. Reversible Gibbs dynamics. Define the reversible nearest-neighbor rates

$$r_k^+ := \gamma \exp\Big(-\frac{\beta R}{2}\big(h((k+1)/R) - h(k/R)\big)\Big), \qquad 0 \le k \le R-1,$$

$$r_k^- := \gamma \exp\Big(-\frac{\beta R}{2}\big(h((k-1)/R) - h(k/R)\big)\Big), \qquad 1 \le k \le R,$$

with $r_R^+ = 0$, $r_0^- = 0$. The generator is $(L_R f)(k) = r_k^+[f(k+1)-f(k)] + r_k^-[f(k-1)-f(k)]$. The Gibbs stationary law

$$\pi_R(k) := Z_R^{-1}\, e^{-\beta R h(k/R)}, \qquad Z_R := \sum_{j=0}^{R} e^{-\beta R h(j/R)},$$

satisfies detailed balance: $\pi_R(k) r_k^+ = \pi_R(k+1) r_{k+1}^-$.

C.2.2. Threshold killing and symmetric tridiagonal form. The killing potential is the same as in Appendix B,

$$\kappa_R(k) = \kappa_0\, \mathbf{1}_{\{k<m\}} + \kappa_0\, \mathbf{1}_{\{R-k<m\}},$$

and the killed generator is $G_R := L_R - \mathrm{diag}(\kappa_R)$, with top eigenvalue $\lambda_{\max}(R) \le 0$ and principal decay rate $a(R) := -\lambda_{\max}(R)$. Because L_R is reversible, G_R is similar to a symmetric tridiagonal matrix: with $D_R := \mathrm{diag}(\sqrt{\pi_R})$,

$$S_R := D_R G_R D_R^{-1}$$

is symmetric, with constant off-diagonal entries $(S_R)_{k,k+1} = \sqrt{r_k^+ r_{k+1}^-} = \gamma$ and diagonal $(S_R)_{kk} = -r_k^+ - r_k^- - \kappa_R(k)$. All numerical calculations below are performed in this symmetric representation. (We reserve plain S_R for this symmetric tridiagonal matrix; the right reactive strip below is written with calligraphic $\mathcal{S}_R$ to avoid notational collision.)

C.3. Strip elimination and the two-channel capacity formula

Let $\mathcal{W} := \{m, m+1, \dots, R-m\}$ be the safe region, and $\mathcal{S}_L := \{0, \dots, m-1\}$, $\mathcal{S}_R := \{R-m+1, \dots, R\}$ the left and right reactive strips. Let $p_L^{(R)}$ be the probability that a trajectory started at $m-1$ is killed before first returning to m, and $p_R^{(R)}$ the analogous probability for the right strip.

Proposition C.1 (Strip elimination; two-channel leakage)**.** *Let $\nu_{\mathcal{W}}^{(R)}$ denote the QSD of the strip-eliminated process restricted to the safe region $\mathcal{W}$. In the metastable one-well regime, the principal decay rate satisfies the effective entrance-leakage formula*

$$a(R) \approx \nu_{\mathcal{W}}^{(R)}(m)\, r_m^-\, p_L^{(R)} + \nu_{\mathcal{W}}^{(R)}(R-m)\, r_{R-m}^+\, p_R^{(R)}.$$

Equivalently, setting $c_{m-1,m}(R) := \pi_R(m) r_m^- = \pi_R(m-1) r_{m-1}^+$ and defining the left/right QSD-local-equilibrium ratios

$$\theta_L^{(R)} := \frac{\nu_{\mathcal{W}}^{(R)}(m)}{\pi_R(m)/\pi_R(\mathcal{W})}, \qquad \theta_R^{(R)} := \frac{\nu_{\mathcal{W}}^{(R)}(R-m)}{\pi_R(R-m)/\pi_R(\mathcal{W})},$$

the symmetric setting gives

$$a(R) \approx \big(\theta_L^{(R)} p_L^{(R)} + \theta_R^{(R)} p_R^{(R)}\big) c_{m-1,m}(R).$$

If the QSD is locally equilibrated at the two strip entrances, so that $\theta_L^{(R)}, \theta_R^{(R)} \to 1$, and if $p_L^{(R)}, p_R^{(R)} \to A_{\text{react}}^{\text{Gibbs}}(\kappa_0, \gamma, m; \beta, h'(0))$, this reduces to the usual two-channel capacity-over-mass expression

$$a(R) \approx 2\, A_{\text{react}}^{\text{Gibbs}}(\kappa_0, \gamma, m; \beta, h'(0)) \cdot c_{m-1,m}(R).$$

At fixed β, the ratios $\theta_{L/R}^{(R)}$ need not converge to 1 as $R \to \infty$; the finite-β QSD boundary layer is the source of the prefactor qualification in Theorem IV.2.

Proof. The standard metastable capacity-over-mass argument, specialized to a one-dimensional killed chain. First eliminate the reactive strips. After elimination, the process on the safe region $\mathcal{W}$ is conservative in the interior but acquires killing at the entrance sites m and $R-m$:

$$m \longrightarrow \text{death at rate } r_m^- p_L^{(R)}, \qquad R-m \longrightarrow \text{death at rate } r_{R-m}^+ p_R^{(R)}.$$

Averaging these entrance killing rates against the QSD of the eliminated safe-region dynamics gives the first displayed formula. Rewriting the QSD entrance masses as local Gibbs-equilibrium masses multiplied by $\theta_L^{(R)}$ and $\theta_R^{(R)}$ gives the second formula. Because h is symmetric about $1/2$, the two entrance conductances coincide: $\pi_R(m)r_m^- = \pi_R(R-m)r_{R-m}^+ = c_{m-1,m}(R)$, and $\pi_R(\mathcal{W}) = 1 + o(1)$. In the low-temperature/local-equilibrium regime the QSD entrance ratios tend to 1, recovering the standard two-channel capacity-over-mass expression. The numerical diagnostics in §C.6.1 show that at fixed β the ratios can instead plateau below 1, producing a β-dependent multiplicative correction to the unit prefactor. □

Remark C.2 (QSD boundary-layer correction)*.* The local-equilibrium replacement $\nu_{\mathcal{W}}^{(R)}(k) \approx \pi_R(k)/\pi_R(\mathcal{W})$ is accurate in the bulk of the metastable well but is most delicate at the safe-side entrance to the killing strip. At fixed β, the QSD is depleted at the entrance because those states have direct access to killing; numerically this depletion persists as R grows. The correction is multiplicative and β-dependent, and it decreases rapidly as β increases. Thus the three-factor expression is prefactor-sharp in the low-temperature/local-equilibrium regime, while at fixed β it should be read as giving the correct barrier factor and WKB scaling up to the QSD boundary-layer factor.

Remark C.3 (Why the baseline reactive factor is not the right Gibbs object)*.* In the Gibbs lift the strip does *not* remain neutral. Detailed balance with the equilibrium measure biases the local strip rates, so the correct reactive factor is $A_{\text{react}}^{\text{Gibbs}}$, not the unbiased Appendix B factor. This is a structural lesson: whenever a killed-process construction is lifted from a uniform baseline to a Gibbs or otherwise symmetry-broken setting, the local boundary physics inherits that bias even though the strip itself is microscopic.

C.4. WKB derivation: Laplace, conductance, three-factor

Theorem C.1 isolates the local factor. The next two results extract the macroscopic and mesoscopic pieces of the entrance-bond conductance $c_{m-1,m}(R)$.

Lemma C.4 (Well normalization)**.** *As $R \to \infty$,*

$$Z_R \sim e^{-\beta R h(1/2)} \sqrt{\frac{2\pi R}{\beta h''(1/2)}}.$$

Equivalently, $\pi_R(k_*) \asymp \sqrt{\beta h''(1/2)/(2\pi R)}$ for $k_* \approx R/2$.

Proof. The minimum of h is nondegenerate and unique at $1/2$; the discrete sum is a Riemann-sum Laplace-method integral,

$$Z_R = \sum_{j=0}^{R} e^{-\beta R h(j/R)} \sim R \int_0^1 e^{-\beta R h(x)}\, dx \sim e^{-\beta R h(1/2)} \sqrt{\frac{2\pi R}{\beta h''(1/2)}}.$$

□

Proposition C.5 (Endpoint conductance asymptotic)**.** *For fixed m,*

$$c_{m-1,m}(R) \sim \gamma \exp\left(-\frac{\beta(2m-1)}{2} h'(0)\right) \sqrt{\frac{\beta h''(1/2)}{2\pi R}}\, e^{-\beta R \Delta h}.$$

Hence the WKB factor is

$$A_{\text{WKB}}(\beta, h, R) := \gamma \exp\left(-\frac{\beta(2m-1)}{2} h'(0)\right) \sqrt{\frac{\beta h''(1/2)}{2\pi R}}.$$

Proof. Write the entrance-bond conductance in terms of the Gibbs weight and the detailed-balance rate:

$$c_{m-1,m}(R) = \pi_R(m)\, r_m^- = \frac{\gamma}{Z_R} \exp\left(-\frac{\beta R}{2}\left(h(m/R) + h((m-1)/R)\right)\right).$$

Since m is fixed as $R \to \infty$, expand near the endpoint:

$$\frac{R}{2}\big(h(m/R) + h((m-1)/R)\big) = Rh(0) + \frac{2m-1}{2}h'(0) + O(R^{-1}).$$

Substituting this into the prefactor's exponential contributes $-\beta Rh(0)$ at leading order. From Theorem C.4, $1/Z_R$ contributes $+\beta Rh(1/2)$ at leading order (the Laplace-asymptotic well normalization concentrates at the minimum $x_* = 1/2$). These two combine into a single macroscopic exponential:

$$-\beta Rh(0) + \beta Rh(1/2) \;=\; -\beta R\big(h(0) - h(1/2)\big) \;=\; -\beta R\Delta h,$$

which is the extensive Gibbs concentration factor. Collecting the remaining finite pieces, the full expression is

$$\exp\Big(-\beta R\Delta h - \tfrac{\beta(2m-1)}{2}h'(0) + o(1)\Big),$$

giving the stated formula. □

Remark C.6 (Endpoint versus saddle)*.* The prefactor involves the endpoint slope $h'(0)$ rather than an interior-saddle curvature $|h''(x_s)|$ because the escape set is the boundary layer near $x = 0$ and $x = 1$, not an interior saddle. This is the correct endpoint-exit analogue of Eyring-Kramers in the present one-dimensional geometry.

Theorem C.7 (Three-factor asymptotic with QSD-prefactor qualification; Theorem IV.2 restated)**.** *Assume the one-well conditions and fix m, κ_0, γ, β. Then, formally, the principal decay rate has leading fixed-β structure*

$$\log a(R) = -\beta R\Delta h - \frac{1}{2}\log R + O_\beta(1).$$

In the low-temperature/local-equilibrium regime in which the QSD entrance ratios of Theorem C.1 tend to 1, this sharpens to

$$a(R) \sim 2\,A^{\text{Gibbs}}_{\text{react}}(\kappa_0, \gamma, m; \beta, h'(0)) \cdot A_{\text{WKB}}(\beta, h, R) \cdot e^{-\beta R\Delta h}.$$

At fixed β the same product may be multiplied by a β-dependent QSD boundary-layer factor; the downstream compression-window arguments use the leading exponential and $R^{-1/2}$ structure, not the unit-prefactor sharpness.

Derivation. Combine Theorem C.1 with Theorem C.5. The entrance-bond conductance carries both the macroscopic exponential $e^{-\beta R\Delta h}$ and the endpoint WKB prefactor $R^{-1/2}$. The local probability of death after entering the strip converges to an R-independent Gibbs-biased constant $A^{\text{Gibbs}}_{\text{react}}$, made explicit below. If the QSD entrance ratios in Theorem C.1 tend to 1, the two boundary channels contribute the stated factor of 2 and the prefactor is sharp. If they instead plateau at a fixed-β value below 1, the same derivation gives the displayed exponential/WKB structure with the corresponding multiplicative QSD correction. □

C.5. The Gibbs-biased reactive factor

$A^{\text{Gibbs}}_{\text{react}}$ must be recomputed from scratch in the Gibbs lift: detailed balance with the Gibbs measure biases the local strip rates, so the microscopic death problem already "knows" about the endpoint slope of h.

Proposition C.8 (Endpoint Gibbs bias in the strip)**.** *Fix $m = 2$. Near the left endpoint, the local strip rates converge to*

$$a_+ := \gamma e^{-\beta h'(0)/2}, \qquad a_- := \gamma e^{+\beta h'(0)/2},$$

where a_+ is the inward rate (toward the safe region) and a_- is the outward rate. Since $h'(0) < 0$ in the one-well setting, $a_+ > a_-$.

Proof. For fixed strip index $j = O(1)$,

$$h((j+1)/R) - h(j/R) = \frac{h'(0)}{R} + O(R^{-2}),$$

so $r_j^+ = \gamma \exp(-\beta h'(0)/2 + O(R^{-1}))$. Similarly $h((j-1)/R) - h(j/R) = -h'(0)/R + O(R^{-2})$ gives $r_j^- = \gamma \exp(+\beta h'(0)/2 + O(R^{-1}))$. □

Proposition C.9 (Explicit Gibbs-biased strip factor for $m = 2$)**.** *Let p_0 and p_1 denote the probabilities of being killed before first return to the safe site 2, starting from strip states 0 and 1. In the large-R limit they satisfy*

$$(a_+ + \kappa_0)p_0 = \kappa_0 + a_+ p_1, \qquad (a_- + a_+ + \kappa_0)p_1 = \kappa_0 + a_- p_0.$$

Hence

$$A^{\text{Gibbs}}_{\text{react}}(\kappa_0, \gamma, 2; \beta, h'(0)) = p_1 = \frac{\kappa_0(a_+ + a_- + \kappa_0)}{a_+^2 + 2a_+\kappa_0 + a_-\kappa_0 + \kappa_0^2}.$$

When $h'(0) \to 0$ (so $a_+ = a_- = \gamma$), this reduces to the unbiased Appendix B reactive factor.

Proof. Condition on the first event from each strip state. From state 0, the process is either killed at rate κ_0 or moves inward at rate a_+, yielding $(a_+ + \kappa_0)p_0 = \kappa_0 + a_+ p_1$. From state 1, the process is either killed at rate κ_0, moves outward at rate a_-, or moves inward to the safe site 2 at rate a_+; since $p_2 = 0$, this gives $(a_- + a_+ + \kappa_0)p_1 = \kappa_0 + a_- p_0$. Solving the linear system for p_1 yields the closed form. Setting $a_+ = a_- = \gamma$ recovers the uniform-baseline formula. □

Remark C.10*.* For general fixed m, $A^{\text{Gibbs}}_{\text{react}}$ is obtained by solving the corresponding finite biased strip problem on $\{0, 1, \dots, m\}$. The essential point for the asymptotic is not the closed form but the structure: $A^{\text{Gibbs}}_{\text{react}}$ is local, R-independent, and carries Gibbs information through detailed balance at the strip scale.

C.6. Numerical verification

We test the canonical convex potential $h(x) = 2(x - 1/2)^2$, for which $\Delta h = 1/2$, $h''(1/2) = 4$, $h'(0) = -2$. Unless noted, we use $\gamma = \kappa_0 = 1$ and $m = 2$. The symmetrized tridiagonal form S_R (Section C.2.2) allows exact diagonalization for moderate βR; the eigenvalue becomes exponentially small and eventually falls below double-precision absolute accuracy, so we report ranges on which direct diagonalization remains trustworthy.

C.6.1. Fixed-β structural ratio and QSD boundary layer. The cleanest structural diagnostic is

$$\widetilde{\Xi}_R := \frac{a(R)}{2\, A^{\text{Gibbs}}_{\text{react},R}\, c_{m-1,m}(R)},$$

where $A^{\text{Gibbs}}_{\text{react},R}$ is computed from the exact finite-R local strip equations and $c_{m-1,m}(R)$ is the exact entrance-bond conductance. The unqualified local-equilibrium reading of Theorem C.1 would predict $\widetilde{\Xi}_R \to 1$ as $R \to \infty$ at fixed β. Direct computation shows a more precise picture: at fixed β, $\widetilde{\Xi}_R$ approaches a β-dependent plateau, and those plateaus move rapidly toward 1 as β increases.

β	$R = 40$	$R = 80$	$R = 160$	$R = 200$
0.5	0.7568	0.7811	0.7915	0.7929
1.0	0.9418	0.9500	0.9536	0.9543
2.0	0.9961	0.9970	0.9974	0.9975

Fixed-β structural ratio $\widetilde{\Xi}_R$. Increasing R alone does not drive the ratio to 1 at low or moderate β; instead the ratio approaches a β-dependent QSD boundary-layer plateau. Increasing β drives that plateau toward 1.

The source of the deficit is visible by comparing the actual QSD entrance mass with the local Gibbs equilibrium mass. For $m = 2$, the strip is $\{0, 1\}$ and the safe-side entrance is $k = 2$. The ratio

$$\frac{\nu^{(R)}_{\mathcal{W}}(2)}{\pi_R(2)/\pi_R(\mathcal{W})}$$

matches $\widetilde{\Xi}_R$ to the displayed accuracy:

β	$R = 40$	$R = 80$	$R = 160$
0.5	0.7568	0.7811	0.7915
1.0	0.9417	0.9500	0.9536
2.0	0.9961	0.9970	0.9974

QSD entrance mass divided by local Gibbs equilibrium entrance mass. The same ratios explain the structural-ratio deficit, confirming that the discrepancy is a QSD boundary-layer effect rather than a failure of the barrier or WKB scaling.

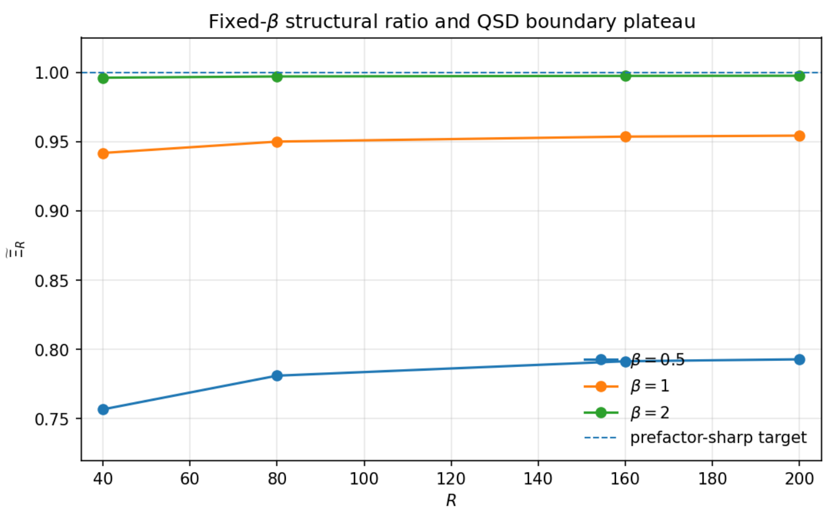

Figure 3: Fixed-β structural ratio $\widetilde{\Xi}_R = a(R)/\big(2\,A^{\mathrm{Gibbs}}_{\mathrm{react},R}\,c_{m-1,m}(R)\big)$ for $\beta \in \{0.5, 1, 2\}$. The ratio approaches a β-dependent plateau as R grows; the plateau itself tends to 1 in the low-temperature direction. Thus the three-factor expression is prefactor-sharp in the local-equilibrium/low-temperature regime, while at fixed β it carries the QSD boundary-layer correction of Theorem C.2.

This revised diagnostic should be read together with the log-linear check below. The boundary-layer correction changes the unit prefactor; it does not change the exponential slope $-\Delta h$ or the $R^{-1/2}$ WKB normalization.

C.6.2. Log-linear suppression. The three-factor formula predicts that $\log a(R)$ should be linear in βR with slope $-\Delta h = -1/2$ (after subtracting the polynomial WKB correction). A least-squares fit of $\log a(R)$ versus βR at $\beta \in \{1, 2, 5\}$, using only the numerically resolved ranges, gives:

β	resolved range of R	fitted slope
1	$10, 15, 20, 25, 30, 35, 40, 45, 50$	-0.513
2	$10, 12, 15, 18, 20, 25, 30$	-0.505
5	$5, 7, 10, 12, 15$	-0.483

Fitted slopes of $\log a(R)$ versus βR. The asymptotic target is $-\Delta h = -0.5$; the fitted values sit very close to this target already at modest βR, even before the polynomial WKB correction has fully settled.

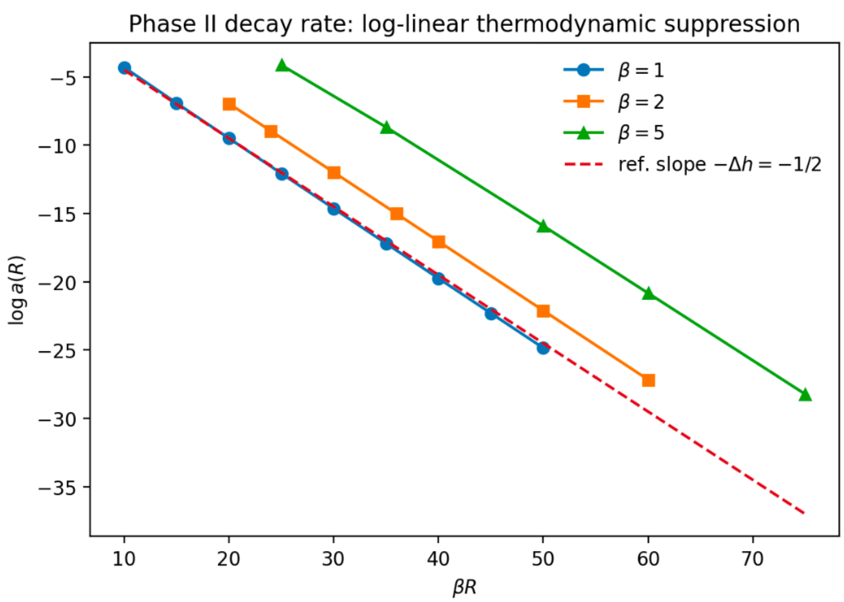

Figure 4: Log-linear thermodynamic suppression of the principal decay rate in the Gibbs-lifted killed-process class. Across all three values of $\beta \in \{1, 2, 5\}$ and the numerically resolved range of R, $\log a(R)$ is approximately linear in βR with slope close to $-\Delta h = -1/2$. The dashed line marks the reference slope $-\Delta h = -1/2$ — the asymptotic slope predicted by Theorem C.1, not a fitted prefactor-normalized prediction; fitted slopes are listed in the preceding table.

A qualitative check confirms the mechanism: at $(\beta, R, m) = (1, 50, 2)$, the quasi-stationary distribution is visually close to the Gibbs law π_R on the bulk of the fiber, but is systematically depleted at the safe-side entrance to the killing strip. This boundary-adjacent depletion is negligible in the low-temperature limit but persists at fixed β, exactly as the structural-ratio table indicates.

C.7. Relation to Appendices B and F

Appendix B exhibited the same killed-process class with uniform fiber law and found $a(R) \sim \gamma\pi^2/R^2$, with mixing and killing both $\Theta(R^2/\gamma)$. In the Gibbs lift of this appendix, the reactive strip carries exponentially small Gibbs mass rather than $O(m/R)$ uniform mass, and $a(R)$ becomes exponentially small with the three-factor structure of Theorem C.7. The two regimes are qualitatively different: baseline protection is diffusive (polynomial in R), Gibbs protection is large-deviation (exponential in R with a polynomial prefactor). The $R^{-1/2}$ factor inside A_{WKB} comes from Laplace asymptotics on the well normalization Z_R, not from a Dirichlet boundary mode; it is a vestige of the underlying continuum diffusion, but the scaling is different from the baseline's R^{-2}.

Appendix F takes the static picture of this appendix and upgrades it to slow-fast co-evolution, where a coarse state z evolves on the hazard timescale while the within-fiber Gibbs-weighted allocation continues to mix rapidly. The averaged coarse process's principal decay rate is then the matrix object diag$(a_1(z))$ in the Feynman-Kac semigroup, with $a_1(z)$ supplied locally by Theorem C.7 at each value of z.

Appendix D. Static Synthesis: Theorem V.1 in Full

D.1. Opening

This appendix supplies the proof-home for Theorem V.1: on a compressed block class of two independent Gibbs-lifted killed-process blocks, any outer evaluator satisfying (A1)–(A5) of Theorem II.1 — with (A5) interpreted through the block-level scalar temporal summaries $c_i(\Delta')$ supplied by spectral compression — is forced to take the weighted geometric form $A_w(c_1, c_2) = c_1^{w_1} c_2^{w_2}$. Spectral compression's role is not to derive (A5) from (A1)–(A4) — the arithmetic-mean counterexample of Theorem A.7 rules that out — but to make (A5) operationally verifiable in the present class by supplying scalar temporal summaries that compose multiplicatively at the block level. The body of Part V stated the theorem and sketched the logic; here we give the full consistency verification of the induced geometric evaluator, the principal-mode normalization of the compressed class that fixes the relative-error representation, the spectral-window analysis that justifies calling compression a thermodynamic rather than kinematic property, and the log-scale synthesis corollary that supplies the body's log-rate coordinate (Section V.4).

The derivation proceeds in three steps. First, a metastable-window argument exhibits a two-sided cycle scale Δ for the Gibbs-lifted class of Appendix C: an upper edge set by the exponentially large killing time $\tau_{\text{kill}} \asymp 1/a_1$ (three-factor suppression of Theorem C.7) and a lower edge set by the polynomial compression time $\tau_{\text{compress}} \asymp 1/(a_2 - a_1)$. The window is exponentially wide in R because the two timescales are supplied by distinct mechanisms: a_1 by large-deviation endpoint escape, a_2 by harmonic relaxation within the well. Second, the compressed block class is the class of two-block configurations whose continuation over every admissible sub-interval inside this window is represented by the principal scalar $e^{-a_1^{(i)}\Delta'}$ up to a uniform relative error. Third, the induced geometric evaluator is verified to satisfy all five axioms on this class, establishing non-vacuous realization of Theorem II.1 in the present setting.

D.2. Setup: two Gibbs-lifted blocks

Fix two blocks indexed by $i \in \{1, 2\}$. Each block is a fixed-fiber system of the Appendix C type, with its own parameters: total resource $R^{(i)} \in \mathbb{Z}_{\geq 0}$, threshold width $m^{(i)} \in \mathbb{Z}_{\geq 1}$, exchange rate $\gamma^{(i)} > 0$, killing rate $\kappa_0^{(i)} > 0$, inverse temperature $\beta^{(i)} > 0$, and a symmetric one-well potential $h^{(i)} : [0, 1] \to \mathbb{R}$ satisfying conditions (i)–(v) of §C.2. Write

$$G^{(i)} := L^{(i)} - \text{diag}(\kappa^{(i)})$$

for the killed generator of block i, with Gibbs stationary law $\pi^{(i)}_{R^{(i)}}(k) \propto e^{-\beta^{(i)} R^{(i)} h^{(i)}(k/R^{(i)})}$ and the usual strip killing potential. Order the decay rates of $G^{(i)}$ as

$$0 < a_1^{(i)} < a_2^{(i)} \leq a_3^{(i)} \leq \cdots,$$

so that $-a_j^{(i)}$ are the eigenvalues of $G^{(i)}$ arranged from closest to zero downward. The two blocks are *externally non-transferable*: after block formation there are no admissible dynamics moving viability-relevant resource from block 1 to block 2 or vice versa. An *exogenous outer schedule* is a weight vector $w = (w_1, w_2)$ with $w_1, w_2 > 0$ and $w_1 + w_2 = 1$, treated as an input to the outer evaluator rather than derived from block thermodynamics. The reading of w as an exposure fraction is operationally consistent but not required; nothing in the appendix pins the weights uniquely to fractions of block runtime.

Remark D.1 (Evaluator vs. conjunction survival). The present appendix is about realizing a normalized outer evaluator on compressed block values, not about the literal conjunction-survival law of a joint system. If one asked for the survival probability of "both blocks survive," the natural object would be the unnormalized product $c_1 c_2$, which is not idempotent. The weighted geometric mean appears here because the outer question is evaluator-valued and comes with exogenous weights w satisfying $w_1 + w_2 = 1$.

D.3. Spectral compression and the metastable window

For a finite killed reversible chain, survival profiles admit a spectral decomposition in the symmetrized form $S^{(i)} := D^{(i)}G^{(i)}(D^{(i)})^{-1}$ where $D^{(i)} := \mathrm{diag}(\sqrt{\pi^{(i)}})$ (Appendix C, §C.2.2). Any initial profile evolves as a sum of exponentials $e^{-a_j^{(i)}t}$; the principal mode controls long-enough times. This motivates the following exact definition.

Definition D.2 (Spectral compression on the outer cycle)**.** Block i is *spectrally compressed* on the cycle timescale Δ if $(a_2^{(i)} - a_1^{(i)})\Delta \gg 1$, equivalently $\exp(-(a_2^{(i)} - a_1^{(i)})\Delta) \ll 1$. Define the compression timescale $\Delta_{\text{comp}}^{(i)} := C/(a_2^{(i)} - a_1^{(i)})$ for a fixed large constant C (chosen so that $e^{-C} \ll 1$). The compression is *uniform on all admissible sub-intervals* if $(a_2^{(i)} - a_1^{(i)})\Delta' \gg 1$ holds for every $\Delta' \in [\Delta_{\text{comp}}^{(i)}, \Delta]$.

The uniform-on-sub-intervals form is what the axiom verification will require (temporal interface sufficiency needs the factorization to hold on any temporal split, not just on the full cycle). In that regime the block is dominated by its principal killed mode, and admits an effective scalar interface value $c_i(\Delta) \approx e^{-a_1^{(i)}\Delta}$ with approximation error controlled below.

D.3.1. Harmonic-well scaling for a_2. For the Gibbs lift of Appendix C, the principal rate a_1 is exponentially small in R because escape requires endpoint activation (Theorem C.7). The second rate a_2 is different: it measures relaxation *within* the well, not escape from it. Near the minimum $x_* = 1/2$, expand

$$h(x) = h(x_*) + \tfrac{1}{2}h''(x_*)(x - x_*)^2 + \cdots .$$

The Gibbs well has width of order $|k - R/2| \sim \sqrt{R/(\beta h''(1/2))}$ in the occupation coordinate k. On that scale, the fast redistribution dynamics is approximated by a discrete Ornstein–Uhlenbeck process with restoring curvature $\beta\gamma h''(1/2)/R$, suggesting the scaling

$$a_2(R) \asymp \frac{\gamma\beta h''(1/2)}{R}$$

up to a numerical constant depending on the local well approximation. A full asymptotic theorem for a_2 is deferred.

D.3.2. Numerical verification. For the canonical test potential $h(x) = 2(x - 1/2)^2$ with $h''(1/2) = 4$, $m = 2$, $\gamma = \kappa_0 = 1$, direct diagonalization of $S^{(i)}$ for $\beta \in \{1, 2\}$ shows that $a_2(R)R$ stabilizes rapidly near $4\beta\gamma$, matching the harmonic-well prediction $\gamma\beta h''(1/2) = 4\beta\gamma$ exactly (Figure 5).

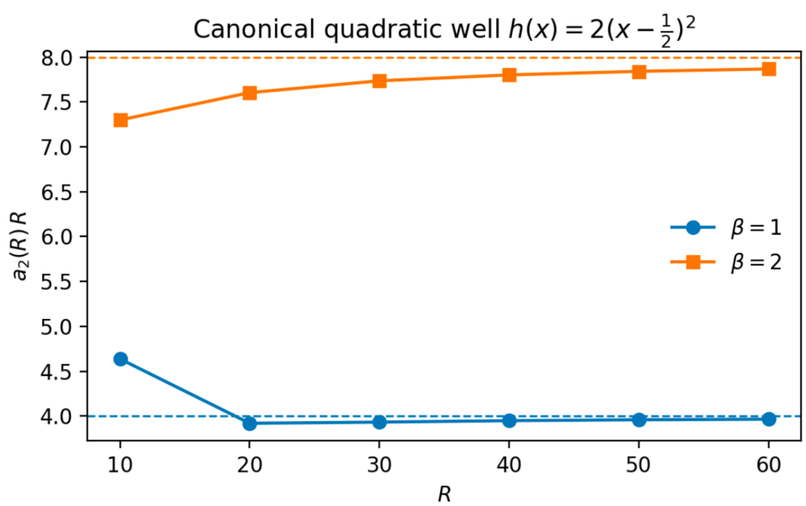

Figure 5: Numerical check of the second decay rate in the canonical quadratic well. For $h(x) = 2(x - 1/2)^2$, $m = 2$, $\gamma = 1$, $\kappa_0 = 1$, direct diagonalization of $S^{(i)}$ gives $a_2(R)R \to 4\beta\gamma$ for $\beta \in \{1, 2\}$, supporting the harmonic-well scaling $a_2(R) \asymp \gamma\beta h''(1/2)/R$ used throughout Theorem D.4. The dashed lines mark the asymptotic targets 4 (for $\beta = 1$) and 8 (for $\beta = 2$).

D.3.3. The metastable window. Combining Theorem C.7 for a_1 with the harmonic-well scaling for a_2 (heuristic, numerically supported, not yet a proved asymptotic) yields, in the canonical harmonic-well regime, the two-sided cycle window

$$\tau_{\text{compress}} \asymp \frac{1}{a_2 - a_1} \asymp \frac{R}{\gamma\beta h''(1/2)}, \qquad \tau_{\text{kill}} \asymp \frac{1}{a_1} \asymp \sqrt{R}\, e^{\beta R \Delta h}$$

(the $\sqrt{R}$ prefactor in τ_{kill} comes from A_{WKB}, the $1/R$ factor in τ_{compress} from the Ornstein–Uhlenbeck restoring curvature). The lower edge is polynomial in R; the upper edge is exponentially large. The window $\Delta \in [\tau_{\text{compress}}, \tau_{\text{kill}}]$ is therefore exponentially wide in R, and admits cycle lengths Δ for which every $\Delta' \in [\Delta_{\text{comp}}, \Delta]$ satisfies Theorem D.2.

D.4. The compressed block class

Definition D.3 (Compressed block class)**.** Fix an outer cycle length $\Delta > 0$ and a compression parameter $C > 0$ large enough that e^{-C} is small. For each block i, set $\Delta_{\text{comp}} := C/(a_2^{(i)} - a_1^{(i)})$ (block-implicit; read with index i in context). Block i belongs to the *compressed class at scale* Δ if the following five conditions hold.

(i) *Ideal compressed scalar.* The block admits an *ideal* (principal-mode) scalar interface value

$$c_i(\Delta') := e^{-a_1^{(i)}\Delta'}$$

for every $\Delta' \in [\Delta_{\text{comp}}, \Delta]$. The ideal scalar composes exactly: $c_i(\Delta_1 + \Delta_2) = c_i(\Delta_1)\, c_i(\Delta_2)$.

(ii) *Exact continuation with residual.* The block also admits an *exact* finite-time continuation factor

$$\hat{c}_i(\Delta') = c_i(\Delta')\big(1 + r_i(\Delta')\big) = e^{-a_1^{(i)}\Delta'}\big(1 + r_i(\Delta')\big),$$

where $r_i(\Delta')$ records residual nonprincipal-mode contribution.

(iii) *Regularity.* $|r_i(\Delta')| \le \frac{1}{2}$, so that $1 + r_i$ is bounded away from zero on the class.

(iv) *Uniform subleading control.* $|r_i(\Delta')| \lesssim e^{-(a_2^{(i)} - a_1^{(i)})\Delta'}$ with a fixed implicit constant, uniformly for $\Delta' \in [\Delta_{\text{comp}}, \Delta]$. This is the exponentially small estimate the axiom verification uses.

(v) *Spectral compression and uniform initial-microstate normalization.* $(a_2^{(i)} - a_1^{(i)})\Delta_{\text{comp}} = C$, so that $|r_i(\Delta')| \lesssim e^{-C} \ll 1$ uniformly on $[\Delta_{\text{comp}}, \Delta]$. The continuation factor $\hat{c}_i(\Delta')$ is taken *past* the initial microstate-dependent transient: we assume the block's initial distribution has a strictly positive lower-bounded principal-mode projection coefficient $\langle \phi_1, \mu_0 \rangle \ge \delta > 0$, uniform over admissible initial microstates, so the constant in (iv) does not blow up under different initial conditions. Raw survival from an arbitrary pre-compression initial microstate is a different object and not what we call the interface value.

The compressed block class at scale Δ is the set of pairs (c_1, c_2) — equivalently $(\hat{c}_1, \hat{c}_2)$ via (ii) — with both blocks satisfying (i)–(v).

Notation convention. The discipline c_i = ideal/principal-mode scalar (exact composition), $\hat{c}_i$ = exact finite-time continuation (approximate composition with residual r_i) is the same as Appendix G's. The body's Definition V.2 adopts the same convention. Theorem D.4's (A5) verification is stated for $\hat{c}_i$; the ideal-scalar reading reduces it to exact (A5).

Remark on the normalization convention. The representation $\hat{c}_i(\Delta') = c_i(\Delta')(1 + r_i(\Delta'))$ is not invariant under arbitrary rescaling $\hat{c}_i \mapsto K\hat{c}_i$; that operation changes the remainder to $K(1 + r_i) - 1$, which is not small for $K \not\approx 1$. Condition (v) above identifies $\hat{c}_i$ with the principal-mode continuation factor obtained after the initial transient, at which point the scalar continuation factor *is* $e^{-a_1^{(i)}\Delta'}$ to leading order. This is the same normalization used in Part V, Definition V.2, and is the reason r_i in (iv) is a principal-mode relative error rather than raw-survival drift. The uniform initial-microstate hypothesis $\langle \phi_1, \mu_0 \rangle \ge \delta > 0$ ensures the relative error bound holds uniformly and does not degenerate as the initial condition approaches a node of ϕ_1.

D.5. Main theorem: Theorem V.1 restated

The main theorem is a realization theorem. Proposition D.6 (established in §D.6) gives the block-level multiplicative factorization of continuation factors on the compressed class — which makes the (A5) functional equation *well-posed* on the class. The theorem below then verifies that the induced geometric evaluator $A_w^{\text{geom}}(c_1, c_2) = c_1^{w_1} c_2^{w_2}$ satisfies all five axioms on this class. As emphasized in Theorem A.7, compression does *not* force A5 for arbitrary (A1)–(A4) evaluators; the arithmetic mean satisfies (A1)–(A4) on classes admitting coordinate-wise multiplication and nonetheless fails A5. What compression does is make A5 formulable, and the realization theorem exhibits a specific evaluator satisfying it. Uniqueness in the sense of Theorem A.8 requires the all-arity (A4) regrouping family on the compressed class — closed by Theorem G.7 — combined with the present two-block realization; the two-block theorem alone does not pin the exponent, since at $n = 2$ the (A4) axiom is vacuous and the $n = 3$ regrouping step in Theorem A.8's proof is exactly what forces the exponent to coincide with the weight.

Theorem D.4 (Weighted geometric outer evaluation on the compressed class; Theorem V.1 restated, formal)**.** *Consider the compressed block class of Theorem D.3 for two independent, externally non-transferable Gibbs-lifted killed-process blocks, with exogenous weights $w = (w_1, w_2)$ summing to 1 with $w_1, w_2 > 0$. The induced geometric evaluator*

$$A_w^{\text{geom}}(c_1, c_2) := c_1^{w_1}\, c_2^{w_2}$$

satisfies axioms (A1)–(A5) of Theorem II.1 on this class: exactly on the ideal scalars (c_1, c_2), and on the exact continuations $(\hat{c}_1, \hat{c}_2)$ to within uniform subleading error $|\rho_i| \lesssim e^{-C}$ in the (A5) functional equation. (A4) is vacuous at $n = 2$; the all-arity (A4) realization on the compressed class is verified in Theorem G.7.

Proof. (A1) Continuity of $(c_1, c_2) \mapsto c_1^{w_1} c_2^{w_2}$ on $(0, \infty)^2$ is immediate. (A2) Strict monotonicity in each coordinate for $w_1, w_2 > 0$. (A3) Idempotence: $A_w^{\text{geom}}(c, c) = c^{w_1 + w_2} = c$ from $w_1 + w_2 = 1$. (A4) Spatial regrouping invariance is vacuous at $n = 2$; the substantive test on heterogeneous n-block compositions is deferred to Appendix G. (A5) For the ideal scalars, $c_i(\Delta_1 + \Delta_2) = c_i(\Delta_1)\, c_i(\Delta_2)$ identically by Definition V.2(i), so $A_w^{\text{geom}}\big(c(\Delta_1 + \Delta_2)\big) = A_w^{\text{geom}}\big(c(\Delta_1)\big) A_w^{\text{geom}}\big(c(\Delta_2)\big)$ exactly. For the exact continuations, Proposition D.6 (established in §D.6) gives $\hat{c}_i(\Delta_1 + \Delta_2) = \hat{c}_i(\Delta_1)\, \hat{c}_i(\Delta_2)(1 + \rho_i)$ with $|\rho_i| \lesssim e^{-C}$ uniformly. Applying A_w^{geom} pointwise:

$$A_w^{\text{geom}}\big(\hat{c}(\Delta_1 + \Delta_2)\big) = \prod_i \big(\hat{c}_i(\Delta_1)\, \hat{c}_i(\Delta_2)(1 + \rho_i)\big)^{w_i} = A_w^{\text{geom}}\big(\hat{c}(\Delta_1)\big)\, A_w^{\text{geom}}\big(\hat{c}(\Delta_2)\big) + O(e^{-C}),$$

which is (A5) with $F_w(s, t) = st$ to uniform subleading error.

(A4) is vacuous at $n = 2$; the all-arity (A4) realization on the compressed class is closed by Theorem G.7, which verifies that A_w^{geom} commutes with n-block regrouping on compressed interface values. Combined with the present two-block realization, the all-arity package then makes Theorem A.8 applicable in the exact-axiom limit, where A_w^{geom} is the unique evaluator satisfying (A1)–(A5). ☐

Remark D.5 (Role of spectral compression in Theorem II.1's hypotheses)*.* Spectral compression makes the (A5) functional equation operationally well-posed on the compressed-block class: after compression, each block admits an exact scalar continuation factor $\hat{c}_i(\Delta')$ composing multiplicatively up to $O(e^{-C})$, with the ideal scalar c_i composing exactly, so temporal concatenation at the evaluator level is a meaningful coordinate-wise operation. Compression does not derive (A5) from (A1)–(A4) — the arithmetic-mean evaluator of Appendix A satisfies (A1)–(A4) on a class admitting coordinate-wise multiplication and nonetheless fails (A5). The content of the present theorem is therefore a realization result: the specific induced geometric evaluator satisfies all five axioms on this class. Uniqueness among (A1)–(A5) evaluators is not established by the two-block theorem alone (cf. Theorem A.8's use of $n = 3$ regrouping); it follows in the exact-axiom limit only after combining the present two-block realization with the all-arity (n-block) realization of Theorem G.7.

D.6. Proof of Proposition D.5 and detail verification

Theorem D.4's proof invoked Proposition D.6 forward. We now establish the proposition, which supplies the block-level factorization on which the (A5) verification in Theorem D.4 depends.

Proposition D.6 (Block-level scalar factorization underlying (A5))**.** *Under the uniform compressed-sub-interval hypothesis of Theorem D.3, for any $\Delta = \Delta_1 + \Delta_2$ with $\Delta_1, \Delta_2 \in [\Delta_{\mathrm{comp}}, \Delta]$ and $i \in \{1,2\}$: the ideal scalars compose exactly,*

$$c_i(\Delta_1 + \Delta_2) = c_i(\Delta_1)\, c_i(\Delta_2),$$

and the exact continuations satisfy

$$\hat{c}_i(\Delta_1 + \Delta_2) = \hat{c}_i(\Delta_1)\, \hat{c}_i(\Delta_2)\big(1 + \rho_i(\Delta_1, \Delta_2)\big), \qquad |\rho_i(\Delta_1, \Delta_2)| \lesssim e^{-(a_2^{(i)} - a_1^{(i)})\Delta_1} + e^{-(a_2^{(i)} - a_1^{(i)})\Delta_2},$$

where

$$1 + \rho_i(\Delta_1, \Delta_2) := \frac{1 + r_i(\Delta_1 + \Delta_2)}{(1 + r_i(\Delta_1))(1 + r_i(\Delta_2))}.$$

Proof. The first identity is immediate from $c_i(\Delta') = e^{-a_1^{(i)}\Delta'}$. For the second, apply Theorem D.3 at each of Δ_1, Δ_2, and $\Delta_1 + \Delta_2$:

$$\hat{c}_i(\Delta_j) = e^{-a_1^{(i)}\Delta_j}(1 + r_i(\Delta_j)) \qquad (j = 1,2), \qquad \hat{c}_i(\Delta_1 + \Delta_2) = e^{-a_1^{(i)}(\Delta_1+\Delta_2)}(1 + r_i(\Delta_1 + \Delta_2)).$$

Multiplying the first two gives $\hat{c}_i(\Delta_1)\, \hat{c}_i(\Delta_2) = e^{-a_1^{(i)}(\Delta_1+\Delta_2)}\,(1 + r_i(\Delta_1))(1 + r_i(\Delta_2))$. Dividing $\hat{c}_i(\Delta_1 + \Delta_2)$ by this,

$$\frac{\hat{c}_i(\Delta_1 + \Delta_2)}{\hat{c}_i(\Delta_1)\, \hat{c}_i(\Delta_2)} = \frac{1 + r_i(\Delta_1 + \Delta_2)}{(1 + r_i(\Delta_1))(1 + r_i(\Delta_2))} =: 1 + \rho_i(\Delta_1, \Delta_2).$$

The denominator is bounded below by $(1/2)^2 = 1/4$ by the smallness hypothesis $|r_i| \le 1/2$ in Theorem D.3, so division is legitimate. Expanding to first order in the three r-terms (each $\lesssim e^{-(a_2^{(i)} - a_1^{(i)})\Delta'}$ for the appropriate Δ'), and using $|r_i(\Delta_1 + \Delta_2)| \le \max(|r_i(\Delta_1)|, |r_i(\Delta_2)|)$ when $a_2^{(i)} > a_1^{(i)}$ (monotone-decreasing exponential bound in Δ'), yields the stated bound on ρ_i. ☐

The evaluator-level (A5) verification using this factorization is folded into the proof of Theorem D.4 above; we do not repeat it here. The remaining axioms (A1)–(A4) for the induced geometric evaluator are also verified inside that proof: continuity, strict monotonicity, and idempotence are immediate for $A_w^{\mathrm{geom}}(c_1, c_2) = c_1^{w_1} c_2^{w_2}$ on $(0, \infty)^2$, and (A4) spatial regrouping is vacuous at $n = 2$. The substantive test of (A4) — closure of the compressed-interface-value class under heterogeneous n-block regrouping, with the induced cluster value $B_\alpha = \prod_{i \in C_\alpha} c_i^{w_i/W_\alpha}$ — is the content of Appendix G.

Remark D.7. (A5) as a condition on the evaluator is not derivable from (A1)–(A4) even when the block supplies scalar temporal summaries: Theorem A.7 exhibits an (A1)–(A4) evaluator (arithmetic mean) that fails (A5) even on a positive profile class with coordinate-wise multiplicative composition. What compression supplies is that the (A5) functional equation becomes well-posed on the compressed class (coordinate-wise temporal concatenation is defined up to $O(e^{-C})$) and that the specific induced geometric evaluator A_w^{geom} satisfies it. Before compression, block continuation depends on microstate details that no single scalar captures, so (A5) is not even well-posed in a principal-mode sense.

D.7. Log-scale synthesis

Substituting the compressed representation $c_i(\Delta) = e^{-a_1^{(i)}\Delta}(1 + r_i(\Delta))$ into Theorem D.4 yields the appendix's main corollary.

Corollary D.8 (Log-scale synthesis; body Section V.4 restated)**.** *On the compressed block class, the cycle-level outer viability satisfies*

$$V_\Delta = A_w(c_1, c_2) = c_1^{w_1} c_2^{w_2} \approx \exp\Big(-\Delta\big(w_1 a_1^{(1)} + w_2 a_1^{(2)}\big)\Big).$$

Equivalently,

$$\log V_\Delta \approx -\Delta\left(w_1 a_1^{(1)} + w_2 a_1^{(2)}\right),$$

so outer geometric aggregation of block survival factors is linear in the log-rate coordinate, with additive weighted-arithmetic-mean structure on decay rates.

Proof. Direct substitution into Theorem D.4 and taking logarithms; the relative-error bound from Theorem D.3 becomes an additive error on $\log V_\Delta$ of size $\sum_i w_i |r_i(\Delta)|$, hence $\lesssim \max_i e^{-(a_2^{(i)} - a_1^{(i)})\Delta}$. □

Theorem D.8 is the first place in the program where the outer Theorem II.1 and the additive-coordinate representation of the log-Feynman-Kac coda (Appendix J) visibly meet: geometric aggregation across non-transferable blocks becomes literal arithmetic aggregation in the log-rate coordinate, with the additive primitive being the weighted mean of principal decay rates. This is the concrete content of the body's Section V.4.

D.8. Negative side: when the outer evaluator is not realized

The positive result of Theorem D.4 is timing-sensitive: the weighted geometric evaluator is realized only on a class of block summaries admitting uniform principal-mode dominance. Two control cases mark the boundary.

D.8.1. Diffusive baseline (Appendix B). In the uniform fiber law with fixed threshold width, Appendix B shows that $a_1(R) \sim \gamma\pi^2/R^2$ and the relevant mixing time is $\tau_{\text{mix}} \asymp R^2/\gamma$, so the two scales coincide:

$$\tau_{\text{compress}} \asymp R^2/\gamma, \qquad \tau_{\text{kill}} \asymp R^2/\gamma.$$

The window $\Delta \in [\tau_{\text{compress}}, \tau_{\text{kill}}]$ collapses in scaling, and no asymptotically robust cycle window exists on which the compressed block class of Theorem D.3 is non-empty uniformly in R. (At any finite R one can of course still pick times; the point is that the window does not persist as $R \to \infty$.) The weighted geometric evaluator is therefore not concretely realized in the uniform baseline class at scaling. This is not a failure of the geometric form once its hypotheses hold — it is a failure of the hypotheses themselves, specifically of (A5). The block never reaches a regime in which a single scalar already says enough.

D.8.2. Short-cycle Gibbs observation. Even within the Gibbs-lifted class of Appendix C, if $\Delta < \tau_{\text{compress}}$ then the principal mode has not yet dominated: block continuation remains sensitive to initial microstate, and temporal interface sufficiency fails for the same reason as in Section III.2's timing warning, namely that the chosen scalar summary does not yet contain enough information about what comes next. The difference from the baseline is that here the failure is only local in scale: the Gibbs lift supplies a broad window in which compression eventually does occur, whereas the baseline supplies no such window.

Both controls identify the same structural fact: in the killed-process realization, (A5) is a spectral condition that does not hold outside the compressed block class. The thermodynamic leg of the program (Appendix C) is what opens the compression window; without it, (A5) cannot be verified on any concrete scale, and the outer theorem is not concretely realized.

Appendix E. Abstract Slow-Fast Bridge: Proposition III.2 in Full

E.1. Opening

Why you are reading this appendix, given that Appendix F covers the dynamic realization Part V uses. At the structural level, the discrete rare-event skeleton studied here is the exact analogue of the continuous-time adiabatic elimination carried out in Appendix F: in both, a fast variable mixes to a stationary law while rare events (hazard arrivals, coarse-state jumps) happen on a slower scale; the fast mixing is averaged away, leaving an effective dynamics on the coarse state. What differs is the *packaging* of that averaged content. Appendix E packages it as a branching Bellman operator whose fixed point answers the infinite-horizon eventual-hazard question; Appendix F packages it as a killed-continuation operator K_ε whose iteration on a rare-event scale yields the finite-horizon fixed-cycle Feynman-Kac semigroup that Part V uses directly. The two are related by a rare-event scaling limit: §F.10 / Theorem F.12 shows that Appendix F's finite-horizon semigroup emerges as the continuum limit of the killed-continuation form of Appendix E's one-cycle averaging step, iterated on the rare-event scale. Reading Appendix E first gives the clean abstract existence theorem and the branching-Bellman perspective; reading Appendix F directly gives the object Part V uses. Both readings are legitimate; the two are complementary, not redundant.

This appendix supplies the formal statement and proof of Proposition III.2 — the existence of an approximate dynamic interface value in a slow-fast regime where the block's coarse state evolves alongside hazard arrival while internal allocation continues to mix rapidly on each fixed-total fiber. In Part III we stated the proposition as an abstract existence result, deferred to here; Appendix F gives the continuous-time Gibbs-lifted killed-process instantiation. The present appendix is the abstract rung of that ladder: a contraction-based closeness theorem in a discrete-time rare-event skeleton, in which the construction is transparent and the error controlled by a single gap-weighted mixing quantity.

Scope note. The canonical-memo versions of the slow-fast bridge (Memos 4 and 5) also contain two stochastic side-results that this appendix does not lift: an approximate expected-total representation under approximate affinity in the total resource, and a zero-resource normalization lemma controlling the affine intercept of the block's pooled value. Neither is used in the present monograph's proof architecture — the synthesis of Parts IV and V relies on the averaged closeness theorem and the geometric-gap corollary below, not on their expected-total refinements — and both are accordingly omitted rather than restated. Readers interested in the stochastic ranking-threshold machinery should consult the original memos directly.

E.2. Rare-event model with fast within-fiber mixing

Fix a block and a scalar conserved total resource $R(u) := \mathbf{1}^\top u$. For each attainable $R \geq 0$, let

$$\Omega_R := \{u \in \mathbb{R}^d_+ : \mathbf{1}^\top u = R\}$$

be the fixed-total fiber. Let $\mathcal{Z}$ be the coarse state space. We regard $u \in \Omega_R$ as the fast internal allocation and $z \in \mathcal{Z}$ as the slow coarse variable.

The model is a *rare-event skeleton*. Starting from (z, u) with total resource R, one cycle proceeds:

(1) A gap length G is drawn from a law $\nu_{z,R}$ on $\{0, 1, 2, \dots\}$.

(2) During the gap, the fast variable evolves under a Markov kernel $K_{z,R}$ on Ω_R while the coarse state is frozen. After the gap, $\tilde{U} \sim K^G_{z,R}(u, \cdot)$.

(3) At the end of the gap, a rare event occurs. With probability $\pi_{\text{haz}}(z, R) \in (0, 1]$ the rare event is hazard and the block survives it with probability $s_{z,R}(\tilde{U}) \in [0, 1]$.

(4) With complementary probability $1-\pi_{\text{haz}}(z,R)$ the rare event is a slow coarse-state jump: $Z' \sim L_{z,R}(\tilde{U},\cdot)$, and the next cycle begins from $(Z',\tilde{U})$.

Total resource remains fixed during the pre-interface phase. The fast variable re-equilibrates between rare events; the slow state changes only at those rare events. This is the simplest discrete analogue of a slow-fast process with many fast updates between slow coarse changes or hazard arrival.

E.3. Averaged objects

For each (z,R), assume $K_{z,R}$ admits a stationary distribution $\pi_{z,R}$ on Ω_R.
Definition E.1 (Averaged survival payoff and averaged slow kernel)**.**

$$\bar{\Phi}(z,R) := \int_{\Omega_R} s_{z,R}(u)\,\pi_{z,R}(du), \qquad \bar{L}_{z,R}(A) := \int_{\Omega_R} L_{z,R}(u,A)\,\pi_{z,R}(du)$$

for measurable $A \subseteq \mathcal{Z}$.
Definition E.2 (Gap-weighted fiber mixing error)**.** For each (z,R) and $t \geq 0$,

$$d_{z,R}(t) := \sup_{u\in\Omega_R} \|K^t_{z,R}(u,\cdot) - \pi_{z,R}\|_{\mathrm{TV}}, \qquad \delta(z,R) := \sum_{t\geq 0} \nu_{z,R}(t)\,d_{z,R}(t),$$

and $\delta_* := \sup_{z\in\mathcal{Z}} \delta(z,R)$ whenever finite.

$\delta(z,R)$ is small when the typical gap between rare events is long enough for the fast chain on the current fiber to nearly forget its initial allocation before the next rare event.

E.4. Exact and averaged Bellman operators

Let $\mathscr{B}(\mathcal{Z}\times\Omega_R)$ denote the bounded measurable functions $V : \mathcal{Z}\times\Omega_R \to \mathbb{R}$ with the sup norm, and $\mathscr{B}(\mathcal{Z})$ the bounded measurable functions $f : \mathcal{Z} \to \mathbb{R}$. Define the exact Bellman operator

$$(T_R V)(z,u) := \sum_{t\geq 0} \nu_{z,R}(t) \int_{\Omega_R} \Big[\pi_{\text{haz}}(z,R)s_{z,R}(x) + (1-\pi_{\text{haz}}(z,R)) \int_{\mathcal{Z}} V(z',x)L_{z,R}(x,dz')\Big] K^t_{z,R}(u,dx) \tag{2}$$

and the averaged Bellman operator

$$(\bar{T}_R f)(z) := \pi_{\text{haz}}(z,R)\bar{\Phi}(z,R) + (1-\pi_{\text{haz}}(z,R)) \int_{\mathcal{Z}} f(z')\,\bar{L}_{z,R}(dz'). \tag{3}$$

T_R is the exact pre-interface recursion: after a within-fiber gap, either hazard occurs (evaluated at the mixed-but-not-necessarily-stationary fast state) or a slow jump occurs and the recursion continues. $\bar{T}_R$ is the same recursion after replacing the fast variable by its stationary fiber law at each rare event.

Write $\pi_{\text{haz}*} := \inf_{z\in\mathcal{Z}} \pi_{\text{haz}}(z,R)$: a lower bound on the probability that the next rare event is hazard rather than another slow transition.
Lemma E.3 (Contraction)**.** *Assume $\pi_{\text{haz}*} > 0$. Then T_R is a contraction on $\mathscr{B}(\mathcal{Z}\times\Omega_R)$ and $\bar{T}_R$ is a contraction on $\mathscr{B}(\mathcal{Z})$, both with modulus at most $1-\pi_{\text{haz}*}$. Consequently each has a unique fixed point.*

Proof. Let $V,W \in \mathscr{B}(\mathcal{Z}\times\Omega_R)$. By (2) the hazard term cancels when subtracting T_RV and T_RW, so

$$|(T_RV)(z,u) - (T_RW)(z,u)| \leq (1-\pi_{\text{haz}}(z,R))\|V-W\|_\infty.$$

Taking the supremum gives $\|T_RV - T_RW\|_\infty \leq (1-\pi_{\text{haz}*})\|V-W\|_\infty$. The same calculation with (3) gives $\|\bar{T}_Rf - \bar{T}_Rg\|_\infty \leq (1-\pi_{\text{haz}*})\|f-g\|_\infty$. Since $1-\pi_{\text{haz}*} < 1$, both operators are contractions and admit unique fixed points by Banach. □

Denote the unique fixed point of T_R by V_R and that of $\bar{T}_R$ by $\bar{c}_R$. $V_R(z,u)$ is the exact pre-interface survival value starting from coarse state z and internal allocation u on the fixed-total fiber. $\bar{c}_R(z)$ is the candidate averaged interface value — the object Proposition III.2 asserts exists.

E.5. Main closeness theorem

Lemma E.4 (One-cycle averaging error). *Let $f \in \mathscr{B}(\mathcal{Z})$ satisfy $0 \le f \le 1$, and let $\tilde{f}(z,u) := f(z)$ be its lift to $\mathcal{Z} \times \Omega_R$. Then for every (z,u),*

$$\big|(T_R\tilde{f})(z,u) - (\bar{T}_R f)(z)\big| \le \delta(z,R).$$

Proof. Set $g_f^{z,R}(x) := \pi_{\mathrm{haz}}(z,R)s_{z,R}(x) + (1-\pi_{\mathrm{haz}}(z,R))\int_{\mathcal{Z}} f(z')L_{z,R}(x,dz')$. Since $0 \le s_{z,R}, f \le 1$, also $0 \le g_f^{z,R} \le 1$. Then

$$(T_R\tilde{f})(z,u) = \sum_{t\ge 0}\nu_{z,R}(t)\int_{\Omega_R} g_f^{z,R}(x)\,K_{z,R}^t(u,dx), \qquad (\bar{T}_R f)(z) = \int_{\Omega_R} g_f^{z,R}(x)\,\pi_{z,R}(dx).$$

For each fixed t, the standard total-variation bound for a $[0,1]$-valued test function gives

$$\left|\int g_f^{z,R}\,dK_{z,R}^t(u,\cdot) - \int g_f^{z,R}\,d\pi_{z,R}\right| \le \|K_{z,R}^t(u,\cdot) - \pi_{z,R}\|_{\mathrm{TV}} \le d_{z,R}(t).$$

Averaging over t with weights $\nu_{z,R}(t)$ gives the claim. □

Theorem E.5 (Approximate dynamic interface value). *Fix a total resource R and assume:*

- (E1) *for each $z \in \mathcal{Z}$, the fast kernel $K_{z,R}$ admits a stationary law $\pi_{z,R}$ on Ω_R;*
- (E2) *the hazard payoff satisfies $0 \le s_{z,R}(u) \le 1$ for all (z,u);*
- (E3) $\pi_{\mathrm{haz}*} := \inf_z \pi_{\mathrm{haz}}(z,R) > 0$;
- (E4) $\delta_* = \sup_z \delta(z,R) < \infty$.

(Local to Appendix E — not *the global evaluator axioms (A1)–(A5) of Theorem II.1.) Then the exact value V_R and the averaged interface value $\bar{c}_R$ satisfy*

$$\sup_{z\in\mathcal{Z}}\sup_{u\in\Omega_R}\big|V_R(z,u) - \bar{c}_R(z)\big| \le \frac{\delta_*}{\pi_{\mathrm{haz}*}}.$$

In particular, for any fixed z and any two initial allocations $u,u' \in \Omega_R$,

$$|V_R(z,u) - V_R(z,u')| \le \frac{2\delta_*}{\pi_{\mathrm{haz}*}}.$$

Proof. Since $0 \le s_{z,R} \le 1$, monotone contraction iteration gives $0 \le V_R \le 1$ and $0 \le \bar{c}_R \le 1$. Let $\tilde{c}_R(z,u) := \bar{c}_R(z)$. Since $\bar{c}_R = \bar{T}_R\bar{c}_R$, Theorem E.4 with $f = \bar{c}_R$ yields

$$\|T_R\tilde{c}_R - \tilde{c}_R\|_\infty \le \delta_*.$$

Using Theorem E.3,

$$\begin{aligned}\|V_R - \tilde{c}_R\|_\infty &= \|T_RV_R - \tilde{c}_R\|_\infty \\ &\le \|T_RV_R - T_R\tilde{c}_R\|_\infty + \|T_R\tilde{c}_R - \tilde{c}_R\|_\infty \\ &\le (1-\pi_{\mathrm{haz}*})\|V_R - \tilde{c}_R\|_\infty + \delta_*.\end{aligned}$$

Rearranging gives $\|V_R - \tilde{c}_R\|_\infty \le \delta_*/\pi_{\mathrm{haz}*}$. The two-allocation bound is the triangle inequality. □

Remark E.6 (The $1/\pi_{\mathrm{haz}*}$ factor). The factor $1/\pi_{\mathrm{haz}*}$ is the natural amplification by the expected number of remaining rare events before hazard. If hazard is unlikely at each rare event, a one-cycle averaging error accumulates over many subsequent slow jumps before final hazard. The theorem cleanly separates the two ingredients: fiber mixing δ_* controls the error made on each cycle; the hazard lower bound $\pi_{\mathrm{haz}*}$ controls how many such cycles can accumulate.

E.6. Geometric-gap corollary

The bound $\delta_*/\pi_{\mathrm{haz}*}$ becomes concrete under the standard slow-fast scaling: long gaps between rare events, geometric fiber mixing.

Corollary E.7 (Geometric gaps and geometric mixing)**.** *Assume the hypotheses of Theorem E.5. Suppose in addition that for some constants $C_R \geq 1$ and $\rho_R \in (0, 1)$,*

$$d_{z,R}(t) \leq C_R \rho_R^t \qquad \text{for all } z \in \mathcal{Z},\ t \geq 0,$$

and that each gap distribution $\nu_{z,R}$ is geometric with parameter $p(z, R) \in (0, 1]$. Then

$$\delta(z, R) \leq \frac{C_R\, p(z, R)}{1 - (1 - p(z, R))\rho_R} \qquad \text{for every } z \in \mathcal{Z}.$$

Consequently, if

$$\delta_*^{\mathrm{geom}} := \sup_{z \in \mathcal{Z}} \frac{C_R\, p(z, R)}{1 - (1 - p(z, R))\rho_R} < \infty,$$

then Theorem E.5 applies with $\delta_* \leq \delta_*^{\mathrm{geom}}$, so

$$\sup_{z \in \mathcal{Z}} \sup_{u \in \Omega_R} \left| V_R(z, u) - \bar{c}_R(z) \right| \leq \frac{\delta_*^{\mathrm{geom}}}{\pi_{\mathrm{haz}*}}.$$

Proof. Substitute the geometric bound on $d_{z,R}(t)$ into the definition of $\delta(z, R)$:

$$\delta(z, R) \leq \sum_{t \geq 0} p(z, R)(1 - p(z, R))^t C_R \rho_R^t = \frac{C_R\, p(z, R)}{1 - (1 - p(z, R))\rho_R}.$$

The error bound for $V_R - \bar{c}_R$ then follows from Theorem E.5. □

Remark E.8. When $p(z, R)$ is small, the mean gap between rare events is of order $1/p(z, R)$. The corollary then expresses exactly the intended slow-fast regime: many fast mixing updates occur between successive rare events, and the dependence of pre-interface viability on the initial internal microstate becomes correspondingly small.

E.7. Relation to Appendix F

Theorem E.5 is the abstract existence result asserted in Proposition III.2: under slow-fast timescale separation with scalar conserved total resource and rapid within-fiber mixing, the block admits an approximate dynamic interface value $\bar{c}_R(z)$, uniform in the initial microstate, with explicit error $\delta_*/\pi_{\mathrm{haz}*}$. The construction is general: any fast kernel with a stationary fiber law, any coarse transition kernel, any gap distribution.

Appendix F is the continuous-time spectral realization of the same abstract pattern. The contraction theorem here proves, in a discrete-time rare-event skeleton, that fast within-fiber mixing reduces pre-interface viability to the value function of an averaged coarse process. Appendix F proves the analogous reduction directly in the Gibbs-lifted killed-process class, by spectral compression and adiabatic elimination rather than by the discrete Bellman contraction; rapid fiber compression is expressed there by the uniform spectral gap and the condition $q_* \ll g_*$, and the averaged coarse process becomes the Feynman-Kac semigroup $e^{\Delta(Q - \mathrm{diag}(a_1))}$.

The two appendices are not independent but complementary, and are related by a rare-event scaling limit. §F.10, Theorem F.12, establishes that Appendix F's finite-horizon Feynman-Kac semigroup is the continuum limit of the killed-continuation operator M_ε naturally associated with the one-cycle averaging step of the present appendix, iterated $\lfloor \Delta/\varepsilon \rfloor$ times on the rare-event scale $\varepsilon \to 0$. The distinction is structural: the Bellman operator $\bar{T}_R$ of Theorem E.5 is a *branching* form of the averaged content, suited to the infinite-horizon eventual-hazard question, whose own fixed-point scaling yields a resolvent-type equation; M_ε is the *killed continuation* form of the same averaged content, suited to the finite-horizon fixed-cycle survival question, whose iterated scaling yields the semigroup $e^{\Delta(Q - \mathrm{diag}(a_1))}$. Both packagings are valid uses of the same within-fiber averaging; they answer different questions. Appendix E supplies the averaging step that removes the fast fiber, Appendix F is the continuous-time finite-horizon semigroup obtained by iterating that averaged step in its killed-continuation form, and §F.10 bridges them.

Appendix F. Dynamic Synthesis: Slow-Fast Feynman-Kac Interface Values

F.1. Opening

This appendix supplies the proof-home for Theorems V.2 and V.3: the outer geometric synthesis of Appendix D persists under slow-fast co-evolution, provided the block's own coarse state moves only on a timescale slower than spectral compression. The body of Part V stated the two theorems and sketched the logic; here we carry out the adiabatic elimination that replaces each block's frozen-state principal-mode survival factor $e^{-a_1^{(i)}(z)\Delta}$ by the Feynman-Kac expectation $\mathcal{A}_i(\Delta, z_i) = \mathbb{E}_z[\exp(-\int_0^\Delta a_1^{(i)}(Z_s)\,ds)]$ of the averaged coarse process, verify the axioms of Theorem II.1 on the resulting dynamic compressed-block class, derive the log-scale synthesis that supplies Section V.4's additive coordinate, and check the conclusions numerically in a curvature-modulated family.

The analysis proceeds in three steps. First, the slow-fast setup: each block carries a finite coarse state space $Z^{(i)}$ with its own generator $Q^{(i)}$, and for each z a fixed-z Gibbs-lifted killed process of the Appendix C type. Uniform spectral compression requires both (a) a uniform gap $g_*^{(i)} = \min_z(a_2^{(i)}(z) - a_1^{(i)}(z)) > 0$ across coarse states, and (b) coarse motion slow relative to compression, $q_*^{(i)} \ll g_*^{(i)}$. Second, the Feynman-Kac interface value replaces Appendix D's frozen scalar: once the fast fiber has been adiabatically eliminated, each block is represented by the value function of the averaged coarse process, which is precisely the continuous-time realization of the slow-fast object Appendix E (the abstract bridge of Proposition III.2) identifies in the abstract setting. Third, Theorem II.1 applies to the dynamic compressed class via the same consistency verification as Appendix D, with temporal interface sufficiency now carried by the Feynman-Kac semigroup composition law rather than by coordinatewise exponential factorization.

F.2. Setup: z-dependent Gibbs blocks

Consider two externally non-transferable blocks $i \in \{1, 2\}$. Each block has:

- a finite coarse state space $Z^{(i)}$;
- a fixed total resource $R^{(i)}$ and fast fiber $\Omega_{R^{(i)}}^{(i)} = \{0, 1, \dots, R^{(i)}\}$;
- for each $z \in Z^{(i)}$, a Gibbs-lifted killed generator $G_z^{(i)}$ on $\Omega_{R^{(i)}}^{(i)}$ of the Appendix C type, with parameters $(h_z^{(i)}, m^{(i)}, \gamma^{(i)}, \kappa_0^{(i)}, \beta^{(i)})$ where the symmetric one-well potential $h_z^{(i)} : [0,1] \to \mathbb{R}$ satisfies the conditions of §C.2 for each z;
- a slow coarse-state generator $Q^{(i)}$ on $Z^{(i)}$, with coarse process $Z_s^{(i)}$.

For each fixed z, let $0 < a_1^{(i)}(z) < a_2^{(i)}(z) \le \cdots$ denote the ordered decay rates of the killed generator $G_z^{(i)}$. Define

$$g_*^{(i)} := \min_{z \in Z^{(i)}} \big(a_2^{(i)}(z) - a_1^{(i)}(z)\big), \qquad q_*^{(i)} := \max_{z \in Z^{(i)}} \sum_{z' \neq z} Q_{zz'}^{(i)}.$$

The quantity $g_*^{(i)}$ is the uniform spectral gap across coarse states; $q_*^{(i)}$ is the maximal coarse jump rate. Block i is said to be in the *slow-fast regime* if $q_*^{(i)} \ll g_*^{(i)}$. As in Appendix D, outer weights $w = (w_1, w_2)$ with $w_1 + w_2 = 1$ are exogenous: they represent externally assigned scheduling or exposure information, not quantities derived from block thermodynamics, and they do not depend on coarse state z.

F.3. Uniform z-dependent spectral compression

For fixed z, Appendix D showed that a block compresses on the cycle timescale when $(a_2(z) - a_1(z))\Delta \gg 1$. Once z moves, the condition needs a moving-landscape version: the cycle must be long enough for principal-mode formation at *any* coarse state the block could occupy, and the coarse landscape must not move so fast that the block is dragged into a new coarse state before the principal mode has formed.

Definition F.1 (Uniform z-dependent spectral compression)**.** Block i is *uniformly spectrally compressed* on the cycle timescale Δ if

$$g_*^{(i)}\Delta \gg 1, \qquad q_*^{(i)} \ll g_*^{(i)}.$$

The first is a lower bound on Δ (longer than worst-case local compression $1/g_*$); the second is an upper bound on coarse motion (slower than compression at every coarse state).

F.3.1. Heuristic gap scaling in the curvature-modulated family. The full z-dependence of $a_2(z) - a_1(z)$ is not proved here; we only identify the scaling needed for the window to exist. Take a curvature-modulated family

$$h_z(x) = c(z)\big(x - \tfrac{1}{2}\big)^2, \qquad z \in Z,$$

with $c(z) > 0$. Near the well minimum $h_z''(1/2) = 2c(z)$. The Gibbs well has width $\sim \sqrt{R/(\beta h_z''(1/2))}$ in occupation space, and within-well relaxation is again discrete-Ornstein-Uhlenbeck-like (Theorem D.2's harmonic-well argument applied locally at each z). This gives the heuristic scaling

$$a_2(z) - a_1(z) \asymp \frac{\gamma\beta h_z''(1/2)}{R} = \frac{2\gamma\beta c(z)}{R}$$

up to a numerical constant. The principal rate $a_1(z)$ remains barrier-controlled, exponentially small in the barrier height $\Delta h_z = h_z(0) - h_z(1/2) = c(z)/4$ (the large-deviation structure of Theorem C.7 applied at fixed z). Curvature modulation therefore changes both scales simultaneously: the compression scale as $1/c(z)$, the killing scale as $\exp(\beta R c(z)/4)$.

F.4. Dynamic block interface value and dynamic compressed class

After adiabatic elimination of the fast fiber (justified in F.5 below under Theorem F.1), each block is represented on the cycle timescale by a scalar that depends on the current coarse state and the coarse dynamics over the cycle.

Definition F.2 (Dynamic block interface value)**.** For block i, cycle length Δ, and initial coarse state $z \in Z^{(i)}$, the *dynamic block interface value* is

$$\mathcal{A}_i(\Delta, z) := \mathbb{E}_z\Big[\exp\Big(-\int_0^\Delta a_1^{(i)}(Z_s^{(i)})\,ds\Big)\Big] = \big[e^{\Delta(Q^{(i)} - \mathrm{diag}(a_1^{(i)}))}\,\mathbf{1}\big]_z,$$

where $\mathbf{1}$ is the all-ones vector on $Z^{(i)}$. The coarse process $Z_s^{(i)}$ is generated by $Q^{(i)}$ and $a_1^{(i)}(\cdot)$ is the fixed-z principal killed rate evaluated along the path.

Remark F.3 (Continuous-time analogue of Appendix E). Theorem E.5 (Proposition III.2 in full, Appendix E) identified the value function of an averaged coarse process on a discrete rare-event skeleton as the abstract slow-fast interface value. The Feynman-Kac expression of Theorem F.2 is the continuous-time analogue: the same averaging content, repackaged for a different question. Appendix E's object is a *branching* Bellman operator, suited to the infinite-horizon eventual-hazard question with a geometric rare-event recursion; Theorem F.2 is instead the finite-horizon scaling limit of a *killed continuation* operator K_ε that applies a survival factor and a coarse transition at each small step. Both package the same within-fiber averaging and reduce to the same generator $Q - \mathrm{diag}(a_1)$ in the continuum limit, but they are distinct finite-ε objects answering different questions (infinite-horizon eventual-hazard vs. finite-horizon fixed-cycle survival). §F.10 establishes the scaling-limit equivalence formally.

Definition F.4 (Dynamic compressed-block class)**.** Fix a cycle length $\Delta > 0$ satisfying Theorem F.1 for both blocks. The *dynamic compressed-block class* consists of pairs $(\mathcal{A}_1, \mathcal{A}_2)$ of dynamic interface values together with their underlying *Feynman-Kac continuation families*

$$\Delta' \mapsto e^{\Delta' M_i}\,\mathbf{1} \in \mathbb{R}_{>0}^{Z^{(i)}}, \qquad M_i := Q^{(i)} - \mathrm{diag}(a_1^{(i)}),$$

such that the block's exact survival from initial coarse state $z_i \in Z^{(i)}$ and any admissible fiber microstate in the metastable well satisfies the *relative-error representation*

$$S_i^{\text{exact}}(\Delta, z_i, u) = \mathcal{A}_i(\Delta, z_i)\,\big(1 + \varepsilon_i^{\text{dyn}}(\Delta, z_i, u)\big), \qquad |\varepsilon_i^{\text{dyn}}(\Delta, z_i, u)| \lesssim e^{-g_*^{(i)}\Delta} + \frac{q_*^{(i)}}{g_*^{(i)}},$$

uniformly in admissible u. The scalar $\mathcal{A}_i(\Delta, z_i)$ is the evaluation of the continuation vector $e^{\Delta M_i}\mathbf{1}$ at the current coarse state z_i; the continuation family is the segment-summary object required for temporal splitting. The first error term is the subleading spectral projection error at fixed z; the second is the coarse-transition penalty during the compression window.

F.5. Dynamic persistence: Theorem V.2 restated

Theorem F.5 (Dynamic persistence of compressed interface values; Theorem V.2 restated, schematic error bound, formal)**.** *Under the slow-fast setup of §F.2 and the uniform compression hypothesis Theorem F.1, for each block i, any initial coarse state $z \in Z^{(i)}$, and any admissible initial fiber microstate u in the metastable well,*

$$S_i^{\text{exact}}(\Delta, z, u) = \mathcal{A}_i(\Delta, z)\left(1 + O\big(e^{-g_*^{(i)}\Delta} + q_*^{(i)}/g_*^{(i)}\big)\right),$$

uniformly in admissible u, where $\mathcal{A}_i(\Delta, z)$ is the dynamic interface value of Theorem F.2. In particular the pair $(\mathcal{A}_1, \mathcal{A}_2)$, together with their continuation families $\Delta' \mapsto e^{\Delta' M_i}\mathbf{1}$, lies in the dynamic compressed-block class of Theorem F.4.

Proof sketch. For fixed z, the fast killed semigroup $e^{tG_z^{(i)}}$ admits a spectral decomposition whose principal mode dominates after times of order $1/(a_2(z) - a_1(z))$. Under the uniform gap hypothesis of Theorem F.1, this dominance is uniform across the finite coarse state space: after time $\gtrsim 1/g_*^{(i)}$, any admissible initial fiber microstate has relaxed to a profile close (in $L^2(\pi_z^{(i)})$) to the principal eigenvector of $G_z^{(i)}$, with residual suppressed by $e^{-g_*^{(i)}\Delta}$.

The cycle length hypothesis $g_*^{(i)}\Delta \gg 1$ ensures the cycle is long enough for this local compression to occur. The slow-motion hypothesis $q_*^{(i)} \ll g_*^{(i)}$ ensures the coarse landscape does not change appreciably on the compression timescale: the probability of a coarse transition during one compression time is $\lesssim q_*^{(i)}/g_*^{(i)}$, which is the second error term.

Given these two hypotheses, one can adiabatically eliminate the fast fiber: on the cycle timescale, the block's state factors as the current principal eigenvector of $G_z^{(i)}$ (depending on the current coarse state z) times a slowly varying amplitude $u(t, z)$. The amplitude evolves by the reduced equation

$$\partial_t u = \left(Q^{(i)} - \text{diag}(a_1^{(i)})\right) u, \qquad u(0, \cdot) = \mathbf{1},$$

whose solution is exactly $\mathcal{A}_i(\Delta, z)$ of Theorem F.2. The two error terms are the schematic remnants of adiabatic elimination: the first is the subleading spectral projection error at fixed z, the second is the coarse-transition penalty during the compression window. □

F.6. Dynamic geometric outer evaluation: Theorem V.3 restated

With Theorem F.5 in hand, each block is represented on the cycle timescale by a positive scalar value $\mathcal{A}_i(\Delta, z_i)$ together with its Feynman-Kac continuation family $\Delta' \mapsto e^{\Delta' M_i}\mathbf{1}$. The type of the interface-value object has shifted from the frozen scalar of Theorem D.4 to a scalar-valued function of the coarse state z, and temporal concatenation is now the semigroup composition $e^{(\Delta_1+\Delta_2)M_i}\mathbf{1} = e^{\Delta_1 M_i}e^{\Delta_2 M_i}\mathbf{1}$ on the continuation vector. The structure parallels Appendix D: Proposition F.6 records the dynamic block-level scalar factorization that makes (A5) well-posed; Theorem F.7 below is the corresponding realization theorem for the induced dynamic geometric evaluator.

Proposition F.6 (Dynamic block-level scalar factorization underlying (A5))**.** *Under Theorem F.1 (uniform z-dependent spectral compression with $q_*^{(i)} \ll g_*^{(i)}$), the Feynman-Kac semigroup on each block satisfies the exact identity*

$$e^{(\Delta_1+\Delta_2)\,M_i}\,\mathbf{1} = e^{\Delta_1 M_i}\left(e^{\Delta_2 M_i}\,\mathbf{1}\right), \qquad M_i := Q^{(i)} - \text{diag}\big(a_1^{(i)}\big),$$

at the continuation-vector level. Scalar dynamic interface values $\mathcal{A}_i(\Delta, z) = [e^{\Delta M_i}\mathbf{1}]_z$ admit approximate multiplicative composition

$$\mathcal{A}_i(\Delta_1 + \Delta_2, z) = \mathcal{A}_i(\Delta_1, z)\,\mathcal{A}_i(\Delta_2, z)\left(1 + O(\epsilon_{\text{dyn}})\right), \qquad \epsilon_{\text{dyn}} \lesssim e^{-g_*^{(i)}\Delta} + \frac{q_*^{(i)}}{g_*^{(i)}},$$

in two regimes: the quasi-static *regime* ($q_*^{(i)}\Delta \ll 1$*,* z *nearly frozen over the cycle) and the* fast-z *regime* ($q_*^{(i)} \gg \max_z a_1^{(i)}(z)$*,* $q_*^{(i)} \ll g_*^{(i)}$*), with the additional* segment-scale condition *in the fast-*z *case that each sub-interval satisfies*

$$q_*^{(i)}\,\Delta_j \gg 1 \qquad (j=1,2),$$

ensuring each segment is long enough for coarse-chain self-averaging; without this condition the vector-level semigroup identity remains exact but the scalar factorization fails on the shorter sub-interval.

Proof. The semigroup identity is immediate: M_i is time-independent, so $e^{(\Delta_1+\Delta_2)M_i} = e^{\Delta_1 M_i}\,e^{\Delta_2 M_i}$ by the standard semigroup law. Evaluating at z and applying the tower property gives

$$\mathcal{A}_i(\Delta_1+\Delta_2, z) = \mathbb{E}_z\left[e^{-\int_0^{\Delta_1} a_1^{(i)}(Z_s)\,ds} \cdot \mathcal{A}_i\big(\Delta_2, Z_{\Delta_1}\big)\right].$$

In the quasi-static regime with $q_*^{(i)}\Delta_1 \ll 1$, $Z_{\Delta_1} \approx z$ (no appreciable coarse motion on scale Δ_1), so $\mathcal{A}_i(\Delta_2, Z_{\Delta_1}) \approx \mathcal{A}_i(\Delta_2, z)$ with error $O(q_*^{(i)}/g_*^{(i)})$; the outer Feynman-Kac exponential reduces to $e^{-a_1^{(i)}(z)\Delta_1} \approx \mathcal{A}_i(\Delta_1, z)$ via Theorem F.5. In the fast-z regime with $q_*^{(i)}\Delta_j \gg 1$ for each j, on each sub-interval Z mixes to the stationary law $\pi_Z^{(i)}$, so $\mathcal{A}_i(\Delta_j, \cdot) \approx e^{-\bar{a}_1^{(i)}\Delta_j}$ uniformly in the starting state, and the factorization follows; without the segment-scale condition, one sub-interval may be too short for the coarse chain to self-average even when the total Δ is in the fast regime, and scalar factorization fails for that split. Both regimes together give the stated bound. In the genuinely intermediate regime, the vector-level semigroup identity remains exact but scalar factorization carries an additional Jensen-gap correction analyzed in Theorem F.11 and §F.7. □

What Proposition F.6 does and does not do. Like Proposition V.0' (body), the proposition establishes block-level scalar factorization for the *inputs* to the outer evaluator in the stated regimes. This makes (A5) operationally well-posed on the dynamic compressed class. It does not establish A5 for arbitrary outer evaluators on the class — the arithmetic-mean counterexample of Theorem A.7 continues to apply. Theorem F.7 below verifies that the induced dynamic geometric evaluator satisfies (A1)–(A5) on the class to the stated subleading error.

Theorem F.7 (Dynamic geometric outer evaluation; Theorem V.3 restated, quasi-static and fast-z regimes, formal)**.** *Consider two independent, externally non-transferable blocks satisfying the hypotheses of Theorem F.5, with exogenous weights* $w = (w_1, w_2)$ *summing to* 1*. The induced dynamic geometric evaluator*

$$A_w^{\text{geom}}(\mathcal{A}_1, \mathcal{A}_2) := \mathcal{A}_1^{w_1}\,\mathcal{A}_2^{w_2}$$

on scalar dynamic interface values $\mathcal{A}_i(\Delta, z_i) = [e^{\Delta M_i}\mathbf{1}]_{z_i}$ *satisfies axioms (A1)–(A5) of Theorem II.1 on the dynamic compressed-block class of Theorem F.4, to within* $O(\epsilon_{\text{dyn}})$ *in the quasi-static and fast-*z *regimes of Theorem F.6. (A4) is vacuous at* $n=2$*; the all-arity dynamic (A4) realization on the compressed class is closed in Theorem G.12. The one-cycle value is*

$$V_\Delta \;=\; \mathcal{A}_1(\Delta, z_1)^{w_1}\,\mathcal{A}_2(\Delta, z_2)^{w_2}.$$

Proof. (A1)–(A3) are immediate for A_w^{geom} on positive scalar inputs. (A4) is vacuous at $n=2$: the substantive n-block A4 realization is Appendix G (Theorem G.7, Theorem G.12), of which the present two-block theorem is the $n=2$ face of the same dynamic evaluator-level class. (A5) follows from Theorem F.6: applying A_w^{geom} pointwise to the approximate scalar factorization,

$$A_w^{\text{geom}}\big(\mathcal{A}(\Delta_1+\Delta_2, z)\big) = \prod_i \big(\mathcal{A}_i(\Delta_1, z)\,\mathcal{A}_i(\Delta_2, z)(1+O(\epsilon_{\text{dyn}}))\big)^{w_i} = A_w^{\text{geom}}\big(\mathcal{A}(\Delta_1, z)\big)\,A_w^{\text{geom}}\big(\mathcal{A}(\Delta_2, z)\big) + O(\epsilon_{\text{dyn}}),$$

which is (A5) with $F_w(s,t) = st$ in the stated regimes. The subleading error inherits from Theorem F.6 together with Theorem F.5. As at $n=2$ in V.3, (A4) is vacuous here on its own; the all-arity dynamic (A4) realization on the compressed class is closed in Theorem G.12 (Theorem A.8 then applies in the exact-axiom limit; see V.1's structural remark on uniqueness for the routing). □

Remark F.8 ((A5) is not automatic in the dynamic setting either). As in the static setting, (A5) is not derivable from (A1)–(A4) for arbitrary evaluators: Theorem A.7 exhibits an (A1)–(A4) arithmetic-mean evaluator that fails (A5) on a class where scalar coordinate-wise composition does hold. What the Feynman-Kac semigroup supplies is that the (A5) functional equation becomes well-posed on the dynamic compressed class (in the quasi-static and fast-z regimes, with the segment-scale condition in the latter), and that the specific induced geometric evaluator A_w^{geom} satisfies it. Without the uniform spectral gap and slow-motion hypotheses of Theorem F.1, the scalar summary $\mathcal{A}_i(\Delta, z)$ ceases to represent block dynamics, and (A5) ceases to be well-posed in a principal-mode sense — the moving-landscape analogue of Appendix D's short-cycle observation.

F.7. Dynamic log-scale synthesis

The outer formula is geometric, but the useful coordinate remains logarithmic.

Corollary F.9 (Dynamic log-scale synthesis; body Section V.4 restated)**.** *Under the hypotheses of Theorem F.7,*

$$\log V_\Delta \approx w_1 \log \mathcal{A}_1(\Delta, z_1) + w_2 \log \mathcal{A}_2(\Delta, z_2).$$

Proof. Take logarithms of Theorem F.7. The only approximation is the same compression error already present in Theorem F.5. □

This is the dynamic generalization of Appendix D's Theorem D.8. The additive primitive in V.4's log-coordinate is no longer the frozen integrated rate $-\Delta\, a_1^{(i)}$, but the log of a Feynman-Kac survival expectation; the Appendix J representational coda is therefore required in its dynamic form.

Remark F.10 (Jensen correction)*.* In general,

$$\log \mathcal{A}(\Delta, z) \neq -\mathbb{E}_z\left[\int_0^\Delta a_1(Z_s)\, ds\right],$$

because the expectation sits inside the exponential. Jensen gives

$$-\log \mathcal{A}(\Delta, z) \leq \mathbb{E}_z\left[\int_0^\Delta a_1(Z_s)\, ds\right],$$

with equality only when the integrated rate is effectively non-random along the coarse path. So the dynamic additive coordinate is not in general the average integrated rate; it is the log of a Feynman-Kac functional. The Jensen gap vanishes in the quasi-static and fast-z limits below and is non-trivial in the intermediate slow-fast regime.

Proposition F.11 (Quasi-static and fast-z limits)**.** *Under the hypotheses of Theorem F.5:*

(i) Quasi-static limit *(coarse state nearly frozen over the cycle; $q_*\Delta \ll 1$):*

$$\mathcal{A}(\Delta, z) \approx \exp\left(-\int_0^\Delta a_1(z(t))\, dt\right),$$

which reduces to $e^{-a_1(z)\Delta}$ when z is truly static, recovering Appendix D.

(ii) Fast-z limit*. Under the triple separation*

$$\max_z a_1(z) \;\ll\; q_* \;\ll\; g_*$$

(killing slower than coarse mixing, coarse mixing still slower than fiber compression so the adiabatic elimination of F.5 applies),

$$\mathcal{A}(\Delta, z) \approx e^{-\bar{a}_1 \Delta}, \qquad \bar{a}_1 := \sum_{z \in Z} \pi_Z(z)\, a_1(z),$$

where π_Z is the stationary law of the coarse chain.

Proof sketch. In the quasi-static limit, the coarse path contributes almost no randomness to the integrated rate, so the Feynman-Kac expectation collapses to the exponential of the path integral. In the fast-z limit, the coarse chain averages to stationarity before appreciable killing occurs, so the integrated rate self-averages to its stationary mean; the Jensen gap of Theorem F.10 closes. □

F.8. Numerical family: curvature modulation

Take the three-state family

$$Z = \{L, M, H\}, \qquad h_z(x) = c(z)\big(x - \tfrac{1}{2}\big)^2, \qquad c(L) = 0.5,\ c(M) = 1.0,\ c(H) = 1.5,$$

with $R = 30$, $m = 2$, $\gamma = 1$, $\kappa_0 = 1$, $\beta = 1$. Direct diagonalization of G_z at each z confirms the expected pattern: the principal rate $a_1(z)$ is barrier-controlled and decreases rapidly with curvature (from $\sim 10^{-3}$ at $c = 0.5$ to $\sim 10^{-5}$ at $c = 1.5$), while the compression gap $a_2(z) - a_1(z)$ remains uniformly open, with $g_* \approx 0.043$. The compression timescale is $1/g_* \approx 23$, while the metastable killing scales $1/a_1(z)$ range from roughly 5×10^2 to 10^5. The moving-landscape window $\Delta \in [1/g_*, 1/\max_z a_1(z)]$ is therefore numerically open across all three coarse states (Figure 6).

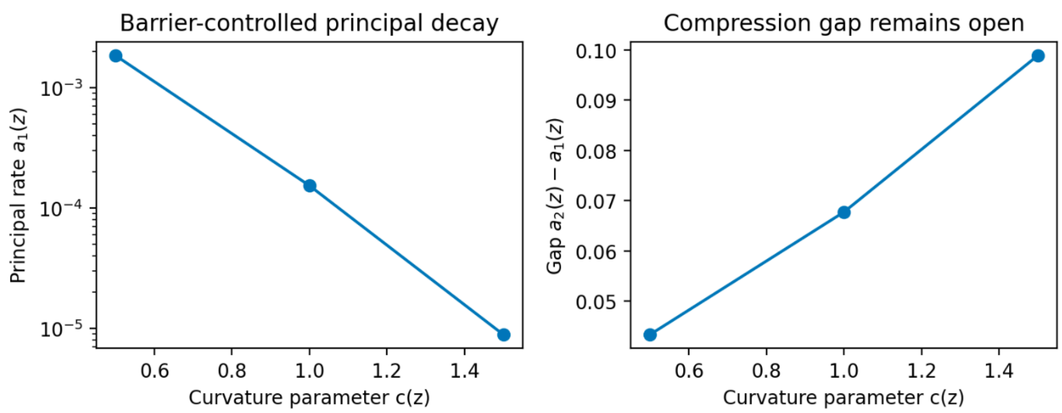

Figure 6: Curvature modulation in the discrete coarse family $h_z(x) = c(z)(x - 1/2)^2$ at $R = 30$, $m = 2$, $\gamma = 1$, $\kappa_0 = 1$, $\beta = 1$. *Left*: the principal decay rate $a_1(z)$ is barrier-controlled and falls rapidly with curvature, by more than two orders of magnitude across $c(z) \in \{0.5, 1.0, 1.5\}$ (log scale). *Right*: the compression gap $a_2(z) - a_1(z)$ remains uniformly open across the same range, with $g_* \approx 0.043$. The separation of scales $1/g_* \approx 23 \ll 1/a_1(z)$ is robust across coarse states, supplying the compression premise of Theorem F.5.

F.8.1. Two-state subfamily. For an explicit coarse-chain Feynman-Kac calculation, restrict to $Z = \{L, M\}$ with $c(L) = 0.5$, $c(M) = 1.0$ and the symmetric two-state coarse generator

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \lambda & -\lambda \end{pmatrix}.$$

Direct matrix-exponential evaluation of Theorem F.2 for three coarse jump rates $\lambda \in \{0.001, 0.005, 0.05\}$ interpolates between the quasi-static approximation (at small λ, reducing to $e^{-a_1(L)\Delta}$ for the L-started chain) and the fast-z approximation (at larger λ approaching $e^{-\bar{a}_1\Delta}$ with $\bar{a}_1 = (a_1(L) + a_1(M))/2$). The intermediate regime $\lambda = 0.005$ exhibits a genuinely distinct Feynman-Kac profile, and the effective dynamic log-rate $-\Delta^{-1} \log \mathcal{A}(\Delta, z)$ interpolates between the frozen-state rate and the stationary average (Figure 7). (The $\lambda = 0.05$ curve sits near the upper edge of the adiabatic window $q_* \ll g_*$; tight quantitative agreement with the fast-z corollary would require λ small relative to $g_* \approx 0.043$ while remaining large relative to the metastable killing scale, a narrow range at the present family's parameters.)

F.9. Negative side: when the landscape moves too fast

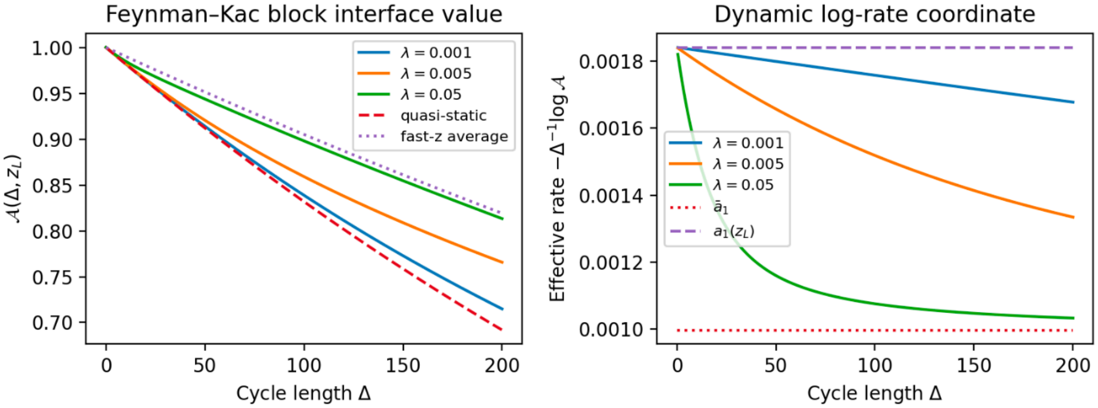

Figure 7: Feynman-Kac interface value $\mathcal{A}(\Delta, z_L)$ and its log-rate coordinate in the two-state subfamily of F.8.1, for three coarse jump rates $\lambda \in \{0.001, 0.005, 0.05\}$. *Left*: $\mathcal{A}(\Delta, z_L)$ lies between the quasi-static approximation $e^{-a_1(L)\Delta}$ (slow limit, upper envelope) and the fast-z approximation $e^{-\bar{a}_1\Delta}$ (fast limit, lower envelope); intermediate λ traces a genuinely distinct profile. *Right*: the dynamic log-rate $-\Delta^{-1}\log\mathcal{A}(\Delta, z_L)$, the additive coordinate of §V.4, interpolates from the frozen-state rate $a_1(L)$ toward the stationary average $\bar{a}_1$ as λ grows. The $\lambda = 0.05$ curve sits near the upper edge of the adiabatic window $q_* \ll g_*$ and is displayed to show approach to the fast-z limit, not as a tight quantitative match.

Appendix D's negative side was fixed-landscape and timing-based: if $\Delta < \tau_{\text{compress}}$, then the block has not yet compressed; in the diffusive baseline of Appendix B there is no asymptotically robust window at all. This appendix adds a new failure mode: even when the cycle is long enough, compression can fail because *the coarse landscape itself moves on the compression scale*.

F.9.1. Structural failure mode. If $q_*^{(i)} \gtrsim g_*^{(i)}$, then the coarse state changes on the same timescale as principal-mode formation. The block does not have time to relax to the current state's principal mode before the state moves again, so the adiabatic elimination of F.5 is no longer justified. The failure is not that the weighted geometric form ceases to be idempotent or some other axiom-level collapse; the failure is that no honest scalar interface value has formed. This is a moving-landscape version of the Section III.2 timing warning: before compression, future continuation remains sensitive to variables the scalar summary has not absorbed.

F.9.2. Numerical anchor. In the two-state family of F.8.1 the good-regime gap is $g_* \approx 0.043$. A coarse jump rate $\lambda = 0.005$ lies comfortably inside the adiabatic window. At $\lambda = 0.08$, the coarse timescale $1/\lambda \approx 12.5$ is already comparable to the compression timescale $1/g_* \approx 23$, so the hypotheses of Theorem F.5 are no longer honest. The Feynman-Kac object $\mathcal{A}(\Delta, z)$ still exists as a value function of the coarse process, but it cannot be claimed as the accurate compressed interface value of the full block dynamics without additional control.

The same moral appears more starkly in Appendix B's diffusive baseline, where $\tau_{\text{compress}} \asymp \tau_{\text{kill}}$ already at the fixed-landscape level and no outer-scale window exists at all. The present moving-landscape failure is genuinely new: the metastable window may exist at each fixed z, yet coarse motion can still destroy it dynamically. Real boundaries that induce the right thermodynamic well (Appendix C) and the right spectral gap (Appendix D) must also induce coarse motion that respects the compression timescale the gap defines — the third condition of Part VI's three-part operational criterion.

F.10. Rare-event scaling and the relation to Appendix E

This closing subsection records the precise relationship between the discrete rare-event bridge of Appendix E and the continuous-time Feynman-Kac interface value developed above. The short version: Appendix F is the finite-horizon continuous-time scaling limit of the averaged one-cycle structure in Appendix E, after fast fiber mixing has already removed the within-fiber microstate. The point is to turn the prose assertion of Theorem F.3 and Appendix E's §E.7 into a scaling-limit theorem.

One point must be kept separate. Appendix E is stated as an infinite-horizon Bellman fixed-point theorem: a rare event is sampled, and if it is not terminal the recursion continues. A literal fixed-point scaling of that object gives a continuous-time resolvent-type equation, not a finite-horizon semigroup. The Feynman-Kac object of Theorem F.2 is a finite-horizon survival semigroup. The equivalence below is therefore stated for the finite-horizon averaged continuation operator naturally associated with Appendix E's one-cycle averaging step, not for the Bellman fixed point itself.

F.10.1. Finite-horizon averaged continuation. Let Z be a finite coarse state space. In Appendix E, fast within-fiber mixing produces an averaged coarse transition kernel and averaged survival payoff. For the continuous-time limit, package those two averaged objects into a small-step killed continuation operator.

Fix a principal-rate function $a_1 : Z \to [0, \infty)$ and a continuous-time coarse generator Q on Z. For each small $\varepsilon > 0$, let P_ε be a Markov kernel on Z and let $s_\varepsilon : Z \to [0, 1]$ be a one-step averaged survival factor. Define

$$(K_\varepsilon f)(z) := s_\varepsilon(z) \sum_{z' \in Z} P_\varepsilon(z, z')\, f(z').$$

The intended reading is that the fast fiber has mixed during the short interval, the coarse state makes one ε-sized move, and the block survives that short interval with averaged probability $s_\varepsilon(z)$. (The order convention — applying the survival factor before or after the transition, i.e. $s_\varepsilon \cdot P_\varepsilon$ versus $P_\varepsilon \cdot s_\varepsilon$ — is immaterial to first order in ε and yields the same generator limit.)

The operator K_ε is not Appendix E's averaged Bellman operator $\overline{T}_R$ itself. It is the finite-horizon killed continuation operator naturally associated with Appendix E's one-cycle averaging step. Once fast within-fiber mixing has produced the averaged transition kernel and the averaged one-step survival factor, those objects can be packaged either as a *branching operator* — Appendix E's Bellman form, suited to the eventual-hazard question with a geometric rare-event recursion — or as a *killed continuation operator*, suited to the finite-horizon survival question with a fixed outer cycle of duration Δ. Both are valid uses of the same averaging content; they answer different questions. Appendix F's finite-horizon semigroup is the scaling limit of the killed-continuation form.

Assume the rare-event scaling

$$P_\varepsilon f = f + \varepsilon\, Qf + o(\varepsilon), \qquad s_\varepsilon(z) = 1 - \varepsilon\, a_1(z) + o(\varepsilon),$$

uniformly for bounded f and uniformly in $z \in Z$. Equivalently, the per-step killing probability is $\varepsilon\, a_1(z) + o(\varepsilon)$, while the coarse transition kernel has generator Q.

F.10.2. Generator convergence.

Proposition F.12 (One-cycle generator limit)**.** *Under the scaling above,*

$$\frac{K_\varepsilon - I}{\varepsilon} f \longrightarrow (Q - \mathrm{diag}(a_1))\, f$$

uniformly on bounded functions $f : Z \to \mathbb{R}$.

Proof. By definition, $K_\varepsilon f(z) = s_\varepsilon(z)(P_\varepsilon f)(z)$. Using the two expansions,

$$K_\varepsilon f(z) = \big(1 - \varepsilon\, a_1(z) + o(\varepsilon)\big)\big(f(z) + \varepsilon\, Qf(z) + o(\varepsilon)\big).$$

Hence

$$K_\varepsilon f(z) = f(z) + \varepsilon\big(Qf(z) - a_1(z)\, f(z)\big) + o(\varepsilon),$$

uniformly in z. This is exactly the claimed generator convergence. □

Theorem F.13 (Rare-event limit to the Feynman-Kac interface value)**.** *Let* $n_\varepsilon := \lfloor \Delta/\varepsilon \rfloor$*. Under the hypotheses above,*

$$K_\varepsilon^{n_\varepsilon}\,\mathbf{1} \longrightarrow e^{\Delta(Q-\operatorname{diag}(a_1))}\,\mathbf{1}$$

uniformly on Z *as* $\varepsilon \to 0$*. Therefore the finite-horizon averaged discrete interface value converges to the Feynman-Kac interface value of Theorem F.2:*

$$\mathcal{A}(\Delta, z) = \left[e^{\Delta(Q-\operatorname{diag}(a_1))}\,\mathbf{1}\right]_z = \mathbb{E}_z \exp\left(-\int_0^\Delta a_1(Z_s)\,ds\right).$$

Proof. Because Z is finite, all operators act on a finite-dimensional normed space. Theorem F.12 gives

$$K_\varepsilon = I + \varepsilon L + o(\varepsilon), \qquad L := Q - \operatorname{diag}(a_1),$$

in operator norm. The finite-dimensional Euler-Chernoff product formula gives

$$\big(I + \varepsilon L + o(\varepsilon)\big)^{\lfloor \Delta/\varepsilon \rfloor} \longrightarrow e^{\Delta L}$$

uniformly for Δ in compact intervals. Applying this to $\mathbf{1}$ yields the stated convergence. The final equality is the standard Feynman-Kac formula for the coarse chain with killing potential a_1. □

F.10.3. Relation to the Appendix E fixed point. Theorem F.13 should not be read as saying that the Appendix E fixed point itself converges directly to the Feynman-Kac semigroup. In the notation of Appendix E, the averaged Bellman operator has the schematic form

$$(\overline{T}f)(z) = h(z)\,\overline{\Phi}(z) + (1 - h(z)) \int f(z')\,\overline{L}_z(dz').$$

If one scales $h_\varepsilon(z) = \varepsilon\, a_1(z) + o(\varepsilon)$ and $\overline{L}_\varepsilon = I + \varepsilon\, Q + o(\varepsilon)$ and then takes a fixed point, the limiting equation is resolvent-like:

$$(a_1 - Q)\, c = a_1\, \overline{\Phi}$$

under the corresponding scaling, not the finite-horizon semigroup equation. This is the correct continuum limit of the eventual-hazard recursion.

The two scaling questions therefore differ in both operator form and horizon. Appendix E's averaged Bellman operator $\overline{T}_R$, scaled as an infinite-horizon fixed point, produces the resolvent-type equation above — the correct continuum limit of the eventual-hazard recursion, where terminal hazard occurs with probability h at each rare event. Appendix F instead asks for survival over a fixed outer cycle $[0, \Delta]$, with killing occurring at rate $a_1(z)$ throughout. The relevant discrete object for that question is K_ε: the same averaged content repackaged as a killed continuation operator and then iterated finitely many times. Once both the operator repackaging and the horizon distinction are made explicit, the two appendices match exactly: Appendix E supplies the averaging step that removes the fast fiber, and Appendix F is the continuous-time finite-horizon semigroup obtained by iterating that averaged step in its killed-continuation form on the rare-event scale.

F.10.4. Error bookkeeping. Theorem F.13 assumes complete fast-fiber averaging inside each ε-cycle. With finite but fast mixing, the Appendix E error term reappears as an additional perturbation of K_ε. If the one-cycle averaging error is δ_ε uniformly in z, then over $n_\varepsilon \simeq \Delta/\varepsilon$ cycles the naive accumulated error is of order

$$O\left(\frac{\Delta}{\varepsilon}\,\delta_\varepsilon\right),$$

up to harmless stability constants. This perturbation vanishes in the rare-event limit only if $\delta_\varepsilon = o(\varepsilon)$; at the borderline scaling $\delta_\varepsilon = O(\varepsilon)$ it contributes an $O(1)$ correction to the limiting semigroup, so Theorem F.13 is not preserved without the stronger condition. The clean limit is exact in the ideal complete-mixing regime $\delta_\varepsilon = 0$.

This is structurally analogous to Appendix F's compression estimate $e^{-g_*\Delta} + q_*/g_*$ of Theorem F.4. The terms are not identical finite-ε quantities: Appendix E's δ_* measures failure of within-fiber averaging against a stationary fiber law in the rare-event skeleton; Appendix F's $e^{-g_*\Delta}$ and q_*/g_* measure, respectively, failure of principal-mode dominance and failure of adiabatic separation in the spectral killed-process realization. In the limiting regime where the fast fiber fully equilibrates to its principal mode before the coarse state moves, these are the same structural obstruction: residual dependence on fast internal state at the interface. For the monograph, the correct claim is therefore convergence of the averaged finite-horizon object plus schematic inheritance of the same compression discipline, not a term-by-term identity of error bounds.

Appendix G. Heterogeneous n-Block Composition and Regrouping Closure

G.1. Opening

This appendix closes the heterogeneous n-block extension of Part V's synthesis. Appendix D realized the outer theorem on two static compressed Gibbs-lifted killed-process blocks; Appendix F extended the realization to two slow-fast blocks with dynamic Feynman-Kac interface values. Both kept the block count at two, which is the minimal case for exhibiting weighted geometric composition but is essentially invisible to the spatial regrouping axiom (A4) of Theorem A.8. A4 is genuinely an n-ary axiom: when coordinates are partitioned into clusters, inner cluster evaluation followed by outer cluster evaluation must agree with direct evaluation. For $n = 2$, the partition is trivial and the test is vacuous. This appendix asks whether the compressed interface-value class is stable under heterogeneous n-block composition and nontrivial regrouping.

Before the technical development, we record the calibration discipline that structures the rest of the appendix. Three distinct layers of n-block composition need to be kept separate; conflating them erases the evaluator-vs-conjunction distinction of §II.2.1.
Remark G.1 (Three layers of n-block composition).

(i) Scalar evaluator layer. Weighted geometric regrouping is *exactly* associative: Theorem G.6.

(ii) Compressed-class layer. Regrouping on compressed interface values is valid up to weighted log-error controlled by constituent compression errors: Theorem G.7.

(iii) Physical product-space layer. Literal product killed or Feynman-Kac super-blocks are *different objects*; in the dynamic setting they additionally exhibit Jensen gaps (§G.7).

Axiom A4 lives at layers (i)–(ii). Layer (iii) is a separate construction available only under stronger hypotheses (independence of fast dynamics, sometimes further) and answering a different question. The appendix's theorems target (i) and (ii); layer (iii) appears for comparison and in the Jensen-gap discussion of §G.7.
Remark G.2 (Routing of uniqueness through Appendix G). The two-block realization theorems of Appendix D and Appendix F (Theorems V.1 and V.3) establish realization of the induced geometric evaluator on the minimal nontrivial case. Theorem G.7 and Theorem G.12 establish that the same induced evaluator is closed under nontrivial n-block regrouping on the compressed class — in GPT 5.5's phrasing, they verify the compressed-class realization of the all-arity (A4) regrouping condition for the induced geometric evaluator, with the stated compression-error bookkeeping. Once this all-arity family is in place, the exact characterization theorem (Theorem A.8) applies to the ideal scalar evaluator at the exact-axiom limit. Approximate uniqueness is not claimed without a separate stability theorem.

Critically: Theorem G.7 verifies closure for the induced evaluator and class; it does *not* re-prove (A4) for arbitrary candidate evaluators. Theorem A.8 remains the characterization theorem; the appendix's role is realization of the all-arity (A4) condition on the compressed class, not exponent-pinning.
Remark G.3 (Uniqueness lives at the evaluator layer, not the product-space layer). The uniqueness routed through Theorem G.2 operates at layer (i)–(ii) of Theorem G.1: it is uniqueness in the space of scalar evaluators acting on compressed interface values. It is *not* a statement that the resulting normalized geometric evaluator value coincides with the survival law of a literal product physical process (layer (iii)). Those are different objects and answer different questions, as separated cleanly in §G.4 and quantified by the Jensen gap of §G.7. The exact-axiom limit characterization of Theorem A.8 therefore tells us about the evaluator value, not about literal joint survival of the constituent processes.

The target of the appendix is therefore not a rederivation of the weighted geometric mean for n variables — that is algebra, once Theorem A.8 has been accepted. The target is that heterogeneous compressed interface values are closed under *evaluator-level* n-block regrouping, with explicit compression-error bookkeeping, and that normalized evaluator super-blocks are separated cleanly from literal product-space survival objects.

The appendix proceeds in three steps. The static half (§G.3) proves that direct n-block evaluation and regrouped evaluation agree exactly on compressed scalar interface values, with log-error controlled by the weighted sum of constituent compression errors. Section G.4 makes the layer (ii) vs (iii) distinction above concrete at the cluster scale, defining the evaluator cluster value B_C and the physical product super-block J_C side-by-side. The dynamic half (§G.5–G.7) carries the same closure into the slow-fast setting, replacing static spectral compression with componentwise uniform gap plus componentwise slow coarse motion, and introduces the n-block Feynman-Kac Jensen gap between evaluator cluster values and product-space Feynman-Kac values with weighted potential. Failure modes and deferred extensions are recorded in §G.8.

G.2. Setup: n heterogeneous Gibbs-lifted blocks

Fix $n \geq 2$. For each block $i \in \{1, \ldots, n\}$, take a fixed-fiber Gibbs-lifted killed-process block of the Appendix C type, with its own parameters

$$\left(h^{(i)}, R^{(i)}, m^{(i)}, \gamma^{(i)}, \kappa_0^{(i)}, \beta^{(i)}\right),$$

so that $h^{(i)} : [0,1] \to \mathbb{R}$ is a symmetric one-well potential satisfying conditions (i)–(v) of §C.2, $R^{(i)} \in \mathbb{Z}_{\geq 0}$ is the fixed total, $m^{(i)} \in \mathbb{Z}_{\geq 1}$ the threshold width, and $\gamma^{(i)}, \kappa_0^{(i)}, \beta^{(i)} > 0$ the exchange rate, base killing rate, and inverse temperature. The blocks are externally non-transferable: after the block interface there are no admissible controls that move viability-relevant resource from one block to another.

Let $G^{(i)}$ denote the killed generator of block i and write its decay rates as

$$0 < a_1^{(i)} < a_2^{(i)} \leq a_3^{(i)} \leq \cdots,$$

so $-a_j^{(i)}$ are the eigenvalues of $G^{(i)}$ ordered from closest to zero downward. Define the block-level compression gap

$$g_i := a_2^{(i)} - a_1^{(i)}.$$

Fix an outer cycle length $\Delta > 0$ and exogenous weights

$$w = (w_1, \ldots, w_n) \in (0,1)^n, \qquad \sum_{i=1}^{n} w_i = 1,$$

as in Appendix D. Weights are evaluator weights: exposure fractions, scheduling, branch prevalence, or another exogenously assigned measure of relative importance. They are not derived here from block thermodynamics (cf. Appendix K, glossary entry "Exogenous schedule / evaluator weights").

Definition G.4 (Static compressed block value)**.** Block i is *statically compressed* on the cycle timescale Δ if $g_i \Delta \gg 1$. Its continuation over one outer cycle is then represented by a positive scalar interface value $c_i(\Delta)$ with relative error $\varepsilon_i(\Delta)$. Schematically,

$$c_i(\Delta) = e^{-a_1^{(i)} \Delta}, \qquad \varepsilon_i(\Delta) = O(e^{-g_i \Delta}).$$

More generally, the abstract compression statement we use is that the exact block continuation equals $c_i(1 + r_i)$ with $|r_i| \leq \varepsilon_i$, uniformly in admissible initial microstates in the metastable well.

The precise principal-mode coefficient depends on normalization conventions and initial-microstate class (as in Appendix D's principal-mode normalization of §D.4). For the regrouping theorem below, what matters is not whether the scalar is written literally as $e^{-a_1^{(i)} \Delta}$ but that the block has compressed to a positive scalar with a controlled relative error.

G.3. Static composition and regrouping with errors

Definition G.5 (Weighted evaluator on compressed n-block values)**.** For positive compressed block values $c = (c_1, \dots, c_n) \in (0, \infty)^n$ and weights $w \in (0,1)^n$ with $\sum_i w_i = 1$, define the induced evaluator

$$A_w(c_1, \dots, c_n) := \prod_{i=1}^{n} c_i^{w_i}.$$

Proposition G.6 (Exact regrouping on scalar values). *Let $\mathcal{C} = \{C_1, \dots, C_k\}$ be a partition of $\{1, \dots, n\}$ into nonempty clusters. Define*

$$W_\alpha := \sum_{i \in C_\alpha} w_i, \qquad \alpha_i := \frac{w_i}{W_\alpha} \ (i \in C_\alpha),$$

and the cluster value

$$B_\alpha := \prod_{i \in C_\alpha} c_i^{\alpha_i}.$$

Then

$$A_w(c_1, \dots, c_n) = A_W(B_1, \dots, B_k), \qquad W = (W_1, \dots, W_k).$$

Proof. Expand the right-hand side:

$$A_W(B_1, \dots, B_k) = \prod_{\alpha=1}^{k} B_\alpha^{W_\alpha} = \prod_{\alpha=1}^{k} \left(\prod_{i \in C_\alpha} c_i^{w_i/W_\alpha} \right)^{W_\alpha} = \prod_{i=1}^{n} c_i^{w_i}.$$

□

The algebraic identity fixes notation; the content enters when the scalar inputs are approximate outputs of heterogeneous compression windows.

Theorem G.7 (Static closure of the compressed class under regrouping). *Suppose block i has exact continuation $\hat{c}_i = c_i(1 + r_i)$ with $c_i > 0$ and $|r_i| \le \varepsilon_i < 1/2$. Then direct and regrouped evaluations agree exactly when applied to the exact continuations:*

$$A_w(\hat{c}_1, \dots, \hat{c}_n) = A_W(\hat{B}_1, \dots, \hat{B}_k), \qquad \hat{B}_\alpha := \prod_{i \in C_\alpha} \hat{c}_i^{w_i/W_\alpha}.$$

Relative to the ideal compressed values,

$$|\log A_w(\hat{c}_1, \dots, \hat{c}_n) - \log A_w(c_1, \dots, c_n)| \le 2 \sum_{i=1}^{n} w_i \varepsilon_i.$$

In particular, in the static spectral-compression regime,

$$\log A_w(\hat{c}) - \log A_w(c) = O\left(\sum_{i=1}^{n} w_i \, e^{-g_i \Delta} \right).$$

Proof. The exact regrouping identity follows from Theorem G.6 applied to $\hat{c}_i$ instead of c_i. For the error bound,

$$\log A_w(\hat{c}) - \log A_w(c) = \sum_{i=1}^{n} w_i \log(1 + r_i).$$

If $|r_i| \le \varepsilon_i < 1/2$ then $|\log(1 + r_i)| \le 2\varepsilon_i$, so

$$\left| \sum_i w_i \log(1 + r_i) \right| \le 2 \sum_i w_i \varepsilon_i.$$

The final statement substitutes $\varepsilon_i = O(e^{-g_i \Delta})$. □

Remark G.8 (What A4 tests here). Theorem G.7 does not make the associativity of multiplication profound; it makes the closure of the input class explicit. Heterogeneous blocks may have different potentials, resources, thresholds, inverse temperatures, principal rates, gaps, and compression errors.

Regrouping remains legitimate because the outer evaluator acts only after each constituent block has already produced a scalar interface value. A4 is therefore tested at the level where it actually belongs: not on raw microstates, but on the compressed interface-value class. The two-block case of Theorem D.4 (Theorem D.4) leaves this test trivial by partition.

Corollary G.9 (Static n-block log-rate synthesis)**.** *In the ideal static spectral-compression regime, with* $c_i(\Delta) \approx e^{-a_1^{(i)}\Delta}$*,*

$$\log A_w(c_1,\dots,c_n) \approx -\Delta \sum_{i=1}^{n} w_i\, a_1^{(i)}.$$

Under a regrouping $\mathcal{C}$, the weights factor through clusters:

$$\sum_{\alpha=1}^{k} W_\alpha \left(\sum_{i\in C_\alpha} \frac{w_i}{W_\alpha}\, a_1^{(i)} \right) = \sum_{i=1}^{n} w_i\, a_1^{(i)}.$$

G.4. Three-layer calibration: evaluator super-blocks versus physical product super-blocks

Theorem G.7 defines a super-block at the evaluator level: a cluster value obtained by applying the normalized weighted geometric evaluator to compressed constituent interface values. When constituent blocks are independent, a different construction is available — the literal product killed process on the product fast state space. This product object has its own survival law, and it is not the same object as the normalized evaluator cluster value. This is the concrete instantiation at cluster scale of the layer (ii) / (iii) distinction flagged at Theorem G.1.

Concretely, for a cluster $C \subseteq \{1,\dots,n\}$, the evaluator-level cluster value is

$$B_C = \prod_{i\in C} c_i^{w_i/W_C}, \qquad W_C = \sum_{i\in C} w_i.$$

If $c_i \approx e^{-a_1^{(i)}\Delta}$ then $B_C \approx \exp\left(-\Delta \sum_{i\in C}(w_i/W_C)\, a_1^{(i)}\right)$. The physical product killed generator for the cluster is

$$G_C = \bigoplus_{i\in C} G^{(i)},$$

interpreted as joint survival of all constituents. Its principal decay rate is

$$a_{1,C} = \sum_{i\in C} a_1^{(i)},$$

and its first compression gap is

$$g_C = \min_{i\in C} g_i,$$

because the first excited product modes are obtained by exciting one constituent block while keeping all others in their principal modes. Independent product dynamics do not produce a cluster gap smaller than every constituent gap; the cluster gap is the slowest constituent compression gap.

Remark G.10 (Appendix D's evaluator-versus-conjunction discipline in n-block form)*.* Appendix D emphasized that the weighted geometric mean on two compressed blocks is not raw conjunction survival. The same discipline is needed here. The object $\prod_{i\in C} c_i$ is the natural unweighted joint-survival product for independent conjunction survival of a cluster. The object $\prod_{i\in C} c_i^{w_i/W_C}$ is the normalized evaluator value used when the cluster is to be represented as one positive scalar for further Theorem A.8 composition. Both are legitimate objects; they answer different questions.

This distinction sharpens in the slow-fast case, where the evaluator-versus-product-space gap acquires a Feynman-Kac Jensen structure (§G.7).

G.5. Dynamic heterogeneous n-block setup

We now generalize Appendix F from two blocks to n heterogeneous blocks. For each block $i \in \{1,\dots,n\}$, let:

- $Z^{(i)}$ be a finite coarse state space;
- $Q^{(i)}$ a continuous-time coarse-state generator on $Z^{(i)}$;
- the coarse chains $Z^{(i)}$ mutually independent by construction: each $Q^{(i)}$ acts on its own state space, with no cross-block coupling;
- for each $z \in Z^{(i)}$, $G_z^{(i)}$ a fixed-z Gibbs-lifted killed generator of the Appendix C type;
- $0 < a_1^{(i)}(z) < a_2^{(i)}(z) \le \cdots$ the ordered fixed-z decay rates of $G_z^{(i)}$.

Define the componentwise uniform compression gap and componentwise coarse jump scale by

$$g_i := \min_{z \in Z^{(i)}} \big(a_2^{(i)}(z) - a_1^{(i)}(z)\big) > 0, \qquad q_i := \max_{z \in Z^{(i)}} \sum_{z' \neq z} Q_{zz'}^{(i)}.$$

These are the per-block analogues of g_* and q_* from Appendix F.

Definition G.11 (Dynamic block interface value)**.** For each block i, the dynamic interface value is

$$\mathcal{A}_i(\Delta, z_i) := \mathbb{E}_{z_i} \exp\Big(-\int_0^\Delta a_1^{(i)}(Z_s^{(i)})\, ds\Big) = \Big[e^{\Delta(Q^{(i)} - \mathrm{diag}(a_1^{(i)}))}\, \mathbf{1}\Big]_{z_i}.$$

The componentwise formulation is part of the theorem below, not merely a warning: each block must compress *locally* before it is admitted to the outer evaluator. The theorem does not require the product coarse state $(Z^{(1)}, \ldots, Z^{(n)})$ to remain approximately fixed as a monolithic object.

G.6. Dynamic n-block theorem with componentwise compression

Theorem G.12 (Dynamic heterogeneous n-block composition; n-block form of Theorem V.3, formal, schematic compression error)**.** *Consider n externally non-transferable blocks as in §G.5, with exogenous weights $w_i > 0$ summing to 1. Assume* componentwise *dynamic compression: for each block i,*

(H1) the fixed-z killed generators $G_z^{(i)}$ lie in the Appendix C metastable class;

(H2) the componentwise uniform gap satisfies $g_i > 0$;

(H3) coarse motion is slow relative to componentwise compression, $q_i \ll g_i$;

(H4) the common outer cycle length Δ lies in the nonempty window

$$\max_i g_i^{-1} \ll \Delta \ll \min_i \big(\sup_z a_1^{(i)}(z)\big)^{-1},$$

so $g_i\Delta \gg 1$ for every block i (compression) and Δ is short relative to each block's metastable killing scale (cycle fits within the metastable regime).

Then block i admits the dynamic scalar interface value $\mathcal{A}_i(\Delta, z_i)$ of Theorem G.11, together with its Feynman-Kac continuation family $\Delta' \mapsto e^{\Delta' M_i}\mathbf{1}$ ($M_i := Q^{(i)} - \mathrm{diag}(a_1^{(i)})$), with schematic compression error

$$\varepsilon_i^{\mathrm{dyn}}(\Delta) \lesssim e^{-g_i\Delta} + q_i/g_i.$$

The induced outer evaluator on the dynamic compressed n-block class is

$$V_\Delta(z_1, \ldots, z_n) = A_w\big(\mathcal{A}_1(\Delta, z_1), \ldots, \mathcal{A}_n(\Delta, z_n)\big) = \prod_{i=1}^n \mathcal{A}_i(\Delta, z_i)^{w_i},$$

up to log-error controlled by

$$O\left(\sum_{i=1}^n w_i \left(e^{-g_i\Delta} + q_i/g_i\right)\right).$$

Proof sketch. For each fixed block i, the proof is the Appendix F argument applied componentwise. The uniform gap g_i controls the time needed for the fast killed fiber to collapse to its principal mode, while

$q_i \ll g_i$ prevents the coarse landscape from moving on that same compression scale. After adiabatic elimination of the fast fiber, the remaining coarse process carries the state-dependent killing potential $a_1^{(i)}(z)$, and the resulting scalar interface value is the Feynman-Kac quantity of Theorem G.11 with the displayed schematic error (Theorem F.5 restated componentwise).

Once each block has been represented by a positive scalar value with its continuation family, the outer problem is evaluator-valued exactly as in Appendix D, and the axiom-verification reasoning of §D.6 applies on the n-block compressed class. Axiom (A5) is carried by the per-block Feynman-Kac continuation families exactly as in §F.6, with the scalar values $\mathcal{A}_i(\Delta, z_i)$ being the evaluations at current coarse states and the continuation families $e^{\Delta' M_i}\mathbf{1}$ being the segment-summary objects required for temporal splitting. Combined with Theorem G.6 and the static error-control argument of Theorem G.7 adapted to the dynamic scalar values, this verifies the all-arity (A4) realization of the induced geometric evaluator on the dynamic compressed class. Applied to the ideal scalar evaluator at the exact-axiom limit, Theorem A.8 then characterizes the form as weighted geometric. Taking logarithms of the product transfers componentwise approximation errors into a weighted sum, giving the stated total error. □

Remark G.13 (Why $\sum_i q_i \ll \min_i g_i$ is strictly stronger than needed). A naive product-coarse-state formulation would put the cluster coarse process on $Z^{(1)} \times \cdots \times Z^{(n)}$. If the component chains are independent, its total jump scale is of order $q_{\text{prod}} \sim \sum_i q_i$, while its compression gap is $g_{\text{prod}} = \min_i g_i$. Treating the whole product process as a monolithic moving landscape would therefore suggest the condition

$$\sum_i q_i \ll \min_i g_i.$$

That condition is sufficient for the monolithic product picture, but it is *strictly stronger* than what evaluator-level regrouping needs. The right hypotheses are componentwise: $q_i \ll g_i$ for each block. Each block forms its own scalar before the outer evaluator acts; requiring the entire product coarse state to remain fixed is an artifact of treating a block-structured evaluator problem as a single physical block. This is the n-block form of layer (ii)-vs-(iii) of Theorem G.1.

Corollary G.14 (Dynamic n-block log-Feynman-Kac synthesis)**.** *Under the hypotheses of Theorem G.12,*

$$\log V_\Delta(z_1, \dots, z_n) = \sum_{i=1}^n w_i \log \mathcal{A}_i(\Delta, z_i) + O\left(\sum_i w_i \varepsilon_i^{\text{dyn}}\right).$$

The additive coordinate in the heterogeneous slow-fast setting is the weighted sum of block-local log-Feynman-Kac interface values.

G.7. Dynamic regrouping and the Feynman-Kac Jensen gap

Let $\mathcal{C} = \{C_1, \dots, C_k\}$ be a partition of the block indices. For each cluster C_α, define

$$W_\alpha := \sum_{i \in C_\alpha} w_i, \qquad \alpha_i := \frac{w_i}{W_\alpha}\ (i \in C_\alpha),$$

and the evaluator-level dynamic cluster value

$$B_\alpha(\Delta, z_{C_\alpha}) := \prod_{i \in C_\alpha} \mathcal{A}_i(\Delta, z_i)^{\alpha_i}.$$

Then exact regrouping on scalar values gives

$$\prod_{i=1}^n \mathcal{A}_i(\Delta, z_i)^{w_i} = \prod_{\alpha=1}^k B_\alpha(\Delta, z_{C_\alpha})^{W_\alpha}.$$

This is the correct dynamic n-block form of A4 on the compressed evaluator class.

But B_α should not be conflated with the literal Feynman-Kac survival value of a product coarse process with weighted cluster killing potential. To see the difference, write

$$X_i := \int_0^\Delta a_1^{(i)}(Z_s^{(i)})\,ds.$$

The evaluator-level cluster value is

$$B_C = \prod_{i\in C} (\mathbb{E}e^{-X_i})^{\alpha_i}.$$

A natural physical product-space Feynman-Kac value with weighted potential $\sum_{i\in C}\alpha_i a_1^{(i)}$ would instead be

$$J_C := \mathbb{E}\exp\left(-\sum_{i\in C}\alpha_i X_i\right).$$

If the coarse processes are independent this factors as $J_C = \prod_{i\in C}\mathbb{E}e^{-\alpha_i X_i}$. In general

$$\prod_{i\in C}(\mathbb{E}e^{-X_i})^{\alpha_i} \neq \prod_{i\in C}\mathbb{E}e^{-\alpha_i X_i}.$$

The power lies outside the expectation in the evaluator value and inside the exponential in the physical product-space Feynman-Kac value.

Proposition G.15 (Jensen gap for dynamic regrouping)**.** *Assume $0<\alpha_i<1$ for all nontrivial constituents of a cluster C, and define*

$$\phi_i(\lambda) := \log \mathbb{E}\,e^{-\lambda X_i}, \qquad 0\le\lambda\le 1.$$

Then ϕ_i is convex with $\phi_i(0)=0$, so $\phi_i(\alpha_i)\le\alpha_i\phi_i(1)$. Consequently, under independence of the coarse chains $Z^{(i)}$,

$$\log J_C = \sum_{i\in C}\phi_i(\alpha_i) \le \sum_{i\in C}\alpha_i\,\phi_i(1) = \log B_C.$$

Equality holds when each relevant integrated rate X_i is effectively deterministic, or in the trivial cases $\alpha_i\in\{0,1\}$.

Proof. ϕ_i is the cumulant-generating function of $-X_i$ restricted to $[0,1]$ and is therefore convex. Since $\phi_i(0)=0$, convexity gives

$$\phi_i(\alpha_i) \le (1-\alpha_i)\phi_i(0) + \alpha_i\,\phi_i(1) = \alpha_i\,\phi_i(1).$$

Summing over $i\in C$ gives the stated inequality. Equality requires X_i to carry no relevant randomness on the interval, or the interpolation to be trivial. □

For small fluctuations, the gap size is visible from the cumulant expansion. If X_i has mean μ_i and variance σ_i^2,

$$\phi_i(\lambda) = -\lambda\mu_i + \frac{\lambda^2}{2}\sigma_i^2 + O(\lambda^3\kappa_{3,i}),$$

so

$$\log B_C - \log J_C \approx \frac{1}{2}\sum_{i\in C}\alpha_i(1-\alpha_i)\,\sigma_i^2.$$

The Jensen gap is small when integrated rates are nearly deterministic and nontrivial when intermediate slow-fast randomness remains.

G.7.1. Numerical anchor: the gap in the curvature-modulated family. For a concrete check, take the curvature-modulated family of §F.8 with three curvature levels

$$c(L)=0.5, \qquad c(M)=1.0, \qquad c(H)=1.5,$$

and parameters $R=30$, $m=2$, $\gamma=1$, $\kappa_0=1$, $\beta=1$. Let block 1 use the two-state subfamily $\{L,M\}$ (as in §F.8.1) and block 2 use $\{M,H\}$. Each coarse chain is symmetric with jump rate λ_i, and the blocks start from L and M respectively. With $w_1=w_2=1/2$ the cluster weights are $\alpha_1=\alpha_2=1/2$, so

$$B_C = \prod_{i=1}^{2}\mathcal{A}_i(\Delta,z_i)^{1/2}, \qquad J_C = \prod_{i=1}^{2}\mathbb{E}_{z_i}\exp(-X_i/2),$$

with both Feynman-Kac quantities computable exactly by the two-state matrix exponential

$$\mathbb{E}_z\, e^{-\theta X_i} = \left[e^{\Delta(Q^{(i)}-\theta\ \mathrm{diag}(a_1^{(i)}))}\mathbf{1}\right]_z.$$

Direct matrix-exponential evaluation confirms the expected behavior at the level of the reduced coarse Feynman-Kac model: $\log B_C - \log J_C$ vanishes in both boundary limits (the quasi-static limit $\lambda_i \to 0$, where each X_i is path-deterministic, and the fast-coarse-mixing limit $\lambda_i \to \infty$, where integrated rates self-average to deterministic effective rates); the gap is maximal in the intermediate slow-fast regime where Appendix F's log-Feynman-Kac correction is also maximal. These limits are statements about the reduced coarse Feynman-Kac object itself. Read back into the full killed-process realization, the fast-coarse-mixing limit must still pass through the slow-fast window: coarse mixing may be fast relative to killing, but it must remain slow relative to fiber compression, so the admissible fast regime for the full block dynamics is the triple separation

$$\max_z a_1^{(i)}(z) \ \ll\ q_i \ \ll\ g_i$$

of Theorem F.11, not an unbounded $\lambda_i \to \infty$. The $\lambda_i \to \infty$ label here is short for "inside the fast-coarse-mixing window of the slow-fast hypotheses." (A plot of $\log B_C - \log J_C$ as a function of Δ for several jump-rate pairs, together with a sweep at fixed $\Delta = 200$ over $(\lambda_1, \lambda_2) = (\lambda, 1.6\lambda)$, is deferred to a later revision.)

Remark G.16 (Appendix F's Jensen correction, now spatial). Theorem F.10 (§F.7) emphasized that

$$\log \mathbb{E}\, e^{-\int_0^\Delta a_1(Z_s)ds} \neq -\mathbb{E}\int_0^\Delta a_1(Z_s)\, ds$$

in the intermediate slow-fast regime — a *temporal* Jensen gap between log-expectation and expected-log. Theorem G.15 is the *spatial-regrouping* analogue: an evaluator-level cluster value is generally not the same as a physical product-space Feynman-Kac value with weighted potential. In the ideal quasi-static boundary limit X_i is deterministic because the path is frozen; in the ideal fast-z boundary limit of the reduced coarse process the integrated rate self-averages to a deterministic effective rate (subject, in the full killed-process model, to the triple separation $\max_z a_1^{(i)}(z) \ll q_i \ll g_i$ of Theorem F.11, not to an unbounded fast limit). The gap vanishes in both limits and is nontrivial in the intermediate regime — the same pattern as Appendix F's temporal Jensen correction.

Remark G.17 (Evaluator super-blocks are the right A4 objects). A4 is an axiom about scalar evaluator values and their admissibility for further composition. It is not an axiom asserting that every regrouped evaluator value must coincide with the literal survival law of a newly constructed product physical process. Theorem G.7 and Theorem G.12 therefore live at the evaluator level (layers (i)–(ii) of Theorem G.1). The product-space Feynman-Kac object (layer (iii)) is useful for calibration because it shows what extra statement would be needed to interpret a cluster as a monolithic physical block, and the Jensen gap of Theorem G.15 measures the discrepancy.

G.8. Failure modes and deferrals

The heterogeneous n-block theorem is a closure result, not a license to ignore heterogeneity. The main failure modes:

G.8.1. Slowest-block control of the global window. For all blocks to be compressed on one outer cycle, the cycle length must satisfy roughly $\Delta \gg \max_i g_i^{-1}$. At the same time it must remain inside the metastable window relevant to the interface interpretation. In the static reading this asks for a nonempty interval of the form

$$\max_i g_i^{-1} \ll \Delta \ll \min_i (a_1^{(i)})^{-1}.$$

Heterogeneity can collapse this window even when each constituent block has a reasonable window in isolation. The obstruction is an intersection obstruction: there may be no common outer cycle scale compatible with every block.

G.8.2. Componentwise motion versus monolithic product motion. Theorem G.12 only requires $q_i \ll g_i$ block by block. A monolithic product-coarse-state theorem would typically require $\sum_i q_i \ll \min_i g_i$, which can fail for large clusters even when every constituent block individually satisfies its slow-motion condition. This is not a contradiction. It says that evaluator-level regrouping and physical product-space compression are different statements (Theorem G.1).

G.8.3. Near-degenerate compression. If $g_i\Delta$ is only order one for some block, the scalar compression error is not small; the block has not fully forgotten its subleading modes on the outer cycle scale, and the appropriate interface object may be vector-valued with principal and near-principal modes retained together. The present appendix keeps explicit error bookkeeping only in the scalar regime. The graceful-degradation theory, where scalar interface values thicken into vector interface structures as gaps close, is the near-degenerate extension (deferred).

G.8.4. Common shocks and coupled coarse processes. The appendix assumes block-local dynamic interface values. If post-interface common uncertainty survives, or the coarse processes are coupled in a way that prevents block-local scalars from being formed independently, then the timing-sensitive classification of §III.2 and the Jensen discipline of Appendix F become active again. Such models may still admit geometric evaluation after conditioning on common variables, but that is a further construction. Weak-coupling perturbations of the scalar regrouping picture are treated in Appendix H. That appendix keeps the evaluator-versus-physical-product distinction used here: scalar compression integrity, raw product Feynman-Kac defects, and normalized-evaluator calibration are separate layers of the weak-coupling error budget.

G.8.5. Other deferrals. Several strategically important directions are deliberately outside scope: multi-well or non-convex potentials with competing basins and basin-dependent quasi-stationary structure; non-reversible or driven dynamics breaking detailed balance; state-dependent schedules $w(z)$ and endogenous weights; and concrete biological modeling mapping candidate units onto the potential/gap/timescale structure. These are arc-scale extensions, not appendix-scale ones.

G.8.6. Appendix-level continuations. The appendix-level deferrals split across the broader program. Appendix H treats weak-coupling degradation while staying inside the scalar-regime question; Appendix I treats the near-degenerate finite-band/vector boundary; the companion typology document [Kim26b] classifies the supplied signatures that result when scalar regrouping data are no longer enough, separating cases of adequate scalarity, basin-witnessed scalarization, partially comparable cone-valued objects, and the formally non-diagnostic case. Appendix G's componentwise compression theorem does not extend to mixed-type or cone-valued composition; that problem belongs to the typology layer.

A note on scope. The representational coda on additive primitives and exponentiated geometric aggregation lives as a standalone Appendix J. Its role is explanatory rather than derivational: it identifies a modeling class — path-additive branch quantities under linear pooling — in which geometric aggregation appears as the exponential-map image of additive-coordinate arithmetic pooling, connecting the program to Peters-style multiplicative-growth and FEP-style surprisal formalisms. See Appendix J for the full treatment.

Appendix H. Weakly Coupled Boundaries and Perturbative Boundary Integrity

H.1. Opening: from exact separation to perturbative integrity

The scalar synthesis developed in Part V and Appendices D–G is an exact-limit result. It assumes that each block has produced a positive scalar interface value before the outer evaluator acts, and that the blocks are externally non-transferable in the sense relevant to that evaluator. These assumptions are correct for the theorem class, but too binary for many biological applications. Real boundaries are often semipermeable: metabolites, signals, pathogens, nutrients, common environmental disturbances, and control variables can cross them.

This appendix records a finite-state perturbative extension of the scalar synthesis. It does not change the outer characterization theorem, and it does not replace the finite-band analysis of Appendix I. Instead, it answers a robustness question:

> When exact block separation is weakly violated, which quantities determine whether the scalar AM–GM synthesis remains approximately valid?

There are two perturbative channels.

Channel 1: scalar compression integrity. Does each block still compress to a scalar interface value, or does coupling excite unresolved nonprincipal modes before the outer evaluator acts?

Channel 2: composition integrity. Once scalar interface values exist, do the physical product-space Feynman–Kac objects still factor approximately as products of marginal block values, or does coupling create cross-block correlation defects?

A third object is not a new perturbative channel but an accounting layer.

Calibration layer. The normalized outer evaluator is not raw conjunction survival. In dynamic settings, a physical product-space Feynman–Kac value and the normalized weighted evaluator can differ even at zero coupling. Appendix G names this as the Feynman–Kac Jensen gap. Weak coupling can perturb that gap, but the perturbation is controlled by the same tilted-sensitivity formula that controls Channel 2.

The finite-state results below make explicit how the exact scalar synthesis degrades under weak coupling. The point is not that every porous boundary remains scalar. The point is that, in the scalar regime, porosity has to be measured in the right coordinates: spectrally, by how much it leaks the block out of its principal interface mode, and pathwise, by how much it changes the Feynman–Kac continuation under the survival-biased bridge. If scalar compression fails, the interface object exits the scalar regime and must be treated by the finite-band machinery of Appendix I.

In the companion typology document [Kim26b], the same exit from the scalar regime is given a typological reading: weak coupling may preserve a rank-one signature, perturb it within tolerance, or push the supplied signature toward a finite-band or vector classification. The present appendix supplies scalar-regime perturbative diagnostics; it does not prove cone-order or basin-witness stability for the broader typology, and it does not resolve the weak-coupling cone/basin-stability arc of that document.

H.2. Channel 1: scalar compression integrity

Fix one candidate block. Let S be a finite state space and let π be a strictly positive reference measure. We work in $\mathcal{H} = L^2(\pi)$ with operator norm $\|\cdot\|$ induced by the $L^2(\pi)$ norm.

Let G_0 be a finite reversible killed generator on S, self-adjoint on $L^2(\pi)$ after the usual reversible representation. Write its eigenvalues as

$$-a_1 > -a_2 \ge -a_3 \ge \cdots, \qquad 0 < a_1 < a_2 \le a_3 \le \cdots,$$

and assume that the principal eigenvalue $-a_1$ is simple. Let P_1 be the orthogonal spectral projection onto the principal eigenspace, let $Q_1 = I - P_1$, and define the compression gap

$$g := a_2 - a_1 > 0.$$

A weakly coupled or externally forced block is represented by a time-dependent generator

$$G_t = G_0 + E_t,$$

where E_t is bounded and measurable in time. The perturbation may represent weak coupling across the boundary, external forcing, or coarse-state motion. It need not be self-adjoint.

Different projections of E_t do different jobs. Define the principal-to-complement leakage rate

$$\lambda_\perp(t) := \|Q_1 E_t P_1\|.$$

This is a scalarity-defect source term: it measures how quickly the perturbation injects amplitude from the principal killed mode into nonprincipal modes. It is not a gap-survival estimate. Separately, let δ_{gap} denote whatever bound is used to ensure a surviving relative complement decay rate, and write the surviving compression gap as $\widetilde{g} > 0$. In a self-adjoint time-independent perturbation one may use Weyl-type estimates such as $\widetilde{g} \ge g - 2\sup_t \|E_t\|$, but the theorem below only needs the relative decay hypothesis stated explicitly.

Finally, let $q(t)$ denote moving-landscape forcing. In a fixed spectral frame, $q(t) = 0$. If the coarse state or potential landscape changes in time, the principal/complement splitting itself moves; $q(t)$ is any rate bound on the induced forcing into the complement, for example a bound of the form $\|\dot{Q}_t P_t\|$ in a moving spectral frame. This is the weak-coupling analogue of the slow-landscape term in Appendix F.

Let $U(t,s)$ be the evolution family generated by G_t. For a survival profile $u(t) = U(t,0)u_0$, define

$$p(t) := P_1 u(t), \qquad y(t) := Q_1 u(t),$$

and assume $P_1 u_0 \ne 0$. We measure scalarity defect relative to the principal decay scale by

$$\epsilon_{\text{scalar}}(t) := \frac{e^{a_1 t}\|Q_1 u(t)\|}{\|P_1 u_0\|}.$$

Theorem H.1 (Scalar compression under leaky weak coupling)**.** *Assume the following finite-dimensional estimates hold on* $[0,\Delta]$.

(H1) Relative complement decay. The homogeneous evolution $W_Q(t,s)$ on $Q_1\mathcal{H}$ generated by the complement block of the perturbed dynamics satisfies

$$\left\| e^{a_1(t-s)} W_Q(t,s) \right\| \le M_Q e^{-\widetilde{g}(t-s)} \qquad (0 \le s \le t \le \Delta)$$

for constants $M_Q \ge 1$ and $\widetilde{g} > 0$.

(H2) Principal amplitude control. There is $M_P \ge 1$ such that

$$e^{a_1 s}\|P_1 u(s)\| \le M_P \|P_1 u_0\| \qquad (0 \le s \le \Delta).$$

(H3) Moving-frame forcing, if present, contributes an additional complement source bounded by $q(s)e^{-a_1 s}\|P_1 u_0\|$.

Then

$$\epsilon_{\text{scalar}}(\Delta) \le M_Q \rho_0 e^{-\widetilde{g}\Delta} + M_Q \int_0^\Delta e^{-\widetilde{g}(\Delta-s)}\big(M_P \lambda_\perp(s) + q(s)\big)\, ds,$$

where

$$\rho_0 := \frac{\|Q_1 u_0\|}{\|P_1 u_0\|}$$

is the initial nonprincipal loading. In particular, after absorbing M_P into constants, the scalarity defect has the schematic form

$$\epsilon_{\text{scalar}}(\Delta) \lesssim e^{-\widetilde{g}\Delta} + \int_0^\Delta e^{-\widetilde{g}(\Delta-s)}\big(\lambda_\perp(s) + q(s)\big)\, ds.$$

Proof. In the fixed spectral frame, the complementary component satisfies

$$\frac{d}{dt} y(t) = Q_1(G_0 + E_t)Q_1 y(t) + Q_1 E_t P_1 p(t),$$

with the additional moving-frame source included when the frame varies. Variation of constants gives

$$y(\Delta) = W_Q(\Delta, 0)y(0) + \int_0^\Delta W_Q(\Delta, s)Q_1 E_s P_1 p(s)\, ds + \int_0^\Delta W_Q(\Delta, s)b_q(s)\, ds,$$

where $\|b_q(s)\| \le q(s)e^{-a_1 s}\|P_1 u_0\|$. Multiply by $e^{a_1\Delta}$ and apply (H1). The initial term is bounded by

$$M_Q e^{-\widetilde{g}\Delta}\|Q_1 u_0\|.$$

For the leakage term,

$$\begin{aligned} e^{a_1\Delta}\, \|W_Q(\Delta, s)Q_1 E_s P_1 p(s)\| &\le M_Q e^{-\widetilde{g}(\Delta-s)} e^{a_1 s}\|Q_1 E_s P_1\|\, \|p(s)\| \\ &\le M_Q M_P e^{-\widetilde{g}(\Delta-s)}\lambda_\perp(s)\|P_1 u_0\|. \end{aligned}$$

The moving-frame term is bounded in the same way with $q(s)$ replacing $M_P\lambda_\perp(s)$. Divide by $\|P_1 u_0\|$ to obtain the displayed estimate. □

If $\lambda_\perp(t) \le \lambda_\perp$ and $q(t) \le q$ are constant bounds, Theorem H.1 gives

$$\epsilon_{\text{scalar}}(\Delta) \le M_Q \rho_0 e^{-\widetilde{g}\Delta} + M_Q \frac{M_P\lambda_\perp + q}{\widetilde{g}}\big(1 - e^{-\widetilde{g}\Delta}\big).$$

Thus in the compression window $\widetilde{g}\Delta \gg 1$, scalar interface integrity is controlled by

$$\frac{\lambda_\perp + q}{\widetilde{g}} \ll 1,$$

up to the displayed constants. This generalizes the slow-fast condition $q_*/g_* \ll 1$ from Appendix F by adding a boundary-leakage channel.

Remark H.2 (Gap drift is not leakage). The leakage rate $\lambda_\perp = \|Q_1 E_t P_1\|$ controls contamination of the scalar interface mode. It does not by itself control gap closure. A perturbation can commute with P_1 and still move secondary eigenvalues. Therefore weak-coupling boundary integrity has two separate scalar requirements: a surviving complement decay rate $\widetilde{g} > 0$, and small leakage/forcing relative to that rate.

Remark H.3 (What Channel 1 does not prove). Theorem H.1 is a finite-state scalar-compression theorem. It does not prove stability of the Gibbs three-factor asymptotic of Appendix C under perturbation, and it does not prove cone or basin-label stability in finite-band regimes. It says only that, once a finite block has a simple principal killed mode and a surviving complement decay estimate, scalar compression is robust to leakage small on the compression scale.

Corollary H.4 (Finite Gibbs-block scalarity at fixed size)**.** *Consider a finite-R Gibbs-lifted killed block of the type used in Appendix C, with finite state space, reversible killed generator G_0, simple principal decay rate a_1, and compression gap $g = a_2 - a_1 > 0$. If a weak perturbation satisfies the hypotheses of Theorem H.1 with surviving relative gap $\widetilde{g} > 0$, then the block still produces an approximate scalar continuation over the cycle:*

$$c_\epsilon(\Delta) \approx e^{-a_{1,\epsilon}\Delta},$$

with scalarity defect controlled by the bound in Theorem H.1. In particular, in the steady-rate compres-

sion window the finite block remains in the scalar interface regime when

$$\frac{\lambda_\perp + q}{\widetilde{g}} \ll 1.$$

This corollary is a finite-state statement. It does not assert that the three-factor Gibbs/Kramers asymptotic of Appendix C is stable under perturbation; that stronger metastability claim requires additional analysis.

H.3. A minimal leaky two-mode block

Theorem H.1 is just variation of constants, but a two-mode block shows why the leakage-to-gap ratio is the right scalar-regime diagnostic. Work in the eigenbasis of the unperturbed killed generator and write

$$G_0 = \begin{pmatrix} -a_1 & 0 \\ 0 & -a_2 \end{pmatrix}, \qquad g = a_2 - a_1 > 0.$$

Let the perturbation inject the principal mode into the nonprincipal mode at rate $\lambda(t)$:

$$E_t = \begin{pmatrix} 0 & 0 \\ \lambda(t) & 0 \end{pmatrix}.$$

If $u(t) = (u_1(t), u_2(t))^\top$ and $u_2(0) = 0$, then $u_1(t) = e^{-a_1 t} u_1(0)$ and

$$u_2(\Delta) = \int_0^\Delta e^{-a_2(\Delta - s)} \lambda(s) e^{-a_1 s} u_1(0)\, ds.$$

After rescaling by the principal decay, the exact scalarity defect is

$$\frac{e^{a_1 \Delta} |u_2(\Delta)|}{|u_1(0)|} = \int_0^\Delta e^{-g(\Delta - s)} |\lambda(s)|\, ds.$$

For constant leakage this becomes

$$\epsilon_{\text{scalar}}(\Delta) = \frac{\lambda}{g}\bigl(1 - e^{-g\Delta}\bigr).$$

Thus compression first kills any initial nonprincipal tail on the scale g^{-1}, and then settles to a leakage-forced plateau of order λ/g. Figure 8 displays this plateau across leakage regimes. When $\lambda/g \ll 1$, a scalar interface value survives with a small steady defect. When $\lambda/g = O(1)$, the block is continually uncompressed as fast as the gap restores it, and the scalar approximation is no longer reliable.

H.4. Channel 2: raw Feynman–Kac product defects

Now assume Channel 1 has succeeded: each block has already compressed to a scalar dynamic interface value on its coarse state space. For block i, let

$$H_i := Q_i - \text{diag}(a_i)$$

be the block Feynman–Kac generator, where Q_i is the coarse-state generator and a_i is the local principal killed rate inherited from the fast fiber. For nonnegative initial and terminal readouts (ℓ_i, r_i), define

$$A_i(\Delta) = \ell_i^\top e^{\Delta H_i} r_i > 0.$$

The usual survival readout is $\ell_i = e_{z_i}$ and $r_i = \mathbf{1}_i$, giving $A_i(\Delta, z_i) = [e^{\Delta H_i} \mathbf{1}_i]_{z_i}$.

On the product coarse state space $Z = Z_1 \times \cdots \times Z_n$, define the uncoupled product Feynman–Kac generator

$$H_0 = \sum_{i=1}^n I_1 \otimes \cdots \otimes H_i \otimes \cdots \otimes I_n.$$

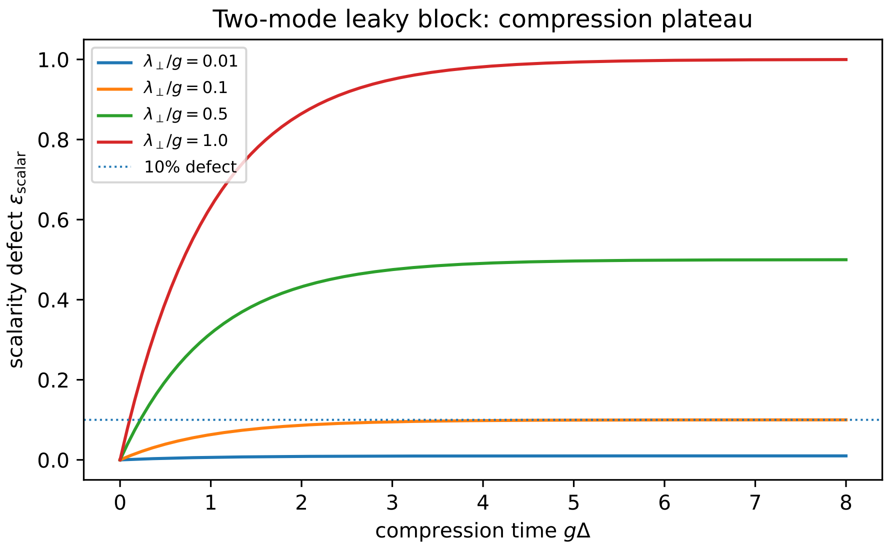

Figure 8: Two-mode leaky block. In the scaled time variable $g\Delta$, constant leakage produces a scalarity defect $(\lambda_\perp/g)(1-e^{-g\Delta})$. The long-cycle plateau is controlled by the leakage-to-gap ratio rather than the raw leakage rate alone.

Let $\ell = \ell_1 \otimes \cdots \otimes \ell_n$ and $r = r_1 \otimes \cdots \otimes r_n$. Since exponentials of Kronecker sums factor,

$$A_0(\Delta) := \ell^\top e^{\Delta H_0} r = \prod_{i=1}^n A_i(\Delta).$$

A weakly coupled product-space model has

$$H_\epsilon = H_0 + \epsilon B, \qquad A_\epsilon(\Delta) = \ell^\top e^{\Delta H_\epsilon} r,$$

where B is a bounded product-space coupling. The raw product-correlation defect is

$$\epsilon_{\mathrm{prod}}(\epsilon; \Delta, \ell, r) := \left| \log A_\epsilon(\Delta) - \sum_{i=1}^n \log A_i(\Delta) \right|.$$

This defect vanishes at $\epsilon = 0$. It measures failure of physical product Feynman–Kac factorization, not failure of the normalized evaluator theorem.

Definition H.5 (Feynman–Kac tilted coupling sensitivity)**.** For $s \in [0, \Delta]$, define the signed tilted coupling sensitivity

$$\mathcal{K}_\epsilon^{\ell,r}(s;\Delta) := \frac{\ell^\top e^{(\Delta-s)H_\epsilon} B e^{sH_\epsilon} r}{\ell^\top e^{\Delta H_\epsilon} r}.$$

Also define the positive tilted magnitude

$$\|B\|_{\mathrm{FK},\epsilon,s}^{\ell,r} := \frac{\left|\ell^\top e^{(\Delta-s)H_\epsilon} B e^{sH_\epsilon} r\right|}{\ell^\top e^{\Delta H_\epsilon} r}.$$

Theorem H.6 (Tilted-sensitivity identity and product-defect bound)**.** *Assume $A_\epsilon(\Delta) > 0$ on an interval containing 0. Then*

$$\frac{d}{d\epsilon} \log A_\epsilon(\Delta) = \int_0^\Delta \mathcal{K}_\epsilon^{\ell,r}(s;\Delta)\, ds.$$

Consequently, for $\epsilon > 0$,

$$\epsilon_{\mathrm{prod}}(\epsilon; \Delta, \ell, r) \le \epsilon \int_0^\Delta \sup_{0 \le \theta \le \epsilon} \|B\|_{\mathrm{FK},\theta,s}^{\ell,r}\, ds.$$

The corresponding first-order expansion is

$$\log A_\epsilon(\Delta) - \log A_0(\Delta) = \epsilon \int_0^\Delta \mathcal{K}_0^{\ell,r}(s;\Delta)\, ds + O(\epsilon^2).$$

Proof. Duhamel's formula for matrix exponentials gives

$$\frac{d}{d\epsilon} e^{\Delta H_\epsilon} = \int_0^\Delta e^{(\Delta - s)H_\epsilon} B e^{sH_\epsilon}\, ds.$$

Multiplying on the left by $\ell^\top$ and on the right by r, then dividing by $A_\epsilon(\Delta)$, yields the derivative identity. Integrating the derivative from 0 to ϵ and taking absolute values yields the non-asymptotic bound. The first-order formula follows by evaluating the derivative at $\epsilon = 0$ and using differentiability of finite-dimensional matrix exponentials. □

The quotient in Theorem H.5 is the important object. A crude operator-norm estimate gives a bound of order $\epsilon\Delta\|B\|$, but that bound ignores killing. In a killed process, surviving trajectories are not sampled from the raw process; they are biased away from high-killing regions. The tilted quotient measures coupling under this survival-biased bridge. Coupling placed where surviving paths concentrate can have large AM–GM significance; coupling of the same raw operator norm placed near high-killing boundary states can be strongly suppressed.

Remark H.7 (Same raw norm, different AM–GM significance). Theorem H.6 explains why a blanket or boundary cannot be assessed by raw coupling norm alone. Two couplings B_1 and B_2 may have the same operator norm. If B_1 acts primarily in metastable low-killing states while B_2 acts primarily in states from which survival to the terminal readout is unlikely, then their Feynman–Kac tilted sensitivities can differ by orders of magnitude. The AM–GM-relevant coupling is coupling seen by surviving paths.

H.5. A two-state product example

The tilted sensitivity is easiest to see in the smallest dynamic product model. Let each compressed block have coarse state space $\{M, L\}$, where M is a low-killing metastable state and L is a high-killing state. Take

$$Q = \begin{pmatrix} -\alpha & \alpha \\ \beta & -\beta \end{pmatrix}, \qquad a(M) < a(L), \qquad H = Q - \mathrm{diag}(a(M), a(L)).$$

For two identical blocks, the uncoupled product generator is

$$H_0 = H \otimes I + I \otimes H,$$

with initial readout $\ell = e_{(M,M)}$ and terminal readout $r = \mathbf{1}$. Now compare three diagonal perturbations of equal raw operator norm:

$$B_{MM} = -\mathbf{1}_{(M,M)}, \qquad B_{ML} = -\mathbf{1}_{(M,L)}, \qquad B_{LL} = -\mathbf{1}_{(L,L)}.$$

Each perturbation has the same raw size, but they act in regions seen very differently by the survival-biased bridge. With representative parameters $\alpha = 0.22$, $\beta = 0.70$, $a(M) = 0.03$, $a(L) = 0.95$, and $\Delta = 5$, the first-order tilted sensitivities have magnitudes approximately

$$4.38 \quad (M, M), \qquad 0.30 \quad (M, L), \qquad 0.023 \quad (L, L).$$

The crude bound $\epsilon\Delta\|B\|$ is the same for all three. The Feynman–Kac tilted bound is not. Figure 9 shows the exact product defect and its first-order tilted approximation as coupling strength varies.

This example is the finite-state content of the blanket-porosity slogan. A boundary can be porous in locations that surviving paths almost never visit, or porous precisely where the persistent ensemble lives. Only the latter strongly affects the AM–GM product structure.

H.6. Finite coarse-Gibbs specialization

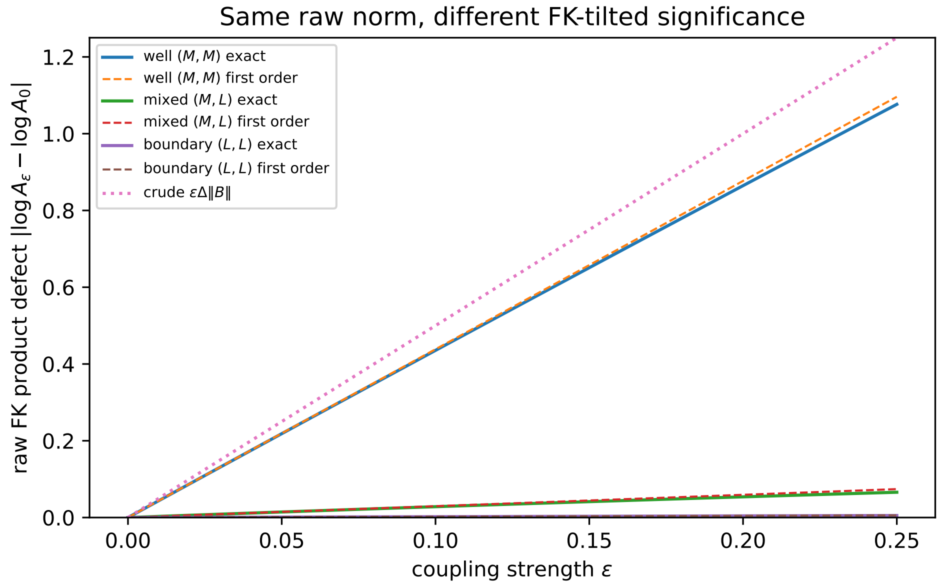

Figure 9: Two coupled two-state Feynman–Kac blocks. The three perturbations have equal raw operator norm, but their product defects differ because the survival-biased bridge spends most of its weight near the low-killing well. Coupling in (M, M) is amplified; coupling in (L, L) is suppressed.

The preceding example is not external to the monograph's killed-process synthesis. It is the finite coarse-state form of the dynamic interface value from Appendix F. There, after fast spectral compression, a block with coarse state z is represented by

$$A(\Delta, z) = \left[e^{\Delta(Q-\mathrm{diag}(a_1))}\mathbf{1}\right]_z,$$

where $a_1(z)$ is the principal killed rate of the fast fiber at coarse state z. Thus the $H = Q - \mathrm{diag}(a)$ used above is exactly the compressed coarse generator, with $a = a_1$.

For a two-state curvature-modulated coarse family, one may read M as the coarse state with lower effective killing and L as the state with higher effective killing. The Feynman–Kac tilted sensitivity is then computable directly from the same quantities already used in the dynamic synthesis:

$$Q, \qquad a_1(M), a_1(L), \qquad B, \qquad \Delta.$$

No new Gibbs asymptotic is required for this finite calculation. What is not proved here is a large-R theorem identifying the asymptotic scaling of $\mathcal{K}_\epsilon$ in terms of the three-factor Gibbs/Kramers decomposition of Appendix C. The finite coarse-Gibbs point is narrower and sufficient for the present appendix: once the fast fiber has compressed, weak cross-block composition is governed by a Feynman–Kac bridge on the compressed coarse state space.

H.7. The calibration layer: evaluator values are not product survivals

The previous subsection controls raw product Feynman–Kac factorization. The monograph's outer theorem, however, characterizes a normalized evaluator on interface values. This distinction is central in Section II.2.1 and Appendix G. Weak coupling does not erase it.

Let the normalized evaluator target be

$$\log V_0^w(\Delta) := \sum_{i=1}^{n} w_i \log A_i(\Delta), \qquad \sum_i w_i = 1.$$

A physical product-space Feynman–Kac object with weighted killing potential is instead of the form

$$A^w_{\epsilon,\text{phys}}(\Delta) = \ell^\top e^{\Delta H^w_\epsilon} r, \qquad H^w_\epsilon = Q_\epsilon - \operatorname{diag}\left(\sum_i w_i a_i\right),$$

when such a physical comparison object has been declared. Even at zero coupling, generally

$$\log A^w_{0,\text{phys}}(\Delta) \neq \sum_i w_i \log A_i(\Delta).$$

This is the Feynman–Kac Jensen/regrouping gap of Appendix G. It is not caused by weak coupling; it is a calibration difference between two mathematical objects.

Define the baseline calibration defect

$$\epsilon^w_{\text{calib},0} := \left|\log A^w_{0,\text{phys}}(\Delta) - \sum_i w_i \log A_i(\Delta)\right|.$$

If the weighted physical object is itself weakly coupled, then

$$\log A^w_{\epsilon,\text{phys}}(\Delta) - \sum_i w_i \log A_i(\Delta) = \underbrace{\left(\log A^w_{0,\text{phys}}(\Delta) - \sum_i w_i \log A_i(\Delta)\right)}_{\text{baseline calibration/Jensen gap}} + \underbrace{\left(\log A^w_{\epsilon,\text{phys}}(\Delta) - \log A^w_{0,\text{phys}}(\Delta)\right)}_{\text{coupling correction to calibration}}.$$

The second term is not a new channel. It is another application of Theorem H.6, now with the weighted physical generator H^w_ϵ. The baseline calibration gap and the weak-coupling correction must be distinguished because only the latter vanishes at $\epsilon = 0$.

Remark H.8 (Why there is no universal coupled evaluator). There is no canonical single quantity V^w_ϵ until the comparison target has been declared. One may evaluate the normalized geometric mean of perturbed scalar block values; one may study raw product-space conjunction survival; or one may study a weighted physical Feynman–Kac object. These are different mathematical objects. This appendix supplies error terms for moving between declared objects, not a universal definition that collapses them.

H.8. Perturbative scalar synthesis schema

The scalar weak-coupling picture is now a three-layer bookkeeping statement.

First, the block-level scalar values must survive leakage. Let $\widehat{A}_i(\Delta)$ denote the actual scalar value produced by block i after weak perturbation, and let $A_i(\Delta)$ be the ideal uncoupled compressed value. Suppose

$$\left|\log \widehat{A}_i(\Delta) - \log A_i(\Delta)\right| \le \epsilon_{\text{scalar},i}.$$

Then the normalized evaluator on perturbed scalar block values satisfies

$$\left|\log \prod_i \widehat{A}_i(\Delta)^{w_i} - \sum_i w_i \log A_i(\Delta)\right| \le \sum_i w_i \epsilon_{\text{scalar},i}.$$

This is the clean scalar-evaluator statement. It uses only Channel 1.

Second, if the modeler also wants a physical raw product-space Feynman–Kac object to factor as a product of marginal values, then Channel 2 contributes

$$\epsilon_{\text{prod}}(\epsilon) = \left|\log A_\epsilon^{\text{raw}}(\Delta) - \log A_0^{\text{raw}}(\Delta)\right|,$$

with ϵ_{prod} controlled by Theorem H.6.

Third, if the declared physical product-space object is being compared to the normalized evaluator, the calibration defect contributes

$$\epsilon_{\text{calib}}^w(\epsilon) := \left|\log A_{\epsilon,\text{phys}}^w(\Delta) - \sum_i w_i \log A_i(\Delta)\right|,$$

which splits into the baseline Jensen gap plus the tilted weak-coupling correction from §.

Putting the layers together gives the schematic finite-state weak-coupling error budget

$$\boxed{\text{total log error} \lesssim \sum_i w_i \epsilon_{\text{scalar},i} + \epsilon_{\text{prod}} + \epsilon_{\text{calib}}^w}$$

with the important convention that the right-hand side depends on which physical/evaluator comparison has been declared. This is an error-budget schema, not a new universal outer theorem.

H.9. FEP bridge: which blankets matter?

Free-energy and Markov-blanket frameworks identify candidate statistical boundaries, but the scalar AM–GM synthesis asks a different question: whether those boundaries support viability aggregation. Recent work on weak Markov blankets [Sak22] replaces the perfectly sealed conditional-independence interface with a quantitative measure of boundary porosity in high-dimensional sparsely-coupled systems. The present appendix supplies the corresponding viability-side projection: weak coupling provides the first quantitative bridge from porosity to AM–GM significance.

A boundary can be porous in many ways. AM–GM-relevant porosity is not merely raw coupling strength. It is coupling projected through the interface-producing dynamics. For scalar integrity, the relevant projected quantity is the leakage-to-gap ratio

$$\frac{\lambda_\perp}{\widetilde{g}}.$$

For composition integrity, the relevant projected quantity is the Feynman–Kac tilted sensitivity

$$\int_0^\Delta \|B\|_{\text{FK},\epsilon,s}^{\ell,r}\, ds.$$

Thus a Markov blanket is structurally significant for AM–GM viability aggregation when its porosity is small relative to the compression gap and small under the survival-biased bridge. A blanket that is weak in an information-theoretic or raw-coupling sense [Sak22] may still be AM–GM-significant if it protects the principal interface mode and suppresses Feynman–Kac product defects; conversely, a small raw coupling may be significant if it acts precisely where surviving paths concentrate.

H.10. Failure interpretation and empirical targets

The perturbative quantities in this appendix are scalar-regime diagnostics. They refine the rank-one synthesis of this monograph; they do not replace the finite-band/vector analysis of Appendix I.

Three failures should be kept separate.

1. If $\epsilon_{\text{scalar},i}$ becomes large, block i has not produced a reliable scalar interface value on the cycle timescale. The block leaves the scalar regime and must be treated as finite-band or vector-valued interface structure, as in Appendix I.
2. If ϵ_{prod} becomes large, the old partition may still contain scalar blocks, but their physical product-space continuation no longer behaves like independent outer branches. The problem is composi-

tion integrity, not scalarity.

3. If $\epsilon^{w}_{\mathrm{calib}}$ is large, the selected physical product-space object was not close to the normalized weighted evaluator. This may be a modeling or calibration issue rather than a boundary-integrity failure.

For empirical work, the lesson is equally sharp. One should not estimate boundary porosity only by raw cross-boundary coupling. The relevant measurements are spectrally and path-weighted: leakage into nonprincipal modes, surviving compression gaps, moving-landscape rates, and coupling sensitivity under the Feynman–Kac tilted bridge. These are the perturbative observables that let the scalar AM–GM boundary become a graded empirical claim rather than a binary idealization.

Finally, none of the finite-state perturbative results here identify a new higher-level block once the old partition fails. They diagnose loss of the old scalar partition. Finding a new partition requires non-perturbative search: spectral clustering, approximate lumpability, large-deviation landscape reconstruction, and finite-band interface analysis.

The typology document [Kim26b] records the first failure mode of §H.10 as a possible type transition in the supplied interface signature: a system whose scalarity defect grows under perturbation may move from a clean rank-one signature to a finite-band/vector classification, with the corresponding change in admissible composition data. The empirical extraction document [Kim26a] is where the perturbative observables of this appendix — leakage rates, surviving gaps, moving-landscape forcing, and Feynman-Kac tilted product sensitivities — become measurement targets in concrete systems. The present appendix supplies the diagnostic coordinates; how to estimate them from data, and how to interpret a transition between rank-one and finite-band signatures empirically, belongs to those documents.

H.11. Closing

The exact scalar synthesis remains the core theorem of this monograph. This appendix adds its weak-coupling robustness layer. Boundary leakage threatens scalar interface formation through nonprincipal modes. Cross-block coupling threatens physical product factorization through tilted Feynman–Kac product defects. Evaluator calibration remains a separate Jensen layer. In the clean limit all three defects vanish or are declared by calibration. Away from the clean limit, they provide quantitative coordinates for graded scalar individuality.

Appendix I. Near-Degenerate Spectral Compression and Vector Interface Structure

I.1. Opening

Appendices D, F, and G realize the outer theorem on blocks that have compressed to positive scalar interface values. The shared compression hypothesis is *rank-one*: on the outer cycle timescale, each block's killed semigroup is dominated by its principal mode. This appendix studies the boundary of that scalar regime. When the compression product

$$(a_2 - a_1)\Delta$$

is large, the scalar interface value is valid. When it is order one, the principal mode has not erased the near-principal modes, and the block has not collapsed to a scalar. The correct interface object is then finite-dimensional: a retained band of spectral modes, with scalar compression recovered as the special rank-one case.

The core result is a static finite-band approximation theorem for reversible killed-process blocks: if modes $1, \dots, K$ are retained and the next gap satisfies $(a_{K+1} - a_1)\Delta \gg 1$, the survival profile over the cycle is approximated by its K-mode projection with error controlled by $e^{-(a_{K+1}-a_1)\Delta}$. A scalarity-defect bound then quantifies exactly when the principal scalar remains adequate: the defect is controlled by the near-principal modal contributions

$$\sum_{r=2}^{K} \rho_r \, e^{-(a_r - a_1)\Delta}.$$

Near-degeneracy is unresolved internal modal memory at the interface; the failure of estimates is a symptom, not the underlying phenomenon.

The appendix also diagnoses why Theorem II.1 does not automatically extend to vector interface values. The weighted geometric mean is canonical on positive scalars. A finite-mode interface object, by contrast, depends on a retained spectral subspace, basis/readout conventions, and possibly signed higher modes. Spatial regrouping invariance (A4) is therefore not naturally formulated until the vector object is supplied with additional regrouping-compatible structure. Candidate scalar reductions are catalogued: principal projection, externally chosen positive readout, norm or cone projection, componentwise pooling under common spectral coordinates, and a possible readout induced by the outer aggregation problem itself. Each introduces extra structure not supplied by the block-local killed dynamics alone.

The dynamic slow-fast extension is explicitly deferred. With moving coarse state z, the retained spectral band becomes a vector bundle over Z and the natural continuation is a matrix-valued Feynman-Kac or adiabatic-mode-tracking problem. That extension requires not only new machinery but also fixed-rank and non-crossing hypotheses: if eigenvalue crossings change the retained band or destroy a compatible mode labeling, the finite-dimensional interface object itself is not stable. This appendix therefore extends the synthesis by locating the boundary of the scalar compressed regime, not by replacing Theorem II.1 with a general vector-valued outer theorem.

This appendix is the monograph's finite-band boundary analysis. It identifies the type shift: scalar interface values become finite-mode interface profiles when rank-one compression is not complete on the outer cycle. The companion typology document [Kim26b] takes this type shift as its starting point and classifies which additional supplied data — basin witnesses, admissible readout classes, cone orders, or composition-compatibility data — restore scalar evaluation, support partial comparison, or leave the formal cone non-diagnostic. The finite-mode vocabulary used here (retained band, scalarity defect, candidate scalar reduction) is preserved throughout; the typology's stratum and signature vocabulary belongs to the companion document.

I.2. Static spectral setup

We work first in the static fixed-z setting. The slow-fast vector problem is deferred to §I.7.

Let Ω be a finite fixed-total fiber and let G be a reversible killed generator on Ω of the Appendix C type. Write π for the reversible measure of the corresponding unkilled fast dynamics. The killed semigroup is

$$Q_t = e^{tG}.$$

Because G is reversible (after the usual symmetrization), we may work in $L^2(\pi)$ and choose an orthonormal eigenbasis

$$\phi_1, \phi_2, \ldots, \phi_N$$

with

$$G\phi_r = -a_r\,\phi_r, \qquad 0 < a_1 < a_2 \le \cdots \le a_N.$$

The principal eigenfunction ϕ_1 may be chosen strictly positive. Higher eigenfunctions need not be positive and typically change sign.

The survival profile over time t is

$$q_t := Q_t\,\mathbf{1}.$$

Expanding $\mathbf{1}$ in the eigenbasis gives

$$q_t = \sum_{r=1}^{N} e^{-a_r t}\, b_r\, \phi_r, \qquad b_r := \langle \mathbf{1}, \phi_r \rangle_\pi.$$

For an initial distribution whose density with respect to π is f, the exact survival scalar is

$$S_f(t) = \langle f, q_t \rangle_\pi = \sum_{r=1}^{N} e^{-a_r t}\, b_r\, \langle f, \phi_r \rangle_\pi.$$

The scalar compressed regime is the case in which the $r = 1$ term dominates for all readouts under consideration. Near-degeneracy is the case in which some of the terms $r \ge 2$ remain comparable on the cycle scale.

Definition I.1 (Retained spectral band)**.** Fix a cycle length Δ and a tolerance parameter $\eta = O(1)$. The near-principal retained band is the set of modes

$$\mathcal{B}_K = \{1, \ldots, K\}$$

chosen so that

$$(a_r - a_1)\Delta \le \eta \quad (1 \le r \le K),$$

while the tail gap satisfies $(a_{K+1} - a_1)\Delta \gg 1$ if $K < N$.

The definition is intentionally cycle-dependent. A mode is not intrinsically part of the interface structure; it is retained if it remains visible at the time the outer interface is reached.

I.3. Finite-mode interface values

Let Π_K denote the orthogonal projection in $L^2(\pi)$ onto

$$E_K := \mathrm{span}\{\phi_1, \ldots, \phi_K\}.$$

Definition I.2 (K-mode interface profile)**.** The K-mode interface profile of the block over cycle length Δ is

$$q_\Delta^{(K)} := \Pi_K\, q_\Delta = \sum_{r=1}^{K} e^{-a_r \Delta}\, b_r\, \phi_r.$$

Equivalently, after choosing the spectral basis, the K-mode interface coordinate vector is

$$\mathbf{c}^{(K)}(\Delta) := \big(e^{-a_1\Delta} b_1,\ e^{-a_2\Delta} b_2,\ \ldots,\ e^{-a_K\Delta} b_K\big).$$

The profile formulation is basis-invariant inside E_K; the coordinate vector is convenient but basis-dependent if eigenvalues within the band are degenerate. This distinction becomes important in §I.7.

Theorem I.3 (Finite-band approximation)**.** *For every $K < N$,*

$$\|q_\Delta - q_\Delta^{(K)}\|_{L^2(\pi)} \le e^{-a_{K+1}\Delta} \, \|(I - \Pi_K)\mathbf{1}\|_{L^2(\pi)}.$$

Equivalently, relative to the principal decay scale,

$$e^{a_1\Delta} \, \|q_\Delta - q_\Delta^{(K)}\|_{L^2(\pi)} \le e^{-(a_{K+1}-a_1)\Delta} \, \|(I - \Pi_K)\mathbf{1}\|_{L^2(\pi)}.$$

Consequently, if $(a_{K+1} - a_1)\Delta \gg 1$, the K-mode interface profile gives a controlled approximation to the full survival profile on the cycle timescale.

Proof. Using the spectral expansion,

$$q_\Delta - q_\Delta^{(K)} = \sum_{r>K} e^{-a_r\Delta} \, b_r \, \phi_r.$$

By orthonormality,

$$\|q_\Delta - q_\Delta^{(K)}\|^2_{L^2(\pi)} = \sum_{r>K} e^{-2a_r\Delta} \, b_r^2 \le e^{-2a_{K+1}\Delta} \sum_{r>K} b_r^2.$$

Since $\sum_{r>K} b_r^2 = \|(I - \Pi_K)\mathbf{1}\|^2_{L^2(\pi)}$, taking square roots gives the first inequality, and multiplying by $e^{a_1\Delta}$ gives the second. □

Corollary I.4 (Readout error bound)**.** *Let $f \in L^2(\pi)$ be an initial-density/readout with $\|f\|_{L^2(\pi)} \le M$. Then*

$$\big| \, S_f(\Delta) - \langle f, q_\Delta^{(K)} \rangle_\pi \, \big| \le M \, e^{-a_{K+1}\Delta} \, \|(I - \Pi_K)\mathbf{1}\|_{L^2(\pi)}.$$

Proof. Apply Cauchy-Schwarz to $S_f(\Delta) - \langle f, q_\Delta^{(K)} \rangle_\pi = \langle f, q_\Delta - q_\Delta^{(K)} \rangle_\pi$ and use Theorem I.3. □

Remark I.5 (What the vector object remembers). The vector interface profile remembers how the block's future survival depends on unresolved low modes. This is not arbitrary high-dimensional microstate dependence; it is low-dimensional modal memory that survives exactly because the near-principal band has not decayed on the interface timescale.

I.4. Scalarity defect

The scalar compressed regime is the $K = 1$ case. Appendix D used precisely this regime: after spectral compression, the block is represented by the principal scalar. This section quantifies the defect of that reduction when $K > 1$.

For a readout f, define the principal scalar approximation

$$S_f^{(1)}(\Delta) := e^{-a_1\Delta} \, b_1 \, \langle f, \phi_1 \rangle_\pi.$$

The K-mode approximation is

$$S_f^{(K)}(\Delta) := \sum_{r=1}^{K} e^{-a_r\Delta} \, b_r \, \langle f, \phi_r \rangle_\pi.$$

Definition I.6 (Readout-relative scalarity defect)**.** Assume $S_f^{(1)}(\Delta) \neq 0$. The readout-relative scalarity defect is

$$\mathcal{D}_K(f, \Delta) := \frac{| \, S_f^{(K)}(\Delta) - S_f^{(1)}(\Delta) \, |}{| \, S_f^{(1)}(\Delta) \, |}.$$

Proposition I.7 (Upper bound on scalarity defect)**.** *For every readout f with $S_f^{(1)}(\Delta) \neq 0$,*

$$\mathcal{D}_K(f, \Delta) \le \sum_{r=2}^{K} \rho_r(f) \, e^{-(a_r - a_1)\Delta},$$

where

$$\rho_r(f) := \left| \frac{b_r \langle f, \phi_r \rangle_\pi}{b_1 \langle f, \phi_1 \rangle_\pi} \right|.$$

The scalar defect is small when every visible nonprincipal modal weight is suppressed on the cycle scale.

Proof. Compute

$$S_f^{(K)} - S_f^{(1)} = \sum_{r=2}^{K} e^{-a_r \Delta} \, b_r \langle f, \phi_r \rangle_\pi.$$

Divide by $|S_f^{(1)}| = e^{-a_1 \Delta} |b_1 \langle f, \phi_1 \rangle_\pi|$ and apply the triangle inequality. □

Proposition I.8 (Lower bound under non-cancellation)**.** *Suppose there exists a mode $r_* \in \{2, \ldots, K\}$ such that*

$$\rho_{r_*}(f) \, e^{-(a_{r_*} - a_1)\Delta} \geq 2 \sum_{\substack{2 \leq r \leq K \\ r \neq r_*}} \rho_r(f) \, e^{-(a_r - a_1)\Delta}.$$

Then

$$\mathcal{D}_K(f, \Delta) \geq \tfrac{1}{2} \, \rho_{r_*}(f) \, e^{-(a_{r_*} - a_1)\Delta}.$$

In particular, if a near-principal mode has non-negligible readout weight and $(a_{r_*} - a_1)\Delta = O(1)$, the scalar approximation is not uniformly reliable.

Proof. Let $z_r := \rho_r(f) \, e^{-(a_r - a_1)\Delta} \, \sigma_r$, where $\sigma_r \in \{\pm 1\}$ is the sign of $b_r \langle f, \phi_r \rangle_\pi / \big(b_1 \langle f, \phi_1 \rangle_\pi\big)$. Then $\mathcal{D}_K(f, \Delta) = |\sum_{r=2}^{K} z_r|$. By the reverse triangle inequality and the dominance assumption,

$$\left| \sum_{r=2}^{K} z_r \right| \geq |z_{r_*}| - \sum_{r \neq r_*} |z_r| \geq \tfrac{1}{2} |z_{r_*}|.$$

□

Remark I.9 (The meaning of the error term). The expression $e^{-g\Delta}$ with $g = a_2 - a_1$ measures unresolved low-mode memory at the interface (the "technical remainder reading is too weak). When $g\Delta \gg 1$, the memory is erased before outer composition. When $g\Delta = O(1)$, the memory survives, and the scalar interface value has degraded into a finite-mode object.

I.5. Why the outer theorem does not automatically extend

Theorem A.8 characterizes evaluators on positive scalar interface values:

$$A_w : (0, \infty)^n \to (0, \infty).$$

The finite-mode object from §I.3 is not in this domain. It is a projected survival profile $q_\Delta^{(K)} \in E_K$, or a coordinate vector after choosing a spectral basis. Higher modes may be signed. Degenerate bands may have no canonical ordered basis. Different readouts of the same vector profile may give different scalar comparisons.

The calibration point is clean: near-degeneracy does not prove that A4 fails on vectors — it breaks the scalar *domain* on which A4 was naturally meaningful. Spatial regrouping invariance says that inner cluster evaluation followed by outer evaluation should agree with direct evaluation. For positive scalars, this is clean because a cluster value is again a positive scalar of the same type. For vector interface structures, one must first specify what type the cluster value has and how constituent vector objects combine. Without such additional structure, A4 is not naturally formulated.

Example I.10 (Readout dependence)**.** Consider two abstract two-mode interface vectors

$$v = (1, 2), \qquad u = (2, 1).$$

The readout $\ell_1(x_1, x_2) = x_1$ ranks u above v, while $\ell_2(x_1, x_2) = x_2$ ranks v above u. A scalar evaluator applied after ℓ_1 and one applied after ℓ_2 are both scalar evaluators, but they answer different questions. The vector object alone has not chosen the scalar by which the block should be represented.

Proposition I.11 (Scalar readouts restore the outer theorem only by adding structure)**.** *Let $\rho_i : E^{(i)}_{K_i} \to (0,\infty)$ be positive scalar readouts for each block (here ρ_i is a readout map indexed by block i, not to be confused with the total resource R of Part IV or with the modal-weight coefficients $\rho_r(f)$ of §I.4 indexed by spectral mode r). Once the readouts $\rho_i(v_i)$ are chosen, Theorem A.8 applies to the resulting positive scalar values; the weighted-geometric form*

$$A^{\rho}_w(v_1,\dots,v_n) := \prod_{i=1}^{n} \rho_i(v_i)^{w_i}$$

is then the forced scalar evaluator on those readout values. But the choice of ρ_i is additional data, not determined by the finite-mode block dynamics alone.

Proof. Once $\rho_i(v_i)$ are positive scalars, Theorem A.8 applied to those scalars forces the weighted-geometric form. The construction inherits the scalar theorem after the readout choice; the theorem does not identify that choice. □

Remark I.12 (No impossibility claim). This appendix does not prove that no natural vector-valued outer theorem can exist. It shows a more modest and more useful fact: the scalar theorem is not type-correct on finite-mode interface structures unless additional regrouping-compatible data are supplied.

The companion typology document [Kim26b] formalizes this added-structure point: scalar evaluation on a finite-band signature is restored only after the supplied signature declares the relevant readout, witness, cone, or compatibility data. The classification distinguishes signatures where scalarity is canonical, where it is basin-witnessed, where it is merely declared as an admissible readout class, or where the cone is formally non-diagnostic.

I.6. Candidate scalar reductions

There are several ways to turn a finite-mode interface object back into a positive scalar. Each is natural in special models; none is supplied by the general killed-process setup alone.

I.6.1. Principal projection. The first candidate is to keep only the principal-mode coefficient, after fixing the principal-mode normalization:

$$q^{(K)}_\Delta \mapsto e^{-a_1\Delta}\, b_1,$$

or, under the Appendix D principal-mode normalization convention, simply $e^{-a_1\Delta}$. Equivalently, for a specified readout f, this gives the scalar $S^{(1)}_f(\Delta) = e^{-a_1\Delta}\, b_1\, \langle f, \phi_1\rangle_\pi$. This is exactly the Appendix D regime. It is justified when the scalarity defect of §I.4 is small. In the near-degenerate regime, this is not a neutral reduction; it discards visible modal memory.

I.6.2. Positive readout map. A second candidate is to choose a positive readout map — defined either on the attainable finite-mode profiles or relative to an externally supplied positive cone — and take the scalar interface value to be $c_\ell := \ell(q^{(K)}_\Delta)$. This can be appropriate if the outer problem supplies a terminal payoff, an initial distribution class, or an experimentally specified readout — but the readout is then part of the model, and different readouts can induce different rankings and different outer aggregates (Theorem I.10). The map is not naturally a linear positive functional on all of E_K, because E_K contains signed directions (higher killed eigenfunctions ϕ_r with $r \geq 2$ are generally signed) and carries no canonical positive cone without extra structure.

I.6.3. Norm, cone, or projective reduction. A third candidate is to use a norm, cone order, or projective metric on the retained subspace. This may be natural if E_K carries a distinguished positive cone and the vectors remain in it — but higher killed modes are generally signed, so a cone or norm reduction imposes geometry not present in the scalar outer theorem.

I.6.4. Componentwise pooling under common spectral coordinates. A fourth candidate is componentwise pooling. If all blocks share compatible K-mode coordinates, one might define

$$\mathbf{A}_w(\mathbf{c}_1,\dots,\mathbf{c}_n)_r := \prod_i c_{i,r}^{w_i}.$$

This is only meaningful under strong compatibility assumptions: common band dimension, common coordinate semantics, positivity of corresponding components, and regrouping rules preserving those coordinates. Generic heterogeneous Gibbs-lifted blocks do not supply this structure.

I.6.5. Readout induced by the outer aggregation problem. A fifth and subtler candidate is that the outer aggregation problem itself could canonicalize the readout. Suppose blocks share spectral structure because they come from a common coarse geometry: the same retained-band cardinality, compatible eigenbasis choices, and a common interpretation of modes across blocks. Then cross-block coherence might select a scalar readout that no block-local dynamics could select in isolation. This candidate is worth naming because it is not mere modeler fiat — it would be a choice forced by the outer problem rather than imposed externally on each block. But the compatibility requirements are strong: a shared retained-band type, a regrouping-compatible way to identify modes across heterogeneous blocks, and a proof that the induced readout is stable under clustering. That is not part of the current killed-process synthesis. At present it is a possible route to a future vector outer theorem, not a result.

These five reductions are not interchangeable. In the typology document [Kim26b] they land in distinct regions depending on whether the scalarization is canonical (recovered when the scalarity defect is small), basin-witnessed (supplied by a partition into metastable regions), merely declared as an admissible readout class, or unsupported by composition-compatible data. The choice between them is therefore part of the supplied signature, not a property of the block dynamics alone.

I.7. Dynamic vector interfaces: deferred structure

The static theorem above stays inside the Gibbs-lifted killed-process machinery already developed. The dynamic slow-fast extension is more serious. Once the coarse state z moves, each fixed-z killed generator has its own retained subspace

$$E_K(z) = \mathrm{span}\{\phi_1(z), \dots, \phi_K(z)\}.$$

The dynamic interface object is then no longer a vector in one fixed subspace; it is a section of a finite-dimensional spectral bundle over the coarse state space.

The natural continuation is a matrix-valued Feynman-Kac or adiabatic-mode-tracking theorem. Schematically, after retaining a K-mode band, the effective evolution should have the form

$$\partial_t u = \big(Q_{\text{coarse}} - A(z) + \mathcal{C}(z)\big)\, u,$$

where $A(z)$ is a matrix of retained decay rates and $\mathcal{C}(z)$ records mode transport, basis rotation, or coupling induced by movement in z. This is the vector analogue of Appendix F's scalar Feynman-Kac object.

That problem requires new machinery, and also new structural hypotheses.

Definition I.13 (Fixed-rank band hypothesis)**.** A moving retained band satisfies the *fixed-rank band hypothesis* if there is a fixed K and a uniform band-tail separation

$$\inf_z \big(a_{K+1}(z) - a_K(z)\big) > 0.$$

This ensures that the K-dimensional retained subspace can be tracked as a subspace across coarse states.

Definition I.14 (Non-crossing coordinate hypothesis)**.** A moving retained band satisfies the *non-crossing coordinate hypothesis* if, in addition, the individual retained modes can be labeled consistently across z without eigenvalue crossings that exchange their identities.

The subspace and coordinate versions should be distinguished. If eigenvalues cross inside the retained band while the band remains separated from the tail, the retained subspace may still be trackable, but componentwise coordinates are not canonical. If the crossing changes which modes belong to the retained band, or if the band-tail gap closes, even the finite-band object can change rank or lose stability. An additional *cycle-scale visibility condition* is also needed: the retained modes must actually be visible on the outer cycle, e.g. $\sup_z (a_K(z) - a_1(z))\,\Delta = O(1)$, or an analogous modewise version. Without such a condition, an isolated but exponentially decayed band would qualify as "retained" under the fixed-rank and non-crossing hypotheses yet carry no interface memory — the near-principal distinction of

§I.2 would be lost.

Remark I.15 (Relation to Appendix G). Appendix G's componentwise compression theorem (Theorem G.12) lives in the rank-one version of these hypotheses. Each block has a clean principal mode, a positive principal gap, and slow coarse motion relative to that gap. In the language of the present appendix, Appendix G assumes that the retained band has $K_i = 1$ for every block, and that no dynamic trajectory enters a state where the rank-one band ceases to be separated. This appendix does not invalidate Appendix G; it identifies the boundary of the regime in which Appendix G is type-correct.

The dynamic vector problem is therefore deferred as its own arc. It should not be hidden inside a static near-degeneracy appendix: it asks for matrix Feynman-Kac machinery, adiabatic projection control, and hypotheses ruling out the bad level-crossing regimes. The matrix-valued Feynman-Kac and mode-tracking problem deferred here is one of the typology document's named technical arcs [Kim26b]; the fixed-rank and non-crossing hypotheses introduced above are the shared admissibility conditions for that arc.

I.8. Implications for falsification and calibration

This appendix upgrades the program's falsification discipline. Earlier scalar-compression failures could be summarized by saying that if $q_*/g_* \sim 1$ or $g_*\Delta$ is not large, the synthesis does not apply. The present results make the failure mode more operational.

If real systems systematically sit in the regime

$$g\Delta \sim 1,$$

then the scalar synthesis does not simply become noisy; it degrades into vector interface structure. The outer geometric evaluator is no longer canonically applicable until a scalar readout or a vector-valued aggregation theory is supplied. Empirically, this predicts residual dependence on low-dimensional internal modes at the interface: two states with the same principal scalar can have different continuation behavior because their near-principal modal coordinates differ.

A system near the AM-GM boundary can fail the scalar theorem in at least two ways: it may fail to compress at all, retaining high-dimensional microstate dependence; or it may compress only to a finite-dimensional near-principal band, producing vector rather than scalar interface structure. The second failure is graceful degradation — not chaos, not absence of structure, but the boundary of scalarity. The scalar interface value is not a generic property of killed-process blocks: it is the rank-one limit of their survival semigroups on the outer cycle timescale.

Appendix J. Representational Coda: Additive Primitives and Exponentiated Geometric Aggregation

J.1. Opening

The monograph's main arc identifies *temporal interface sufficiency* — axiom (A5) of Theorem II.1 — as the genuinely additional principle needed to force geometric aggregation across non-transferable branches. The static killed-process realization of Theorem D.4 and the dynamic Feynman-Kac realization of Theorem F.7 show how that axiom gets operationally supplied by spectral compression. This appendix records a separate, representational observation: in a broad class of path-based models, (A5) is no longer an independent surprise once one works in the right coordinate. If the primitive branch quantity is additive under temporal concatenation, and cross-branch aggregation is linear in that additive coordinate, then exponentiating back to the positive viability domain yields weighted geometric aggregation *automatically*, with temporal compositionality inherited from the additive representation rather than imposed as a separate axiom.

The content is explanatory rather than derivational. It does not replace the outer characterization of Theorem A.8; it identifies a modeling class in which that characterization becomes structurally natural, and clarifies why Peters-style multiplicative-growth formalisms and Friston-style surprisal/free-energy formalisms sit naturally on the geometric side of the AM-GM boundary without needing to import (A5) as an extra postulate.

J.2. Additive path primitives

The elementary representation-theoretic object:

Definition J.1 (Additive path primitive)**.** For each branch i, let $L_i(I) \in \mathbb{R}$ be a quantity assigned to a time interval $I = [t_0, t_1]$ such that whenever $I = I_1 \cup I_2$ with $I_1 = [t_0, t_*]$, $I_2 = [t_*, t_1]$,

$$L_i(I) = L_i(I_1) + L_i(I_2).$$

Call L_i an *additive path primitive*. Examples include cumulative hazard, cumulative log-viability, surprisal-like path cost, and negative log path weight.

Define the associated positive-domain quantity

$$v_i(I) := e^{-L_i(I)}. \tag{4}$$

Proposition J.2 (Exponentiated additivity is multiplicative temporal composition)**.** *Let L_i satisfy Theorem J.1, and define v_i by (4). For every concatenation $I = I_1 \cup I_2$,*

$$v_i(I) = v_i(I_1)\, v_i(I_2).$$

Proof. $v_i(I) = e^{-L_i(I)} = e^{-L_i(I_1)-L_i(I_2)} = e^{-L_i(I_1)}\, e^{-L_i(I_2)} = v_i(I_1)\, v_i(I_2)$. □

The proposition is algebraically trivial but conceptually useful: multiplicative branchwise temporal composition can arise not because multiplication is primitive but because it is the positive-domain image of additivity.

J.3. Linear pooling in the additive coordinate

Temporal additivity of branchwise quantities does not by itself determine cross-branch aggregation. *Linear pooling* is the simplest aggregation rule in the additive coordinate — the additive-coordinate analogue of arithmetic averaging — and is introduced here as a separate structural assumption. A fuller treatment would seek an additive-space characterization theorem establishing linear pooling uniquely under the right analogues of (A3)–(A5); that is not attempted here.

Definition J.3 (Linear pooling in the additive coordinate)**.** Fix weights $w = (w_1, \ldots, w_n)$ with $w_i > 0$ and $\sum_i w_i = 1$. For additive branch quantities $L = (L_1, \ldots, L_n)$, define

$$J_w(L_1, \ldots, L_n) := \sum_{i=1}^{n} w_i L_i. \tag{5}$$

To return to the positive domain, define

$$A_w(v_1, \ldots, v_n) := \exp\big(-J_w(-\ln v_1, \ldots, -\ln v_n)\big), \qquad v_i > 0. \tag{6}$$

Proposition J.4 (Exponentiated linear pooling is weighted geometric)**.** *Under Theorem J.3,*

$$A_w(v_1, \ldots, v_n) = \prod_{i=1}^{n} v_i^{w_i}.$$

Proof. Substituting (5) into (6),

$$A_w(v_1, \ldots, v_n) = \exp\left(-\sum_i w_i(-\ln v_i)\right) = \exp\left(\sum_i w_i \ln v_i\right) = \prod_i v_i^{w_i}.$$

□

The weighted geometric mean is exactly the positive-domain image of the arithmetic-mean pooling rule under the exponential map.

J.4. Temporal compositionality in the transformed representation

The force of Theorem A.8's axiom (A5) is that once two temporal subintervals have each been compressed to scalar summaries, later whole-horizon evaluation must factor through those summaries alone. In the additive-primitive setting, this is not an independent surprise — it falls out of the representation together with linear pooling.

Proposition J.5 (Temporal compositionality in the transformed representation)**.** *Let $v^{(1)}, v^{(2)} \in (0, \infty)^n$ be positive branch profiles associated with two adjacent temporal subintervals, and let whole-interval branch values be given coordinatewise by*

$$v^{(12)} = v^{(1)} \odot v^{(2)}, \qquad v_i^{(12)} = v_i^{(1)} v_i^{(2)}.$$

If A_w is defined by (6), then

$$A_w\big(v^{(12)}\big) = A_w\big(v^{(1)}\big)\, A_w\big(v^{(2)}\big).$$

Proof. Using Theorem J.4,

$$A_w\big(v^{(12)}\big) = \prod_i \big(v_i^{(12)}\big)^{w_i} = \prod_i \big(v_i^{(1)} v_i^{(2)}\big)^{w_i} = \prod_i \big(v_i^{(1)}\big)^{w_i} \prod_i \big(v_i^{(2)}\big)^{w_i} = A_w\big(v^{(1)}\big)\, A_w\big(v^{(2)}\big).$$

□

Remark J.6 (What this explains and what it does not replace). Theorem J.5 does not replace the outer characterization of Theorem A.8. What it does is narrower: once branches are modeled by additive path primitives and cross-branch aggregation is taken linear in the additive coordinate, multiplicative temporal composition of the positive-domain evaluator is automatic. One does not begin with positive viability values and then ask why the evaluator should multiply across temporal segments; one begins with an additive path quantity, and multiplication in the positive domain is simply the induced image of additivity plus additive-coordinate linear pooling.

J.5. Connection to the monograph's body

The representational framework of §J.2–J.4 specializes to the monograph's killed-process realizations under the following identifications. In Appendix D's static compressed setting, each block's scalar interface value $c_i = e^{-a_1^{(i)}\Delta}$ plays the role of v_i, with additive primitive $L_i = a_1^{(i)}\Delta$; log-aggregation $\log A_w(c) = -\Delta \sum_i w_i a_1^{(i)}$ (Theorem D.8) is the instance of Theorem J.4 specialized to frozen-landscape principal rates. In Appendix F's dynamic setting, the Feynman-Kac interface value $\mathcal{A}_i(\Delta, z_i)$ plays the role of v_i, with additive primitive $L_i = -\log \mathcal{A}_i(\Delta, z_i)$; this is *not* the expected integrated rate $\mathbb{E}_z \int_0^\Delta a_1^{(i)}(Z_s)\, ds$ in general — the two differ by the temporal Jensen gap of Theorem F.10, closing only in the quasi-static and fast-z boundary limits.

The heterogeneous n-block synthesis of Theorem G.12 adds one further wrinkle recorded in Theorem G.15: evaluator-level cluster values $B_C = \prod_{i \in C} v_i^{w_i/W_C}$ and physical product-space Feynman-Kac values F_C are different objects in the coordinate where temporal concatenation is multiplicative; the log of the evaluator-level value inherits the exponentiated-linear-pooling structure of the present coda, while the physical product-space value belongs to a distinct layer of Theorem G.1 with its own Jensen gap. The coda therefore lives cleanly at layers (i)–(ii) — the scalar evaluator and compressed-class layers — without reaching layer (iii).

J.6. Peters-style log structure and the FEP bridge

The Peters algebra of multiplicative growth is the most immediate instance. For a multiplicative process with per-period positive returns $r_i(t) > 0$, take the additive cost primitive

$$L_i = -\sum_t \log r_i(t),$$

which is additive under temporal concatenation by construction. Then $v_i = e^{-L_i} = \prod_t r_i(t)$ is the cumulative multiplicative outcome, consistent with the coda's e^{-L} convention, and exponentiated linear pooling of the L_i gives the weighted geometric mean of per-branch multiplicative outcomes — the coarse content of the Kelly/Peters result [Kel56; PG16]. The present coda extracts this as representation rather than as a derivation of a specific semantic claim: the log-domain coordinate is naturally additive when the primitive is a multiplicative process; geometric aggregation is what the exponential map produces on the way back.

The Free Energy Principle invites an analogous reading [PPF22]. Surprisal, negative log-path-weight, and free-energy-like functionals are additive over time by construction: $L_i(I) = \int_I \ell_i(t)\, dt$ for some instantaneous rate. In this representation, an FEP-style formalism does not encounter (A5) in its survival-probability form, because temporal composition has already been linearized away in the additive path-cost coordinate; the multiplicative picture appears only after exponentiation to a path weight or typicality factor. The bridge is structural rather than semantic — an additive surprisal or path-cost functional is not automatically a survival probability or viability value, and connecting the FEP fully to the AM-GM boundary program requires a principled persistence interpretation (killed-process, viability-set exit, or quasi-stationary construction) that the coda does not supply.

J.7. Scope limits

The coda is explanatory, not totalizing. Explicitly:

- It does *not* derive the inner arithmetic Theorem II.2. The question of when a block supports a scalar fungible resource with sufficiently fast internal transfer is separate.
- It does *not* solve the timing-sensitive interface problem of §III.2. Even with an additive primitive, one must still ask when within-block uncertainty has been compressed to an interface value and what residual uncertainty persists after the interface.
- It does *not* show that absorbing states alone force geometric aggregation. The caution of Appendix A's Theorem A.7 still stands: one needs either the (A1)–(A5) axiom set or a model class, such as the present one, in which those axioms become structurally natural.
- It does *not* show that temporal additivity alone forces linear pooling across branches. Linear pooling in the additive coordinate is a further structural assumption, albeit a natural one.

- It does *not* show that every survival-based viability functional admits an additive primitive of the present kind. The coda identifies a broad modeling class, not a universal representation theorem.

The role of the coda is therefore narrow but useful: geometric aggregation appears in the program as the unique outer evaluator forced by non-transferability plus scalar compression (Theorem A.8), and also as the positive-domain image of additive-coordinate linear pooling in path-additive models. Temporal compositionality is sometimes not a mysterious axiom waiting to be justified, but evidence that the more primitive branch quantity may live in an additive coordinate.

Appendix K. Canonical Assumptions Glossary

K.1. Purpose and reference convention

This appendix fixes the vocabulary used in the body and theorem appendices. It is not a theorem catalog. Each entry records the intended meaning of a load-bearing term, the operational condition or usage discipline attached to it, and the place where the term is first used rigorously in the monograph or supporting appendices.

References to appendices use their current labels. Near-degenerate / vector-interface content is proved in Appendix I.

K.2. Glossary

Absorbing failure / absorbing viability.

A viability model is absorbing when failure, death, or dissolution is represented by an absorbing state: after failure occurs, the branch remains failed at all later horizons. Branch viability is then survival probability to the relevant horizon.

Usage. Absorbing viability gives branchwise temporal multiplication by the chain rule for nested survival events. It does not, by itself, force geometric aggregation across branches; the outer evaluator still needs temporal interface sufficiency.

Cross-reference. Part II.1; Appendix A.2–A.5.

Additive primitive / log-rate coordinate.

An additive primitive is a path quantity — such as cumulative hazard or cumulative log-viability — that adds under temporal concatenation. Exponentiating an additive primitive returns a positive multiplicative viability quantity.

Usage. In the scalar compressed regime, outer geometric aggregation is arithmetic after taking logarithms: $\log A_w(c_1, \ldots, c_n) = \sum_i w_i \log c_i$, and the additive primitive is the path-integrated hazard $\int_0^\Delta a_1\, ds$. In the dynamic slow-fast regime, the additive object is the path-integrated hazard *inside* the Feynman-Kac functional; the scalar log-Feynman-Kac value $\log \mathcal{A}_i(\Delta, z_i)$ is the induced log-coordinate *after* expectation, and is not in general equal to the mean integrated rate, nor scalar-additive across temporal segments, except in the quasi-static and fast-z limits where the Jensen gap closes.

Cross-reference. Part II.5; Part V.4; Appendix A.9–A.10; Appendix F.7; Appendix J representational coda.

Block.

A block is a candidate coherent unit whose internal dynamics are to be aggregated arithmetically before the interface and whose external relations to other blocks are to be aggregated geometrically after the interface.

Usage. The word is deliberately conditional: calling something a block does not assert that it has already produced a valid scalar interface value. Parts III–V ask when that candidate block actually compresses to one.

Contrast. A Markov blanket or physical membrane may suggest a candidate block, but blanket structure alone is not the AM-GM block condition; resource transferability, scalarity, and timing also matter.

Cross-reference. Part II.1; Part III.1; Part V.1.

Branch.

A branch is an externally non-transferable survival or viability path to which the outer evaluator is applied after the relevant interface values have formed.

Usage. In Appendix A, branches are primitive inputs to the outer characterization theorem. In the synthesis, compressed blocks supply branch-like positive inputs to the same outer evaluator.

Contrast. Garrabrant-style branches are usually specified decision-theoretically. In this monograph, the central unresolved question is when a dynamical subsystem has compressed enough to function as such a branch.

Cross-reference. Part II.1–II.2; Appendix A.2.

Branching Bellman operator vs. killed continuation operator.

Two inequivalent finite-ε packagings of the same within-fiber averaging content. The *branching Bellman operator* $\bar{T}_R$ of Appendix E has the form $(\bar{T}f)(z) = \pi_{\text{haz}}(z)\bar{\Phi}(z) + (1 - \pi_{\text{haz}}(z)) \int f(z')\bar{L}_z(dz')$: at each rare event, a geometric recursion either terminates (hazard) or continues (coarse transition). The *killed continuation operator* K_ε of §F.10 has the form $(K_\varepsilon f)(z) = s_\varepsilon(z) \sum_{z'} P_\varepsilon(z, z')f(z')$: a survival factor is applied and a coarse transition is taken at each small step.

Usage. Both reduce to the same generator $Q - \text{diag}(a_1)$ in the continuum limit but answer different questions. Scaling $\bar{T}_R$ as a fixed point yields a resolvent-type equation $(a_1 - Q)c = a_1\bar{\Phi}$ — the continuum limit of the infinite-horizon eventual-hazard recursion. Iterating K_ε a finite number $\lfloor \Delta/\varepsilon \rfloor$ of times yields the finite-horizon Feynman-Kac semigroup $e^{\Delta(Q-\text{diag}(a_1))}$. Appendix E uses the branching form; Appendix F uses the scaling limit of the killed-continuation form.

Cross-reference. Appendix E.4–E.5; Appendix F.10.

Cluster.

A cluster is a regrouping of several blocks inside an n-block outer evaluation problem. It is an evaluator-level object unless explicitly realized as a physical product process.

Usage. For a cluster C_α, total weight is $W_\alpha = \sum_{i \in C_\alpha} w_i$, and the evaluator-level cluster value is $B_\alpha = \prod_{i \in C_\alpha} c_i^{w_i/W_\alpha}$.

Cross-reference. Appendix G.

Coarse potential.

A coarse potential is the macroscopic landscape on the fixed-total fiber that determines the Gibbs equilibrium measure for the fast redistribution dynamics in the reversible killed-process setting.

Usage. In the current concrete realization, the coarse potential is an assumed input, typically a one-well function h with Gibbs law $\pi_R(k) \propto e^{-\beta R h(k/R)}$. The program does not yet derive which coarse potentials real systems induce.

Cross-reference. Part IV.3; Appendix C.2; Part VI.2–VI.4.

Componentwise versus monolithic compression.

Componentwise compression requires each block i to satisfy its own spectral gap and slow-motion conditions, such as $g_i > 0$ and $q_i/g_i \ll 1$. Monolithic compression treats a product cluster as one coarse process and imposes a single condition on the whole product state.

Usage. The dynamic regrouping theorem of Appendix G uses componentwise compression. A monolithic condition such as $\sum_i q_i \ll \min_i g_i$ is stricter than needed for evaluator-level regrouping and should not be silently substituted.

Cross-reference. Appendix G.

Compression timescale.

The compression timescale is the time required for the killed semigroup to become dominated by the mode or retained band used as the interface object.

Usage. In the scalar regime the relevant scale is $\tau_{\text{compress}} \sim 1/(a_2 - a_1)$. In the finite-band regime it is governed by the gap after the retained band, $1/(a_{K+1} - a_1)$.

Cross-reference. Part V.1–V.2; Appendix D.3; Appendix I.

Compression window.

A compression window is a two-sided interval of admissible outer cycle lengths: long enough for spectral compression, short enough that the block has not already decayed on the metastable killing scale.

Usage. In the scalar Gibbs-lifted class the intended window is

$$\frac{1}{a_2 - a_1} \ll \Delta \ll \frac{1}{a_1}.$$

The diffusive baseline fails because mixing, compression, and killing remain on comparable scales.

Cross-reference. Part V.2 closing interpretation; Appendix B.5–B.7; Appendix D.3.

Conjunction survival.

Conjunction survival is the literal probability that all constituents in a joint system survive, typically represented by an unnormalized product such as $\prod_i c_i$ under independence.

Usage. Conjunction survival is not the normalized outer evaluator of Theorem II.1. The normalized evaluator is idempotent and weight-dependent: $A_w(c) = \prod_i c_i^{w_i}$.

Cross-reference. Part II.2; Appendix D.2, Remark D.1; Appendix G.

Evaluator super-block versus physical product super-block.

An evaluator super-block is a cluster value obtained by applying the normalized outer evaluator to already-compressed constituent interface values. A physical product super-block is a literal product killed or Feynman-Kac process on a product state space.

Usage. The two objects answer different questions. In the static independent case the physical product principal rate is $\sum_i a_1^{(i)}$ and the first product gap is $\min_i g_i$, while the evaluator cluster value uses normalized weights.

Cross-reference. Appendix G.

Exogenous schedule / evaluator weights.

The exogenous schedule is the vector of positive weights $w = (w_1, \ldots, w_n)$ with $\sum_i w_i = 1$ supplied to the outer evaluator. These weights may be interpreted as exposure fractions, prevalence weights, or scheduling weights, but are not derived from block thermodynamics in the current theory.

Usage. All current synthesis theorems treat w as an input. State-dependent schedules $w(z)$ and endogenous weights are open extensions, not hidden assumptions of the present results.

Cross-reference. Part V.2, Definition V.2; Appendix D.2; Appendix G.

Feynman-Kac interface value.

In the slow-fast dynamic synthesis, the block interface value is the Feynman-Kac expectation

$$\mathcal{A}_i(\Delta, z) = \mathbb{E}_z \exp\left(-\int_0^\Delta a_1^{(i)}(Z_s^{(i)})\, ds\right) = \left[e^{\Delta(Q^{(i)} - \operatorname{diag}(a_1^{(i)}))}\mathbf{1}\right]_z.$$

Usage. This is the continuous-time form of the averaged coarse-process value from the abstract slow-fast bridge. Its logarithm, not generally the negative mean integrated rate, is the additive coordinate used in the dynamic synthesis.

Cross-reference. Part V.3; Appendix E.7; Appendix F, Definition F.2; §F.10 for the scaling limit from Appendix E.

Fixed-total fiber.

A fixed-total fiber is the state space of internal allocations with a conserved arithmetic total. In the two-site killed-process class, it is $\Omega_R = \{0, 1, \ldots, R\}$; in higher-dimensional variants it is the simplex of nonnegative allocations satisfying $\sum_i u_i = R$.

Usage. The fiber is where fast internal redistribution occurs before the interface. Current concrete theorems use the reversible killed-process setting on fixed-total fibers; higher-dimensional fiber geometry is an open extension, not assumed by the existing one-dimensional calculations.

Cross-reference. Part IV.1–IV.3; Appendix B.2; Appendix C.2; Appendix E.2.

Hazard timescale, compression timescale, outer cycle timescale.

The hazard timescale is the scale on which viability-relevant failure or killing occurs; the compression timescale is the scale on which internal dynamics produce the interface object; the outer cycle timescale Δ is the scale at which the outer evaluator is applied.

Usage. The scalar synthesis requires the ordering $\tau_{\text{compress}} \ll \Delta \ll \tau_{\text{kill}}$, with additional slow-coarse-motion ordering in the dynamic setting. A claim about an interface value is always relative to these timescales.

Cross-reference. Part III.4; Part V.1–V.3; Appendix D.3; Appendix E.6.

Interface value.

An interface value is the positive scalar by which a block is represented once internal compensation has finished and external non-transferability begins.

Usage. For the outer theorem to apply, the value must be scalar, ratio-scale, and dynamically produced by compression rather than assigned by external labeling. In the static Gibbs-lifted realization it is approximately $e^{-a_1\Delta}$; in the dynamic realization it is the Feynman-Kac value $\mathcal{A}(\Delta, z)$.

Cross-reference. Part I.3; Part III.1; Part V.2–V.3.

Interface value versus conjunction survival.

The interface value is an admissible scalar input to a normalized outer evaluator. Conjunction survival is the literal probability that several subsystems all survive.

Usage. The weighted geometric mean belongs to the former problem. Confusing it with literal conjunction survival erases the distinction between normalized evaluation and raw joint survival.

Cross-reference. Part II.2; Appendix D.1–D.2; Appendix G.

Jensen gap.

A Jensen gap is the discrepancy created when logarithm, expectation, and exponential are interchanged in nonlinear survival expressions.

Usage. Temporally, Appendix F distinguishes $\log \mathbb{E}[e^{-X}]$ from $-\mathbb{E}X$ for integrated rate X. Spatially, Appendix G distinguishes evaluator cluster values $\prod_i (\mathbb{E}e^{-X_i})^{\alpha_i}$ from physical product-space values $\mathbb{E}e^{-\sum_i \alpha_i X_i}$. The gap vanishes in effectively deterministic boundary regimes and is nontrivial in intermediate slow-fast regimes.

Cross-reference. Part V.3–V.4; Appendix F; Appendix G.

Metastability.

Metastability is the separation between rapid equilibration within a well and rare escape to a killing or transition region. In the Gibbs-lifted killed-process setting, it appears as exponentially suppressed hazard-reaching under a coarse potential.

Usage. A metastable well supplies a long killing scale while preserving a shorter compression scale. The term is used here in the reversible killed-process class unless explicitly extended; multi-well metastability is a further structural extension.

Cross-reference. Part IV.3; Part V.2; Appendix C; Appendix D.3.

Non-transferability.

Non-transferability is the absence of admissible controls that move viability-relevant resource across the relevant boundary after the interface.

Usage. Outside the boundary, one branch's gain cannot compensate another branch's loss before failure. Non-transferability is necessary for the outer geometric story but not sufficient by itself; scalar interface values and temporal sufficiency are also required.

Contrast. Peters-style non-ergodicity concerns trajectory/ensemble mismatch in multiplicative growth. AM-GM non-transferability concerns cross-branch inability to compensate after the interface. The two are aligned in log structure but are not identical.

Cross-reference. Part I.2; Part II.1–II.2; Appendix A.

Quasi-stationary distribution (QSD).

A quasi-stationary distribution is the conditional distribution of a killed process given survival up to a long time, when this conditional law has stabilized before absorption. The survey [MV12] gives a systematic treatment for population processes with absorption and its spectral-theoretic background.

Usage. In the reversible finite killed-process class, the QSD is tied to the principal killed eigenmode and carries the principal-mode-dominated behavior used to justify scalar interface values. In near-degenerate or multi-well settings, basin-conditioned QSDs and multiple visible modes may replace a single global scalar summary.

Cross-reference. Part IV.2–IV.3; Appendix B.6; Appendix C.3; Appendix D.3; Appendix I.

Rank-one compression.

Rank-one compression is the regime in which the killed semigroup has collapsed onto its principal eigenprojection on the interface timescale.

Usage. There are two compatible limits: at fixed spectrum, $\Delta \to \infty$ kills higher modes; at fixed Δ, opening the gap $g = a_2 - a_1 \to \infty$ does the same. The operational parameter is $g\Delta$.

Cross-reference. Part V.1–V.2; Appendix D.3; Appendix I.

Ratio-scale versus affine representation.

A ratio-scale representation has a meaningful zero and supports multiplicative comparison. An affine representation is defined only up to positive slope and additive intercept, so ratios are not invariant.

Usage. The weighted geometric mean is meaningful on ratio-scale positive interface values. A merely affine internal representation cannot be fed into the outer geometric evaluator without an additional zero-resource normalization or readout choice.

Cross-reference. Part III.1–III.3.

Readout.

A readout is a positive functional or projection that maps a vector interface object to a scalar suitable for evaluation.

Usage. Readouts are not part of the scalar theorem. They become necessary in the finite-mode/vector regime, where no canonical weighted geometric evaluator is available until extra regrouping-compatible structure selects a scalar output.

Cross-reference. Appendix I.

Retained band / finite-mode interface profile.

A retained band is the collection of spectral modes whose gaps from the principal mode are not large on the cycle scale. The finite-mode interface profile is the survival profile projected onto that retained band.

Usage. For cycle length Δ, modes $1, \ldots, K$ are retained when $(a_r - a_1)\Delta = O(1)$ for $r \le K$ while $(a_{K+1} - a_1)\Delta \gg 1$. The scalar interface value is the special case $K = 1$.

Cross-reference. Appendix I.

Scalarity.

Scalarity is the property that a block's future continuation, for the purposes of outer evaluation, factors through one positive number at the interface.

Usage. Scalarity is a compression property; writing down a scalar in a model is not itself sufficient. It holds only when unresolved microstate or modal information is irrelevant on the outer cycle timescale.

Cross-reference. Part III.1; Part V.1–V.2; Appendix D.3–D.6.

Scalarity defect.

The scalarity defect measures the error incurred by replacing a finite-mode interface profile with the principal scalar alone.

Usage. In the near-degenerate setting it is controlled by visible nonprincipal modal weights, schematically $\sum_{r=2}^{K} \rho_r e^{-(a_r - a_1)\Delta}$. A large defect means the interface still carries low-mode memory.

Cross-reference. Appendix I.

Slow coarse motion.

Slow coarse motion is the dynamic condition that the coarse state changes slowly compared with fiber compression.

Usage. In the current slow-fast synthesis this is expressed as $q_*/g_* \ll 1$, where $g_* = \min_z(a_2(z) - a_1(z))$ is the uniform spectral gap and q_* is the maximal coarse jump rate. The condition is componentwise in the n-block extension.

Cross-reference. Part V.3; Appendix F; Appendix G.

Spectral compression.

Spectral compression is the dominance of the principal killed mode on the outer cycle timescale.

Usage. In the scalar static regime the operational condition is $(a_2 - a_1)\Delta \gg 1$. In the dynamic regime it is strengthened to a uniform gap and slow coarse motion. In the near-degenerate regime $(a_2 - a_1)\Delta = O(1)$, scalar compression fails and finite-mode structure remains.

Cross-reference. Part V.1–V.3; Appendix D.3; Appendix F; Appendix I.

Temporal interface sufficiency.

Temporal interface sufficiency is the requirement that once temporal subintervals have each been compressed to scalar interface values, later whole-horizon evaluation depends only on those scalar summaries.

Usage. In Appendix A it is Axiom A.5, stated in literal scalar form. In the static killed-process synthesis it is realized concretely by spectral compression: scalar continuation factors $c_i(\Delta')$ compose multiplicatively on the compressed class. In the dynamic slow-fast setting, the segment summary is semigroup-valued before state evaluation — the admissible temporal summary is the Feynman-Kac continuation vector $e^{\Delta' M_i}\mathbf{1}$, and the scalar $\mathcal{A}_i(\Delta, z)$ is obtained only after conditioning on the current coarse state z.

Cross-reference. Part II.2; Appendix A.3–A.4; Appendix D.6; Appendix F.6.

Total-sufficiency.

Total-sufficiency is the property that one-step block viability depends on the component allocation $u = (u_1, \ldots, u_d)$ only through the arithmetic total $R = \sum_i u_i$.

Usage. Under complete internal transfer before the next hazard, deterministic allocations collapse to the total, and the block behaves as if resources were pooled into a single fungible scalar. Expected-total coherence extends the ranking to lotteries when the pooled value map $R \mapsto V(R)$ is affine on the attainable-total domain; nonlinear pooled value leaves Jensen effects. Incomplete internal transfer destroys total-sufficiency by leaving allocation-memory on the relevant timescale.

Cross-reference. Part II.3, Theorem II.2.

Transferability.

Transferability is the availability of admissible dynamics that move viability-relevant resource so that losses in one location can be compensated by gains elsewhere before the next viability-relevant event.

Usage. Inside a block, transferability is the source of arithmetic total-sufficiency; outside the block, non-transferability is the source condition for geometric evaluation. The AM-GM boundary is the dynamical interface between these two regimes.

Cross-reference. Part I.2; Part II.3; Part III.4; Appendix E.

Vector interface structure.

Vector interface structure is the finite-dimensional interface object that replaces a scalar when rank-one compression has not occurred but a low-dimensional retained band remains valid.

Usage. This object is inside the current reversible killed-process machinery only in the static near-degenerate treatment. The dynamic version requires matrix-valued Feynman-Kac or adiabatic mode-tracking machinery and a fixed-rank/non-crossing hypothesis; it is deferred.

Cross-reference. Appendix I.

Zero-resource normalization.

Zero-resource normalization is the condition that the pooled viability value vanishes when the designated scalar viability resource is zero.

Usage. When an affine pooled value $\Phi(z, R) = a(z)R + b(z)$ satisfies $\Phi(z, 0) = 0$, the intercept vanishes and the representation becomes proportional. This can be structurally natural in survival settings but is not automatic if state variables contain uncounted reserves or inherited viability.

Cross-reference. Part III.2.

Acknowledgments

This monograph presents a self-contained synthesis of a research program, conceived of by the author. The program was initially developed as a sequence of technical memos. The memos established the technical results in narrower scope; the present monograph lifts those results into a unified architecture and resolves several inter-memo consistency questions that surfaced during synthesis.

Both the memos and the monograph were developed with extensive use of large language models — primarily Anthropic's Claude (Opus 4.6 and 4.7) and OpenAI's GPT (5.4 and 5.5 Pro) — for proof drafting, exposition refinement, and review. The drafting workflow used multiple independent model instances at each revision stage: typically two to three instances per cold-reader round and per content-fidelity audit, with disagreements between instances triangulated against the memo corpus and the author's structural decisions. The monograph passed multiple rounds of independent cold-reader review and a content-fidelity audit verifying that the synthesis preserves what the memos established. Theorems carrying explicit "formal" status disclose their offstage technical dependencies in §I.5.

Mathematical conjectures, architectural decisions about scope and structure, and final responsibility for errors and overclaims rest with the author.

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